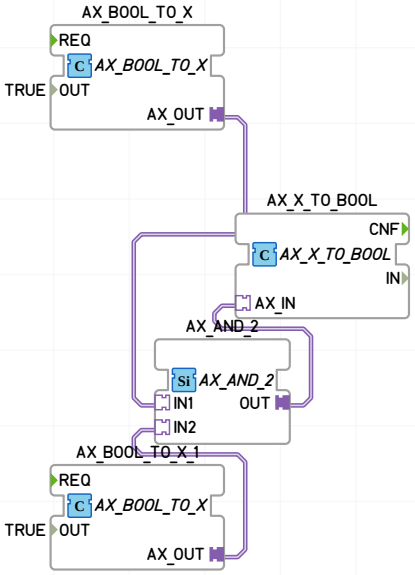


Uebung_000_AX

Einfache Addition



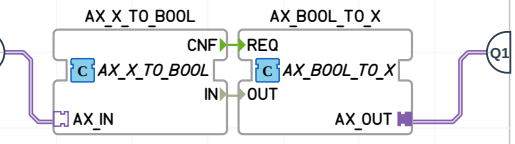
Uebung_000b_AX

AX_AND



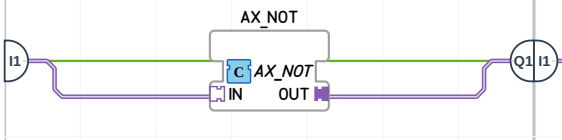
Uebung_001_AX

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



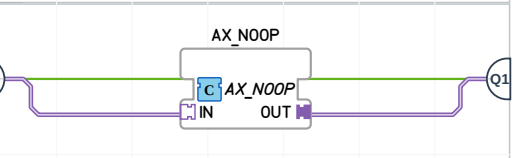
Uebung_001_AX_b

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



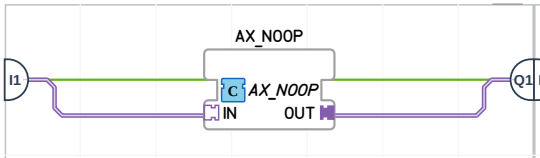
Uebung_001c2_AX

DigitalInput_I1 negiert auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



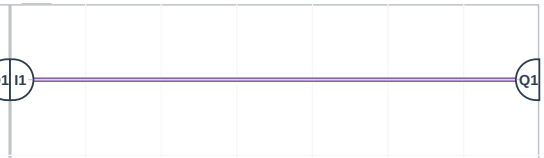
Uebung_001c3_AX

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



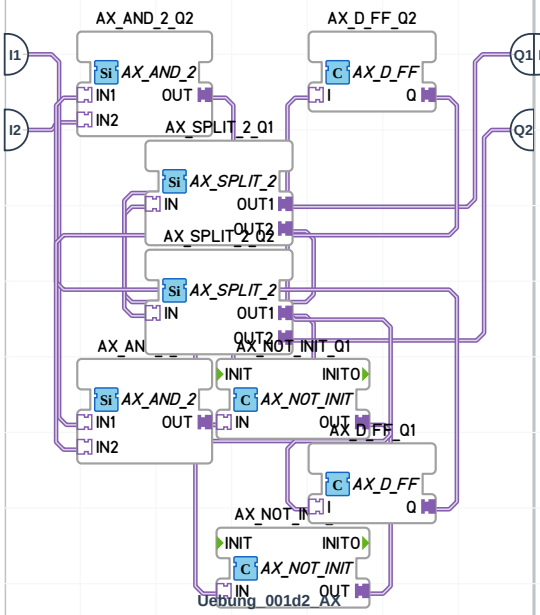
Uebung_001c4_AX

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



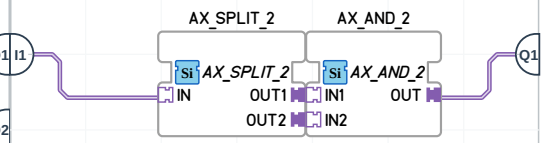
Uebung_001c_AX

DigitalInput_I1 auf DigitalOutput_Q1 --> Eingang abfragen bei Boot.



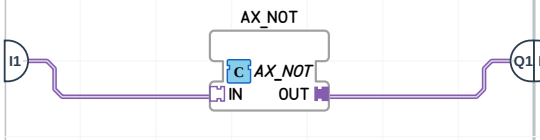
Uebung_001d2_AX

DigitalInput_I1/2 auf DigitalOutput_Q1/2, als Alternative(Verriegelt) mit Plug and Socket, ohne ECC



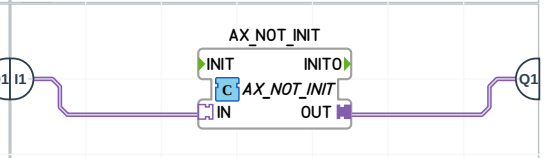
Uebung_001d_AX

DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket



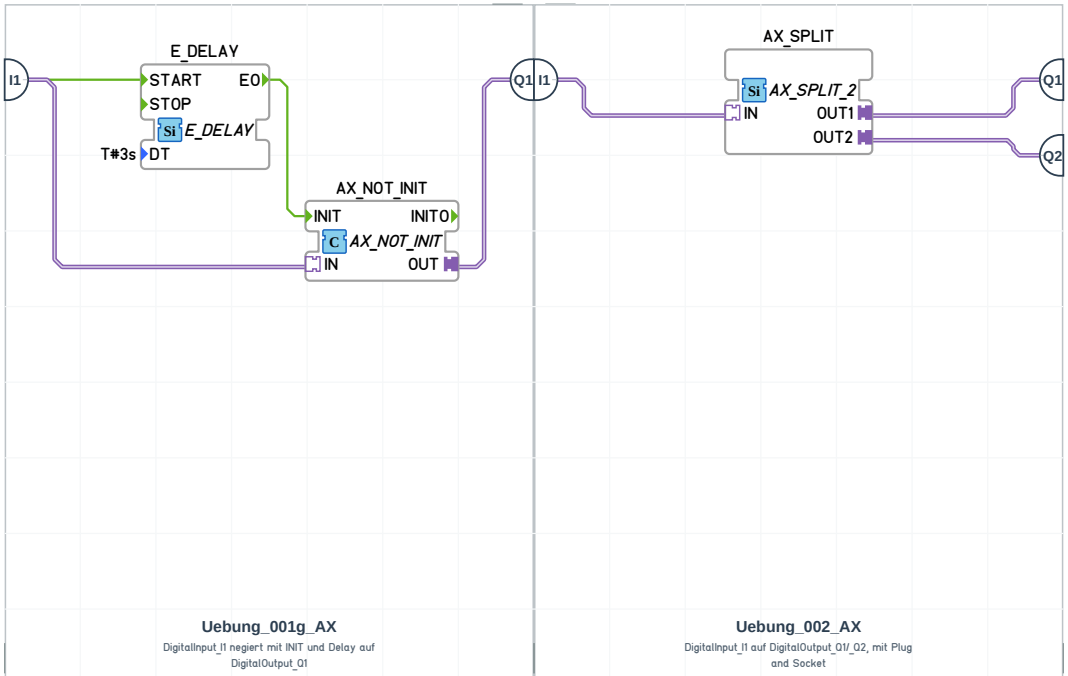
Uebung_001e_AX

DigitalInput_I1 negiert auf DigitalOutput_Q1



Uebung_001f_AX

DigitalInput_I1 negiert mit INIT auf DigitalOutput_Q1

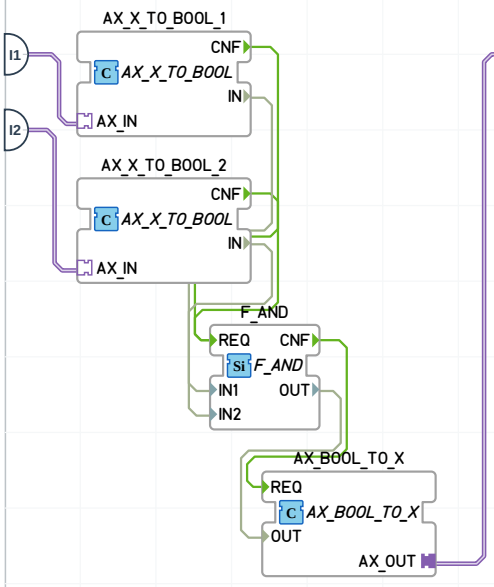


Uebung_001g_AX

DigitalInput_I1 negiert mit INIT und Delay auf DigitalOutput_Q1

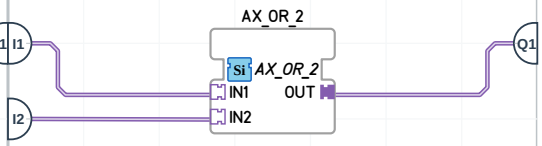
Uebung_002_AX

DigitalInput_I1 auf DigitalOutput_Q1/_Q2, mit Plug and Socket



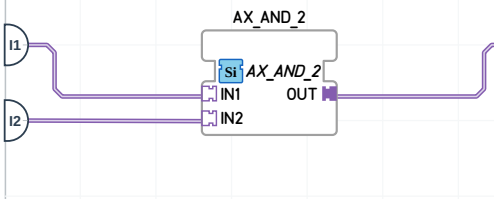
Uebung_002a2_AX

DigitalInput_I1/_I2 mit F_AND (bitwise) auf DigitalOutput_Q1



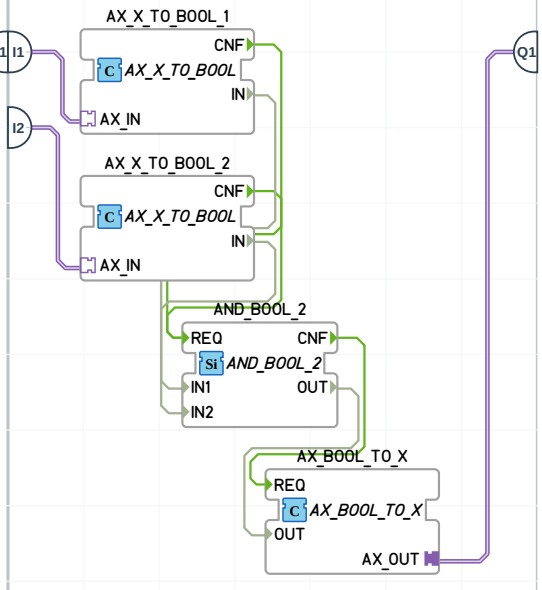
Uebung_002a3_AX

DigitalInput_I1/_I2 mit OR auf DigitalOutput_Q1, mit Plug and Socket



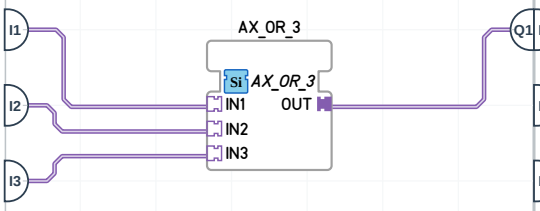
Uebung_002a4_AX

DigitalInput_I1/_I2 mit AX_AND_2 auf DigitalOutput_Q1



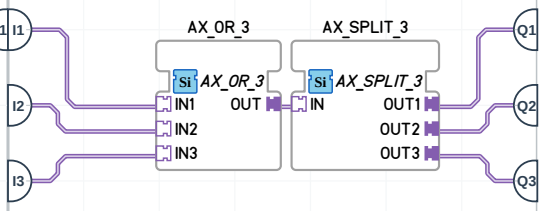
Uebung_002a4b_AX

DigitalInput_I1/_I2 mit AND_BOOL und Negate auf DigitalOutput_Q1



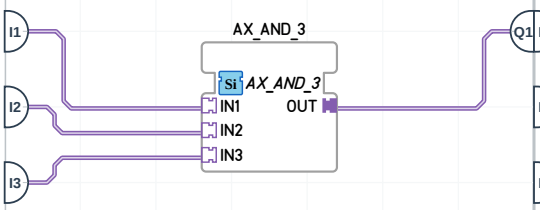
Uebung_002a5_AX

DigitalInput_I1-3 mit OR auf DigitalOutput_Q1, mit Plug and Socket



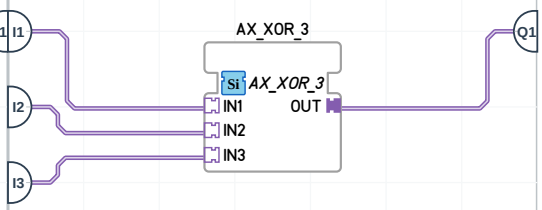
Uebung_002a5b_AX

DigitalInput_I1-3 mit OR auf DigitalOutput_Q1-3, mit Plug and Socket



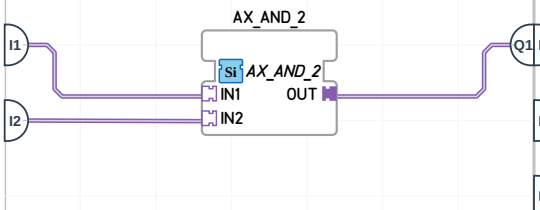
Uebung_002a6_AX

DigitalInput_I1-3 mit AND auf DigitalOutput_Q1, mit Plug and Socket



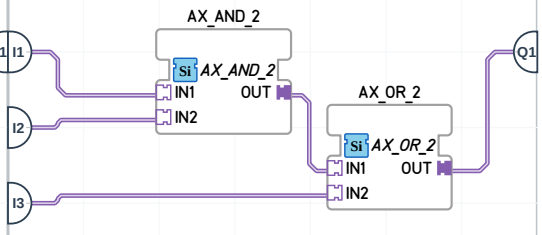
Uebung_002a7_AX

DigitalInput_I1-3 mit XOR auf DigitalOutput_Q1, mit Plug and Socket



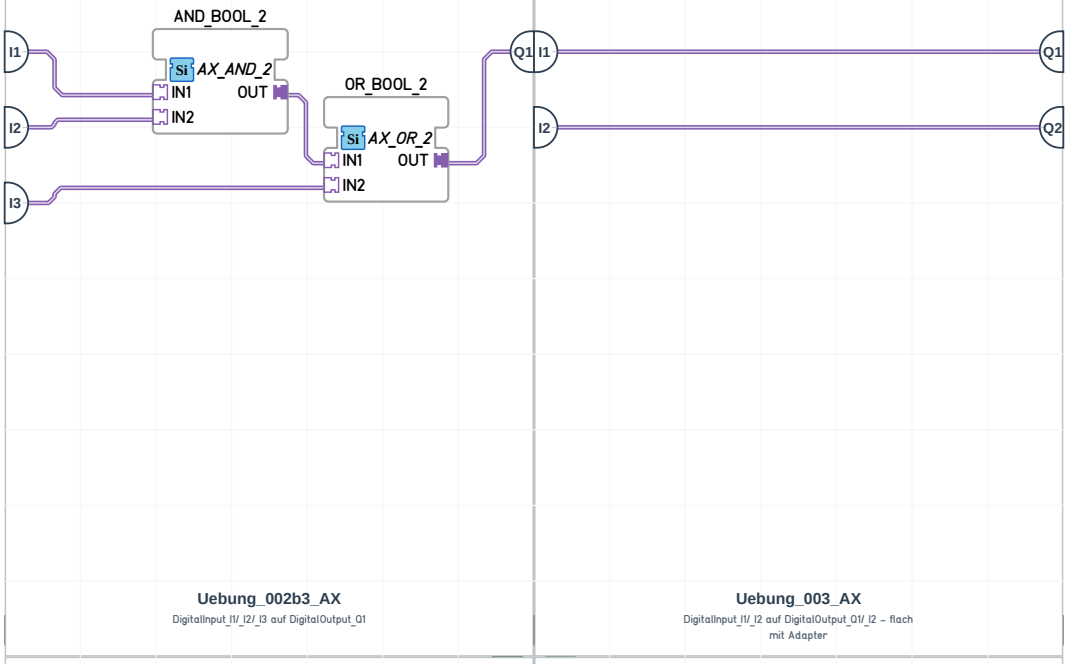
Uebung_002a_AX

DigitalInput_I1/_I2 mit AND auf DigitalOutput_Q1, mit Plug and Socket



Uebung_002b2_AX

DigitalInput_I1/_I2/_I3 auf DigitalOutput_Q1, mit AX Adapter



Uebung_002b3_AX

DigitalInput_I1/_I2/_I3 auf DigitalOutput_Q1

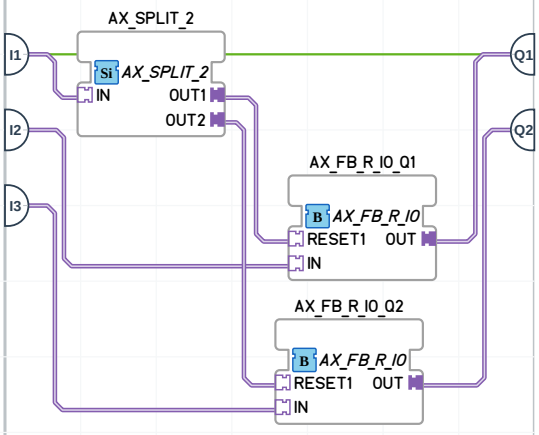
Uebung_003_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – flach mit Adapter



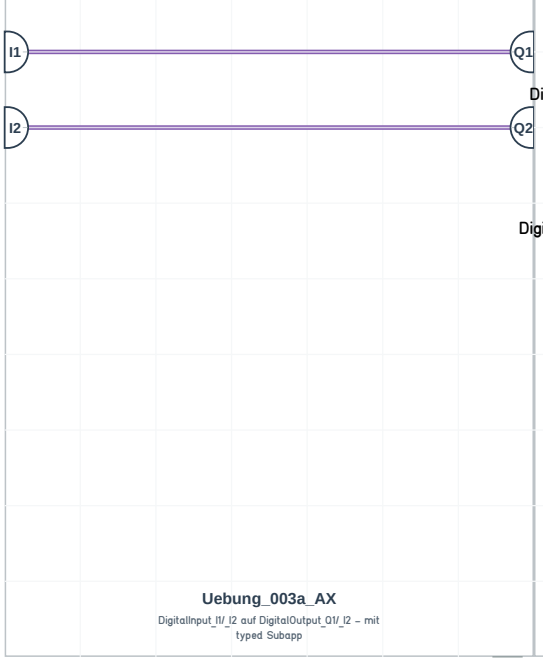
Uebung_003a0_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – mit untyped Subapp



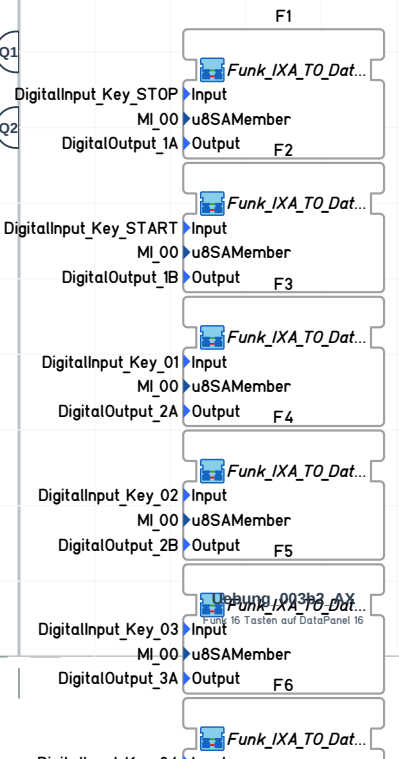
Uebung_003a2b_AX

2x R_I0 mit IXA



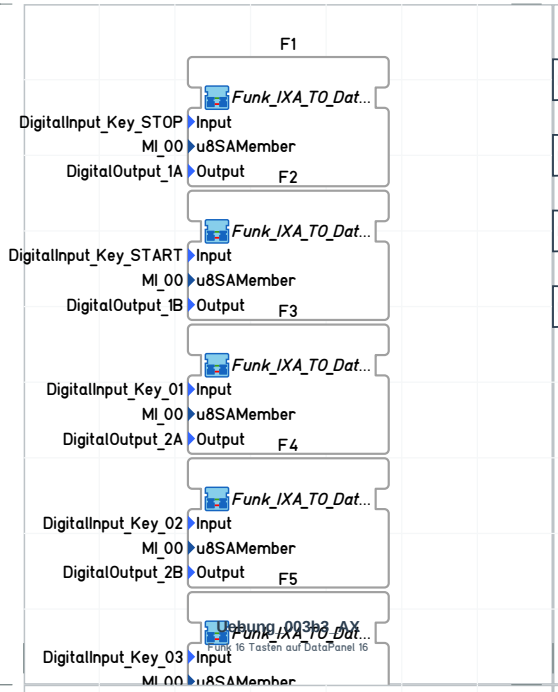
Uebung_003a_AX

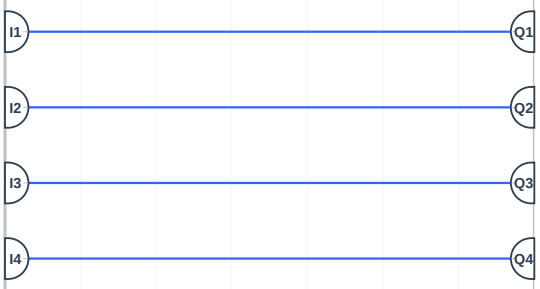
DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2 – mit typed Subapp

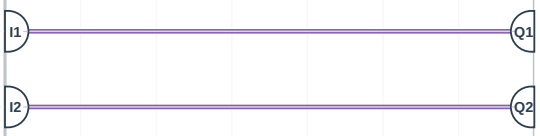


Uebung_003b3_AX

Funk 16 Tasten auf DataPanel 16





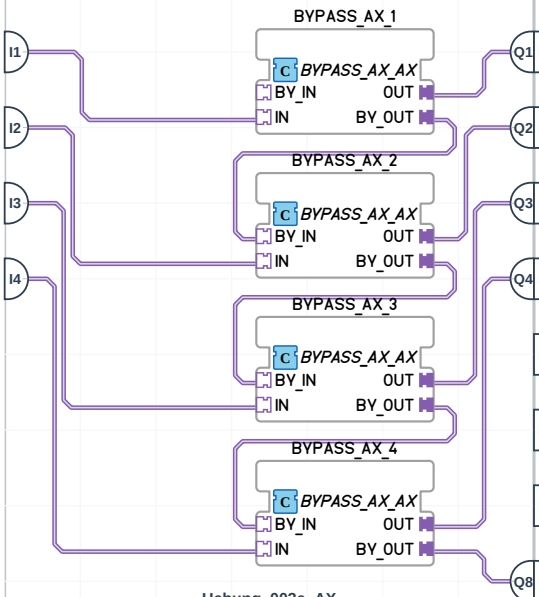


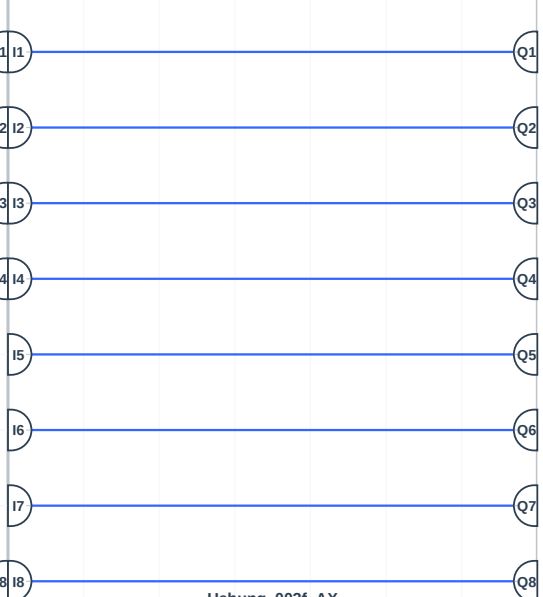
Uebung_003c_AX

DigitalInput_I1-2-3-4 auf AUX

Uebung_003d_AX

DigitalInput_I1/_I2 auf DigitalOutput_Q1/_I2



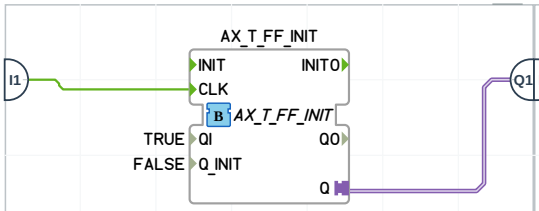


Uebung_003e_AX

DigitalInput_I1-4 auf DigitalOutput_Q1-4 - flach mit Adapter mit Bypass (Druckumlaufventil)

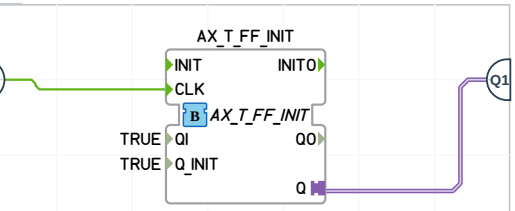
Uebung_003f_AX

DigitalInput_I1-8 auf DigitalOutput_Q1-8



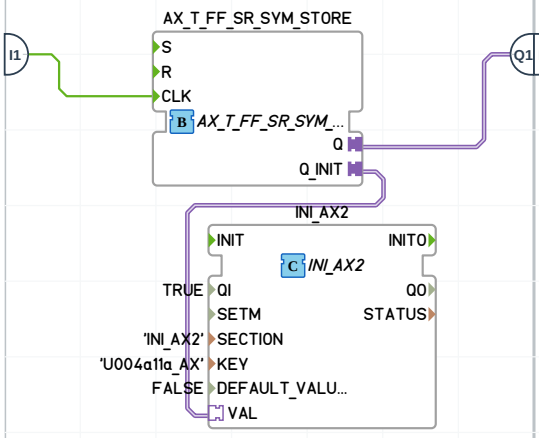
Uebung_004a10a_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und INIT auf FALSE



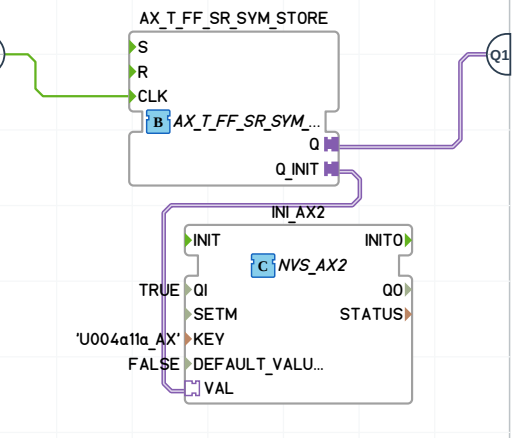
Uebung_004a10b_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und INIT auf TRUE



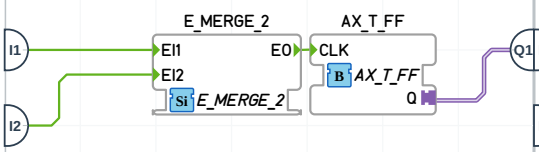
Uebung_004a11a_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und STORE (INI)



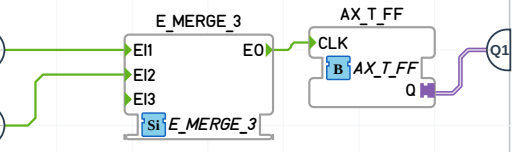
Uebung_004a11b_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK und STORE (NVS)



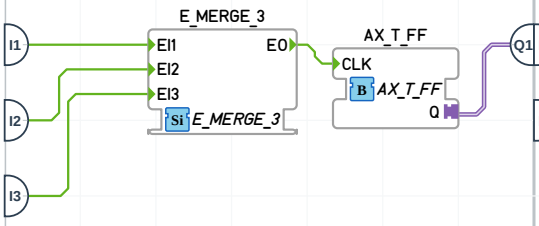
Uebung_004a2_2_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_2



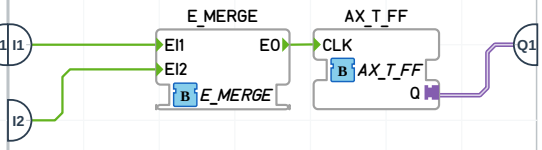
Uebung_004a2_3_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_3



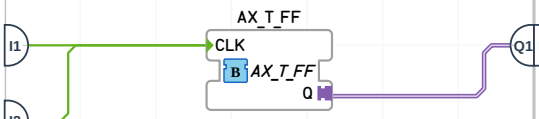
Uebung_004a2_3b_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE_3 (3 Taster)



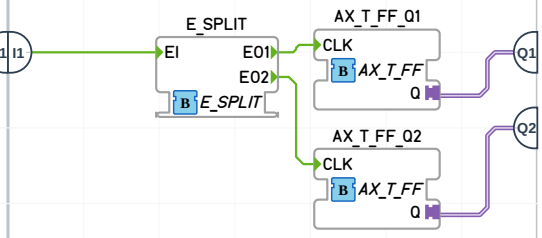
Uebung_004a2_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_MERGE



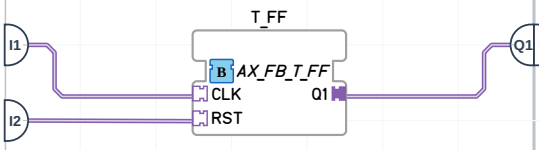
Uebung_004a3_AX

wie Uebung_004a2 aber ohne E_MERGE



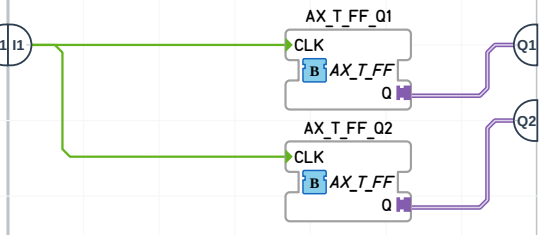
Uebung_004a4_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK mit E_SPLIT



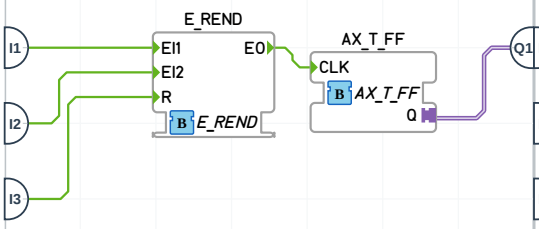
Uebung_004a4_AX_T

Exercise for AX_FB_T_FF (Toggle Flip-Flop with Adapter)



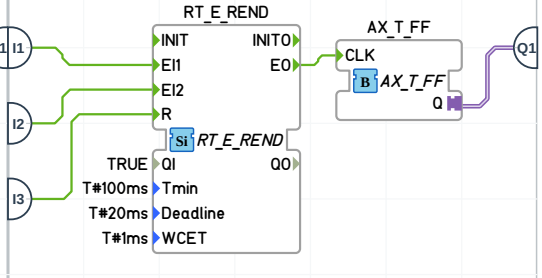
Uebung_004a5_AX

wie Uebung_004a4 aber ohne E_SPLIT



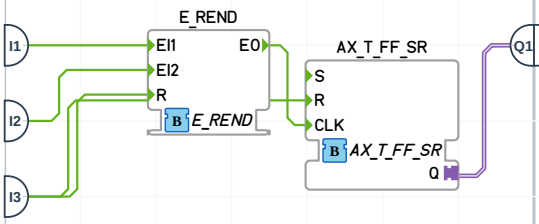
Uebung_004a6_AX

Toggle Flip-Flop mit IE und E_REND (Rendezvous)



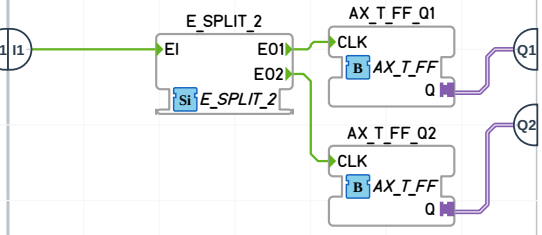
Uebung_004a6a_AX

Toggle Flip-Flop mit IE und E_REND (Rendezvous)



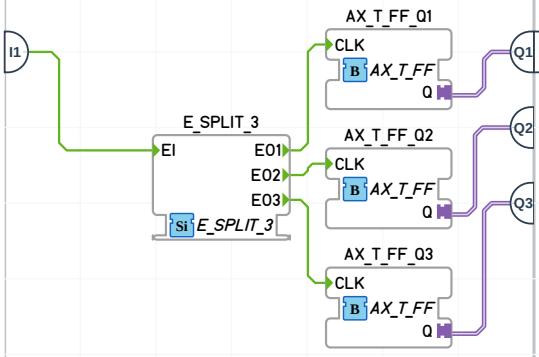
Uebung_004a7_AX

SR und T-Flip-Flop mit IE und E_REND (Rendezvous)



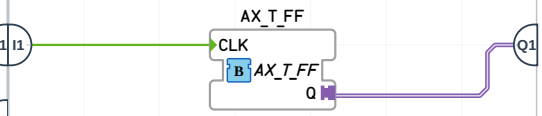
Uebung_004a8_AX

mit E_SPLIT_2



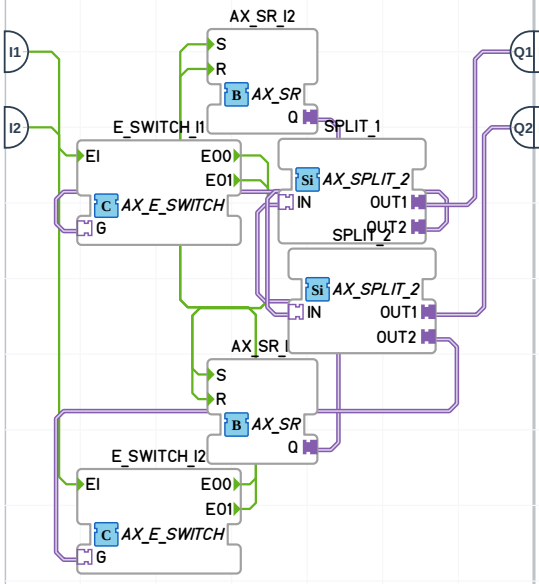
Uebung_004a9_AX

mit E_SPLIT_3



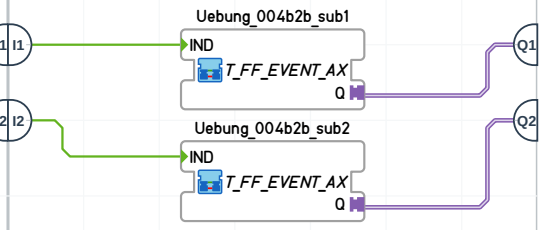
Uebung_004a_AX

Toggle Flip-Flop mit IE mit BUTTON_SINGLE_CLICK



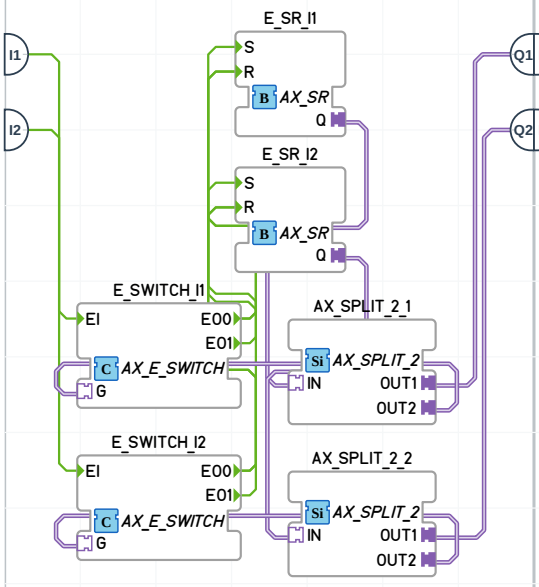
Uebung_004b2_AX

Toggle Flip-Flop mit IE / Split / doppelt (aus AX_E_SWITCH und AX_SR aufgebaut)



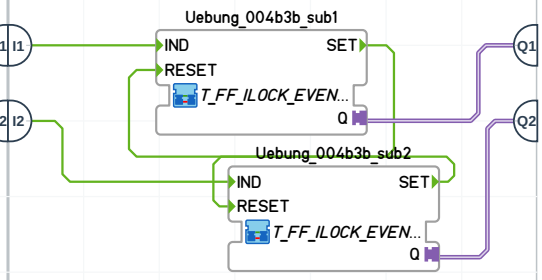
Uebung_004b2b_AX

Zwei unabhängige Toggle-Flip-Flops unter Verwendung von Sub-Applikationen



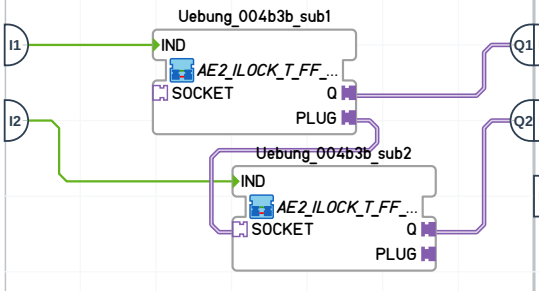
Uebung_004b3_AX

Toggle Flip-Flop mit IE / Split / Verriegelt



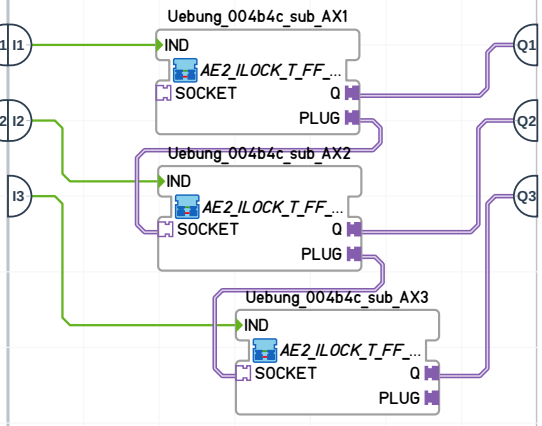
Uebung_004b3b_AX

Zwei gegenseitig verriegelte Toggle-Flip-Flops (Radio-Button-Logik) via Event-Verdrahtung



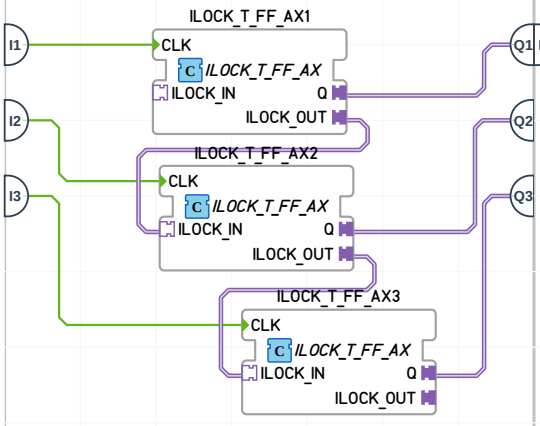
Uebung_004b3c_AX

Zwei gegenseitig verriegelte Toggle-Flip-Flops unter Verwendung von bidirektionalen Adaptern



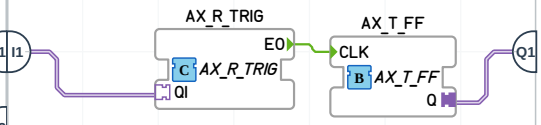
Uebung_004b4c_AX

Drei gegenseitig verriegelte Toggle-Flip-Flops in einer Kette via AE2-Adapter



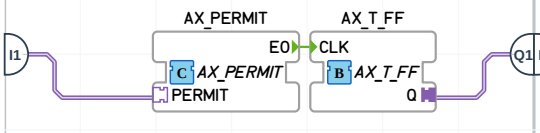
Uebung_004b4d_AX

Drei gegenseitig verriegelte Toggle-Flip-Flops in einer Kette via AE2-Adapter mit ILOCK_T_FF_AX



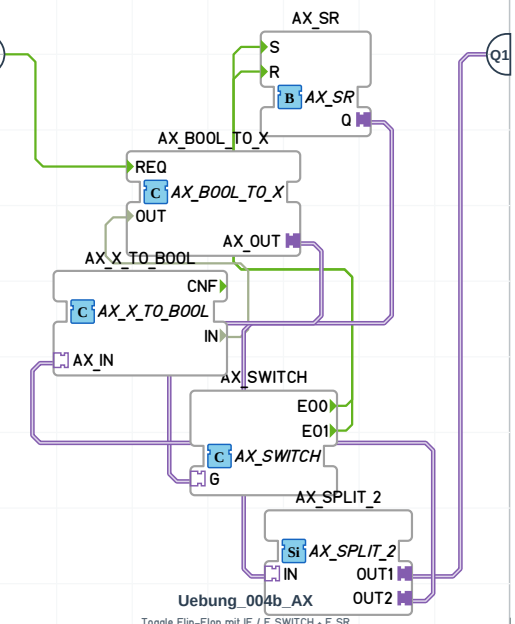
Uebung_004b5_AX

Toggle Flip-Flop mit IXA / AX_R_TRIG + AX_T_FF



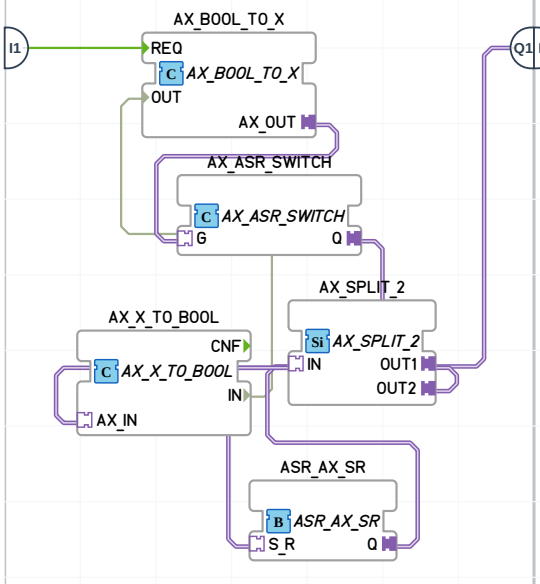
Uebung_004b6_AX

Toggle Flip-Flop mit IXA / AX_PERMIT + AX_T_FF



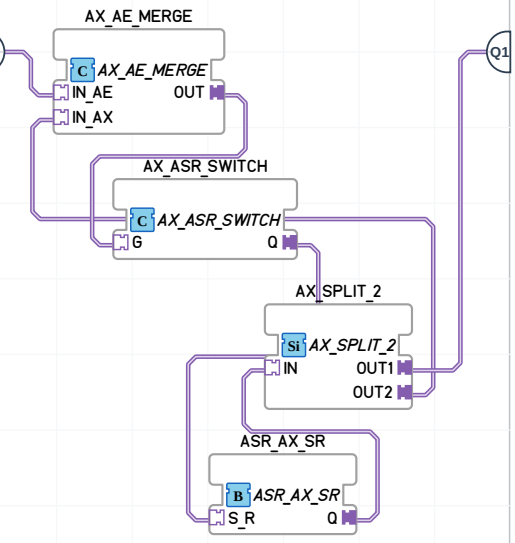
Uebung_004b_AX

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



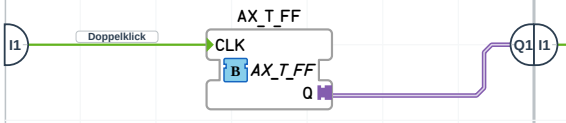
Uebung_004b_AX_ASR

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



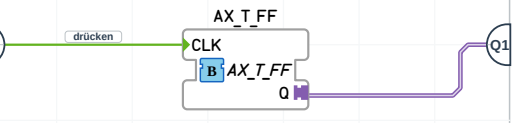
Uebung_004b_AX_ASR_X

Toggle Flip-Flop mit IE / E_SWITCH + E_SR



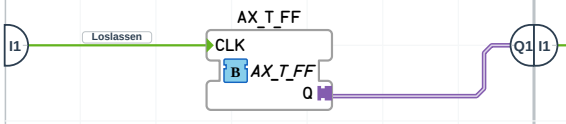
Uebung_004c1_AX

Toggle Flip-Flop mit IE mit BUTTON_DOUBLE_CLICK



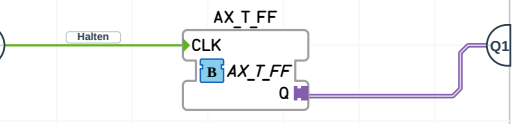
Uebung_004c2_AX

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_START



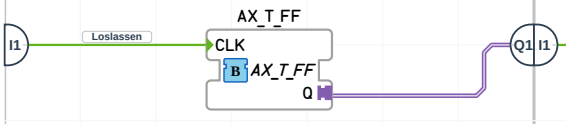
Uebung_004c3_AX

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_UP



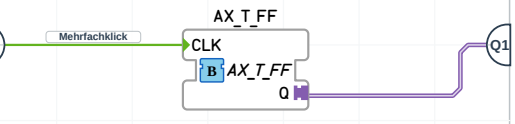
Uebung_004c4_AX

Toggle Flip-Flop mit IE mit BUTTON_LONG_PRESS_HOLD



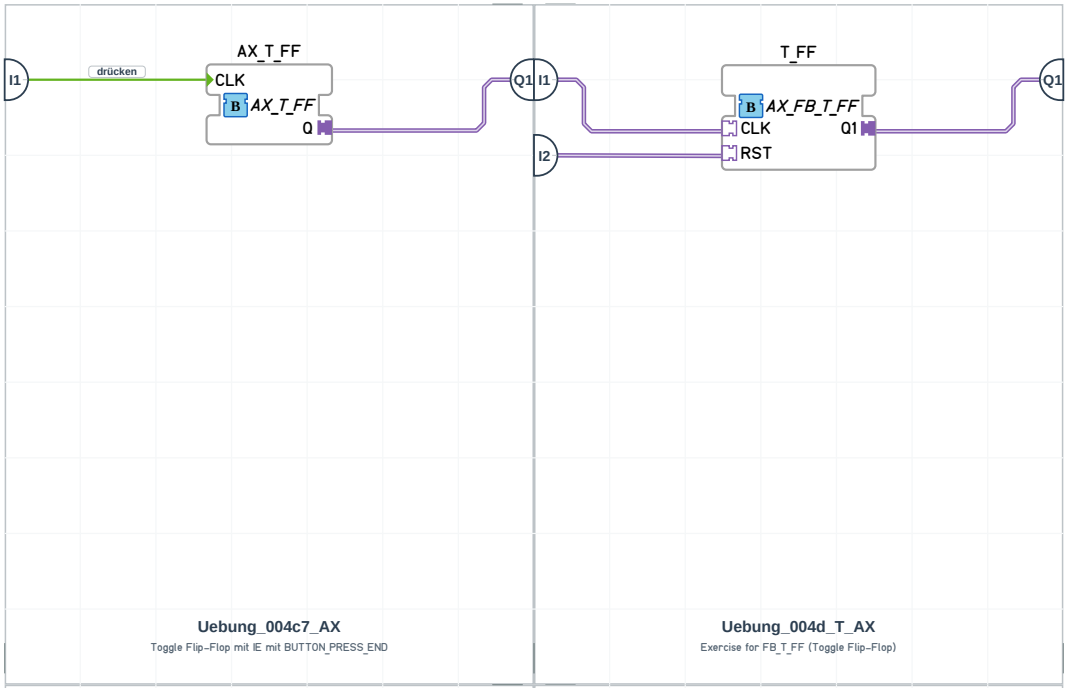
Uebung_004c5_AX

Toggle Flip-Flop mit IE mit BUTTON_PRESS_END



Uebung_004c6_AX

Toggle Flip-Flop mit IE mit BUTTON_MULTIPLE_CLICK
(3-fach Klick)

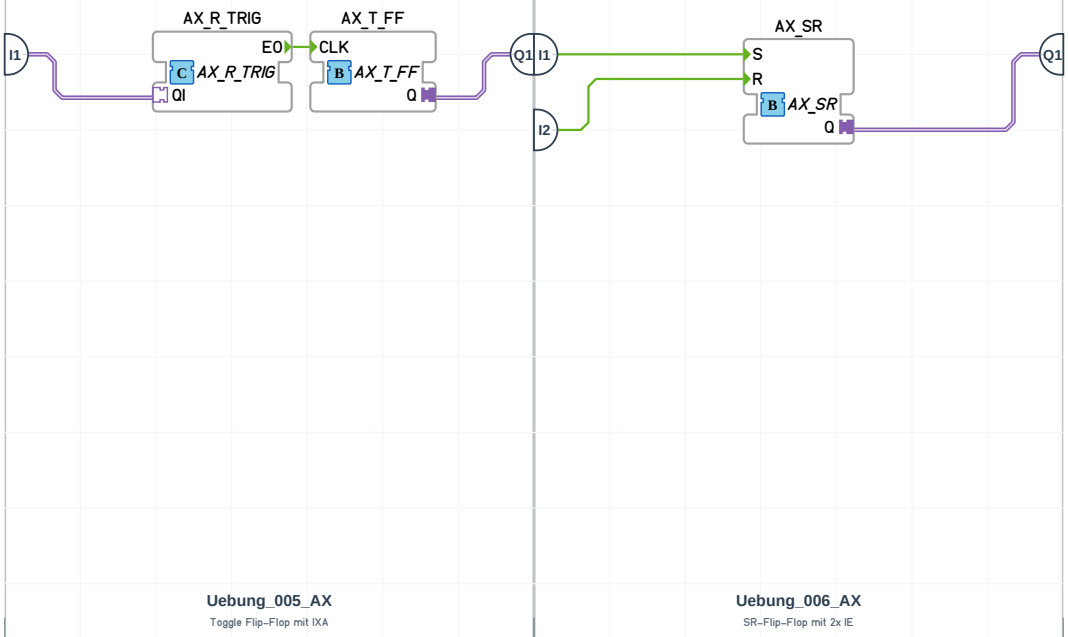


Uebung_004c7_AX

Toggle Flip-Flop mit IE mit BUTTON_PRESS_END

Uebung_004d_T_AX

Exercise for FB_T_FF (Toggle Flip-Flop)

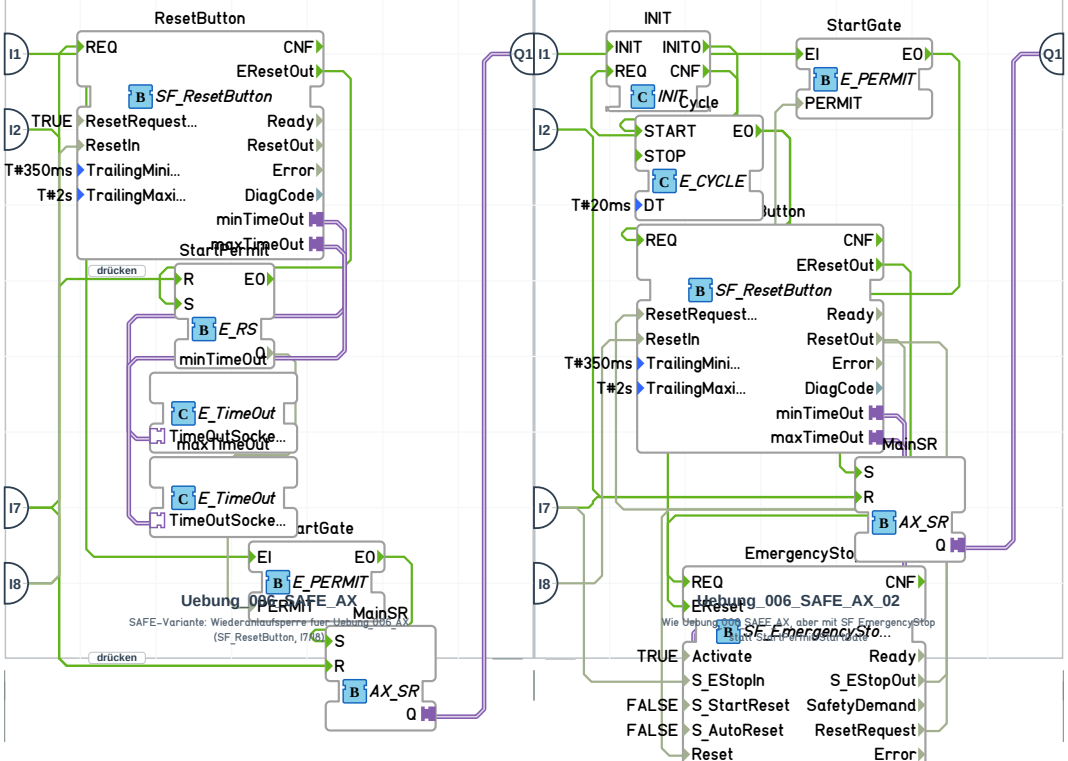


Uebung_005_AX

Toggle Flip-Flop mit IXA

Uebung_006_AX

SR-Flip-Flop mit 2x IE

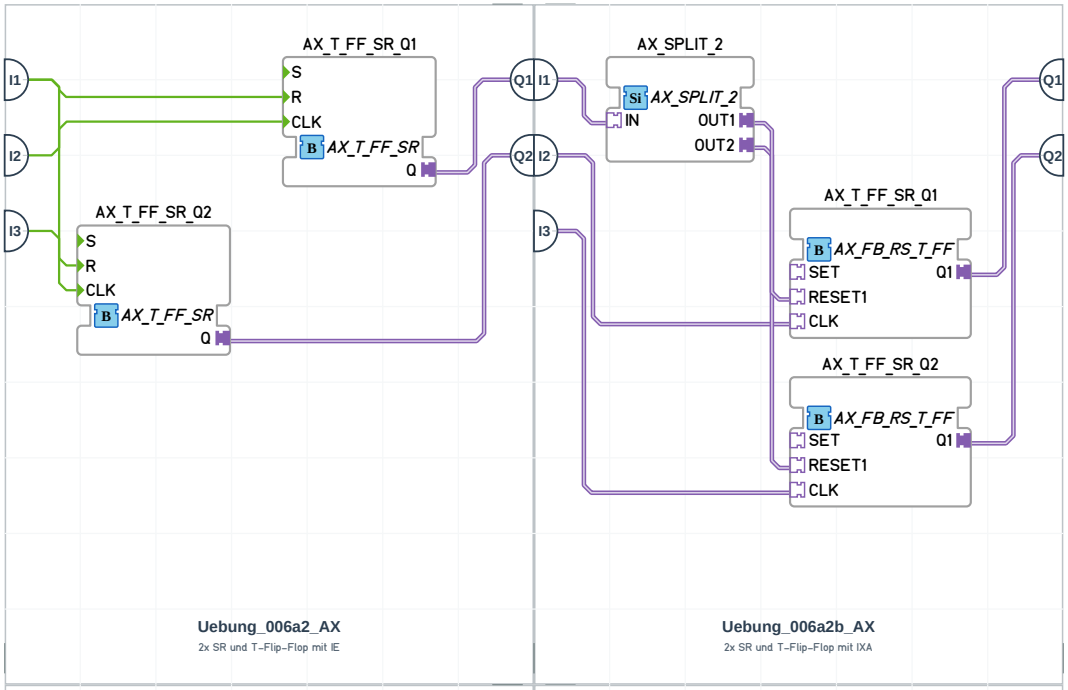


Uebung_006_SAFE_AX

SAFE-Variante: Wiederanlaufsperre fuer Uebung006_AX (SF_ResetButton, I7, Q1)

Uebung_006_SAFE_AX_02

Wie Uebung006_SAFE_AX, aber mit SF_EmergencyStop (inkl. SF_ResetButton)

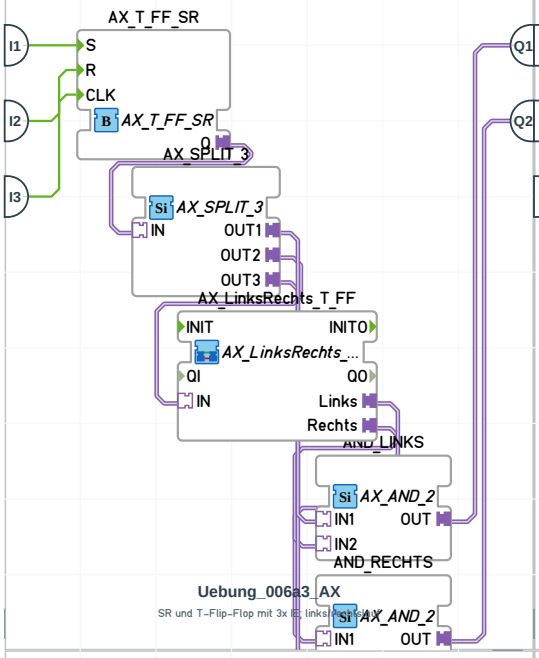


Uebung_006a2_AX

2x SR und T-Flip-Flop mit IE

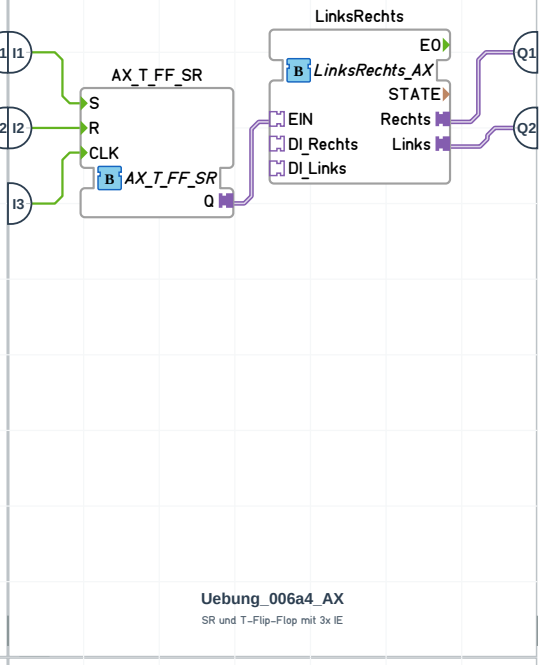
Uebung_006a2b_AX

2x SR und T-Flip-Flop mit IXA



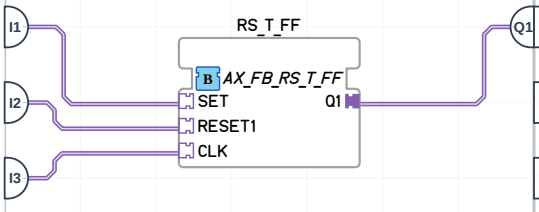
Uebung_006a3_AX

SR und T-Flip-Flop mit 3x IE, links/rechts



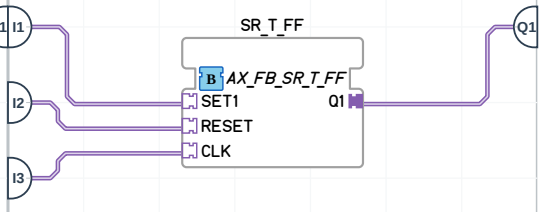
Uebung_006a4_AX

SR und T-Flip-Flop mit 3x IE



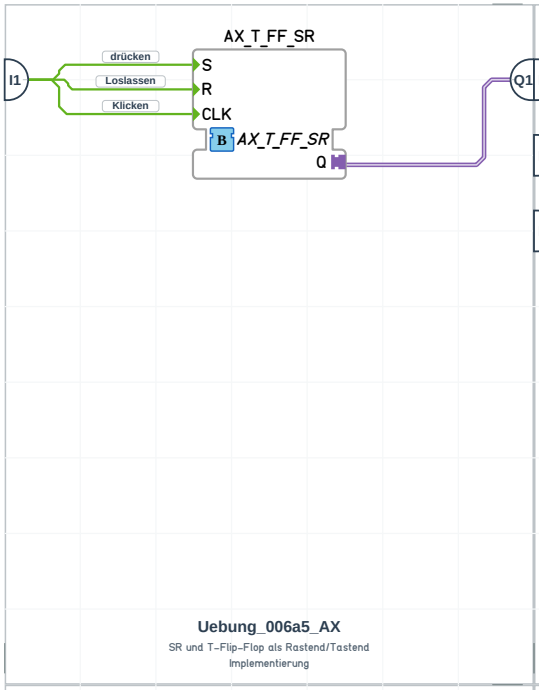
Uebung_006a4_AX_RS

Exercise for AX_FB_RS_T_FF (Reset-Dominant Toggle Flip-Flop with Adapter)



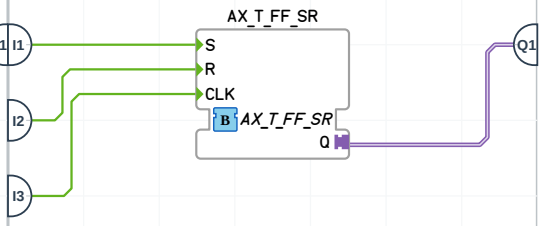
Uebung_006a4_AX_SR

Exercise for AX_FB_SR_T_FF (Set-Dominant Toggle Flip-Flop with Adapter)



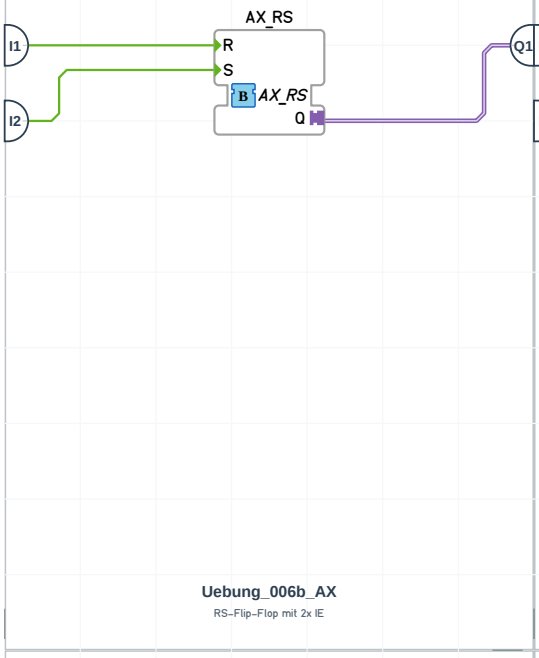
Uebung_006a5_AX

SR und T-Flip-Flop als Rastend/Tastend Implementierung



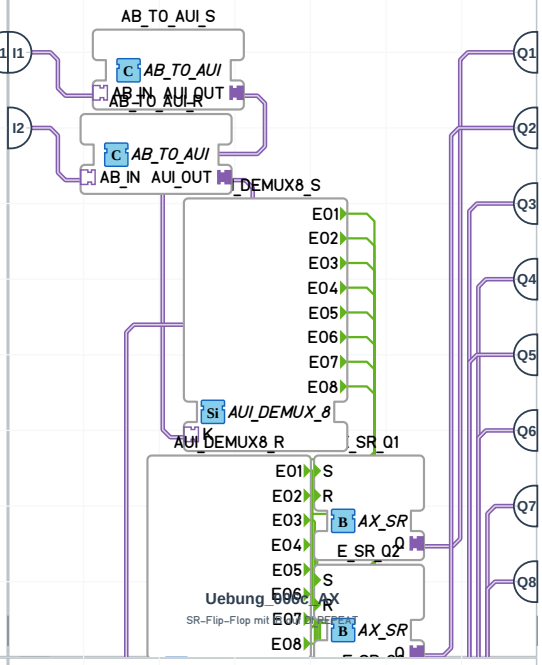
Uebung_006a_AX

SR und T-Flip-Flop mit 3x IE



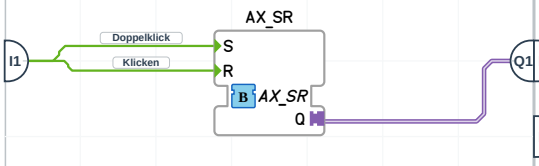
Uebung_006b_AX

RS-Flip-Flop mit 2x IE



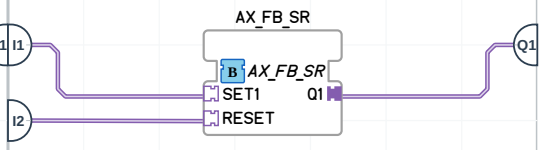
Uebung_006c_AX

SR-Flip-Flop mit 8x IE



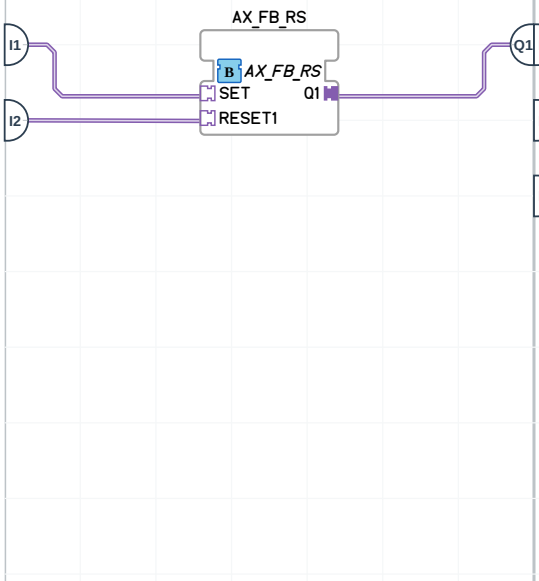
Uebung_006d_AX

SR-Flip-Flop mit 2x IE und Doppelklick.



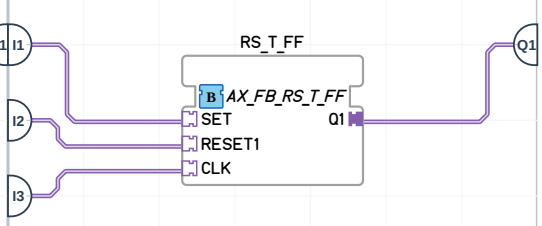
Uebung_006e1_AX

SR-Flip-Flop mit 2x IXA



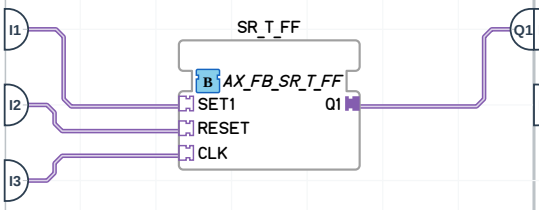
Uebung_006e2_AX

RS-Flip-Flop mit 2x IXA



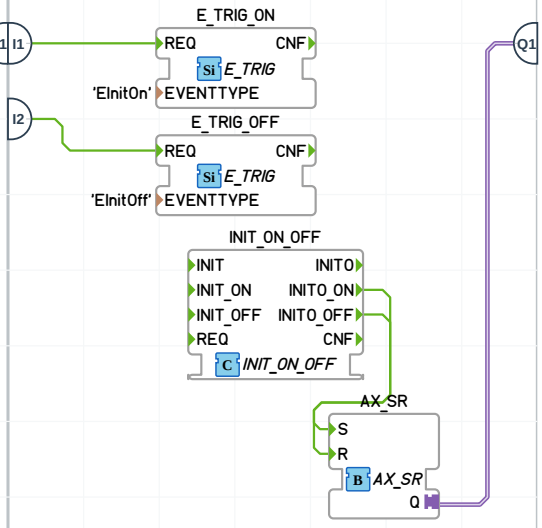
Uebung_006f_RS_AX

Exercise for FB_RS_T_FF (Reset-Dominant Toggle Flip-Flop)



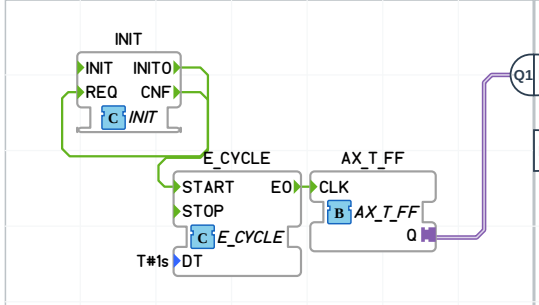
Uebung_006f_SR_AX

Exercise for FB_SR_T_FF (Set-Dominant Toggle Flip-Flop)



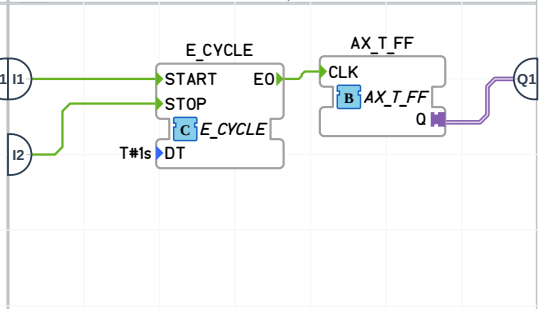
Uebung_006g_AX

SR-Flip-Flop via INIT_ON_OFF, ausgeloest von 2x E_TRIG (EInitOn/EInitOff Broadcast auf unverbundene Events)



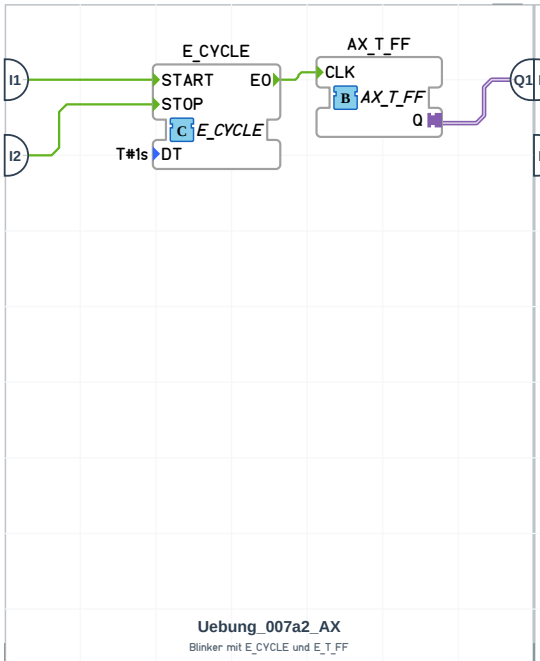
Uebung_007_AX

Blinker mit E_CYCLE und E_T_FF



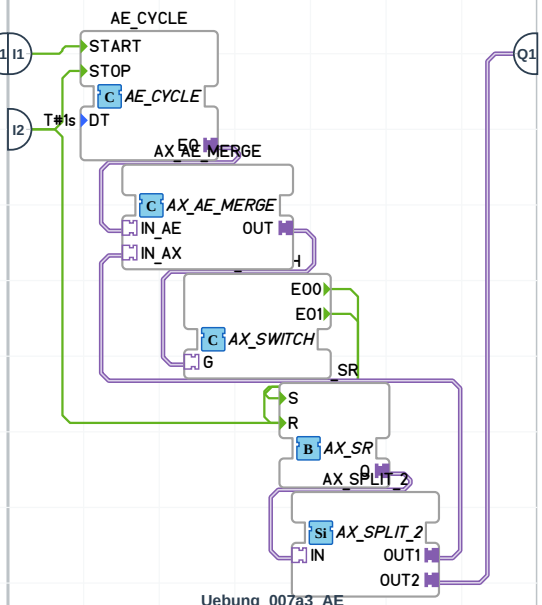
Uebung_007a1_AX

Blinker mit E_CYCLE und E_T_FF



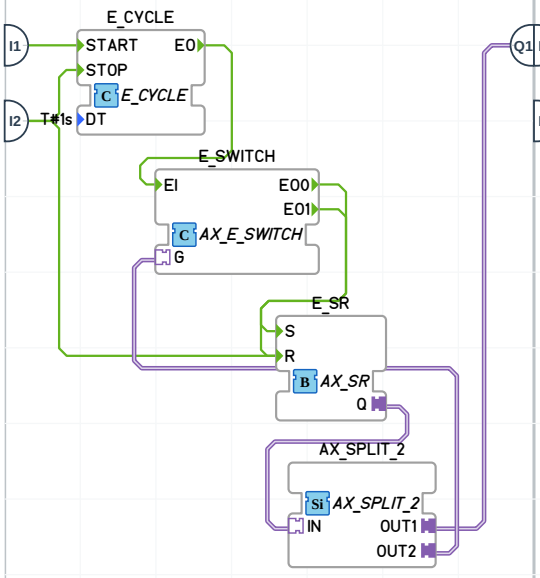
Uebung_007a2_AX

Blinker mit E_CYCLE und E_T_FF



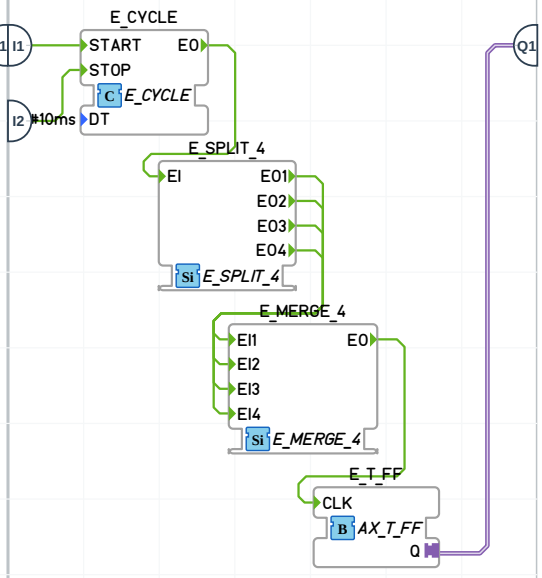
Uebung_007a3_AE

Blinker mit E_CYCLE, E_SWITCH und E_SR



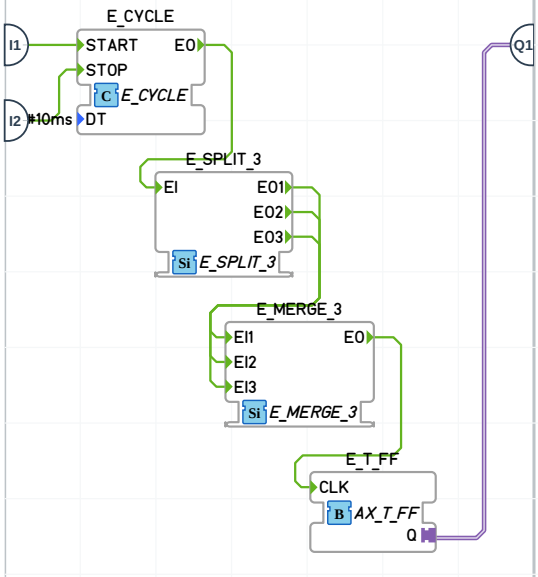
Uebung_007a3_AX

Blinker mit E_CYCLE, E_SWITCH und E_SR



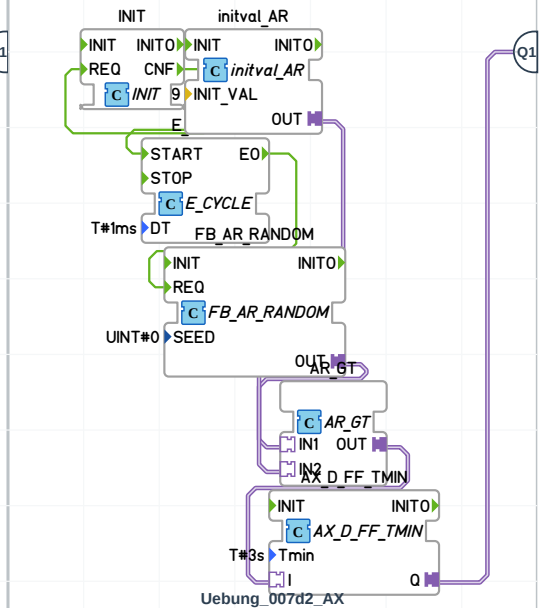
Uebung_007b_AX

Blinker mit E_CYCLE und E_T_FF



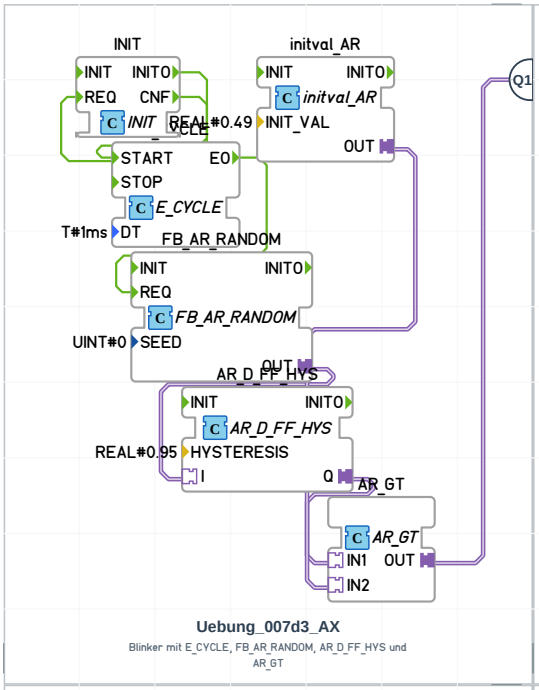
Uebung_007c_AX

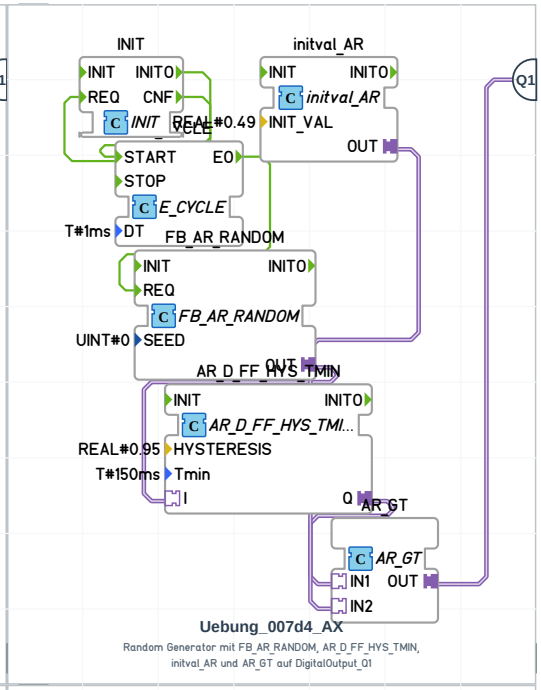
Blinker mit E_CYCLE und E_T_FF

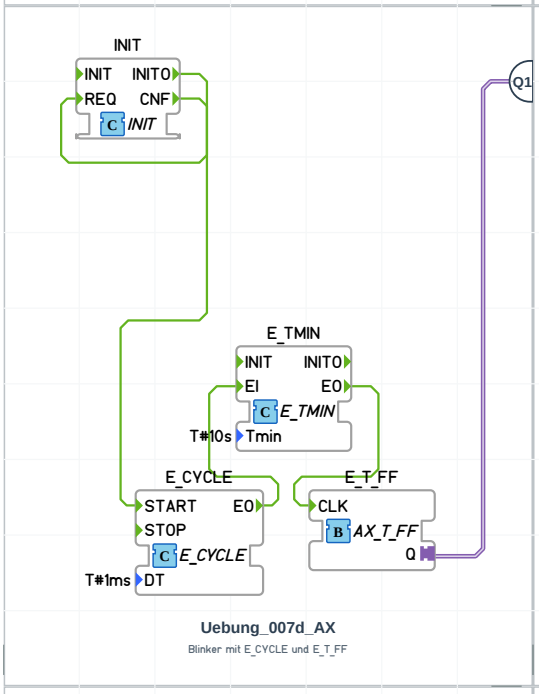


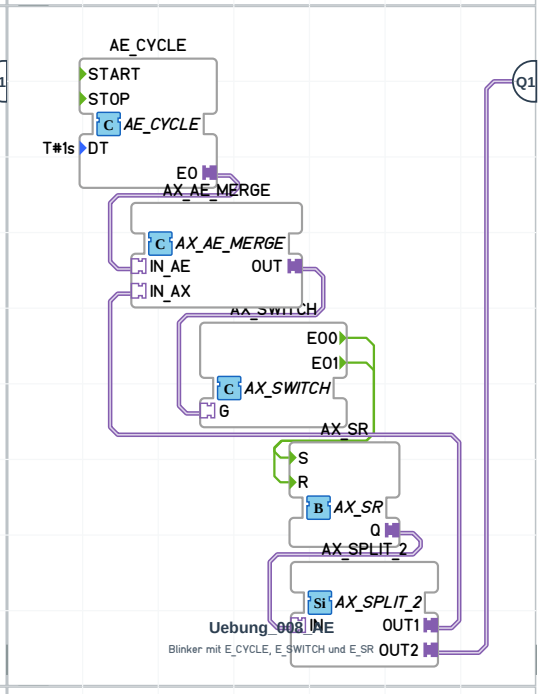
Uebung_007d2_AX

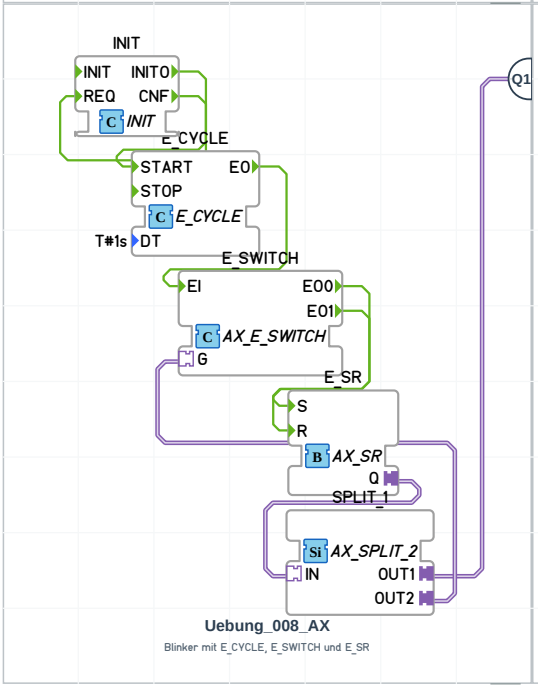
Blinker mit E_CYCLE, FB_AR_RANDOM, AR_GT und AX_D_FF_TMIN

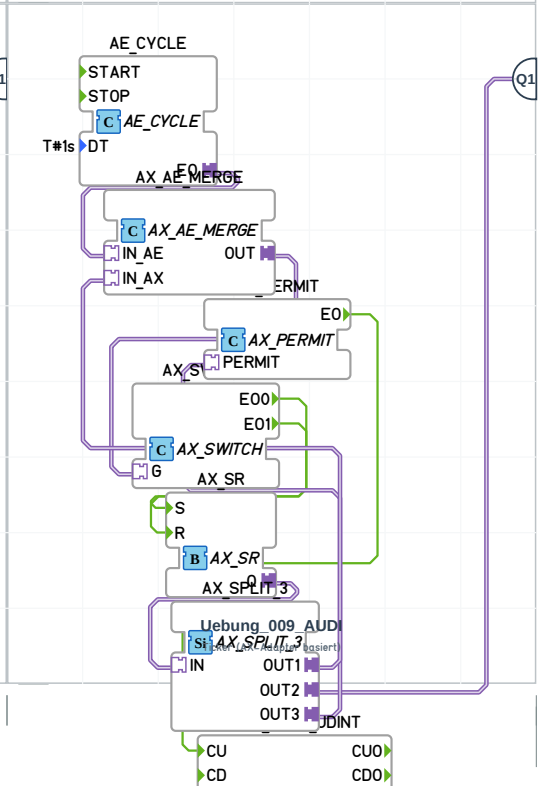

INIT
initval_AR
INIT
INITO
REQ
CNF
C INIT
REAL#0.49
INIT_VAL
OUT
START
EO
STOP
C E_CYCLE
T#1ms
DT
FB_AR_RANDOM
INIT
INITO
REQ
C FB_AR_RANDOM
UINT#0
SEED
OUT
AR_D_FF_HYS
INIT
INITO
C AR_D_FF_HYS
REAL#0.95
HYSTERESIS
I
Q
AR_GT
C AR_GT
IN1
OUT
IN2
Q1
Uebung_007d3_AX
Blinker mit E_CYCLE, FB_AR_RANDOM, AR_D_FF_HYS und AR_GT

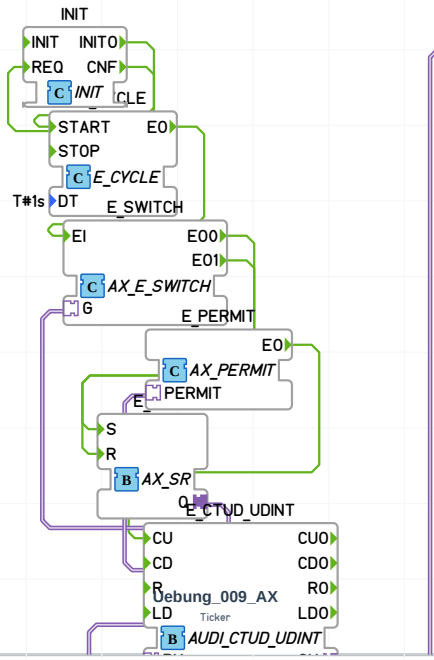

INIT
initval_AR
INIT
INITO
REQ
CNF
C INIT
REAL#0.49
INIT_VAL
OUT
START
EO
STOP
C E_CYCLE
T#1ms
DT
FB_AR_RANDOM
INIT
INITO
REQ
C FB_AR_RANDOM
UINT#0
SEED
OUT
AR_D_FF_HYS_TMIN
INIT
INITO
C AR_D_FF_HYS_TMI...
REAL#0.95
HYSTERESIS
T#150ms
Tmin
I
Q
AR_GT
C AR_GT
IN1
OUT
IN2
Q1
Uebung_007d4_AX
Random Generator mit FB_AR_RANDOM, AR_D_FF_HYS_TMIN, initval_AR und AR_GT auf DigitalOutput_Q1

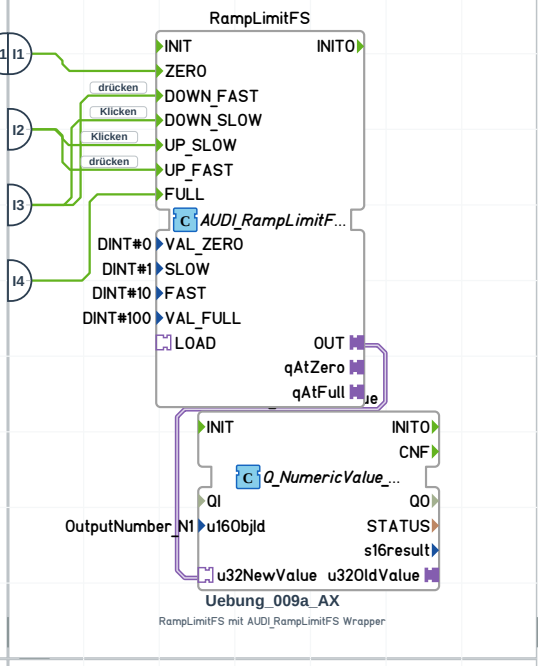

INIT
INIT
INITO
REQ
CNF
C INIT
E_TMIN
INIT
INITO
EI
EO
C E_TMIN
T#10s
Tmin
E_CYCLE
START
EO
STOP
C E_CYCLE
T#1ms
DT
E_T_FF
CLK
B AX_T_FF
Q
Q1
Uebung_007d_AX
Blinker mit E_CYCLE und E_T_FF

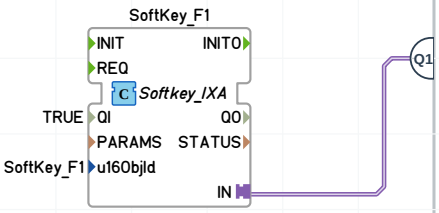

AE_CYCLE
START
STOP
C AE_CYCLE
T#1s
DT
EO
AX_AE_MERGE
C AX_AE_MERGE
IN_AE
OUT
IN_AX
AX_SWITCH
EO0
EO1
C AX_SWITCH
G
AX_SR
S
R
B AX_SR
Q
AX_SPLIT_2
Si AX_SPLIT_2
IN
OUT1
OUT2
Q1
Uebung_008_AX
Blinker mit E_CYCLE, E_SWITCH und E_SR

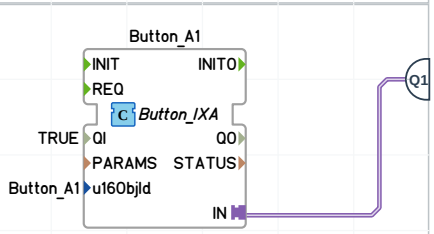

INIT
INIT
INITO
REQ
CNF
C INIT
E_CYCLE
START
EO
STOP
C E_CYCLE
T#1s
DT
E_SWITCH
EI
EO0
EO1
C AX_E_SWITCH
G
E_SR
S
R
B AX_SR
Q
SPLIT_1
Si AX_SPLIT_2
IN
OUT1
OUT2
Q1
Uebung_008_AX
Blinker mit E_CYCLE, E_SWITCH und E_SR

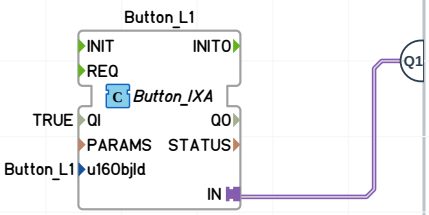

AE_CYCLE
START
STOP
C AE_CYCLE
T#1s
DT
EO
AX_AE_MERGE
C AX_AE_MERGE
IN_AE
OUT
IN_AX
C AX_PERMIT
PERMIT
EO0
EO1
C AX_SWITCH
G
AX_SR
S
R
B AX_SR
Q
AX_SPLIT_3
Si AX_SPLIT_3
IN
OUT1
OUT2
OUT3
CU
CD
CUO
CDO
Q1
Uebung_009_AUDI

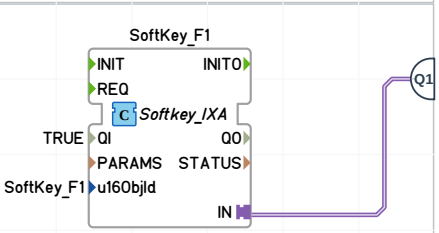

INIT
INIT
INITO
REQ
CNF
C INIT
CLE
START
EO
STOP
C E_CYCLE
T#1s
DT
E_SWITCH
EI
EO0
EO1
C AX_E_SWITCH
G
E_PERMIT
EO
C AX_PERMIT
PERMIT
E
S
R
B AX_SR
Q
E_CTUD_UDINT
CU
CUO
CD
CDO
R
RO
LD
LDO
Ticker
B AUDI_CTUD_UDINT
Uebung_009_AX

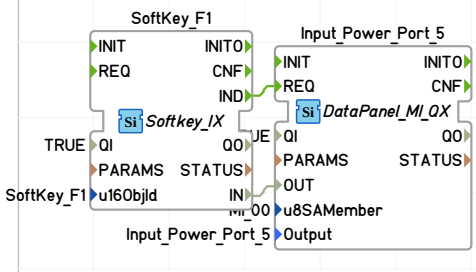

RampLimitFS
Q1
I1
INIT
INITO
ZERO
drücken
DOWN_FAST
Klicken
I2
DOWN_SLOW
Klicken
UP_SLOW
drücken
UP_FAST
I3
FULL
C AUDI_RampLimitF...
DINT#0
VAL_ZERO
I4
DINT#1
SLOW
DINT#10
FAST
DINT#100
VAL_FULL
LOAD
OUT
qAtZero
qAtFull
INIT
INITO
CNF
C Q_NumericValue_...
QI
QO
OutputNumber_IN1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_009a_AX
RampLimitFS mit AUDI_RampLimitFS Wrapper


SoftKey_F1
INIT
INITO
REQ
C Softkey_IXA
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
IN
Q1
Uebung_010_AX
SoftKey_F1 auf DigitalOutput_Q1


Button_A1
INIT
INITO
REQ
C Button_IXA
TRUE
QI
QO
PARAMS
STATUS
Button_A1
u16ObjId
IN
Q1
Uebung_010a2_AX
Button_A1 auf DigitalOutput_Q1

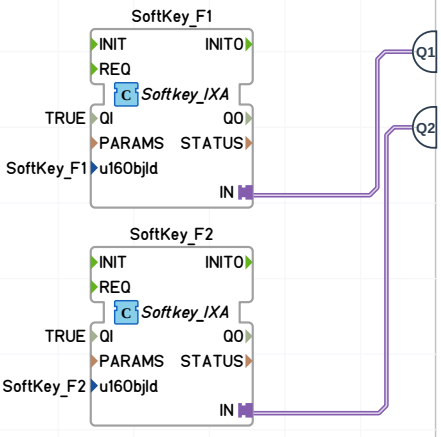

Button_L1
INIT
INITO
REQ
C Button_IXA
TRUE
QI
QO
PARAMS
STATUS
Button_L1
u16ObjId
IN
Q1
Uebung_010a3_AX
Button_L1 (Latching) auf DigitalOutput_Q1


SoftKey_F1
INIT
INITO
REQ
C Softkey_IXA
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
IN
Q1
Uebung_010a4_AX
SoftKey_F1 auf DigitalOutput_Q1 (Datapanel)



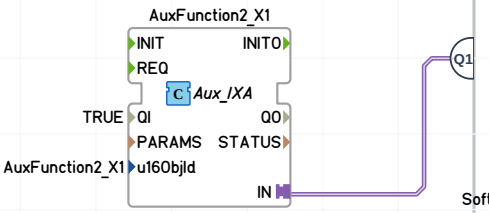
Uebung_010a5_AX

SoftKey_F1 auf DigitalOutput_Q1 (Datapanel)



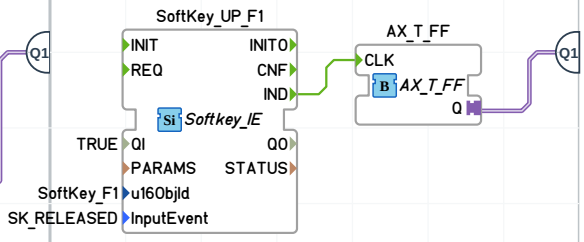
Uebung_010a_AX

SoftKey_F1/_F2 auf DigitalOutput_Q1/_Q2



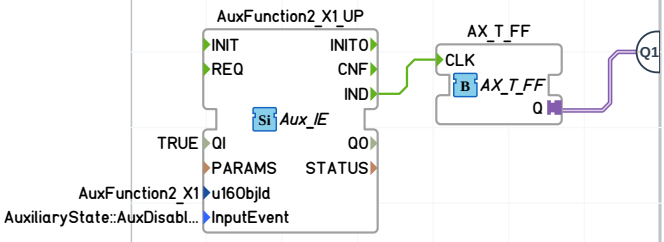
Uebung_010b1_AX

AuxFunction2_X1 auf DigitalOutput_Q1



Uebung_010b2_AX

Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED



Uebung_010b3_AX

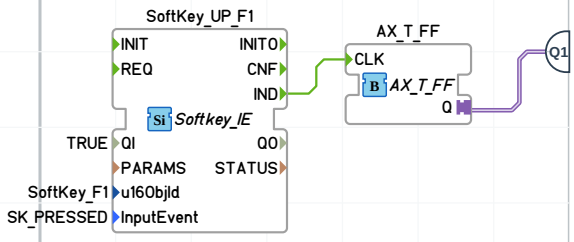
Toggle Flip-Flop mit IE AuxFunction2_X1

Uebung_010b4_AX

SoftKey_F1-8 auf DigitalOutput_Q1-8

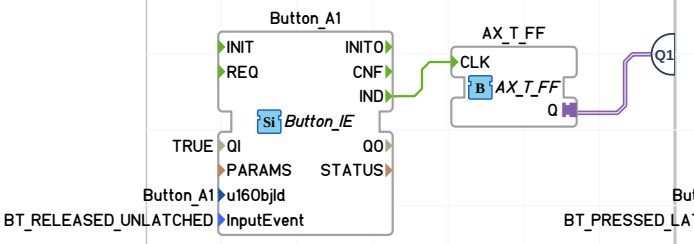
Uebung_010b5_AX

SoftKey_F1-10 auf DigitalOutput_Q1-10



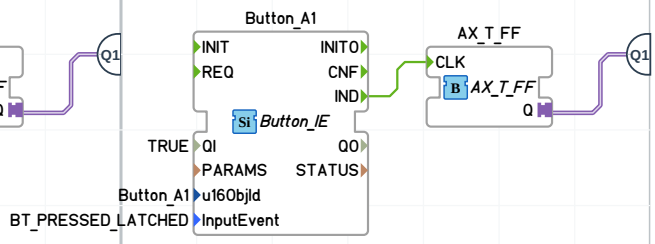
Uebung_010b6_AX

Toggle Flip-Flop mit IE SoftKey_F1 SK_PRESSED



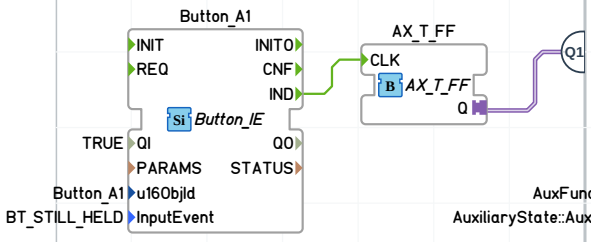
Uebung_010b7_AX

Toggle Flip-Flop mit IE Button_A1 BT_RELEASED_UNLATCHED



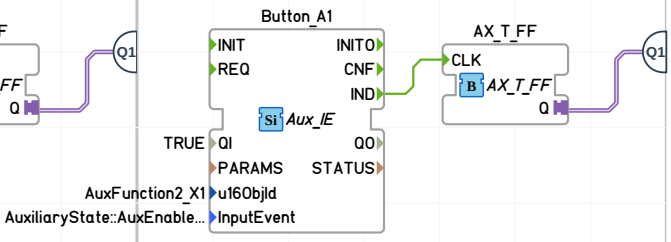
Uebung_010b8_AX

Toggle Flip-Flop mit IE Button_A1 BT_PRESSED_LATCHED



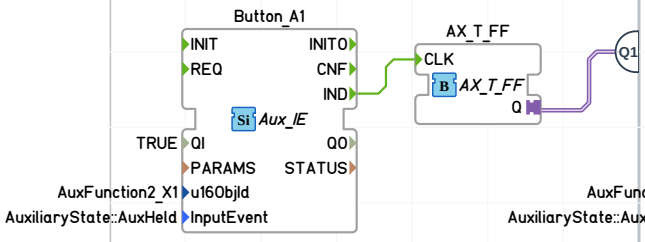
Uebung_010b9_AX

Toggle Flip-Flop mit IE Button_A1 BT_STILL_HELD



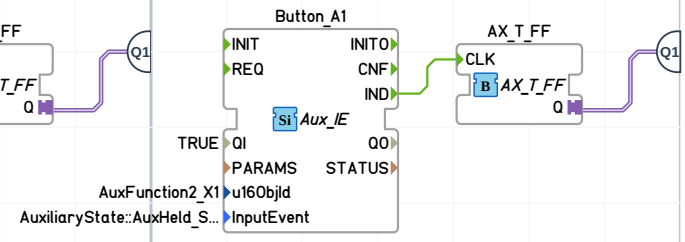
Uebung_010bA2_AX

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxEnabled



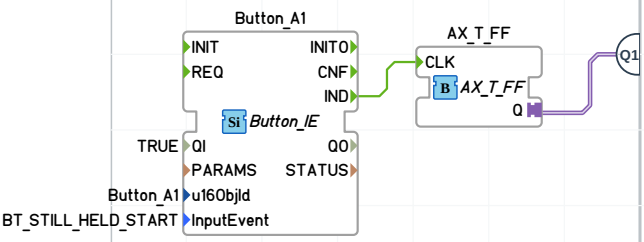
Uebung_010bA3_AX

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxHeld



Uebung_010bA4_AX

Toggle Flip-Flop mit IE AuxFunction2_X1 AuxHeld_START



Uebung_010bA_AX

Toggle Flip-Flop mit IE Button_A1 BT_STILL_HELD



Uebung_010c2_AX

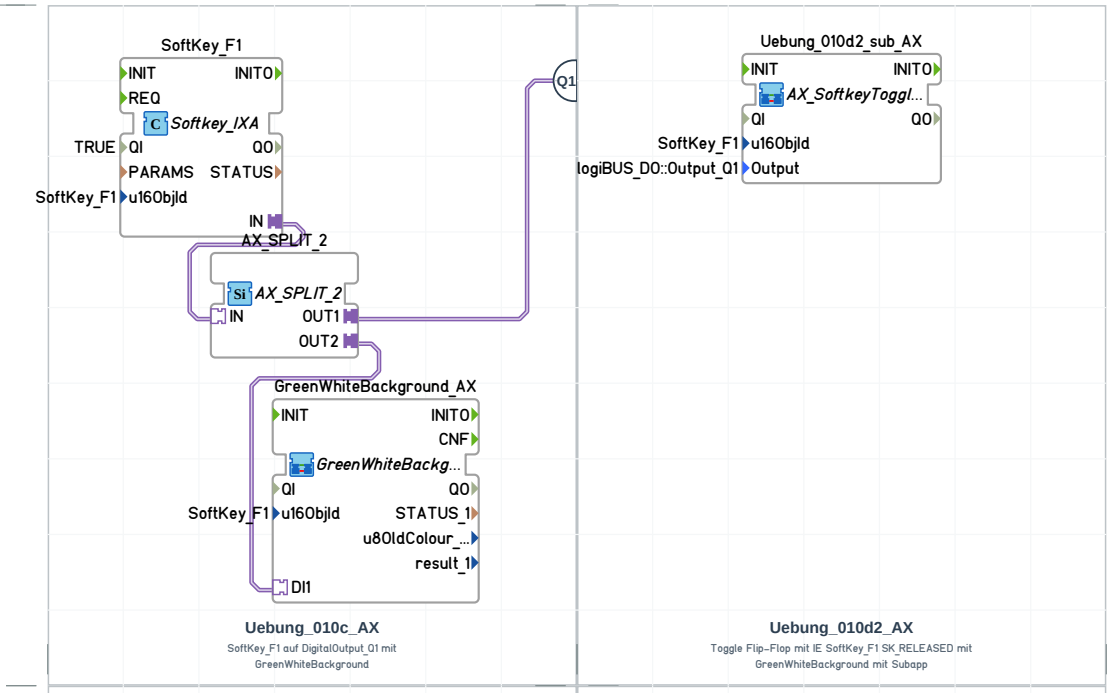
SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Subapp

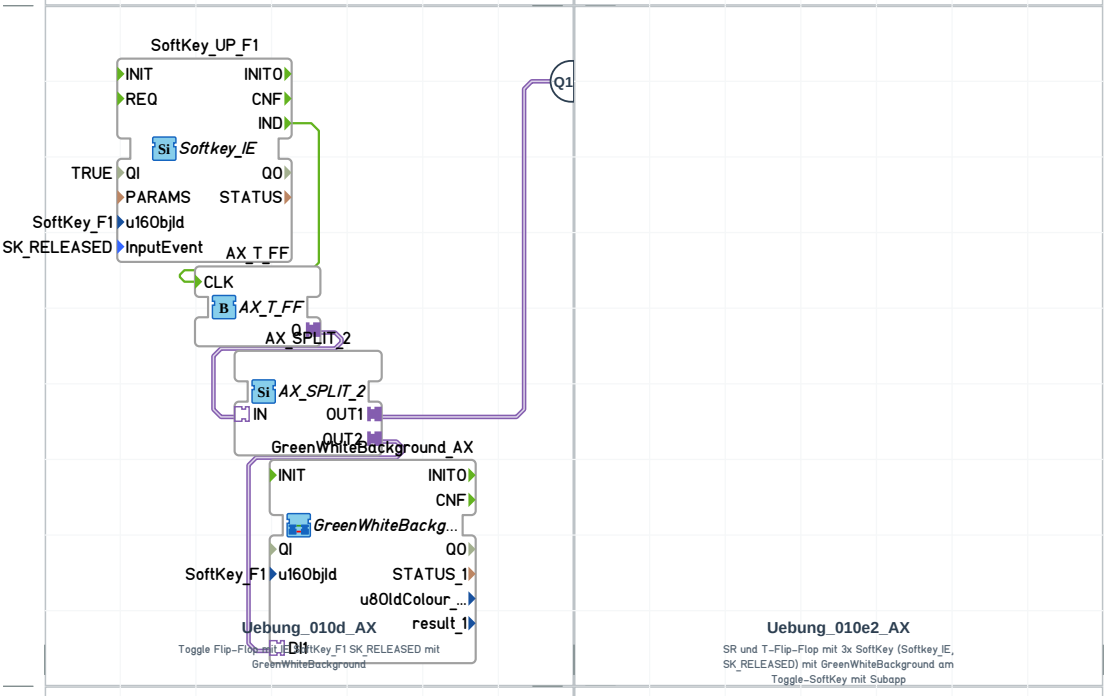
Uebung_010c3_AX

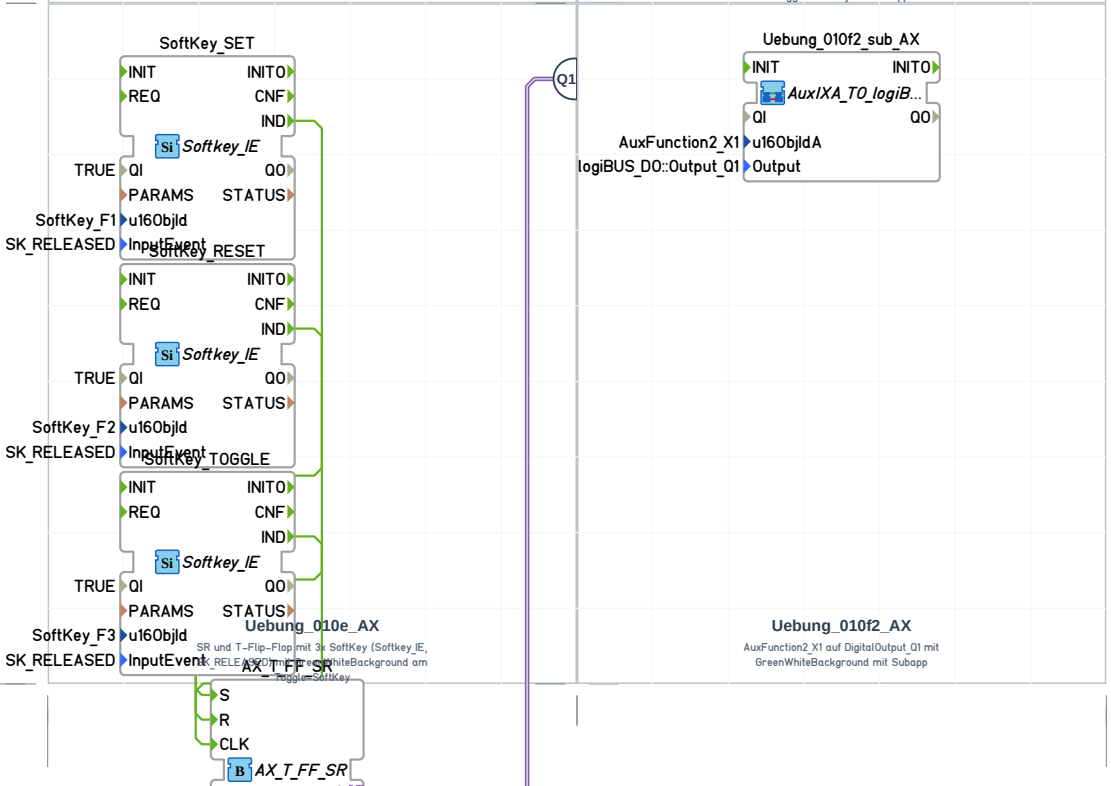
SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Typed Subapp

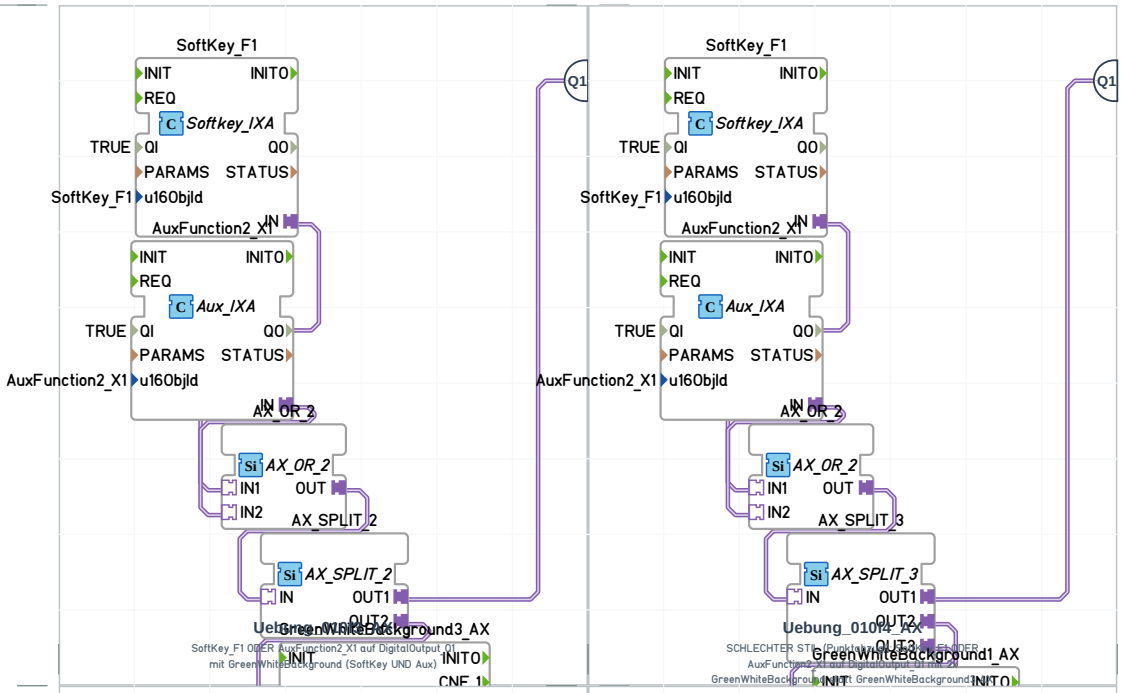
Uebung_010c4_AX

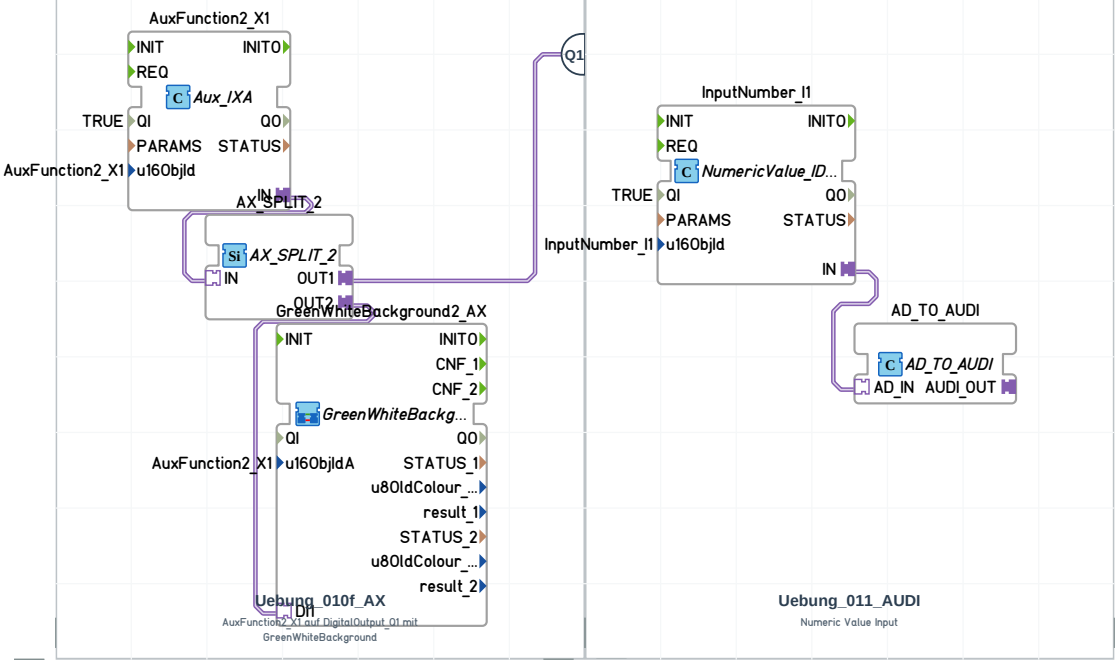
SoftKey_F1/_F2 auf DigitalOutput_Q1/_Q2 mit GreenWhiteBackground mit Typed Subapp

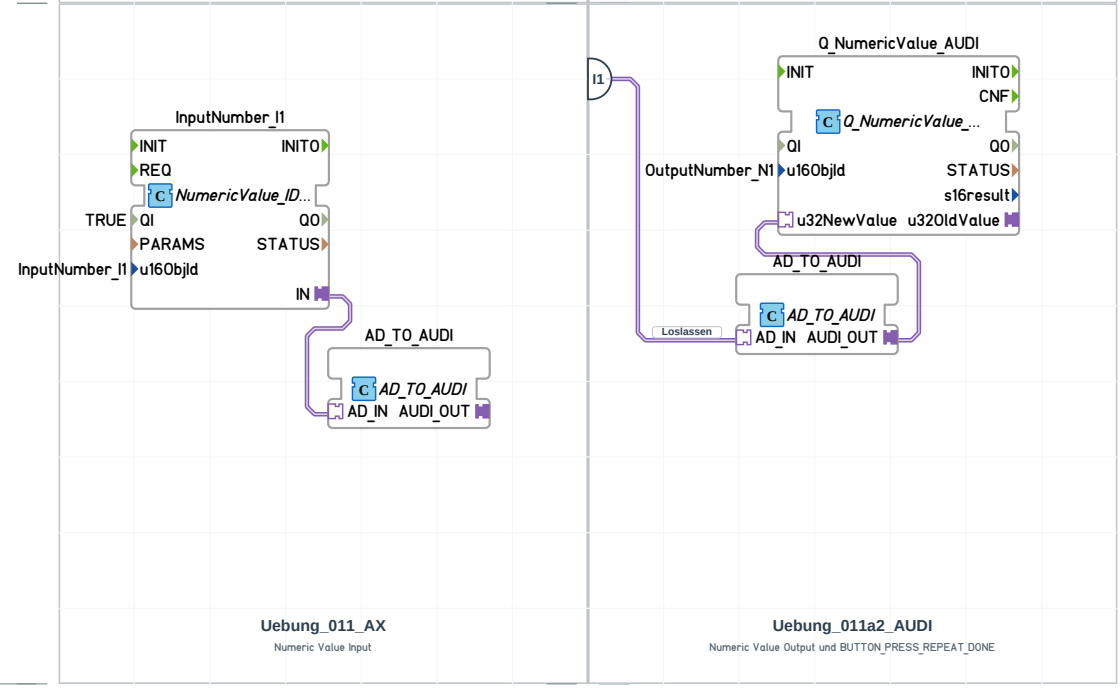

SoftKey_F1
INIT
INITO
REQ
Softkey_IXA
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
IN
AX_SPLIT_2
AX_SPLIT_2
IN
OUT1
OUT2
Q1
GreenWhiteBackground_AX
INIT
INITO
CNF
GreenWhiteBackg...
QI
QO
SoftKey_F1
u16ObjId
STATUS_1
u8OldColour_...
result_1
DI1
Uebung_010d2_sub_AX
INIT
INITO
AX_SoftkeyToggl...
QI
QO
SoftKey_F1
u16ObjId
logiBUS_DO::Output_Q1
Output
Uebung_010c_AX
SoftKey_F1 auf DigitalOutput_Q1 mit GreenWhiteBackground
Uebung_010d2_AX
Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED mit GreenWhiteBackground mit Subapp

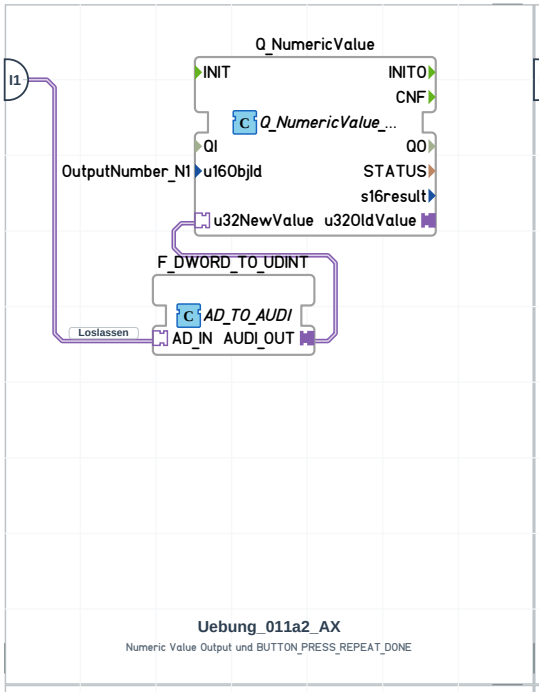

SoftKey_UP_F1
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
AX_T_FF
CLK
AX_T_FF
Q
AX_SPLIT_2
AX_SPLIT_2
IN
OUT1
OUT2
Q1
GreenWhiteBackground_AX
INIT
INITO
CNF
GreenWhiteBackg...
QI
QO
SoftKey_F1
u16ObjId
STATUS_1
u8OldColour_...
result_1
DI1
Uebung_010d_AX
Toggle Flip-Flop mit IE SoftKey_F1 SK_RELEASED mit GreenWhiteBackground
Uebung_010e2_AX
SR und T-Flip-Flop mit 3x SoftKey (Softkey_IE, SK_RELEASED) mit GreenWhiteBackground am Toggle-SoftKey mit Subapp

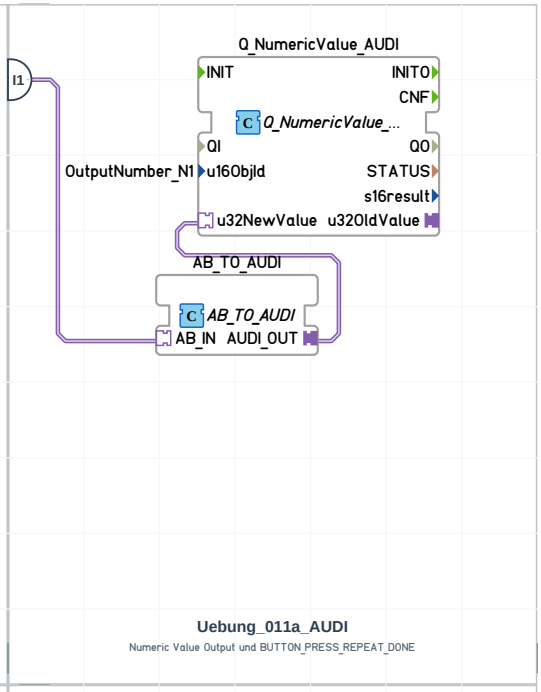

SoftKey_SET
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
SoftKey_RESET
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F2
u16ObjId
SK_RELEASED
InputEvent
SoftKey_TOGGLE
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F3
u16ObjId
SK_RELEASED
InputEvent
AX_T_FF_SR
S
R
CLK
AX_T_FF_SR
Q1
Uebung_010f2_sub_AX
INIT
INITO
AuxIXA_TO_logiB...
QI
QO
AuxFunction2_X1
u16ObjIdA
logiBUS_DO::Output_Q1
Output
Uebung_010e_AX
SR und T-Flip-Flop mit 3x SoftKey (Softkey_IE, SK_RELEASED) mit GreenWhiteBackground am Toggle-SoftKey
Uebung_010f2_AX
AuxFunction2_X1 auf DigitalOutput_Q1 mit GreenWhiteBackground mit Subapp

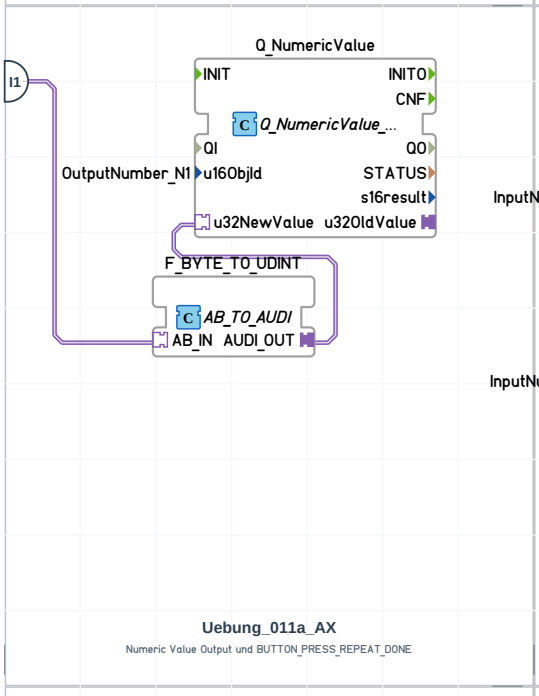

SoftKey_F1
INIT
INITO
REQ
Softkey_IXA
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
AuxFunction2_X1
IN
Aux_IXA
AuxFunction2_X1
AX_OR_2
AX_OR_2
IN1
OUT
IN2
AX_SPLIT_2
AX_SPLIT_2
OUT1
OUT2
Q1
Uebung_010f_AX
SoftKey_F1 ODER AuxFunction2_X1 auf DigitalOutput_01
mit GreenWhiteBackground (SoftKey UND Aux)
GreenWhiteBackground3_AX
CNF_1
SoftKey_F1
INIT
INITO
REQ
Softkey_IXA
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
AuxFunction2_X1
IN
Aux_IXA
AuxFunction2_X1
AX_OR_2
AX_OR_2
IN1
OUT
IN2
AX_SPLIT_3
AX_SPLIT_3
OUT1
OUT2
OUT3
Q1
Uebung_010f_AX
GreenWhiteBackground1_AX
INIT
INITO

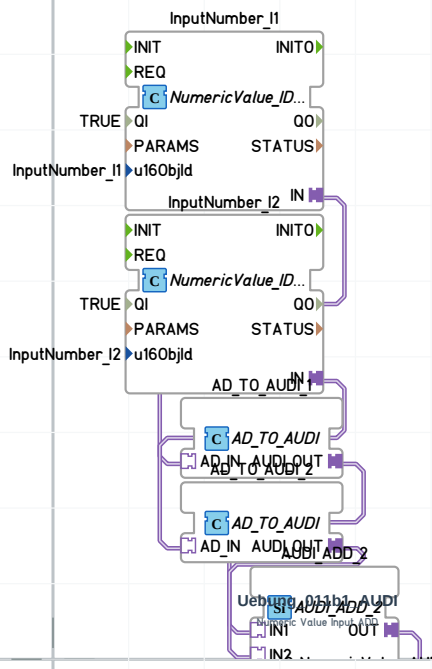

AuxFunction2_X1
INIT
INITO
REQ
Aux_IXA
TRUE
QI
QO
PARAMS
STATUS
AuxFunction2_X1
u16ObjId
IN
AX_SPLIT_2
AX_SPLIT_2
IN
OUT1
OUT2
Q1
GreenWhiteBackground2_AX
INIT
INITO
CNF_1
CNF_2
GreenWhiteBackg...
QI
QO
AuxFunction2_X1
u16ObjIdA
STATUS_1
u8OldColour_...
result_1
STATUS_2
u8OldColour_...
result_2
DI
Uebung_010f_AX
AuxFunction2_X1 auf DigitalOutput_01 mit
GreenWhiteBackground
InputNumber_I1
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
IN
AD_TO_AUDI
AD_TO_AUDI
AD_IN
AUDI_OUT
Uebung_011_AUDI
Numeric Value Input

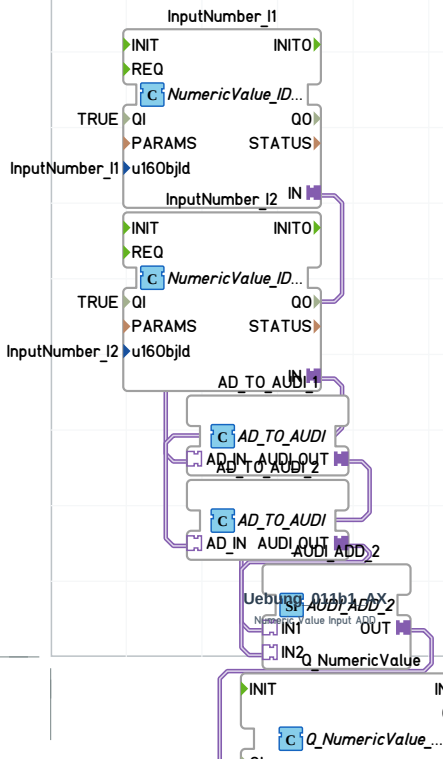

InputNumber_I1
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
IN
AD_TO_AUDI
AD_TO_AUDI
AD_IN
AUDI_OUT
Uebung_011_AX
Numeric Value Input
I1
Q_NumericValue_AUDI
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
AD_TO_AUDI
AD_TO_AUDI
Loslassen
AD_IN
AUDI_OUT
Uebung_011a2_AUDI
Numeric Value Output und BUTTON_PRESS_REPEAT_DONE

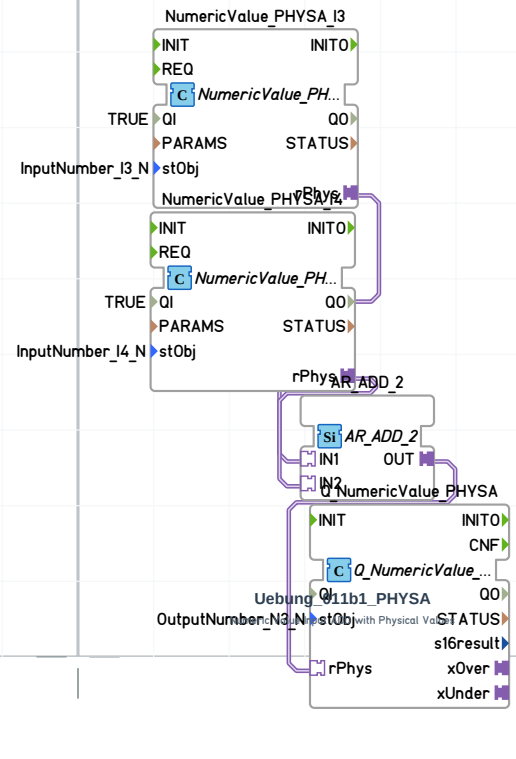

Q_NumericValue
I1
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
F_DWORD_TO_UDINT
AD_TO_AUDI
Loslassen
AD_IN
AUDI_OUT
Uebung_011a2_AX
Numeric Value Output und BUTTON_PRESS_REPEAT_DONE

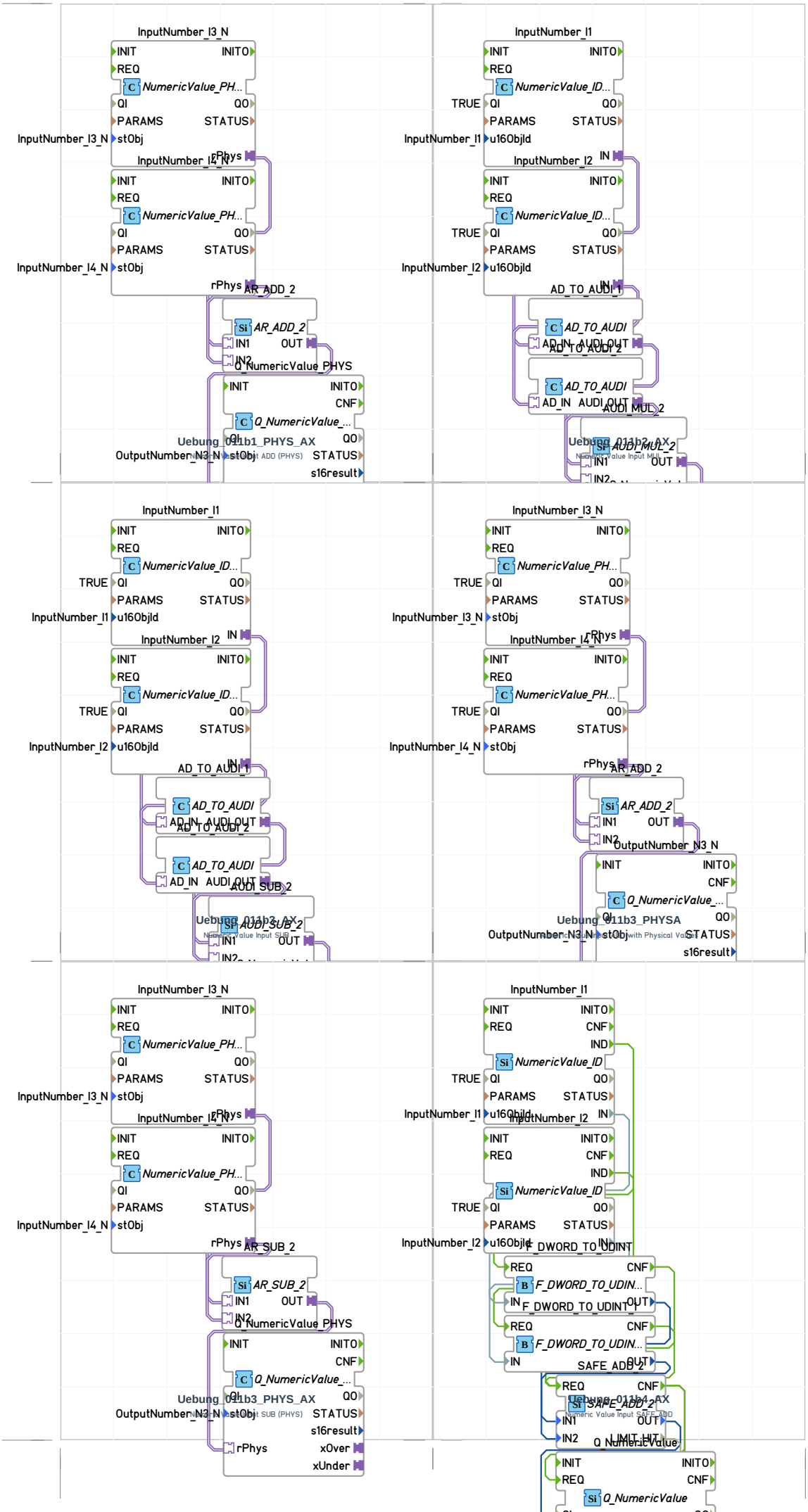

Q_NumericValue_AUDI
I1
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
AB_TO_AUDI
AB_TO_AUDI
AB_IN
AUDI_OUT
Uebung_011a_AUDI
Numeric Value Output und BUTTON_PRESS_REPEAT_DONE

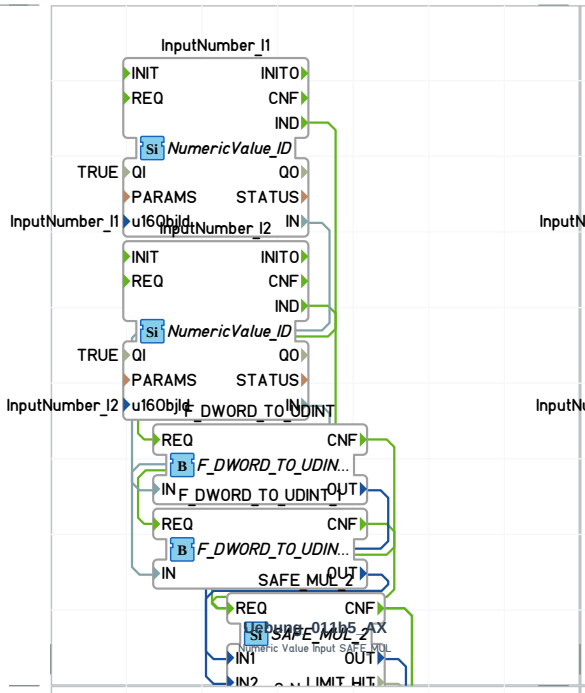

Q_NumericValue
I1
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
F_BYTE_TO_UDINT
AB_TO_AUDI
AB_IN
AUDI_OUT
Uebung_011a_AX
Numeric Value Output und BUTTON_PRESS_REPEAT_DONE

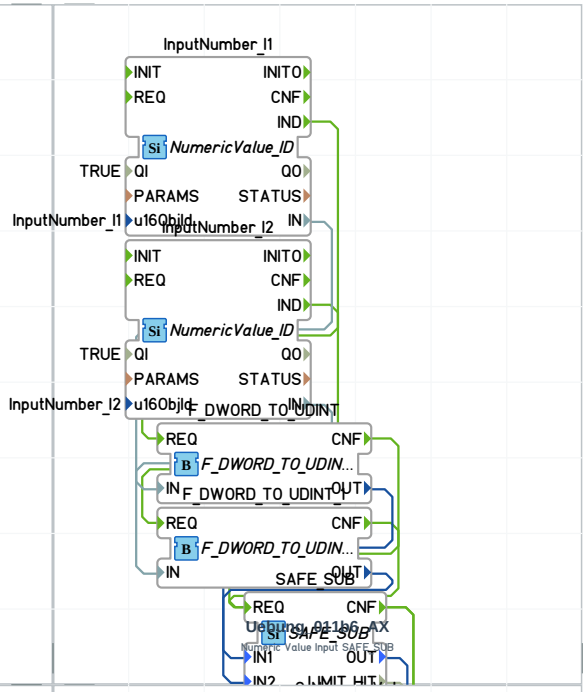

InputNumber_I1
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
InputNumber_I2
IN
AD_TO_AUDI_1
AD_TO_AUDI
AD_IN
AUDI_OUT
AD_TO_AUDI_2
AUDI_ADD_2
AR_ADD_2
IN1
IN2
OUT

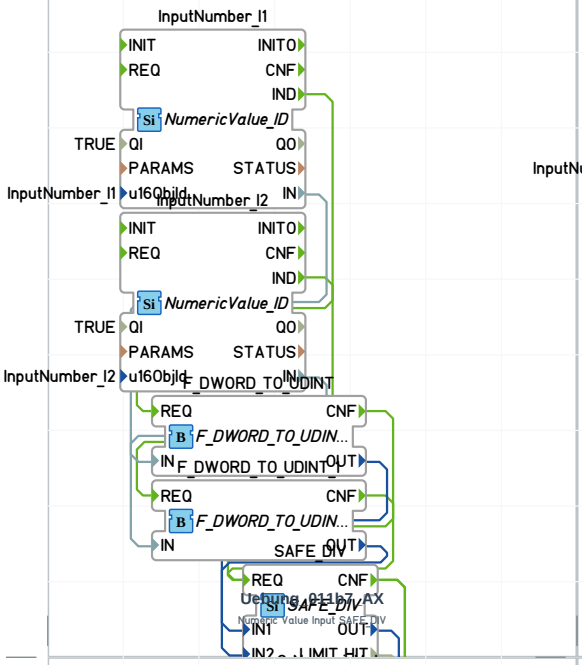

InputNumber_I1
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
InputNumber_I2
IN
AD_TO_AUDI_1
AD_TO_AUDI
AD_IN
AUDI_OUT
AD_TO_AUDI_2
AUDI_ADD_2
AR_ADD_2
IN1
IN2
OUT
Uebung_011b1_AX
Q_NumericValue
INITO
CNF
Q_NumericValue_...

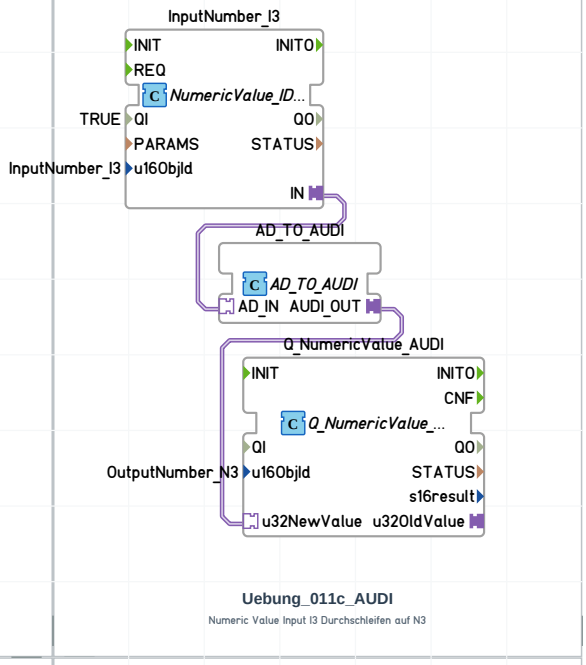

NumericValue_PHYSA_I3
INIT
INITO
REQ
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
NumericValue_PHYSA_I4
InputNumber_I4_N
AR_ADD_2
IN1
IN2
OUT
Q_NumericValue_PHYSA
INITO
CNF
Q_NumericValue_...
QI
QO
STATUS
s16result
rPhys
xOver
xUnder
Uebung_011b1_PHYSA
OutputNumber_N3_N


InputNumber_I3_N
INIT
INITO
REQ
NumericValue_PH...
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
InputNumber_I4_N
INIT
INITO
REQ
NumericValue_PH...
QI
QO
PARAMS
STATUS
InputNumber_I4_N
stObj
rPhys
AR_ADD_2
AR_ADD_2
IN1
OUT
IN2
Q_NumericValue_PHYS
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
Uebung_011b1_PHYS_AX
OutputNumber_N3_N
stObj
STATUS
s16result
InputNumber_I1
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
InputNumber_I2
IN
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I2
u16ObjId
AD_TO_AUDI_1
IN
AD_TO_AUDI
AD_IN
AUDI_OUT
AD_TO_AUDI_2
AD_TO_AUDI
AD_IN
AUDI_OUT
AUDI_MUL_2
Uebung_011b2_AX
AUDI_MUL_2
IN1
OUT
IN2
InputNumber_I1
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
InputNumber_I2
IN
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I2
u16ObjId
AD_TO_AUDI_1
IN
AD_TO_AUDI
AD_IN
AUDI_OUT
AD_TO_AUDI_2
AD_TO_AUDI
AD_IN
AUDI_OUT
AUDI_SUB_2
Uebung_011b2_AX
AUDI_SUB_2
IN1
OUT
IN2
InputNumber_I3_N
INIT
INITO
REQ
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
InputNumber_I4_N
INIT
INITO
REQ
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I4_N
stObj
rPhys
AR_ADD_2
AR_ADD_2
IN1
OUT
IN2
OutputNumber_N3_N
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
Uebung_011b3_PHYSA
OutputNumber_N3_N
stObj
STATUS
s16result
InputNumber_I3_N
INIT
INITO
REQ
NumericValue_PH...
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
InputNumber_I4_N
INIT
INITO
REQ
NumericValue_PH...
QI
QO
PARAMS
STATUS
InputNumber_I4_N
stObj
rPhys
AR_SUB_2
AR_SUB_2
IN1
OUT
IN2
Q_NumericValue_PHYS
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
Uebung_011b3_PHYS_AX
OutputNumber_N3_N
stObj
STATUS
s16result
rPhys
xOver
xUnder
InputNumber_I1
INIT
INITO
REQ
CNF
IND
NumericValue_ID
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
InputNumber_I2
IN
INIT
INITO
REQ
CNF
IND
NumericValue_ID
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I2
u16ObjId
F_DWORD_TO_UDINT
REQ
CNF
F_DWORD_TO_UDIN...
IN
F_DWORD_TO_UDINT
OUT
REQ
CNF
F_DWORD_TO_UDIN...
IN
SAFE_ADD_2
OUT
REQ
CNF
SAFE_ADD_2
IN1
OUT
IN2
LIMIT_HIT
Q_NumericValue
INIT
INITO
REQ
CNF
Q_NumericValue



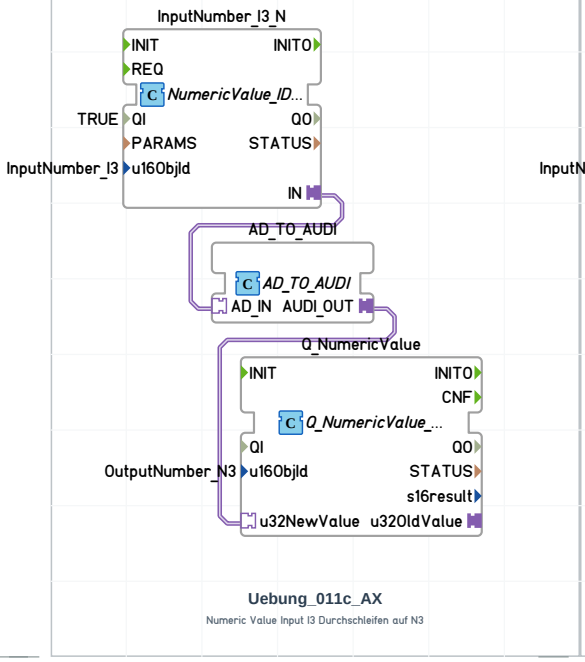






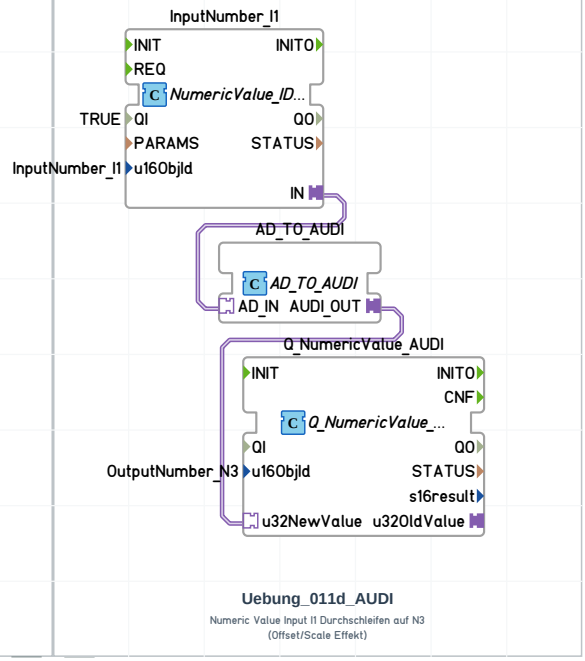
Uebung_011c_AUDI

Numeric Value Input I3 Durchschleifen auf N3



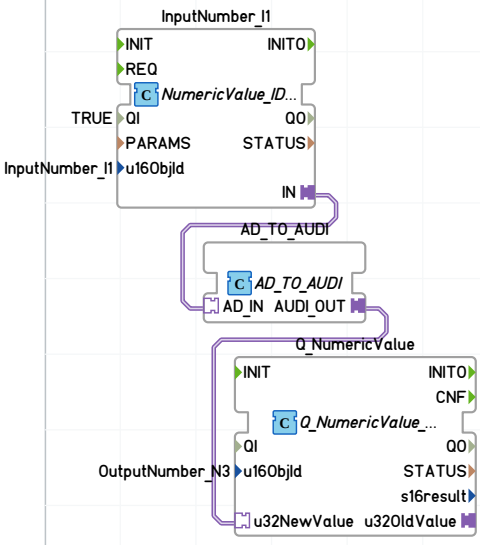
Uebung_011c_AX

Numeric Value Input I3 Durchschleifen auf N3



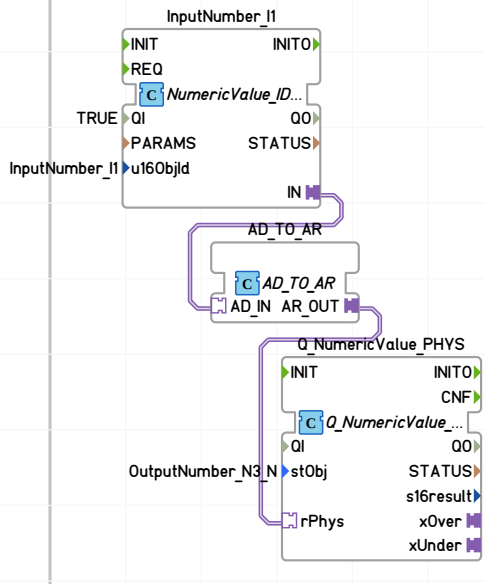
Uebung_011d_AUDI

Numeric Value Input I1 Durchschleifen auf N3
(Offset/Scale Effekt)



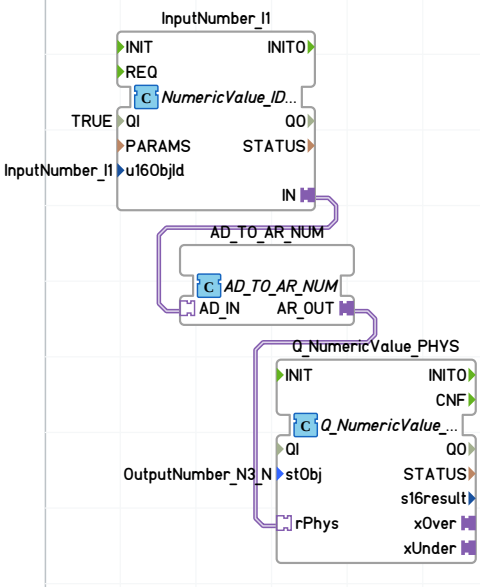
Uebung_011d_AX

Numeric Value Input I1 Durchschleifen auf N3 (Offset/Scale Effekt)



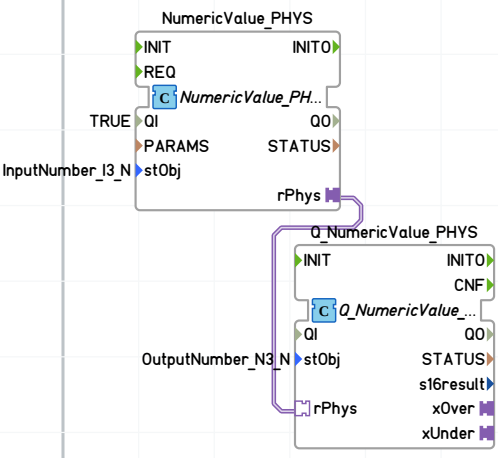
Uebung_011e_MIXA

Numeric Value Input I1 Durchschleifen auf N3 (Software Scale via NumericObjectPool_S) falsch gemischt!



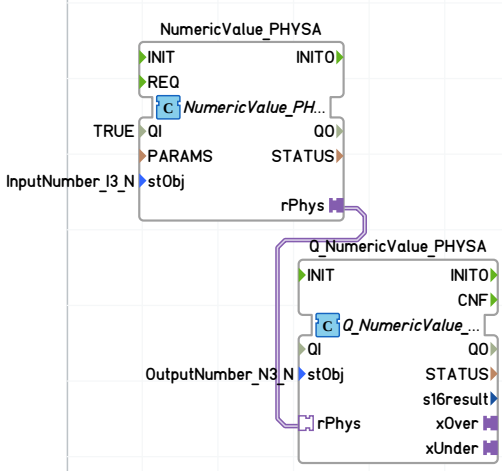
Uebung_011e_MIX_AX

Numeric Value Input I1 Durchschleifen auf N3 (Software Scale via NumericObjectPool_S) falsch gemischt!



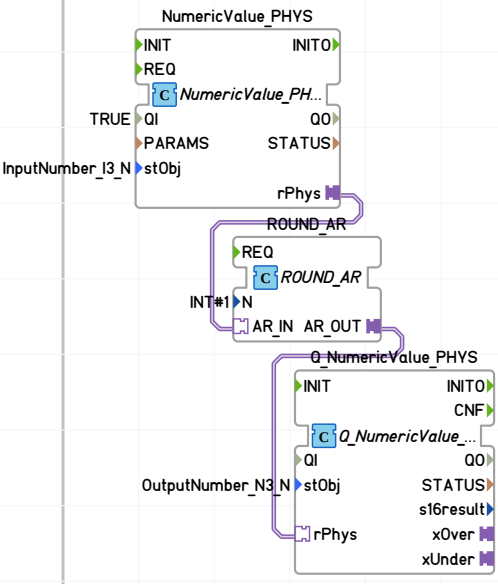
Uebung_011f_AX

Numeric Value Input I3 Durchschleifen auf N3 (Input und Output PHYS via NumericObjectPool_S)



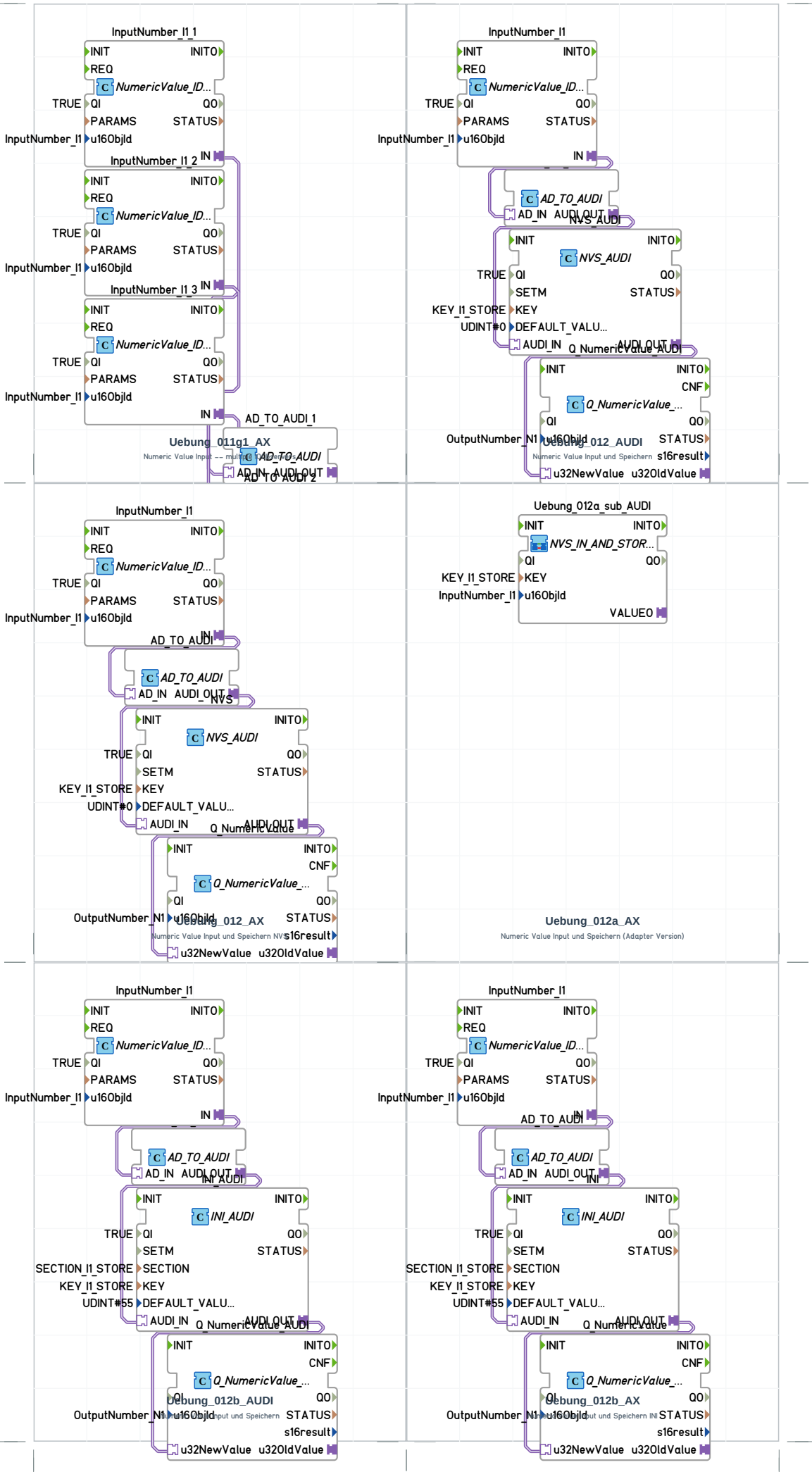
Uebung_011f_PHYSA

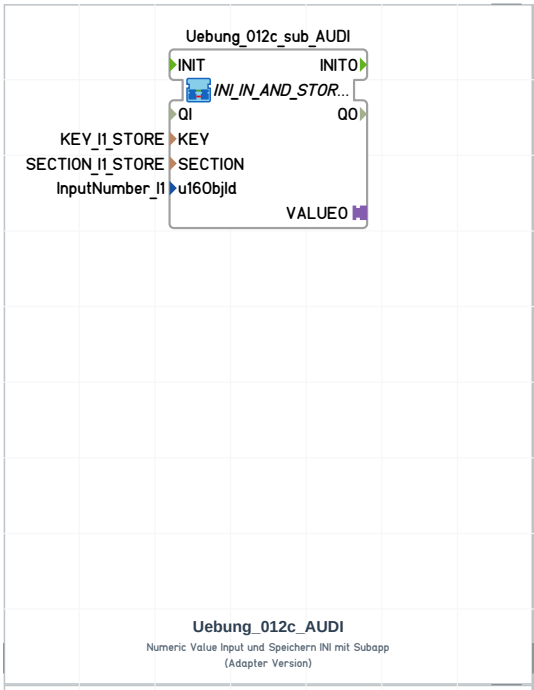
Numeric Value Input I1 Durchschleifen auf N3 mit PHYSA (Physikalische Werte)



Uebung_011f_ROUND_AR_AX

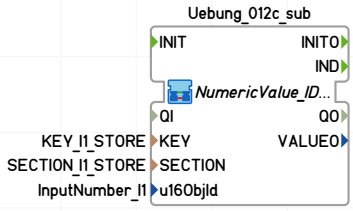
Wie Uebung_011f_AX, aber mit OSCAT ROUND_AR dazwischen (rundet auf N Nachkommastellen)


InputNumber_I1_1
InputNumber_I1_2
InputNumber_I1_3
NumericValue_ID...
INIT
INITO
REQ
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
IN
AD_TO_AUDI_1
AD_TO_AUDI_2
Uebung_011g1_AX
Numeric Value Input -- multiple Instances
AD_TO_AUDI
AD_IN
AUDI_OUT
InputNumber_I1
NVS_AUDI
SETM
KEY
KEY_I1_STORE
UDINT#0
DEFAULT_VALU...
AUDI_IN
AUDI_OUT
Q_NumericValue_...
CNF
OutputNumber_N1
Uebung_012_AUDI
Numeric Value Input und Speichern
s16result
u32NewValue
u32OldValue
Uebung_012a_sub_AUDI
NVS_IN_AND_STOR...
VALUE0
Uebung_012a_AX
Numeric Value Input und Speichern (Adapter Version)
INI_AUDI
SECTION
SECTION_I1_STORE
UDINT#55
Uebung_012b_AUDI
Uebung_012b_AX



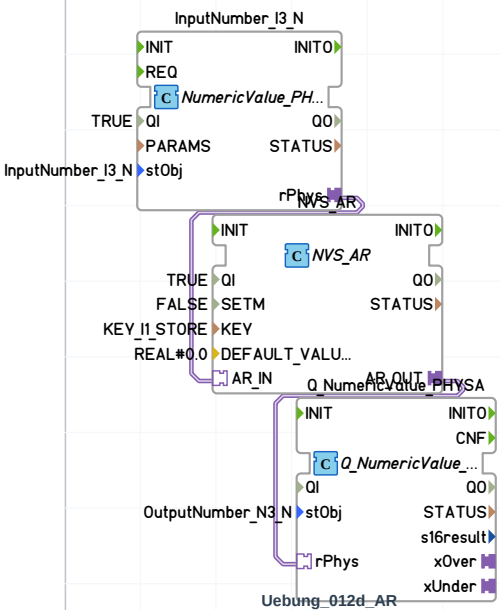
Uebung_012c_AUDI

Numeric Value Input und Speichern INI mit Subapp (Adapter Version)



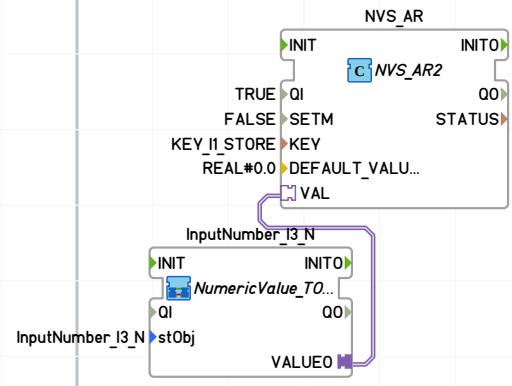
Uebung_012c_AX

Numeric Value Input und Speichern INI mit Subapp



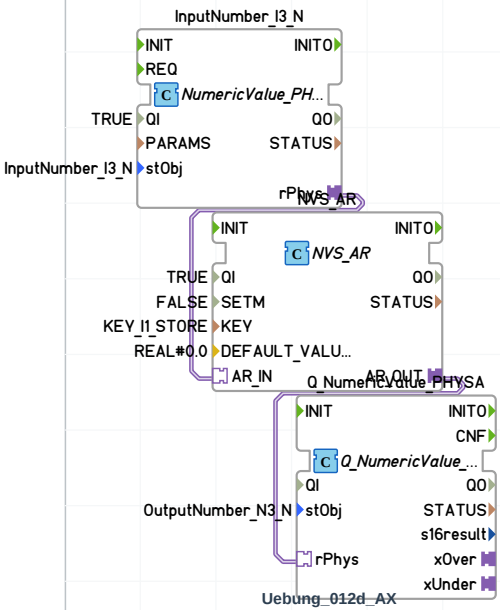
Uebung_012d_AR

Numeric Value Input PHYS und Speichern NVS (AR Adapter, plain)



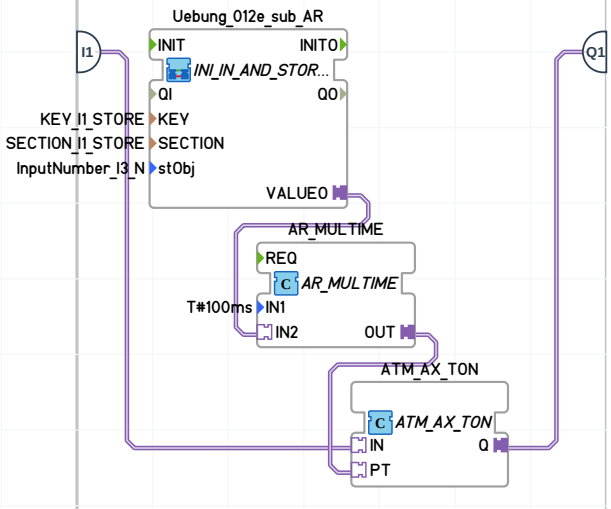
Uebung_012d_AR2

Numeric Value Input PHYS und Speichern NVS via AR2 (bidirektional) – NumericValue_TO_AR2 direkt an ein NVS_AR2 angeschlossen



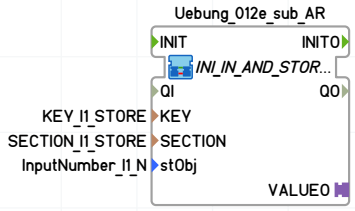
Uebung_012d_AX

Numeric Value Input PHYS und Speichern NVS



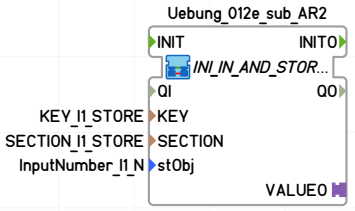
Uebung_012e2_AR

Numeric Value Input PHYS und Speichern INI mit Subapp (AR Adapter)



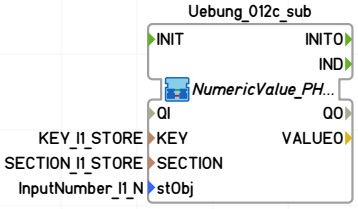
Uebung_012e_AR

Numeric Value Input PHYS und Speichern INI mit Subapp (AR Adapter)



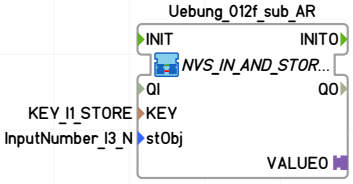
Uebung_012e_AR2

Numeric Value Input PHYS und Speichern INI mit Subapp (AR2 Adapter, echter in sich geschlossener Baustein)



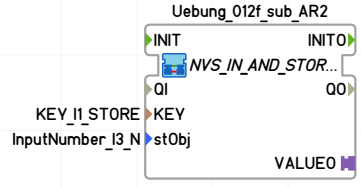
Uebung_012e_AX

Numeric Value Input PHYS und Speichern INI mit Subapp



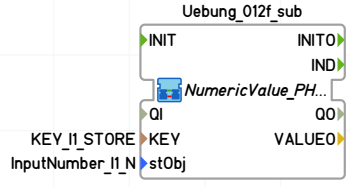
Uebung_012f_AR

Numeric Value Input PHYS und Speichern NVS mit Subapp (AR Adapter)



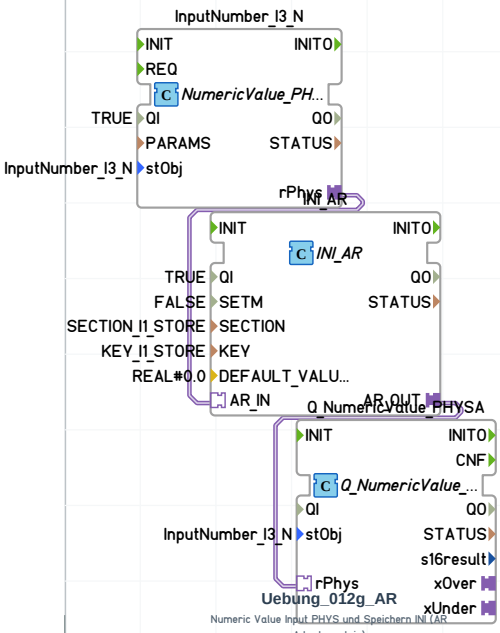
Uebung_012f_AR2

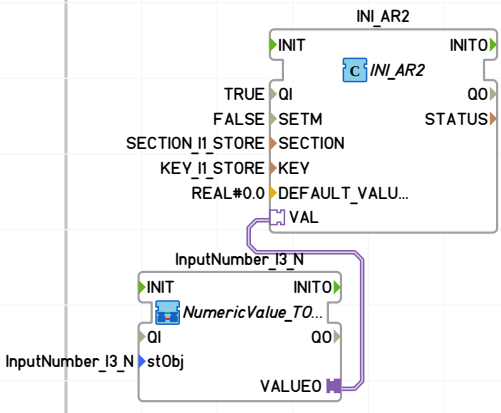
Numeric Value Input PHYS und Speichern NVS mit Subapp (AR2 Adapter, echter in sich geschlossener Baustein)

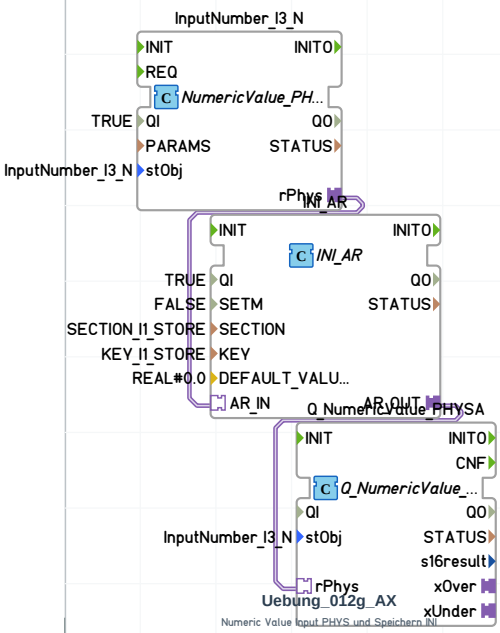


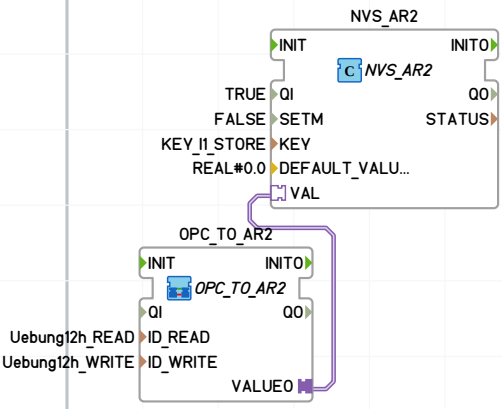
Uebung_012f_AX

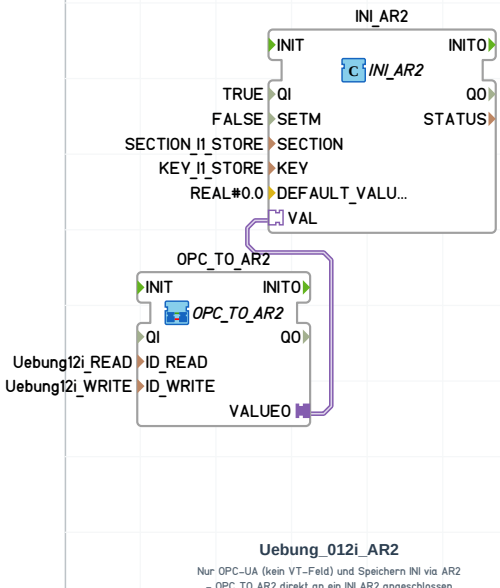
Numeric Value Input PHYS und Speichern NVS mit Subapp

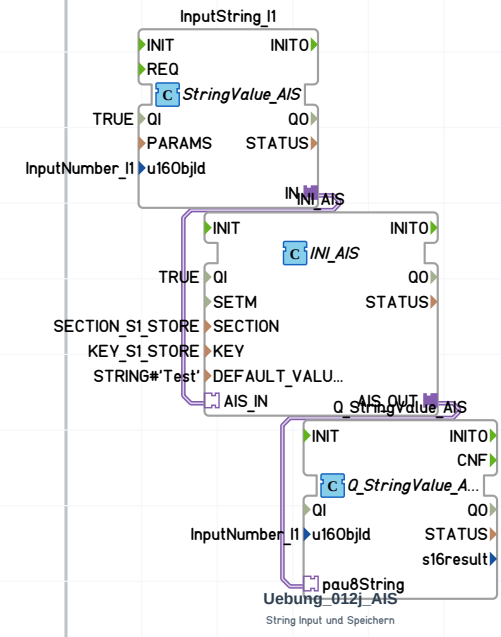

InputNumber_I3_N
INIT
INITO
REQ
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
INI_AR
INIT
INITO
INI_AR
TRUE
QI
QO
FALSE
SETM
STATUS
SECTION_I1_STORE
SECTION
KEY_I1_STORE
KEY
REAL#0.0
DEFAULT_VALU...
AR_IN
AR_OUT
Q_NumericValue_PHYSA
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
InputNumber_I3_N
stObj
STATUS
s16result
rPhys
xOver
xUnder
Uebung_012g_AR
Numeric Value Input PHYS und Speichern INI (AR Adapter, plain)

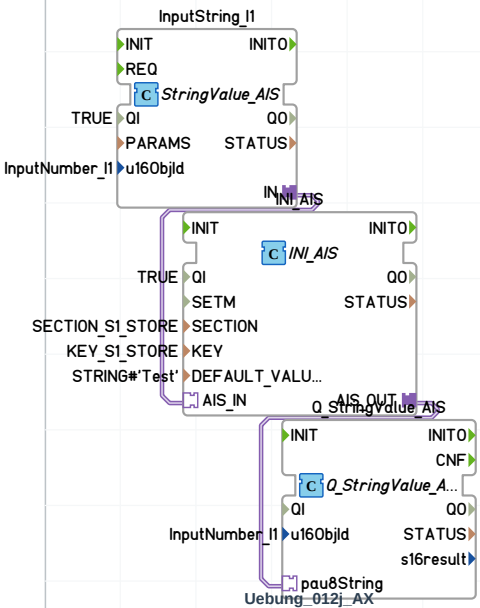

INI_AR2
INIT
INITO
INI_AR2
TRUE
QI
QO
FALSE
SETM
STATUS
SECTION_I1_STORE
SECTION
KEY_I1_STORE
KEY
REAL#0.0
DEFAULT_VALU...
VAL
InputNumber_I3_N
INIT
INITO
NumericValue_TO...
QI
QO
InputNumber_I3_N
stObj
VALUEO
Uebung_012g_AR2
Numeric Value Input PHYS und Speichern INI via AR2 (bidirektional) – NumericValue_TO_AR2 direkt an ein INI_AR2 angeschlossen


InputNumber_I3_N
INIT
INITO
REQ
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
INI_AR
INIT
INITO
INI_AR
TRUE
QI
QO
FALSE
SETM
STATUS
SECTION_I1_STORE
SECTION
KEY_I1_STORE
KEY
REAL#0.0
DEFAULT_VALU...
AR_IN
AR_OUT
Q_NumericValue_PHYSA
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
InputNumber_I3_N
stObj
STATUS
s16result
rPhys
xOver
xUnder
Uebung_012g_AX
Numeric Value Input PHYS und Speichern INI


NVS_AR2
INIT
INITO
NVS_AR2
TRUE
QI
QO
FALSE
SETM
STATUS
KEY_I1_STORE
KEY
REAL#0.0
DEFAULT_VALU...
VAL
OPC_TO_AR2
INIT
INITO
OPC_TO_AR2
QI
QO
Uebung12h_READ
ID_READ
Uebung12h_WRITE
ID_WRITE
VALUEO
Uebung_012h_AR2
Nur OPC-UA (kein VT-Feld) und Speichern NVS via AR2 – OPC_TO_AR2 direkt an ein NVS_AR2 angeschlossen


INI_AR2
INIT
INITO
INI_AR2
TRUE
QI
QO
FALSE
SETM
STATUS
SECTION_I1_STORE
SECTION
KEY_I1_STORE
KEY
REAL#0.0
DEFAULT_VALU...
VAL
OPC_TO_AR2
INIT
INITO
OPC_TO_AR2
QI
QO
Uebung12i_READ
ID_READ
Uebung12i_WRITE
ID_WRITE
VALUEO
Uebung_012i_AR2
Nur OPC-UA (kein VT-Feld) und Speichern INI via AR2 – OPC_TO_AR2 direkt an ein INI_AR2 angeschlossen


InputString_I1
INIT
INITO
REQ
StringValue_AIS
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I1
u16ObjId
IN
INI_AIS
INIT
INITO
INI_AIS
TRUE
QI
QO
SETM
STATUS
SECTION_S1_STORE
SECTION
KEY_S1_STORE
KEY
STRING#'Test'
DEFAULT_VALU...
AIS_IN
AIS_OUT
Q_StringValue_AIS
INIT
INITO
CNF
Q_StringValue_A...
QI
QO
InputNumber_I1
u16ObjId
STATUS
s16result
pau8String
Uebung_012j_AIS
String Input und Speichern



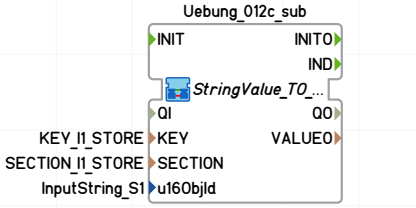
Uebung_012j_AX

String Input und Speichern INI



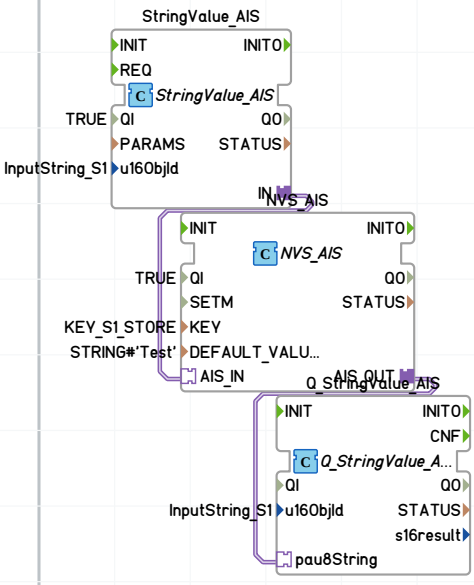
Uebung_012k_AIS

String Input und Speichern INI mit Subapp (AIS Adapter)



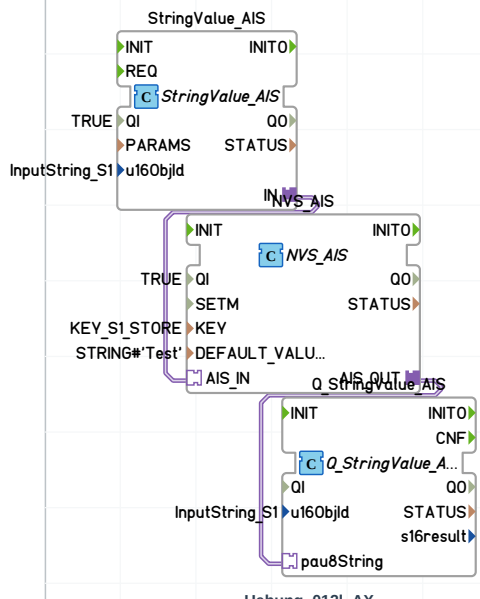
Uebung_012k_AX

String Input und Speichern INI mit Subapp



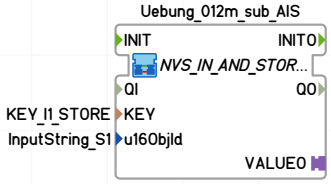
Uebung_012l_AIS

String Input und Speichern NVS (AIS Adapter, plain)



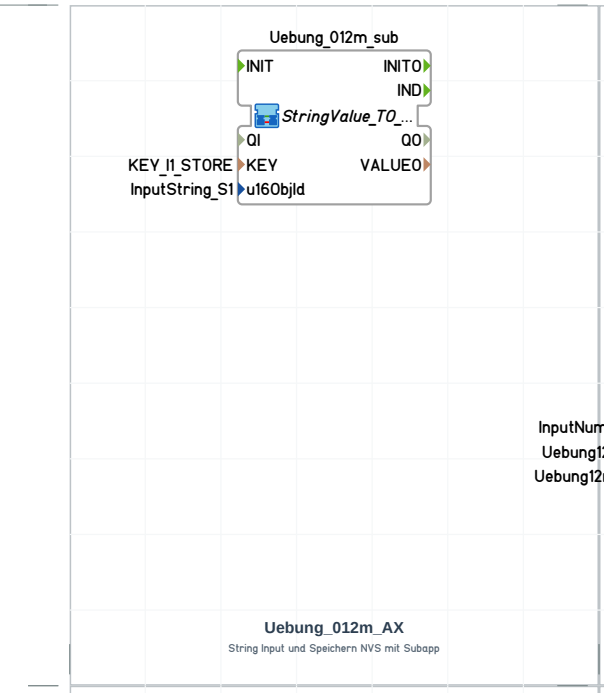
Uebung_012l_AX

String Input und Speichern NVS



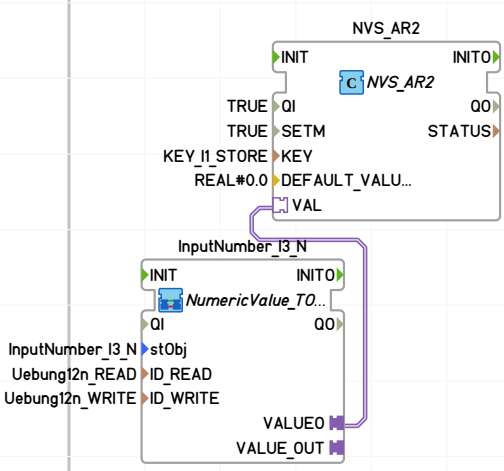
Uebung_012m_AIS

String Input und Speichern NVS mit Subapp (AIS Adapter)



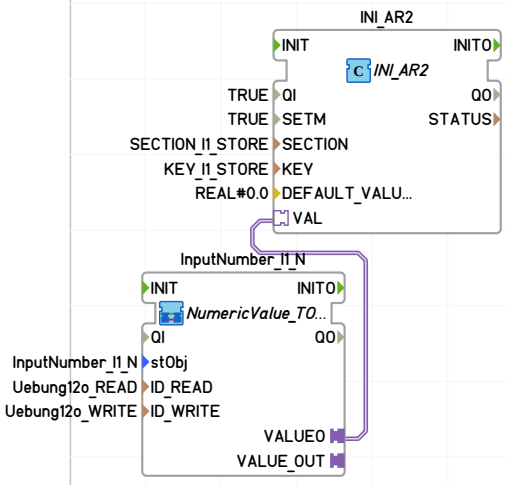
Uebung_012m_AX

String Input und Speichern NVS mit Subapp



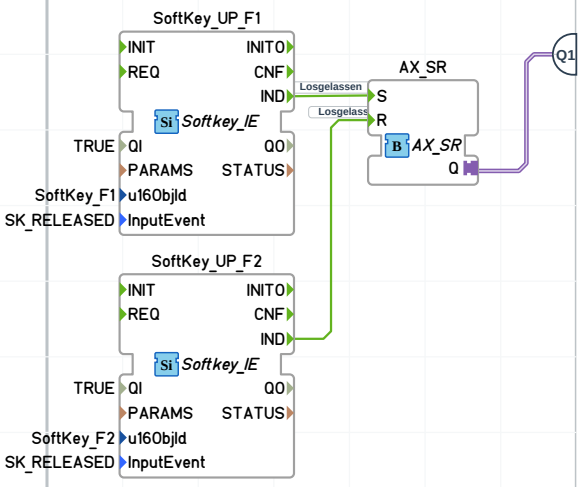
Uebung_012n_AR2

VT-Feld UND OPC-UA (letzter Schreiber gewinnt) und Speichern NVS via AR2 – NumericValue_TO_AR2_OPC direkt an ein NVS_AR2 angeschlossen



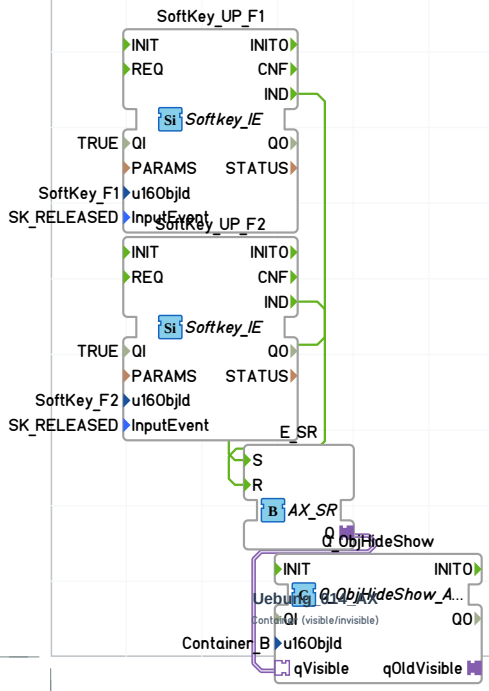
Uebung_012o_AR2

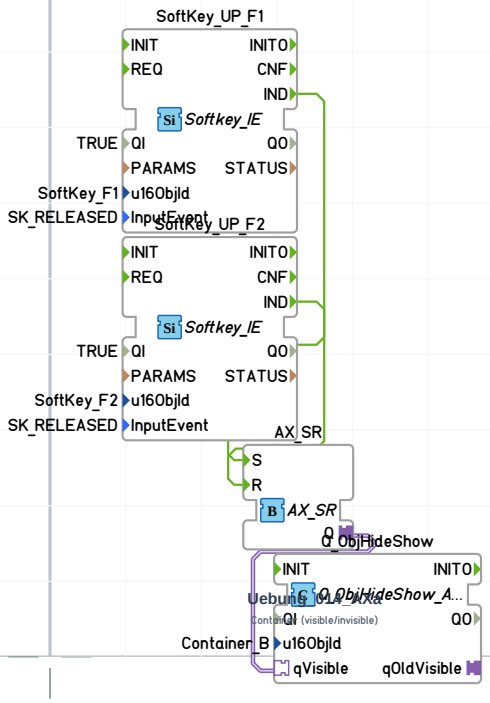
VT-Feld UND OPC-UA (letzter Schreiber gewinnt) und Speichern INI via AR2 – NumericValue_TO_AR2_OPC direkt an ein INI_AR2 angeschlossen

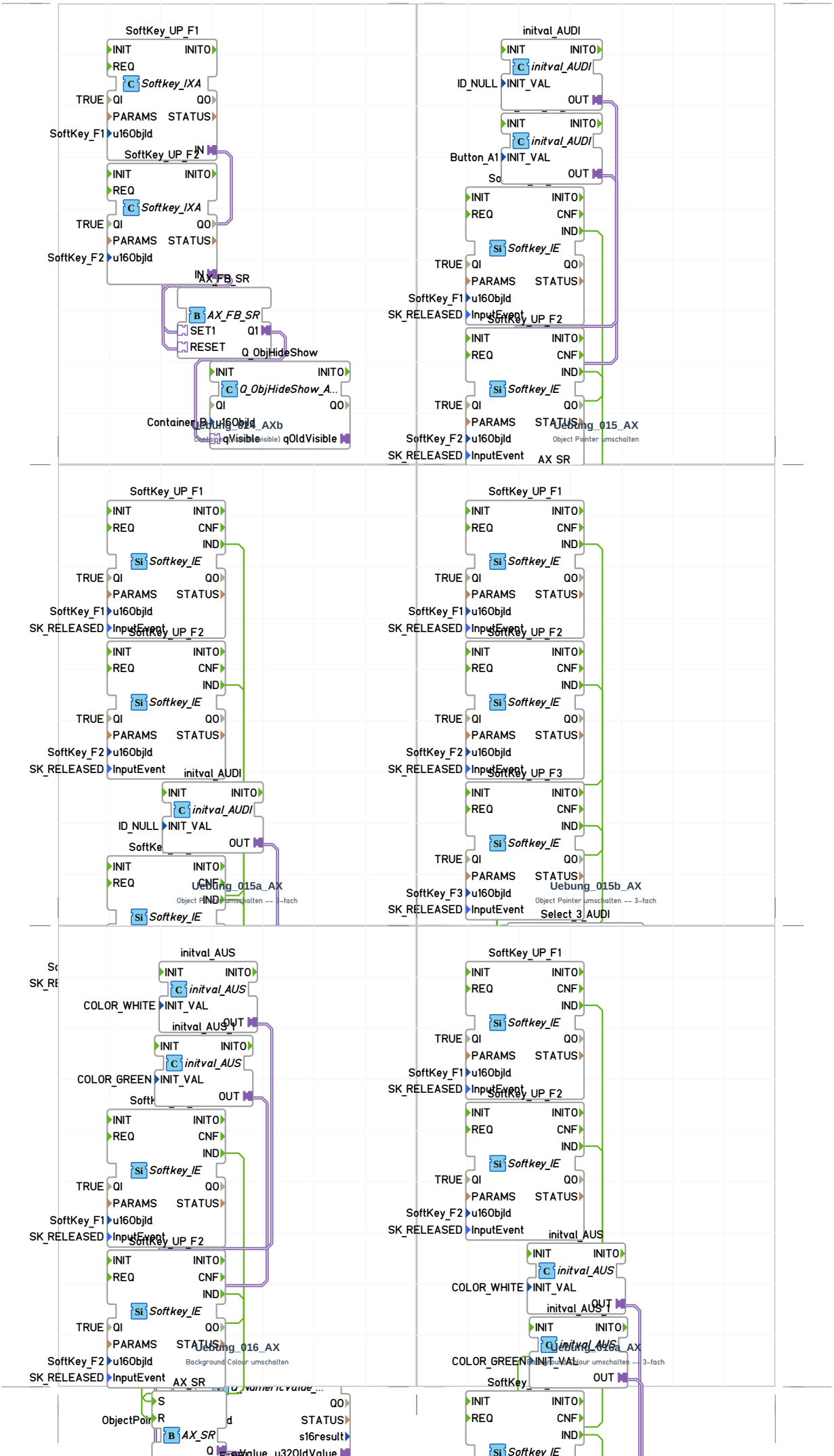


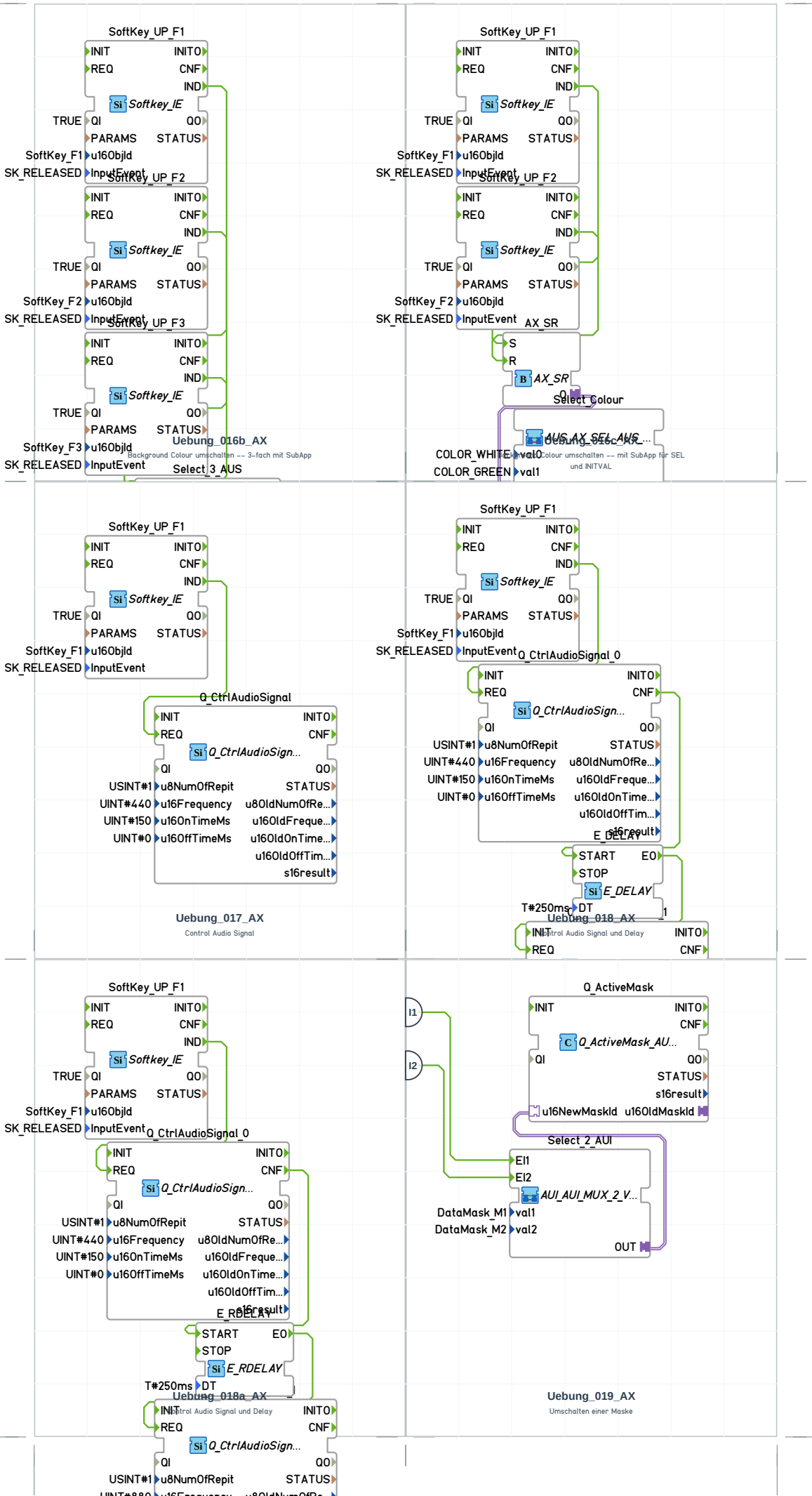
Uebung_013_AX

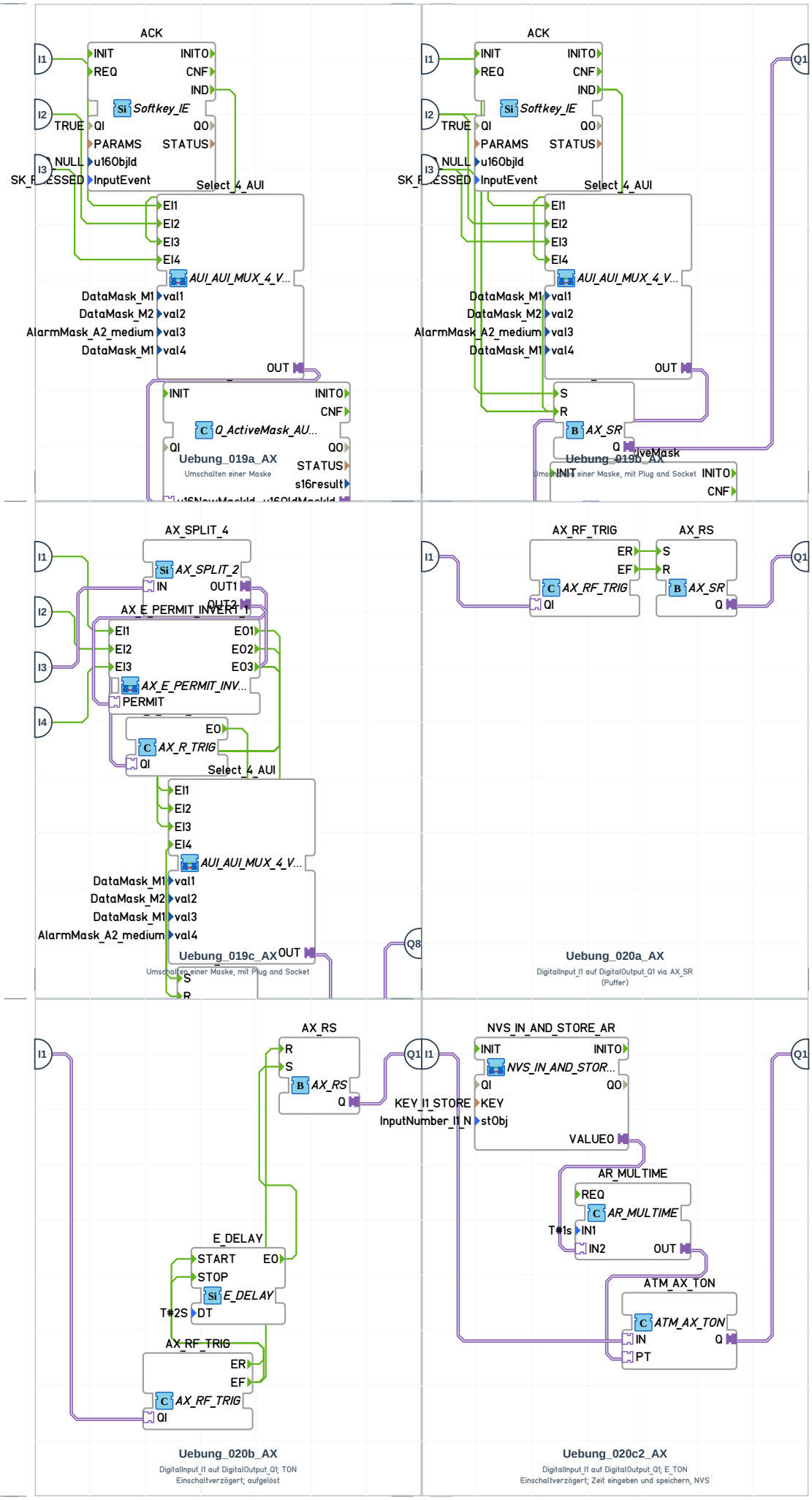
SR-Flip-Flop (Softkey)

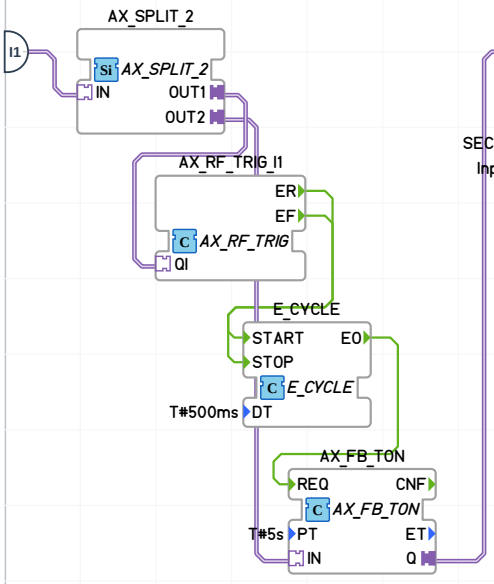





SoftKey_UP_F1
INIT
INITO
REQ
Softkey_IXA
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
IN
SoftKey_UP_F2
SoftKey_F2
AX_FB_SR
AX_FB_SR
SET1
Q1
RESET
Q_ObjHideShow
Q_ObjHideShow_A...
qVisible
qOldVisible
initval_AUDI
initval_AUDI
ID_NULL
INIT_VAL
OUT
Button_A1
Softkey_IE
SK_RELEASED
InputEvent
Uebung_015_AX
Object Pointer umschalten
AX_SR
Uebung_015a_AX
Uebung_015b_AX
Select_3_AUDI
SoftKey_F3
SoftKey_UP_F3
initval_AUS
COLOR_WHITE
COLOR_GREEN
Uebung_016_AX
Background Colour umschalten
ObjectPointer
S
R
AX_SR
STATUS
s16result

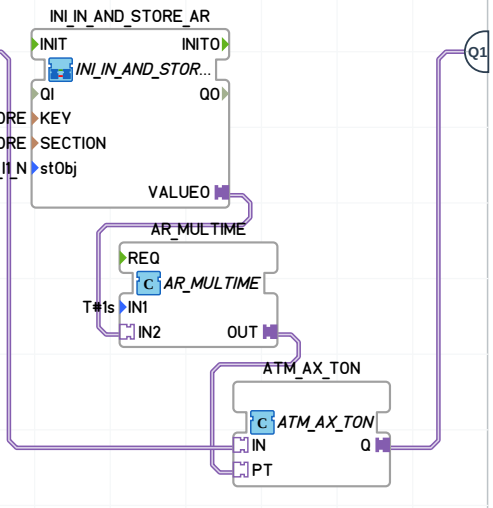

SoftKey_UP_F1
INIT
INITO
REQ
CNF
IND
Softkey_IE
TRUE
QI
QO
PARAMS
STATUS
SoftKey_F1
u16ObjId
SK_RELEASED
InputEvent
SoftKey_UP_F2
SoftKey_F2
SoftKey_UP_F3
SoftKey_F3
Uebung_016b_AX
Background Colour umschalten -- 3-fach mit SubApp
Select_3_AUS
AX_SR
S
R
Q
Select_Colour
COLOR_WHITE
COLOR_GREEN
val0
val1
Uebung_016c_AX
Background Colour umschalten -- mit SubApp für SEL und INITVAL
Q_CtrlAudioSignal
Q_CtrlAudioSign...
USINT#1
u8NumOfRepit
UINT#440
u16Frequency
UINT#150
u16OnTimeMs
UINT#0
u16OffTimeMs
STATUS
u80ldNumOfRe...
u16OldFreque...
u16OldOnTime...
u16OldOffTim...
s16result
Uebung_017_AX
Control Audio Signal
Q_CtrlAudioSignal_0
E_DELAY
START
STOP
EO
T#250ms
DT
Uebung_018_AX
Control Audio Signal und Delay
E_RDELAY
Uebung_018a_AX
Q_ActiveMask
Q_ActiveMask_AU...
u16NewMaskId
u16OldMaskId
Select_2_AUI
EI1
EI2
AUI_AUI_MUX_2_V...
DataMask_M1
DataMask_M2
val1
val2
OUT
Uebung_019_AX
Umschalten einer Maske


ACK
INIT
REQ
INITO
CNF
IND
Softkey_IE
QI
QO
PARAMS
STATUS
u16ObjId
InputEvent
TRUE
NULL
Select_4_AUI
EI1
EI2
EI3
EI4
AUI_AUI_MUX_4_V...
DataMask_M1
DataMask_M2
AlarmMask_A2_medium
val1
val2
val3
val4
OUT
Q_ActiveMask_AU...
STATUS
s16result
Uebung_019a_AX
Umschalten einer Maske
S
R
AX_SR
Q
Uebung_019b_AX
Umschalten einer Maske, mit Plug and Socket
AX_SPLIT_4
AX_SPLIT_2
IN
OUT1
OUT2
AX_E_PERMIT_INVERT_1
EO1
EO2
EO3
AX_E_PERMIT_INV...
PERMIT
EO
AX_R_TRIG
Uebung_019c_AX
Umschalten einer Maske, mit Plug and Socket
AX_RF_TRIG
ER
EF
Uebung_020a_AX
DigitalInput_I1 auf DigitalOutput_Q1 via AX_SR
(Puffer)
AX_RS
E_DELAY
START
STOP
E_DELAY
T#2S
DT
Uebung_020b_AX
DigitalInput_I1 auf DigitalOutput_Q1; TON
Einschaltverzögert; aufgelöst
NVS_IN_AND_STORE_AR
NVS_IN_AND_STOR...
KEY_I1_STORE
InputNumber_I1_N
KEY
stObj
VALUEO
AR_MULTIME
AR_MULTIME
T#1s
IN1
IN2
ATM_AX_TON
ATM_AX_TON
PT
Uebung_020c2_AX
DigitalInput_I1 auf DigitalOutput_Q1; E_TON
Einschaltverzögert; Zeit eingeben und speichern, NVS



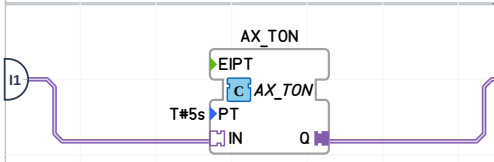
Uebung_020c3_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_FB_TON; Einschaltverzögert



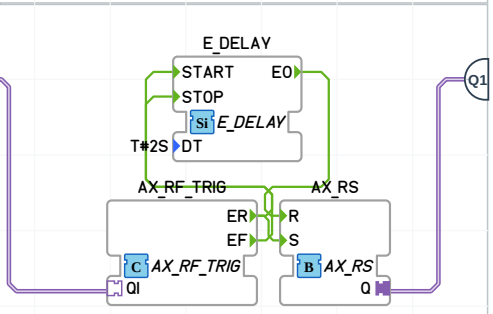
Uebung_020c4_AX

DigitalInput_I1 auf DigitalOutput_Q1; E_TON Einschaltverzögert; Zeit eingeben und speichern, INI



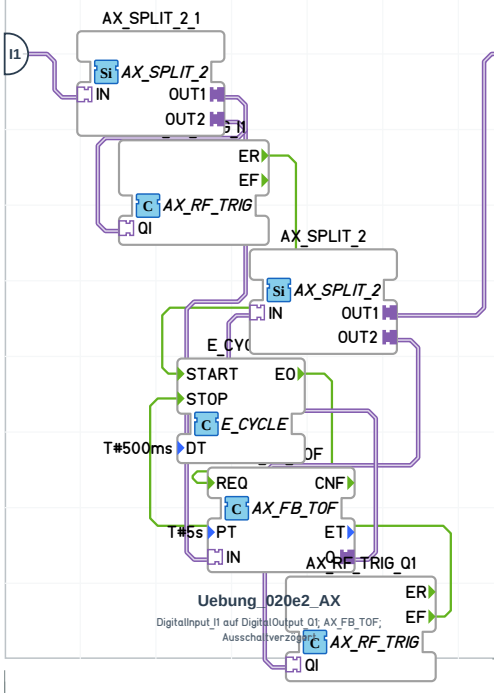
Uebung_020c_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TON; Einschaltverzögert



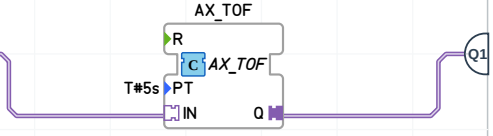
Uebung_020d_AX

DigitalInput_I1 auf DigitalOutput_Q1; TOF Ausschaltverzögert; aufgelöst



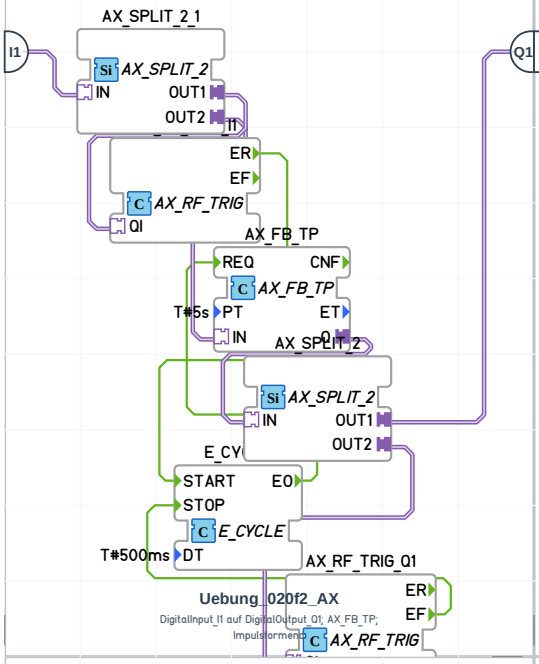
Uebung_020e2_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_FB_TOF; Ausschaltverzögert



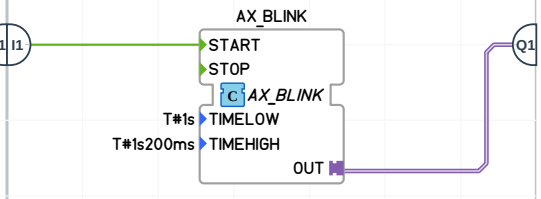
Uebung_020e_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TOF; Ausschaltverzögert



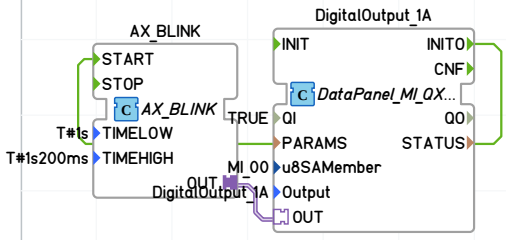
Uebung_020f2_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_FB_TP; Impulsformend



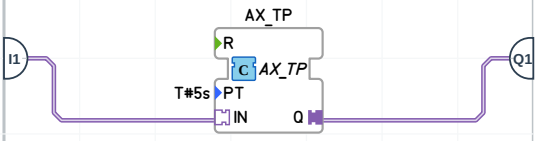
Uebung_020f3_AX

DigitalInput_I1 auf DigitalOutput_Q1; BLINKER



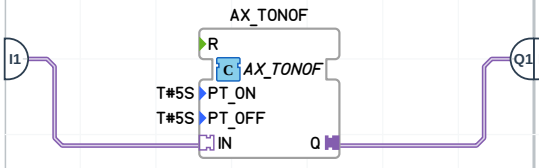
Uebung_020f4_AX

DigitalInput_I1 auf DataPanel_1A; BLINKER



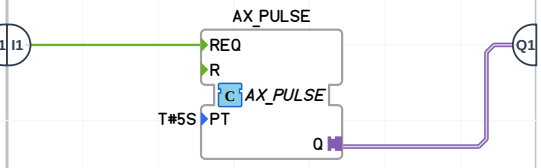
Uebung_020f_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TP; Impulsformend



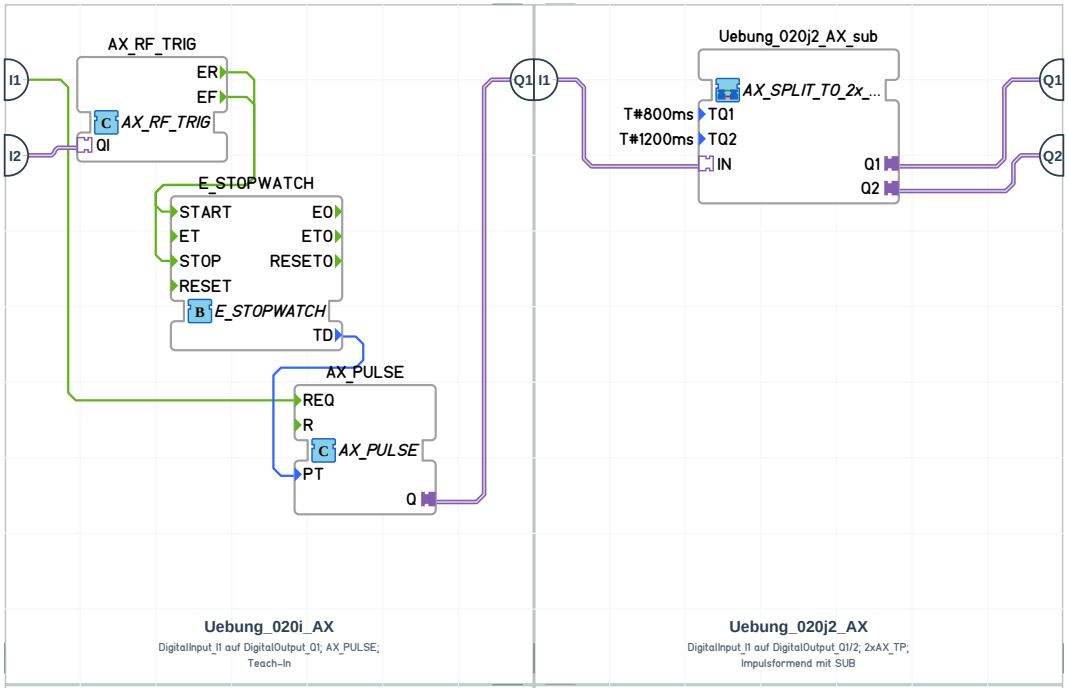
Uebung_020g_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_TONOF; Ein-/Ausschaltverzögert



Uebung_020h_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_PULSE; Impulsformend

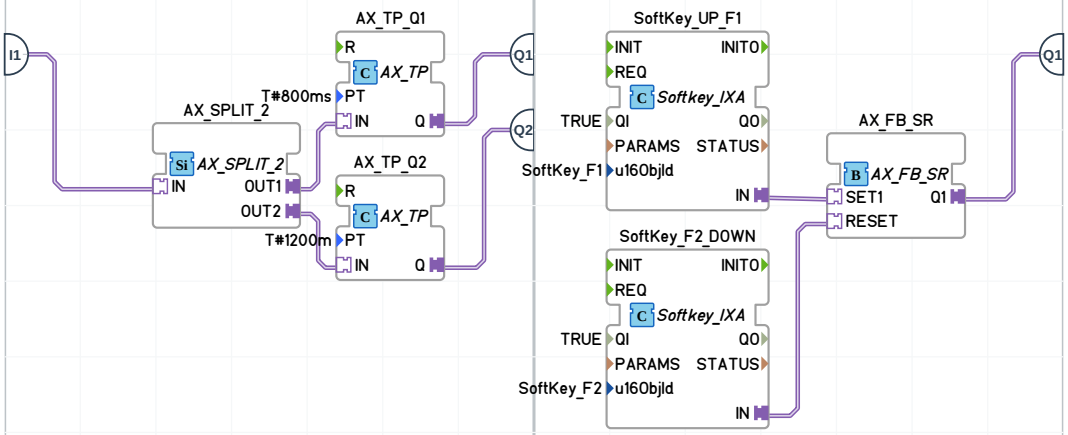


Uebung_020i_AX

DigitalInput_I1 auf DigitalOutput_Q1; AX_PULSE; Teach-In

Uebung_020j2_AX

DigitalInput_I1 auf DigitalOutput_Q1/2; 2xAX_TP; Impulsformend mit SUB

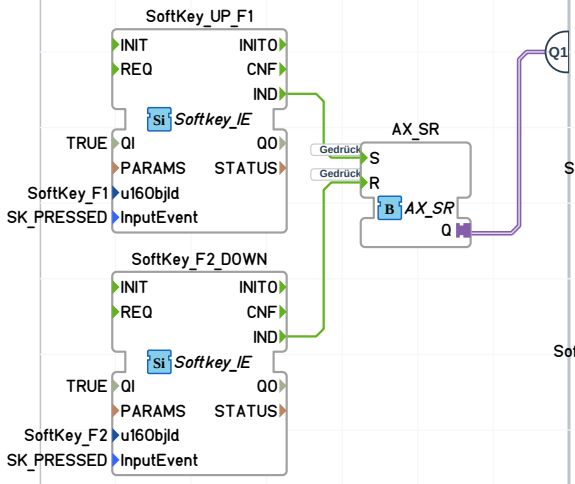


Uebung_020j_AX

DigitalInput_I1 auf DigitalOutput_Q1/2; 2xAX_TP; Impulsformend

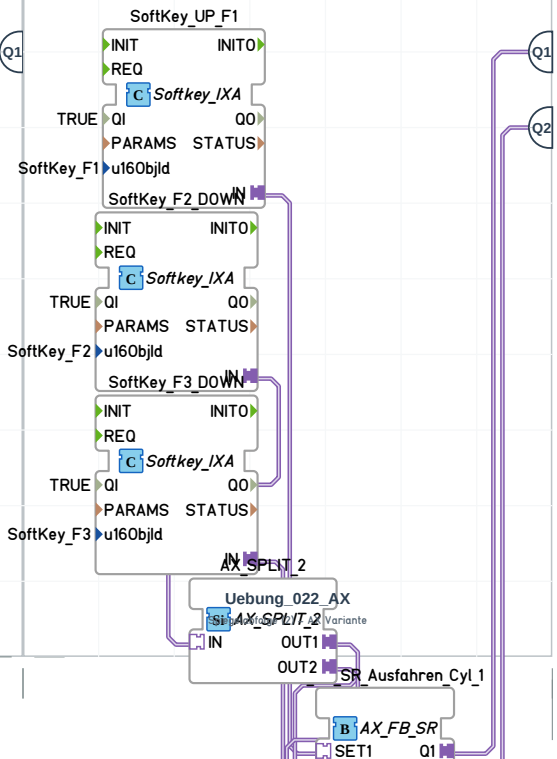
Uebung_021_AX

Spiegelabfolge (1) – AX Variante



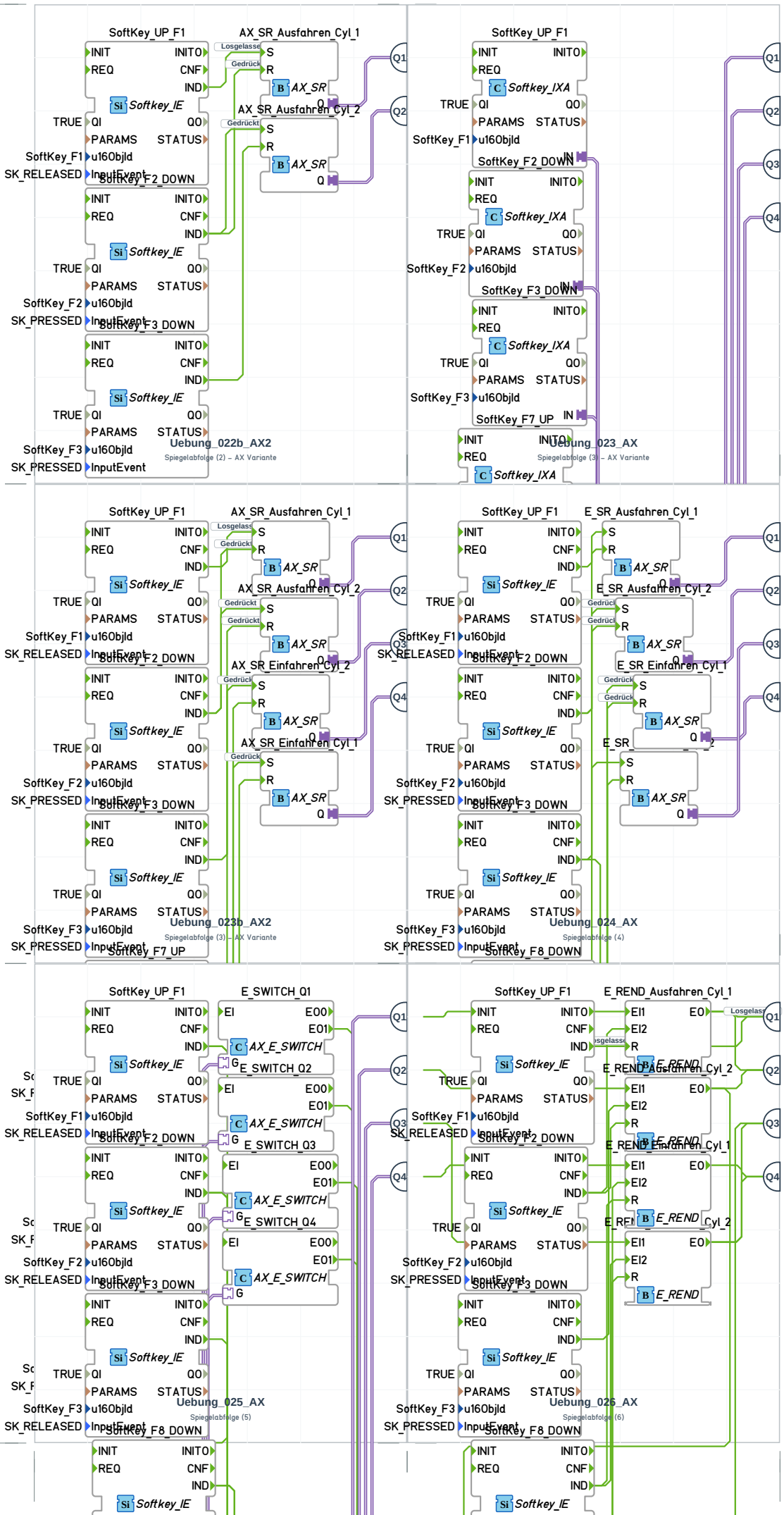
Uebung_021b_AX2

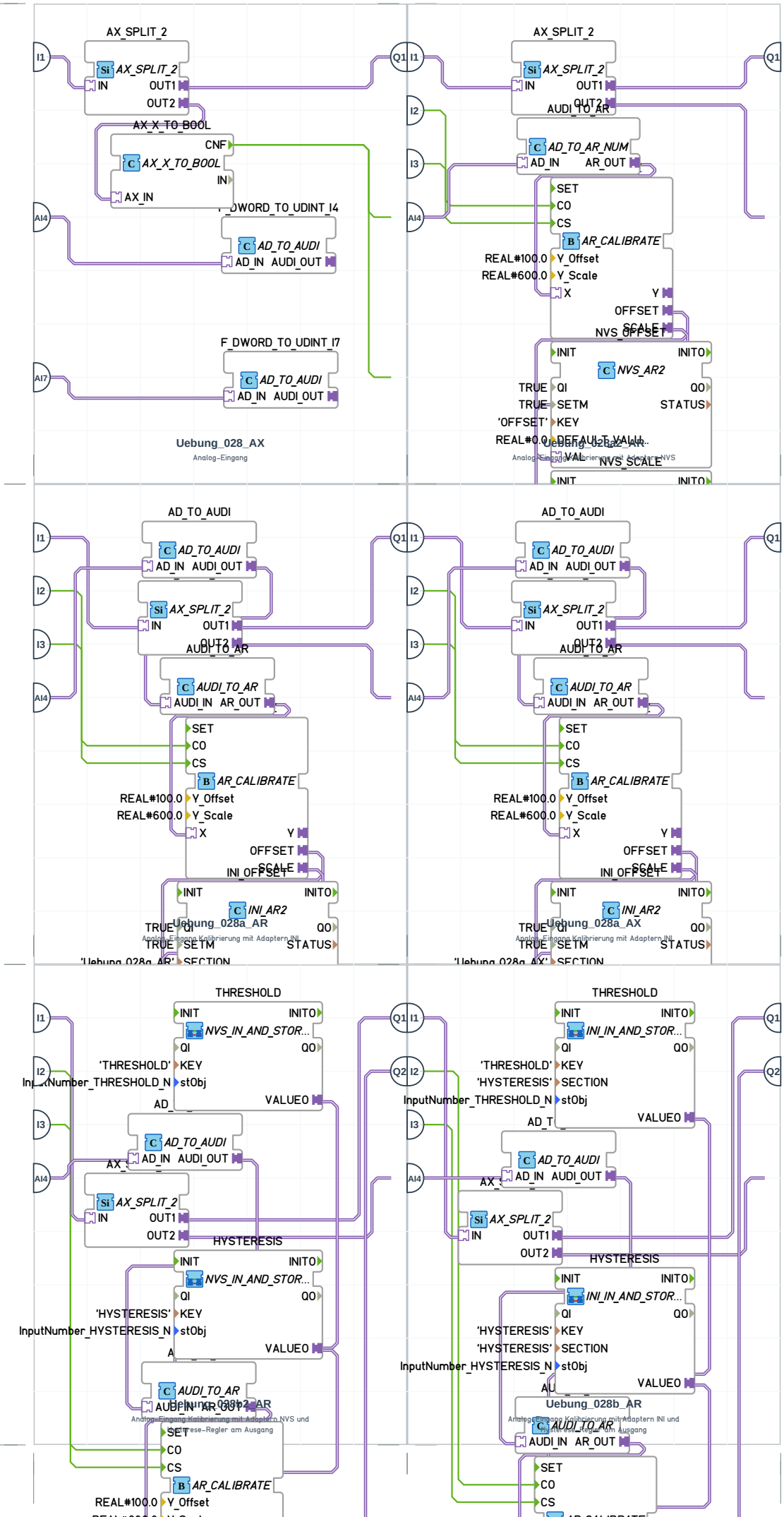
Spiegelabfolge (1) – AX Variante

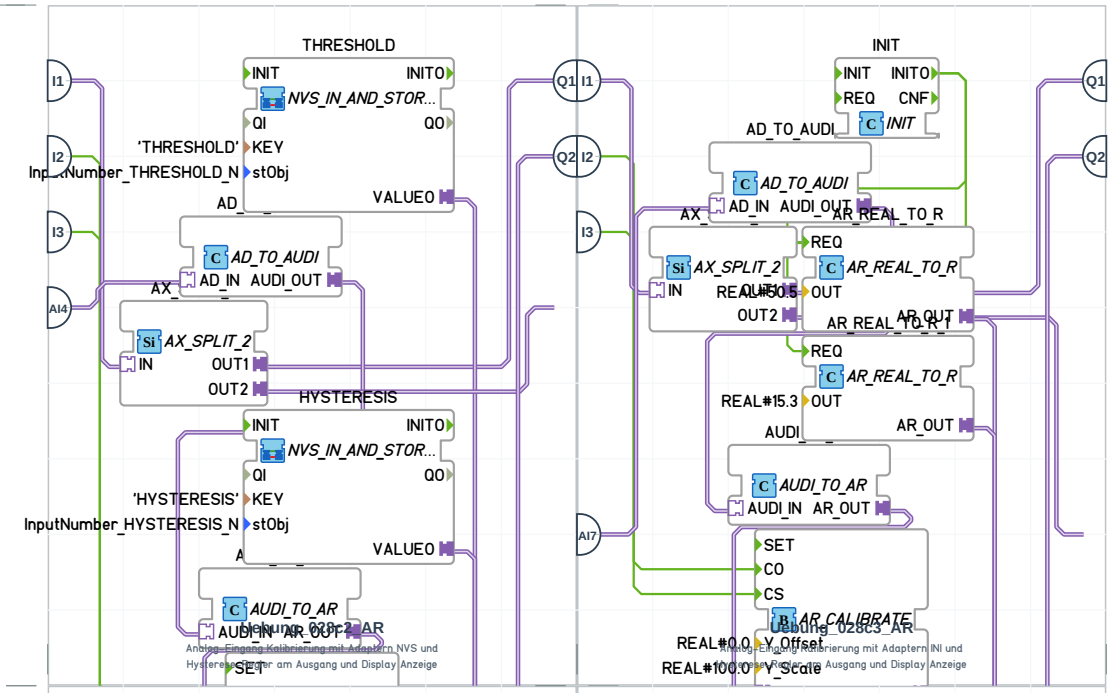


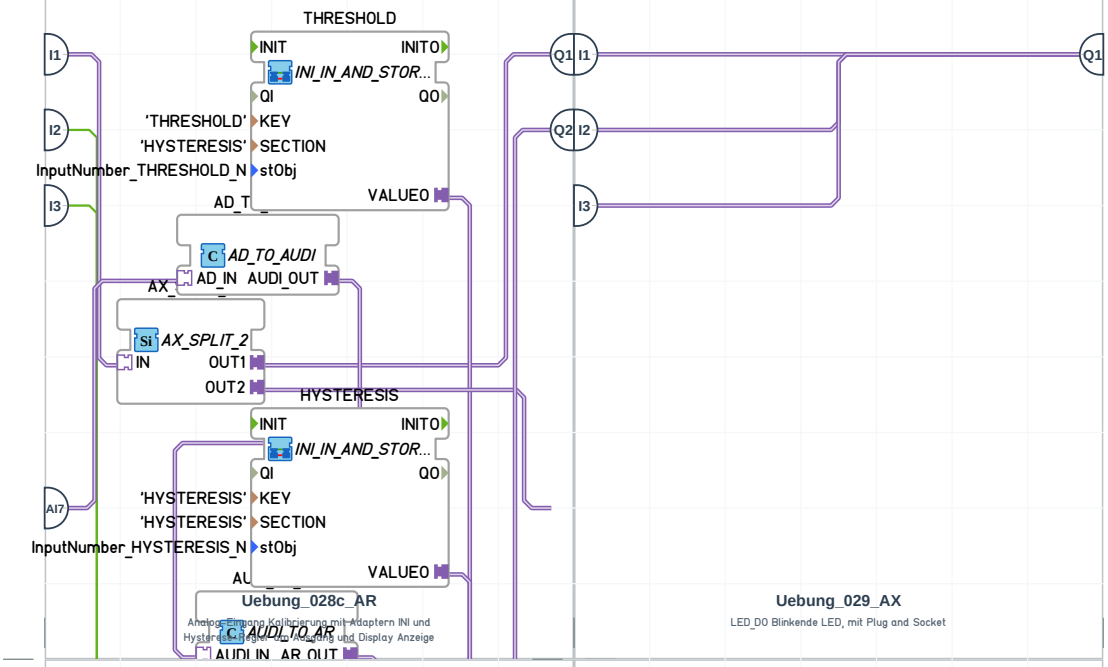
Uebung_022_AX

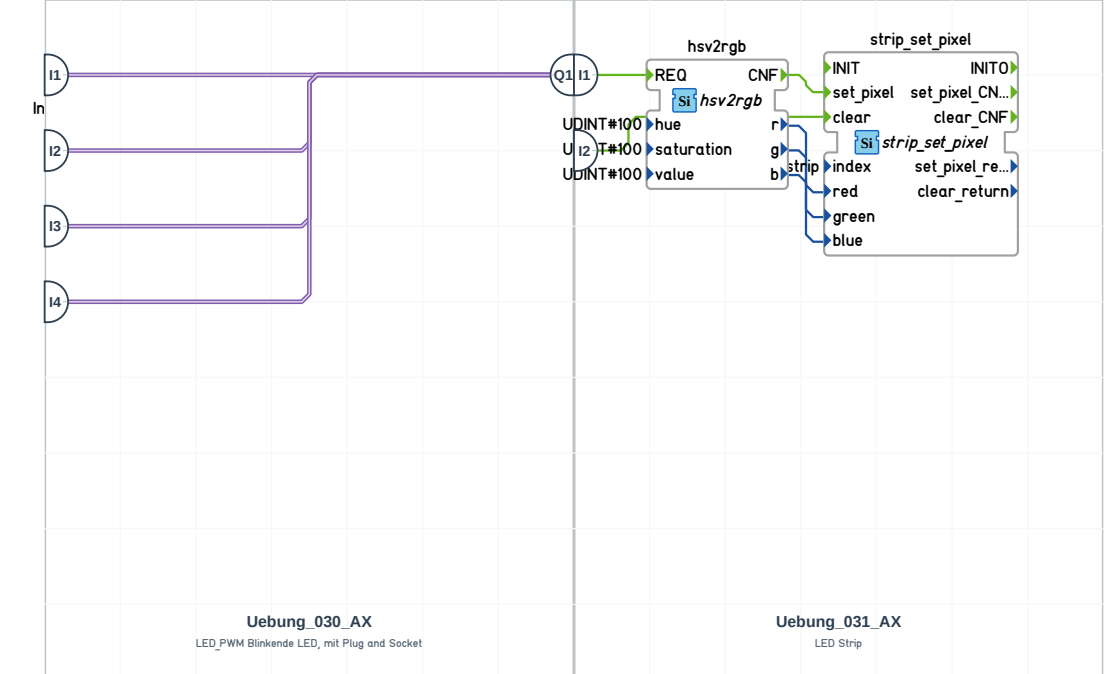
Spiegelabfolge (1) – AX Variante


SoftKey_UP_F1
AX_SR_Ausfahren_Cyl_1
AX_SR_Ausfahren_Cyl_2
Softkey_IE
Softkey_IXA
AX_SR
Uebung_022b_AX2
Spiegelabfolge (2) – AX Variante
Uebung_023_AX
Spiegelabfolge (3) – AX Variante
AX_SR_Einfahren_Cyl_2
AX_SR_Einfahren_Cyl_1
Uebung_023b_AX2
Spiegelabfolge (3) – AX Variante
E_SR_Ausfahren_Cyl_1
E_SR_Ausfahren_Cyl_2
E_SR_Einfahren_Cyl_1
Uebung_024_AX
Spiegelabfolge (4)
E_SWITCH_Q1
AX_E_SWITCH
E_SWITCH_Q2
E_SWITCH_Q3
E_SWITCH_Q4
Uebung_025_AX
Spiegelabfolge (5)
E_REND_Ausfahren_Cyl_1
E_REND_Ausfahren_Cyl_2
E_REND_Einfahren_Cyl_1
E_REND
Uebung_026_AX
Spiegelabfolge (6)

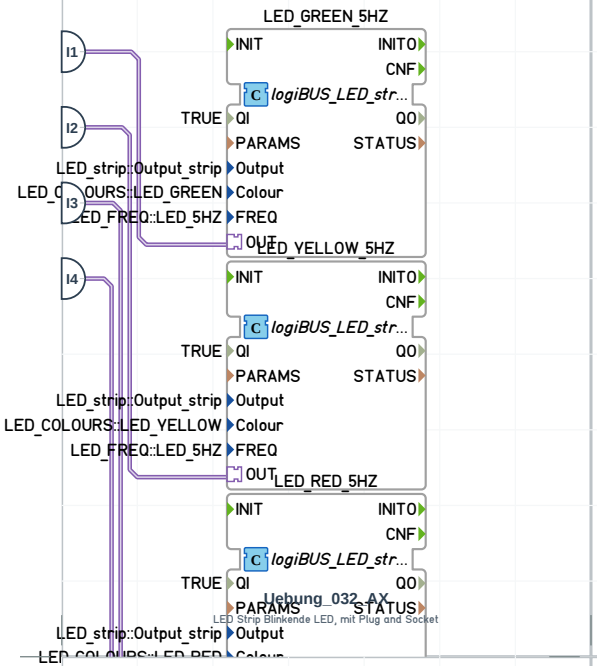

AX_SPLIT_2
AX_SPLIT_2
IN
OUT1
OUT2
AX_X_TO_BOOL
AX_X_TO_BOOL
CNF
IN
AX_IN
F_DWORD_TO_UDINT_I4
AD_TO_AUDI
AD_IN
AUDI_OUT
F_DWORD_TO_UDINT_I7
AD_TO_AUDI
AD_IN
AUDI_OUT
Uebung_028_AX
Analog-Eingang
AUDI_TO_AR
AD_TO_AR_NUM
AD_IN
AR_OUT
AR_CALIBRATE
SET
CO
CS
REAL#100.0
Y_Offset
REAL#600.0
Y_Scale
X
Y
OFFSET
SCALE
NVS_OFFSET
NVS_SCALE
NVS_AR2
INIT
INITO
QI
QO
STATUS
TRUE
'OFFSET'
REAL#0.0
VAL
Uebung_028a_AR
Analog-Eingang Kalibrierung mit Adaptern NVS
AD_TO_AUDI
AUDI_TO_AR
INI_OFFSET
INI_AR2
Uebung_028a_AX
Analog-Eingang Kalibrierung mit Adaptern INI
THRESHOLD
NVS_IN_AND_STOR...
'THRESHOLD'
KEY
stObj
VALUEO
HYSTERESIS
'HYSTERESIS'
InputNumber_HYSTERESIS_N
Uebung_028b_AR
Analog-Eingang Kalibrierung mit Adaptern NVS und Hysterese-Regler am Ausgang
INI_IN_AND_STOR...
SECTION
InputNumber_THRESHOLD_N
Uebung_028b_AX
Analog-Eingang Kalibrierung mit Adaptern INI und Hysterese-Regler am Ausgang

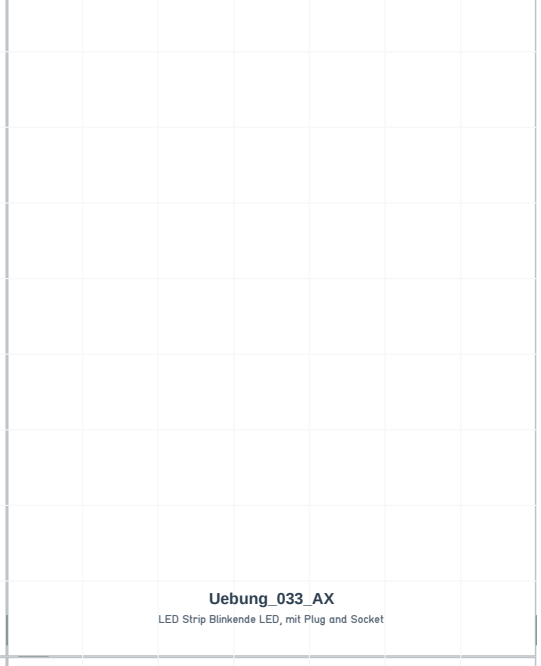

THRESHOLD
INIT
INITO
NVS_IN_AND_STOR...
QI
QO
'THRESHOLD'
KEY
InputNumber_THRESHOLD_N
stObj
VALUEO
AD_TO_AUDI
AD_IN
AUDI_OUT
AX_SPLIT_2
IN
OUT1
OUT2
HYSTERESIS
'HYSTERESIS'
InputNumber_HYSTERESIS_N
AUDI_TO_AR
AUDI_IN
AR_OUT
SET
Uebung_028c2_AR
Analog-Eingang Kalibrierung mit Adaptern NVS und Hysterese
Sensor am Ausgang und Display Anzeige
INIT
REQ
CNF
AR_REAL_TO_R
REAL#50.5
REAL#15.3
AR_CALIBRATE
CO
CS
Y_Offset
Y_Scale
REAL#0.0
REAL#100.0
Uebung_028c3_AR
Analog-Eingang Kalibrierung mit Adaptern INI und Hysterese
Sensor am Ausgang und Display Anzeige

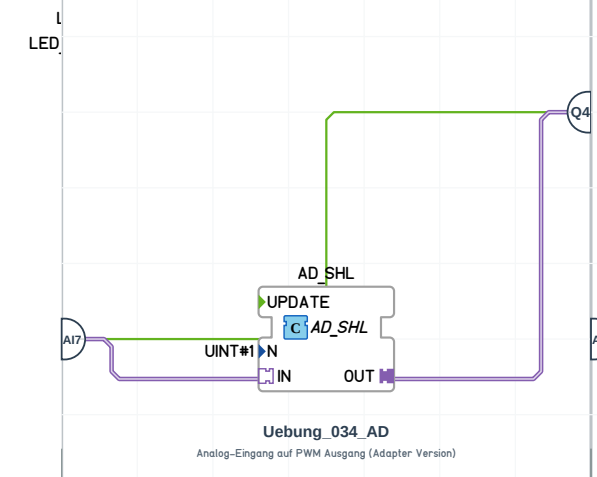

THRESHOLD
INIT
INITO
INI_IN_AND_STOR...
QI
QO
'THRESHOLD'
'HYSTERESIS'
KEY
SECTION
InputNumber_THRESHOLD_N
stObj
VALUEO
AD_TO_AUDI
AD_IN
AUDI_OUT
AX_SPLIT_2
IN
OUT1
OUT2
HYSTERESIS
InputNumber_HYSTERESIS_N
AUDI_TO_AR
AUDI_IN
AR_OUT
Uebung_028c_AR
Analog-Eingang Kalibrierung mit Adaptern INI und Hysterese
Sensor am Ausgang und Display Anzeige
Uebung_029_AX
LED_DO Blinkende LED, mit Plug and Socket

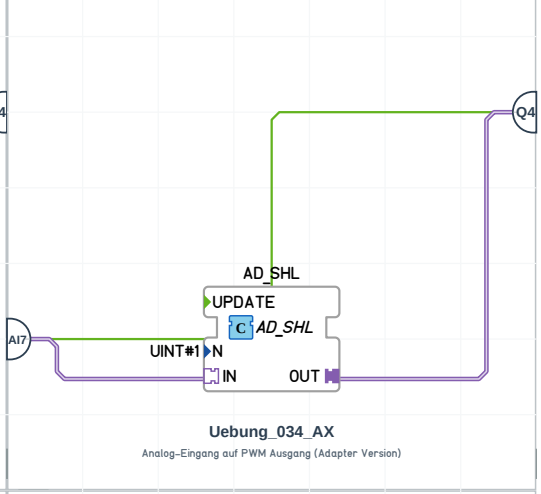

hsv2rgb
REQ
CNF
hsv2rgb
UDINT#100
hue
saturation
value
r
g
b
strip_set_pixel
INIT
INITO
set_pixel
set_pixel_CN...
clear
clear_CNF
strip_set_pixel
index
set_pixel_re...
red
clear_return
green
blue
strip
Uebung_030_AX
LED_PWM Blinkende LED, mit Plug and Socket
Uebung_031_AX
LED Strip

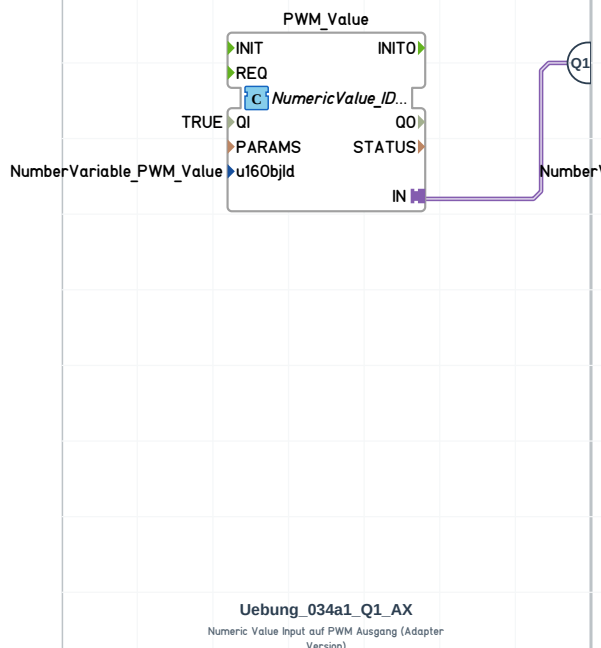

'SCALE'
KEY
REAL#1.0
DEFAULT_VALU...
VAL
Q_NumericValue_PHYSA

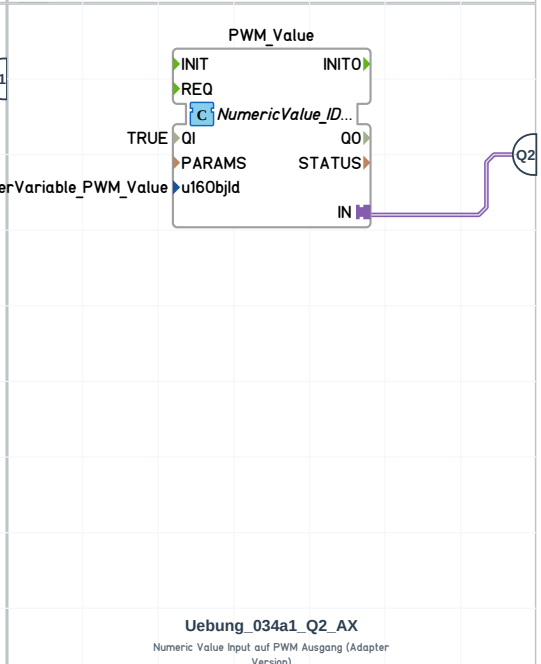

LED_GREEN_5HZ
INIT
INITO
CNF
logiBUS_LED_str...
TRUE
QI
QO
PARAMS
STATUS
I1
I2
I3
I4
LED_strip::Output_strip
Output
LED_COLOURS::LED_GREEN
Colour
LED_FREQ::LED_5HZ
FREQ
OUT
LED_YELLOW_5HZ
LED_COLOURS::LED_YELLOW
LED_RED_5HZ
LED_COLOURS::LED_RED
Uebung_032_AX
LED Strip Blinkende LED, mit Plug and Socket

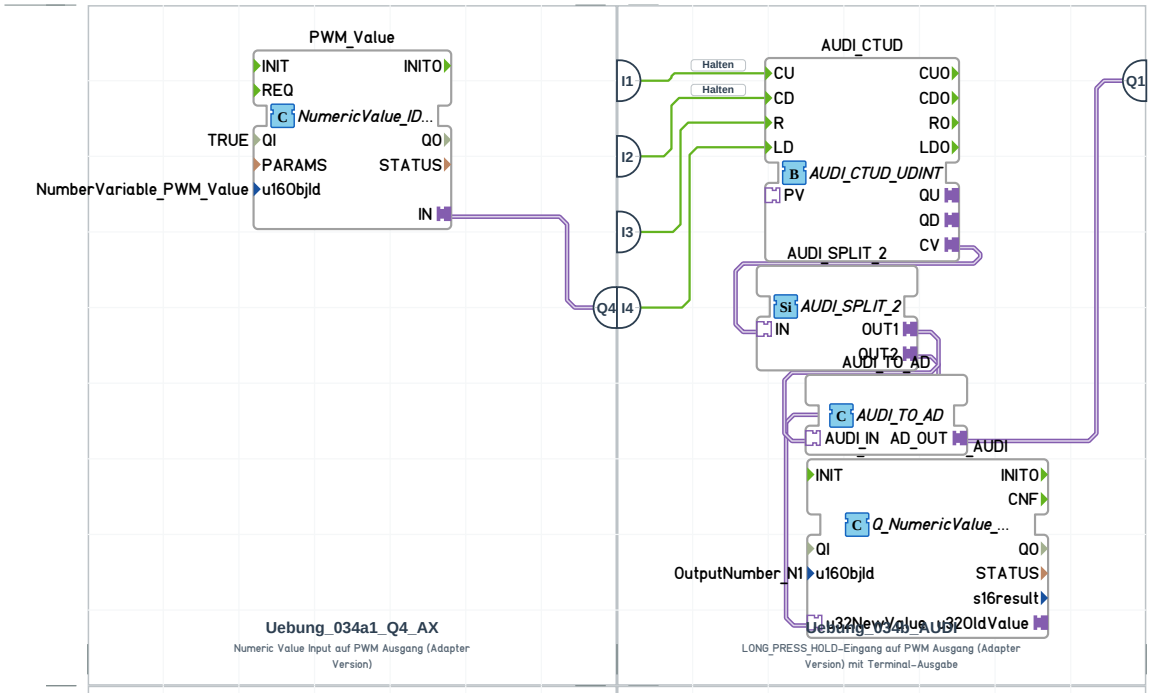

Uebung_033_AX
LED Strip Blinkende LED, mit Plug and Socket

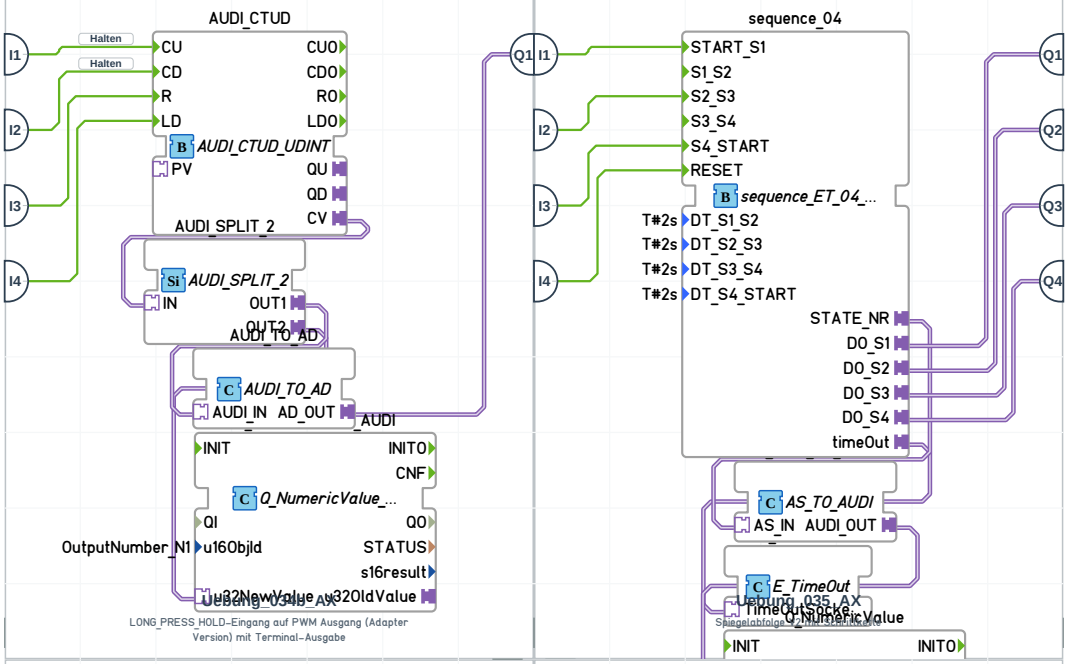

AD_SHL
UPDATE
AD_SHL
AI7
UINT#1
N
IN
OUT
Q4
Uebung_034_AD
Analog-Eingang auf PWM Ausgang (Adapter Version)

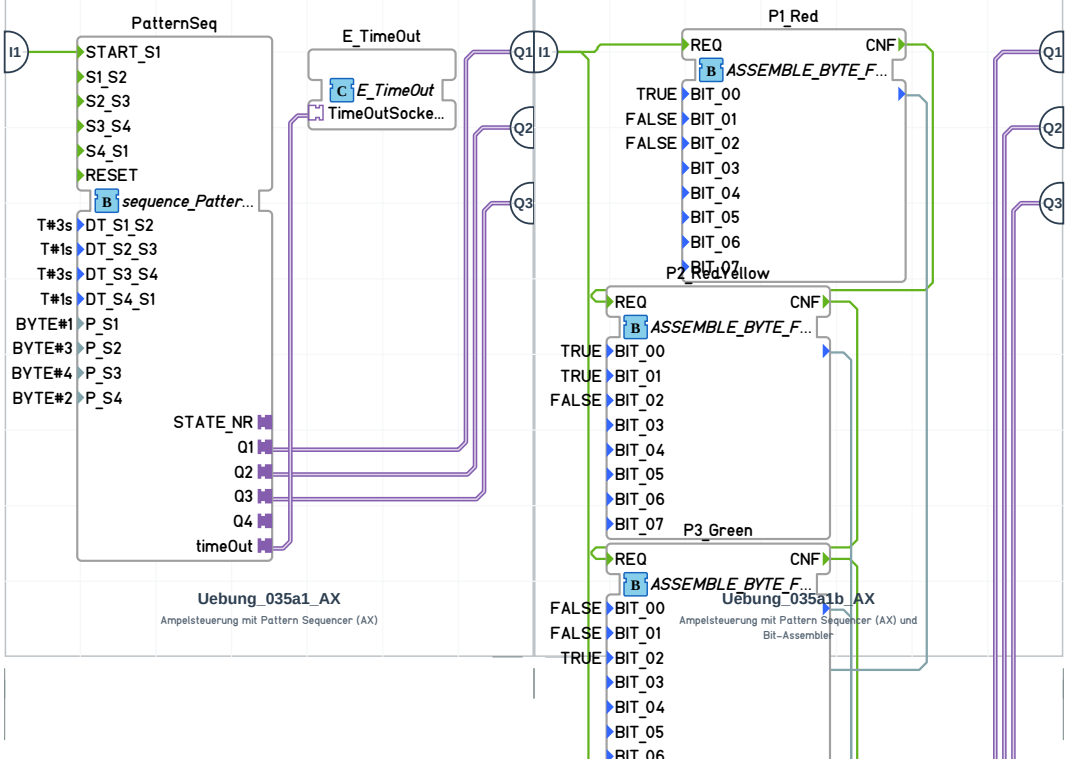

AD_SHL
UPDATE
AD_SHL
AI7
UINT#1
N
IN
OUT
Q4
Uebung_034_AX
Analog-Eingang auf PWM Ausgang (Adapter Version)

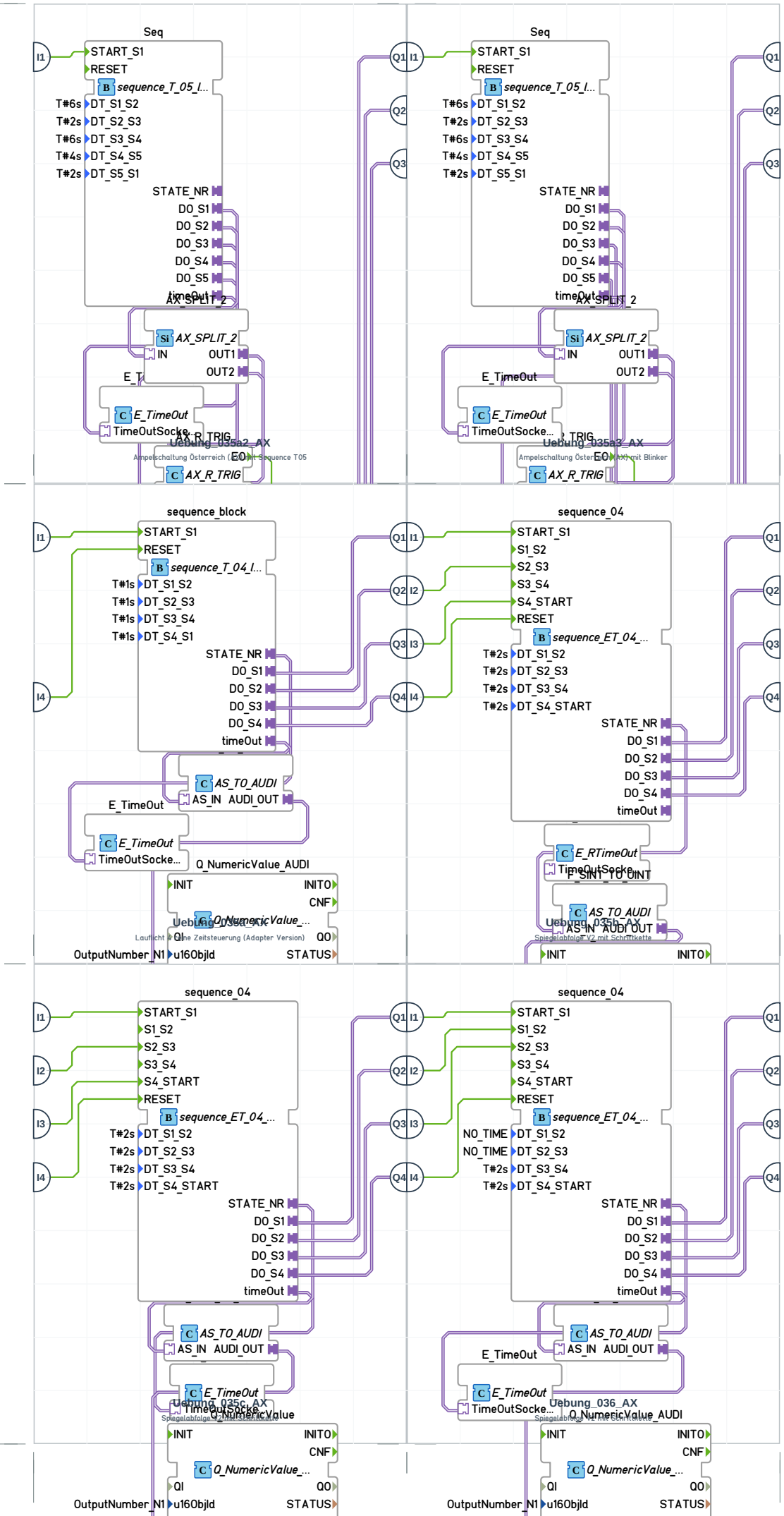

PWM_Value
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
NumberVariable_PWM_Value
u16ObjId
IN
Q1
Uebung_034a1_Q1_AX
Numeric Value Input auf PWM Ausgang (Adapter Version)

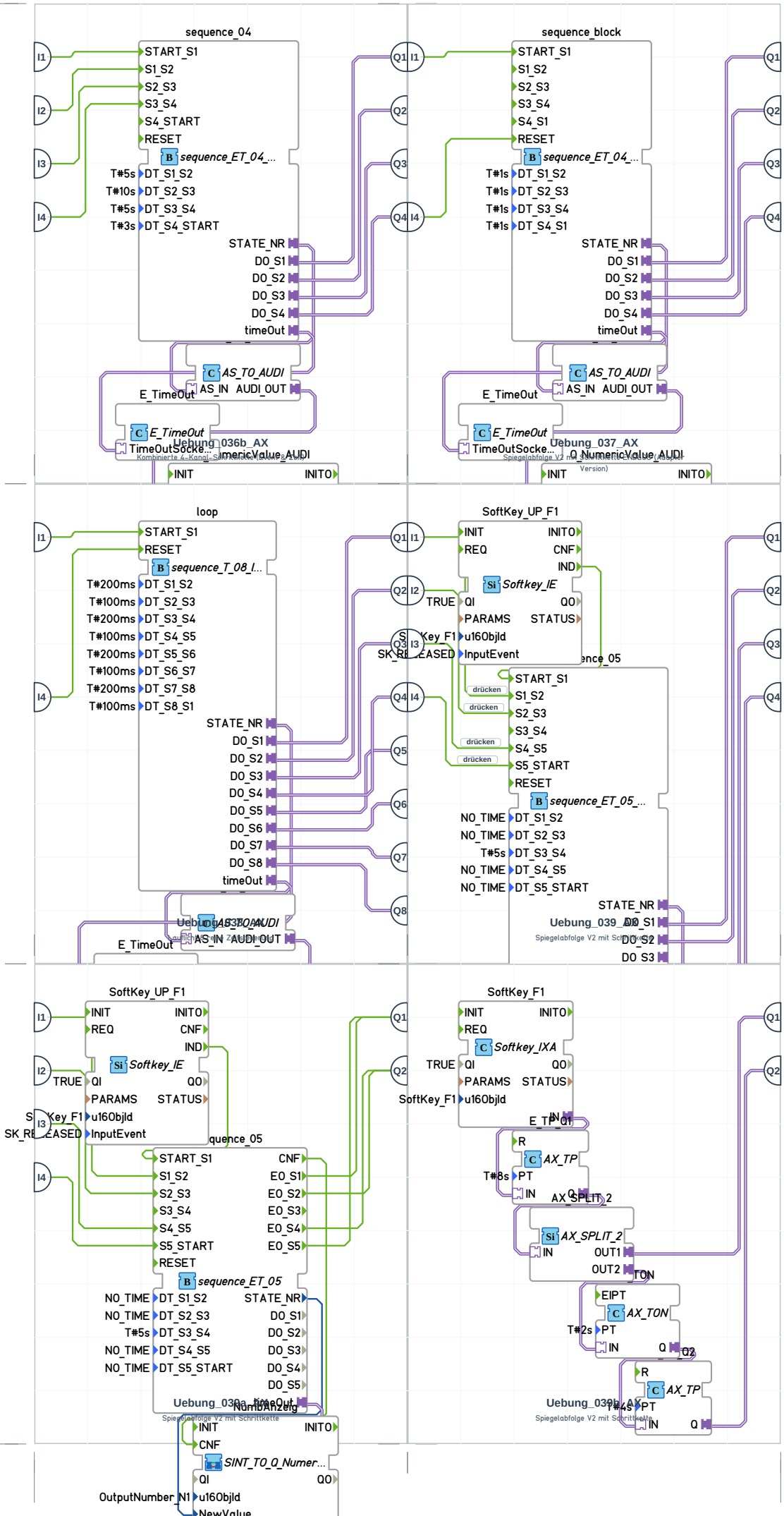

PWM_Value
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
NumberVariable_PWM_Value
u16ObjId
IN
Q2
Uebung_034a1_Q2_AX
Numeric Value Input auf PWM Ausgang (Adapter Version)

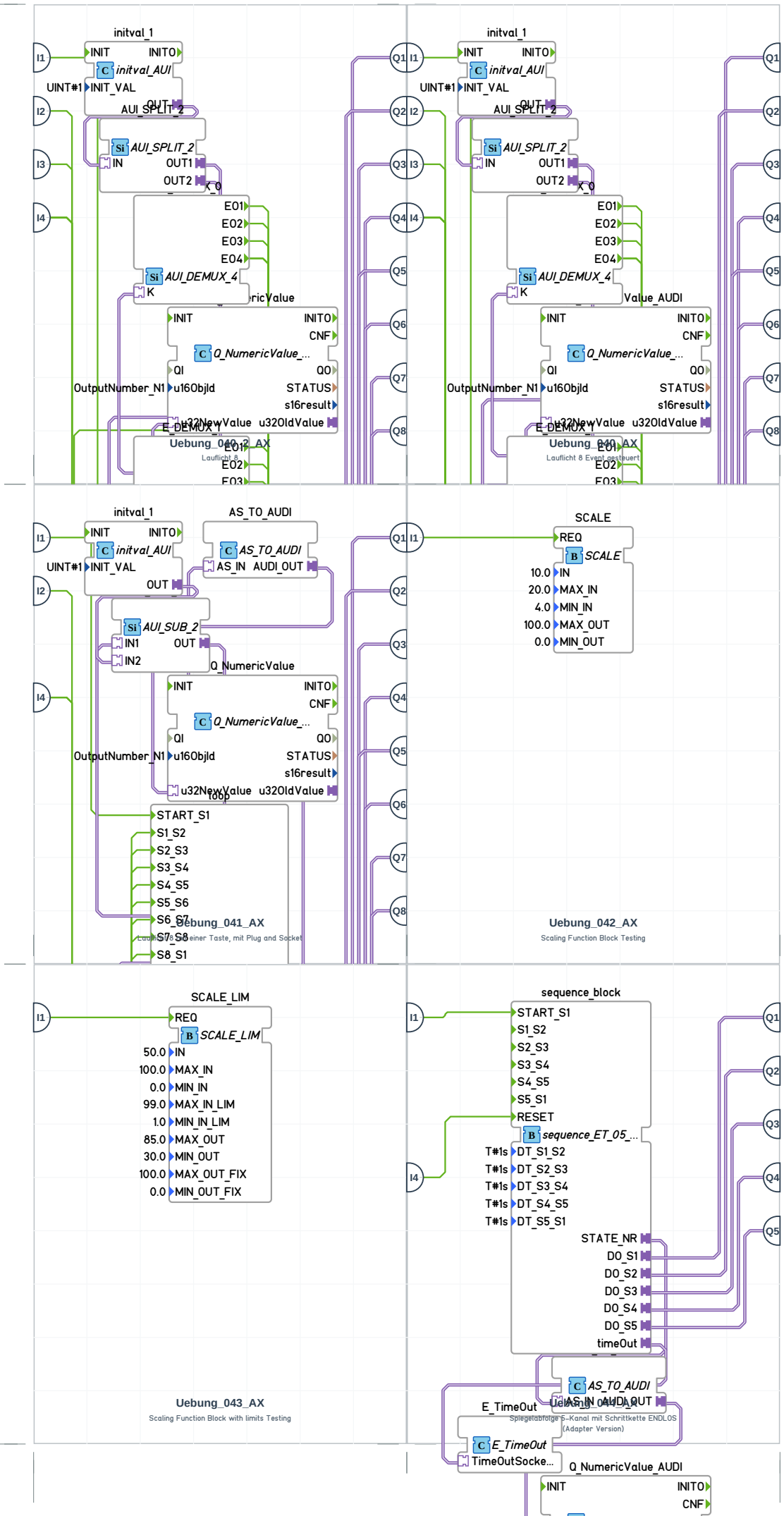

PWM_Value
INIT
INITO
REQ
NumericValue_ID...
TRUE
QI
QO
PARAMS
STATUS
NumberVariable_PWM_Value
u16ObjId
IN
AUDI_CTUD
I1
Halten
CU
CUO
I2
Halten
CD
CDO
R
RO
I3
LD
LDO
AUDI_CTUD_UDINT
PV
QU
QD
CV
Q1
AUDI_SPLIT_2
Q4
I4
AUDI_SPLIT_2
IN
OUT1
OUT2
AUDI_TO_AD
AUDI_TO_AD
AUDI_IN
AD_OUT
AUDI
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_034a1_Q4_AX
Numeric Value Input auf PWM Ausgang (Adapter Version)
Uebung_034a_AUDI
LONG_PRESS_HOLD-Eingang auf PWM Ausgang (Adapter Version) mit Terminal-Ausgabe

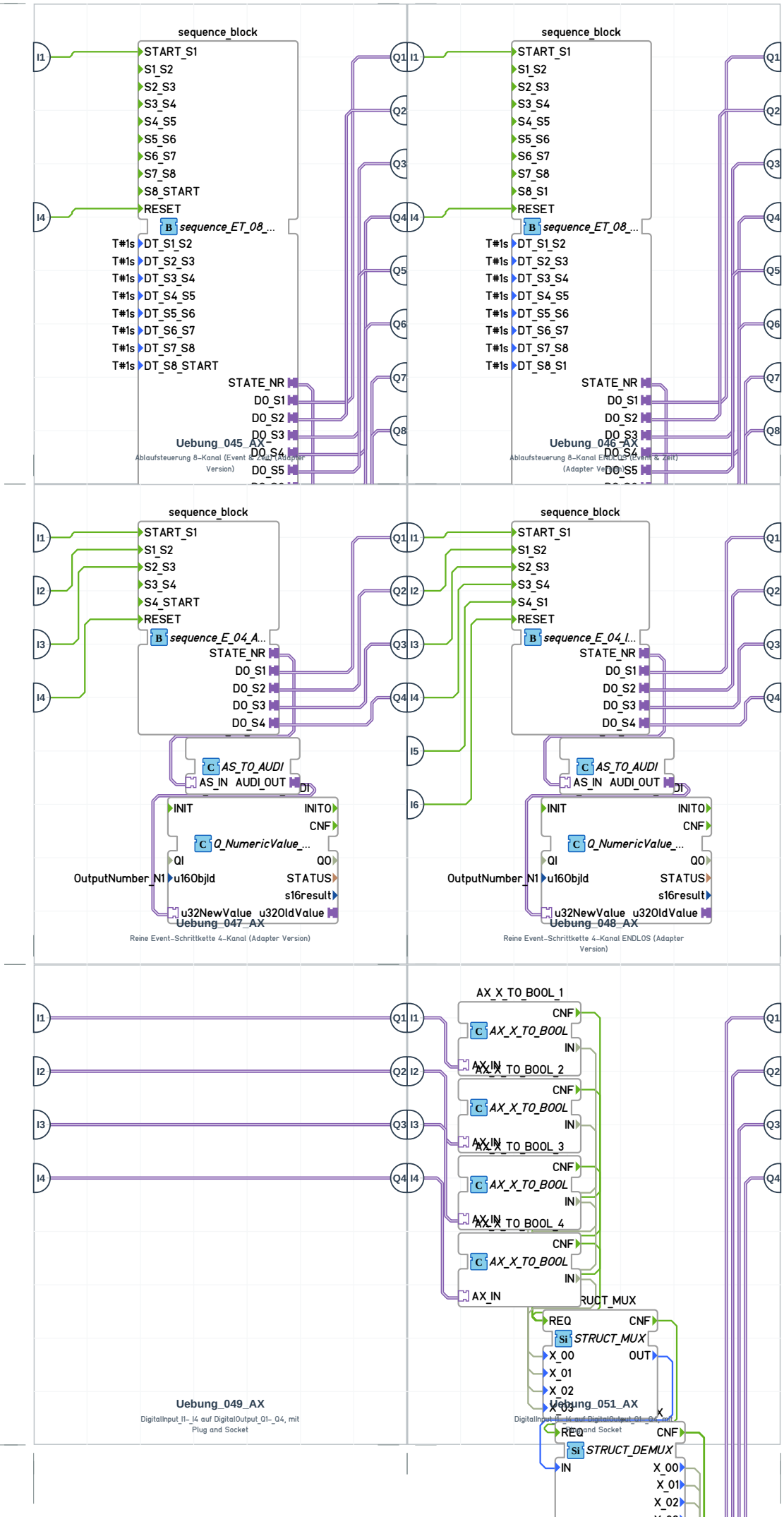

AUDI_CTUD
I1
Halten
CU
CUO
Halten
CD
CDO
I2
R
RO
I3
LD
LDO
AUDI_CTUD_UDINT
PV
QU
QD
CV
AUDI_SPLIT_2
I4
AUDI_SPLIT_2
IN
OUT1
OUT2
AUDI_TO_AD
AUDI_TO_AD
AUDI_IN
AD_OUT
AUDI
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_034a_AX
LONG_PRESS_HOLD-Eingang auf PWM Ausgang (Adapter Version) mit Terminal-Ausgabe
sequence_04
I1
START_S1
Q1
S1_S2
Q2
I2
S2_S3
Q3
I3
S3_S4
Q4
I4
S4_START
RESET
sequence_ET_04_...
T#2s
DT_S1_S2
T#2s
DT_S2_S3
T#2s
DT_S3_S4
T#2s
DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
AS_TO_AUDI
AS_IN
AUDI_OUT
E_TimeOut
TimeOutSocke...
Uebung_035_AX
Ampelsteuerung mit Pattern Sequencer (AX)
INIT
INITO

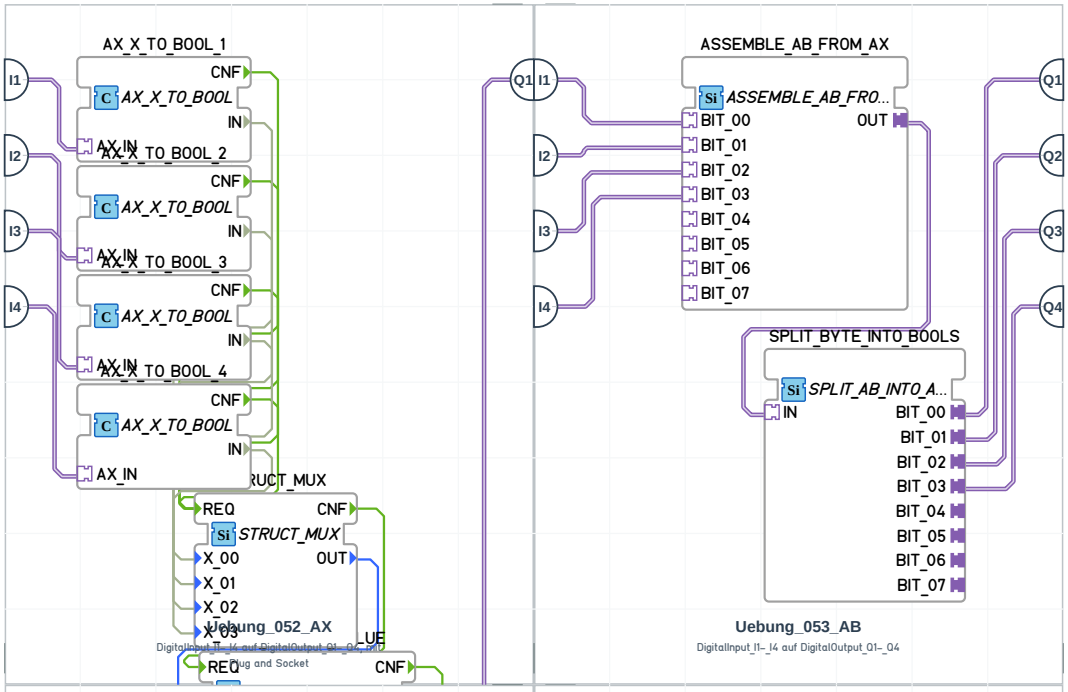

PatternSeq
I1
START_S1
S1_S2
S2_S3
S3_S4
S4_S1
RESET
sequence_Patter...
T#3s
DT_S1_S2
T#1s
DT_S2_S3
T#3s
DT_S3_S4
T#1s
DT_S4_S1
BYTE#1
P_S1
BYTE#3
P_S2
BYTE#4
P_S3
BYTE#2
P_S4
STATE_NR
Q1
Q2
Q3
Q4
timeOut
E_TimeOut
E_TimeOut
TimeOutSocke...
Uebung_035a1_AX
Ampelsteuerung mit Pattern Sequencer (AX)
P1_Red
I1
REQ
CNF
ASSEMBLE_BYTE_F...
TRUE
BIT_00
FALSE
BIT_01
FALSE
BIT_02
BIT_03
BIT_04
BIT_05
BIT_06
BIT_07
P2_RedYellow
REQ
CNF
ASSEMBLE_BYTE_F...
TRUE
BIT_00
TRUE
BIT_01
FALSE
BIT_02
BIT_03
BIT_04
BIT_05
BIT_06
BIT_07
P3_Green
REQ
CNF
ASSEMBLE_BYTE_F...
FALSE
BIT_00
FALSE
BIT_01
TRUE
BIT_02
BIT_03
BIT_04
BIT_05
BIT_06
Uebung_035a1b_AX
Ampelsteuerung mit Pattern Sequencer (AX) und Bit-Assembler

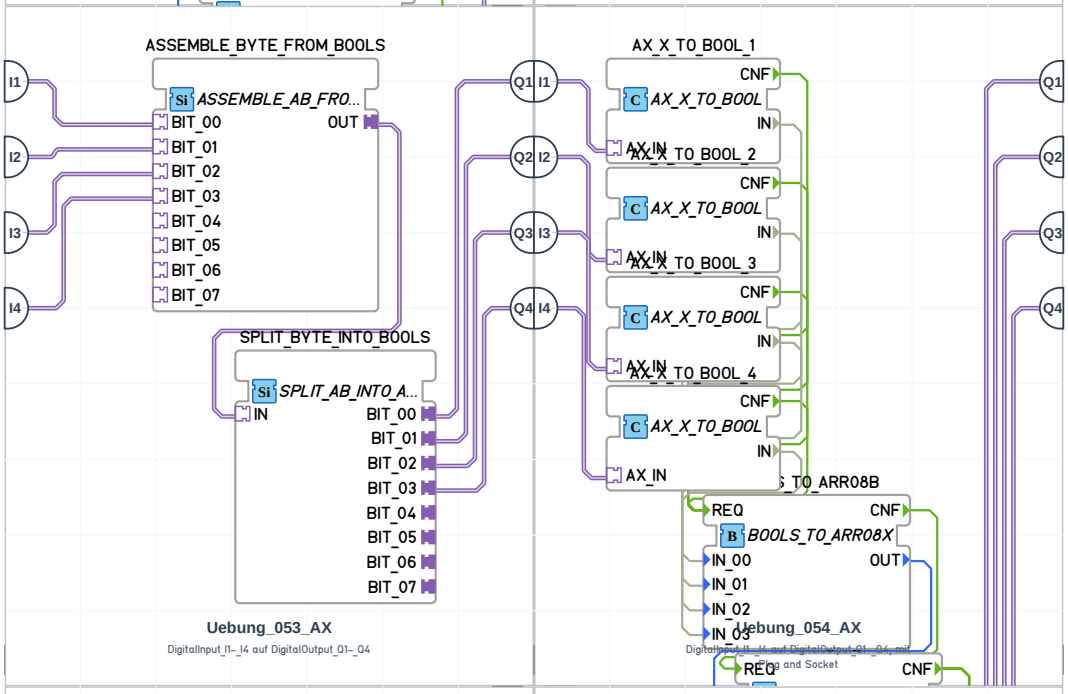

Seq
START_S1
RESET
sequence_T_05_l...
T#6s DT_S1_S2
T#2s DT_S2_S3
T#6s DT_S3_S4
T#4s DT_S4_S5
T#2s DT_S5_S1
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
timeOut
AX_SPLIT_2
IN
OUT1
OUT2
E_TimeOut
TimeOutSocke...
AX_R_TRIG
EO
Uebung_035a2_AX
Ampelschaltung Österreich (Adapter Version) mit Sequence T05
Seq
START_S1
RESET
sequence_T_05_l...
T#6s DT_S1_S2
T#2s DT_S2_S3
T#6s DT_S3_S4
T#4s DT_S4_S5
T#2s DT_S5_S1
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
timeOut
AX_SPLIT_2
IN
OUT1
OUT2
E_TimeOut
TimeOutSocke...
AX_R_TRIG
EO
Uebung_035a3_AX
Ampelschaltung Österreich (Adapter Version) mit Blinker
sequence_block
START_S1
RESET
sequence_T_04_l...
T#1s DT_S1_S2
T#1s DT_S2_S3
T#1s DT_S3_S4
T#1s DT_S4_S1
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
AS_TO_AUDI
AS_IN
AUDI_OUT
E_TimeOut
TimeOutSocke...
Q_NumericValue_AUDI
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u160ObjId
STATUS
sequence_04
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_ET_04_...
T#2s DT_S1_S2
T#2s DT_S2_S3
T#2s DT_S3_S4
T#2s DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
E_RTimeOut
TimeOutSocke...
F_SINT_TO_UINT
AS_TO_AUDI
AS_IN
AUDI_OUT
INIT
INITO
sequence_04
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_ET_04_...
T#2s DT_S1_S2
T#2s DT_S2_S3
T#2s DT_S3_S4
T#2s DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
AS_TO_AUDI
AS_IN
AUDI_OUT
E_TimeOut
TimeOutSocke...
Q_NumericValue
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u160ObjId
STATUS
sequence_04
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_ET_04_...
NO_TIME DT_S1_S2
NO_TIME DT_S2_S3
T#2s DT_S3_S4
T#2s DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
AS_TO_AUDI
AS_IN
AUDI_OUT
E_TimeOut
TimeOutSocke...
Uebung_036_AX
Q_NumericValue_AUDI
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u160ObjId
STATUS

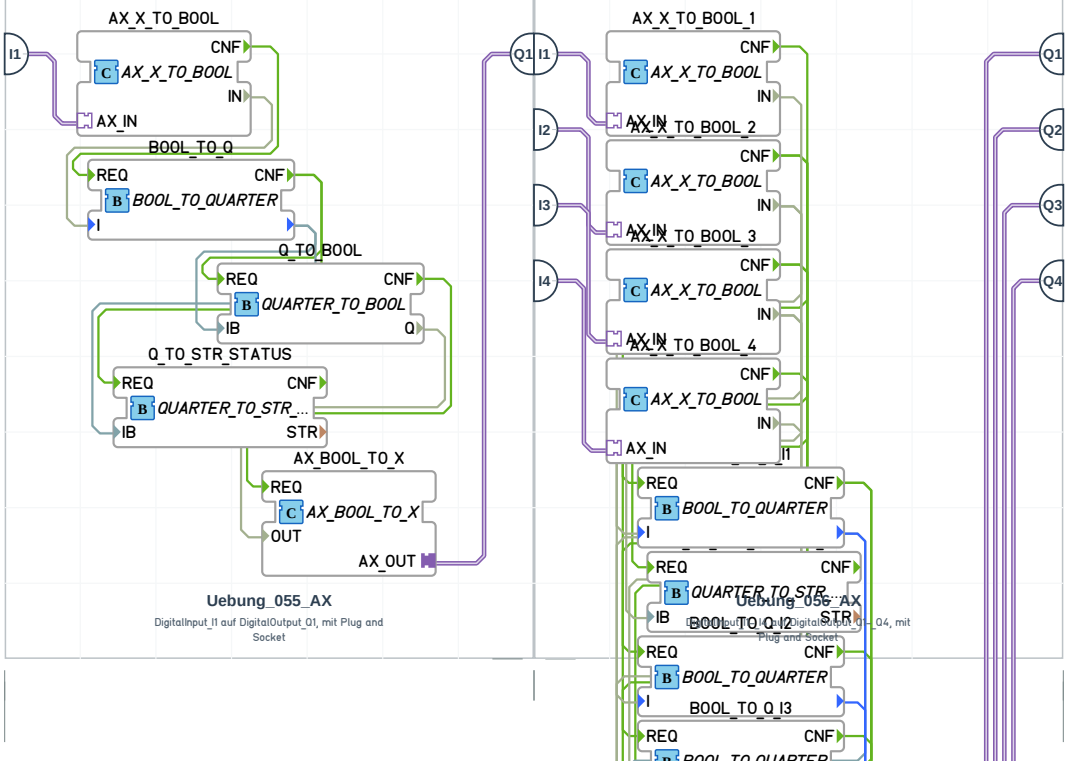

sequence_04
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_ET_04_...
T#5s DT_S1_S2
T#10s DT_S2_S3
T#5s DT_S3_S4
T#3s DT_S4_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
AS_TO_AUDI
AS_IN AUDI_OUT
E_TimeOut
E_TimeOut
TimeOutSocke...
Uebung_036b_AX
INIT INITO
sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_S1
RESET
sequence_ET_04_...
T#1s DT_S1_S2
T#1s DT_S2_S3
T#1s DT_S3_S4
T#1s DT_S4_S1
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
timeOut
AS_TO_AUDI
AS_IN AUDI_OUT
E_TimeOut
E_TimeOut
TimeOutSocke...
Uebung_037_AX
INIT INITO
loop
START_S1
RESET
sequence_T_08_l...
T#200ms DT_S1_S2
T#100ms DT_S2_S3
T#200ms DT_S3_S4
T#100ms DT_S4_S5
T#200ms DT_S5_S6
T#100ms DT_S6_S7
T#200ms DT_S7_S8
T#100ms DT_S8_S1
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
DO_S6
DO_S7
DO_S8
timeOut
AS_TO_AUDI
AS_IN AUDI_OUT
E_TimeOut
SoftKey_UP_F1
INIT INITO
REQ CNF
IND
Softkey_IE
TRUE QI QO
PARAMS STATUS
u16ObjId
InputEvent
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_START
RESET
sequence_ET_05_...
NO_TIME DT_S1_S2
NO_TIME DT_S2_S3
T#5s DT_S3_S4
NO_TIME DT_S4_S5
NO_TIME DT_S5_START
STATE_NR
DO_S1
DO_S2
DO_S3
drücken
SoftKey_UP_F1
INIT INITO
REQ CNF
IND
Softkey_IE
TRUE QI QO
PARAMS STATUS
u16ObjId
InputEvent
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_START
RESET
sequence_ET_05
NO_TIME DT_S1_S2
NO_TIME DT_S2_S3
T#5s DT_S3_S4
NO_TIME DT_S4_S5
NO_TIME DT_S5_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
timeOut
NumbAnzeig
SINT_TO_Q_Numer...
QI QO
OutputNumber_N1 u16ObjId
NewValue
SoftKey_F1
INIT INITO
REQ
Softkey_IXA
TRUE QI QO
PARAMS STATUS
SoftKey_F1 u16ObjId
AX_TP
T#8s PT
IN Q
AX_SPLIT_2
IN OUT1
OUT2
AX_TON
T#2s PT
IN Q
AX_TP
T#4s PT
IN Q

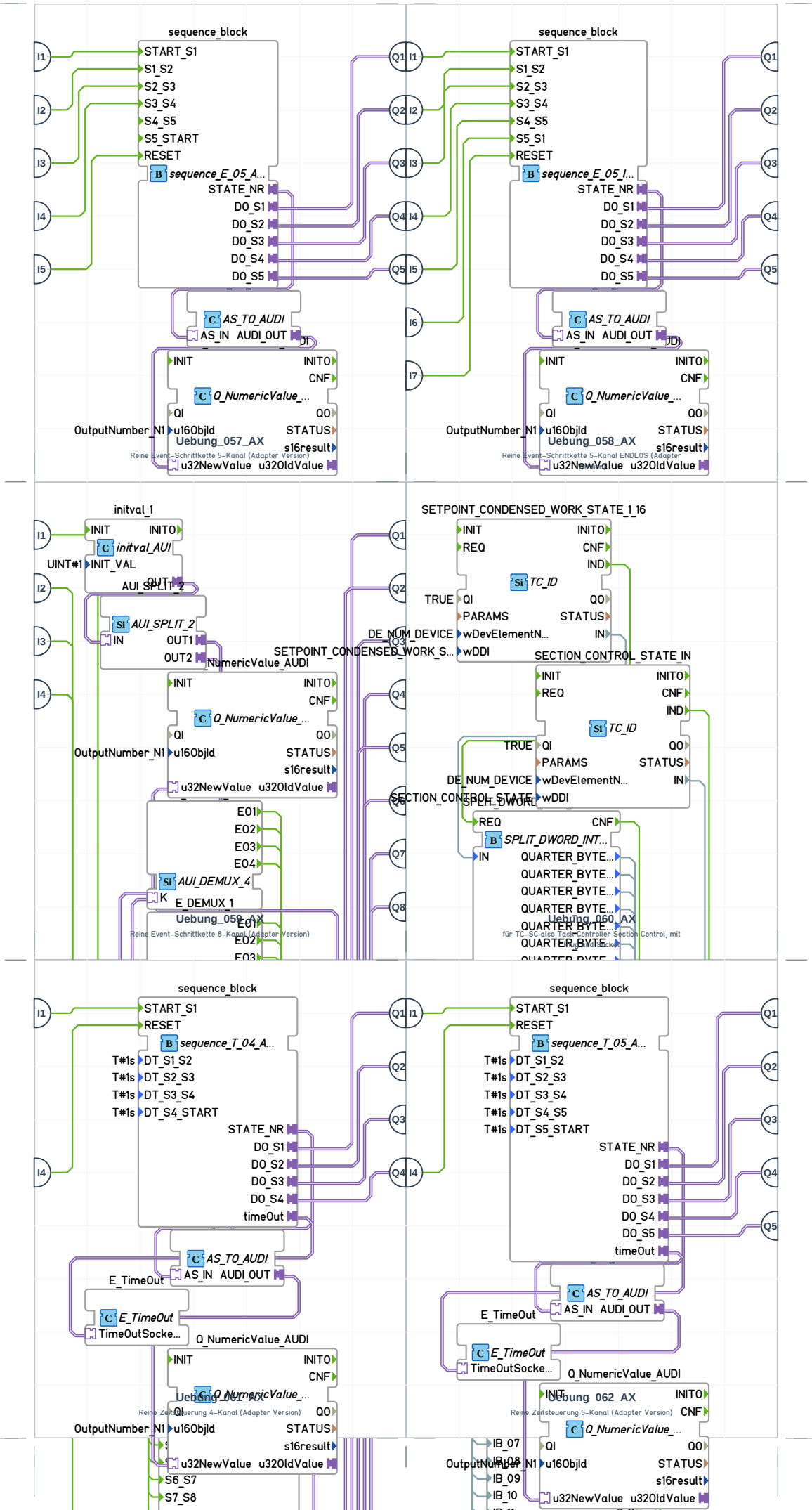

initval_1
INIT
INITO
initval_AUI
UINT#1
INIT_VAL
OUT
I1
I2
I3
I4
AUI_SPLIT_2
IN
OUT1
OUT2
E01
E02
E03
E04
AUI_DEMUX_4
K
Q_NumericValue_...
QI
QO
STATUS
s16result
OutputNumber_N1
u16ObjId
u32NewValue
u32OldValue
CNF
E_DEMUX
Uebung_040_2_AX
Lauflicht 8
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Value_AUDI
Uebung_040_AX
Lauflicht 8 Event gesteuert
AS_TO_AUDI
AS_IN
AUDI_OUT
AUI_SUB_2
IN1
IN2
1000
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_S6
S6_S7
S7_S8
S8_S1
Uebung_041_AX
SCALE
REQ
10.0
20.0
4.0
100.0
0.0
MAX_IN
MIN_IN
MAX_OUT
MIN_OUT
Uebung_042_AX
Scaling Function Block Testing
SCALE_LIM
50.0
100.0
0.0
99.0
1.0
85.0
30.0
100.0
0.0
MAX_IN_LIM
MIN_IN_LIM
MAX_OUT_FIX
MIN_OUT_FIX
Uebung_043_AX
Scaling Function Block with limits Testing
sequence_block
RESET
T#1s
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_S1
STATE_NR
D0_S1
D0_S2
D0_S3
D0_S4
D0_S5
timeOut
E_TimeOut
TimeOutSocke...
Q_NumericValue_AUDI
Uebung_044_AX
Spiegelabfolge 6-Kanal mit Schrittkette ENDLOS
(Adapter Version)

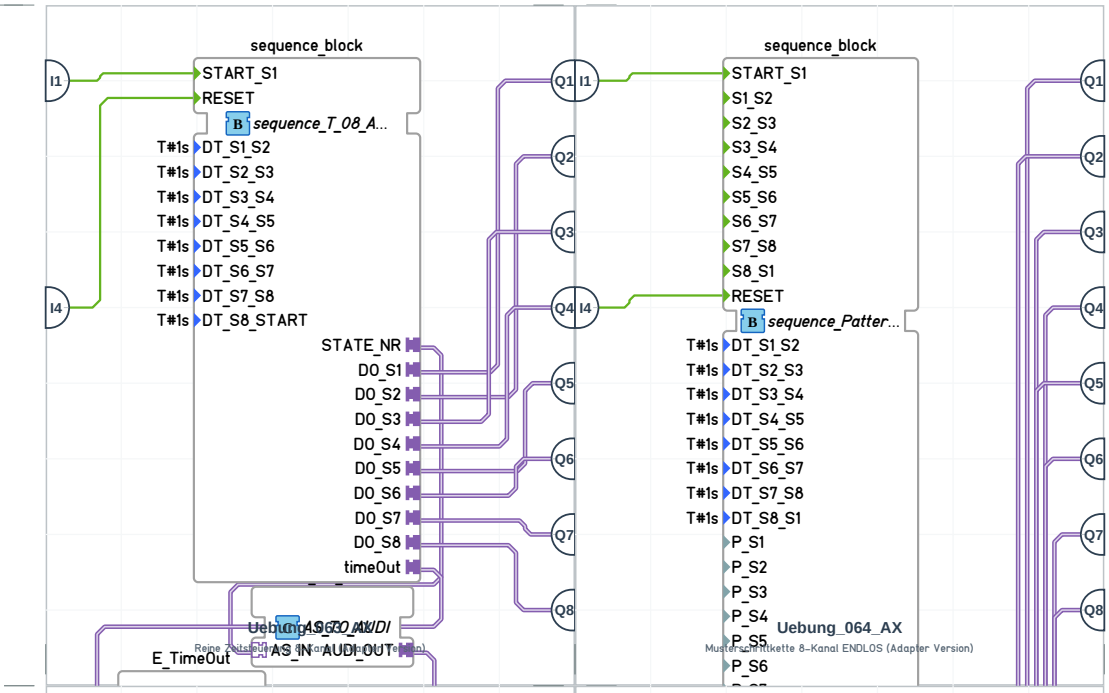

sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_S6
S6_S7
S7_S8
S8_START
RESET
sequence_ET_08_...
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_S6
DT_S6_S7
DT_S7_S8
DT_S8_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
Uebung_045_AX
Ablaufsteuerung 8-Kanal (Event & Zeit) (Adapter Version)
sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_S6
S6_S7
S7_S8
S8_S1
RESET
sequence_ET_08_...
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_S6
DT_S6_S7
DT_S7_S8
DT_S8_S1
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
Uebung_046_AX
Ablaufsteuerung 8-Kanal ENDLOS (Event & Zeit) (Adapter Version)
sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_START
RESET
sequence_E_04_A...
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
AS_TO_AUDI
AS_IN
AUDI_OUT
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_047_AX
Reine Event-Schrittkette 4-Kanal (Adapter Version)
sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_S1
RESET
sequence_E_04_I...
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
AS_TO_AUDI
AS_IN
AUDI_OUT
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_048_AX
Reine Event-Schrittkette 4-Kanal ENDLOS (Adapter Version)
AX_X_TO_BOOL_1
AX_X_TO_BOOL_2
AX_X_TO_BOOL_3
AX_X_TO_BOOL_4
AX_X_TO_BOOL
CNF
IN
AX_IN
STRUCT_MUX
REQ
CNF
OUT
X_00
X_01
X_02
X_03
Uebung_049_AX
DigitalInput_I1-_I4 auf DigitalOutput_Q1-_Q4, mit Plug and Socket
Uebung_051_AX
STRUCT_DEMUX
IN
X_00
X_01
X_02

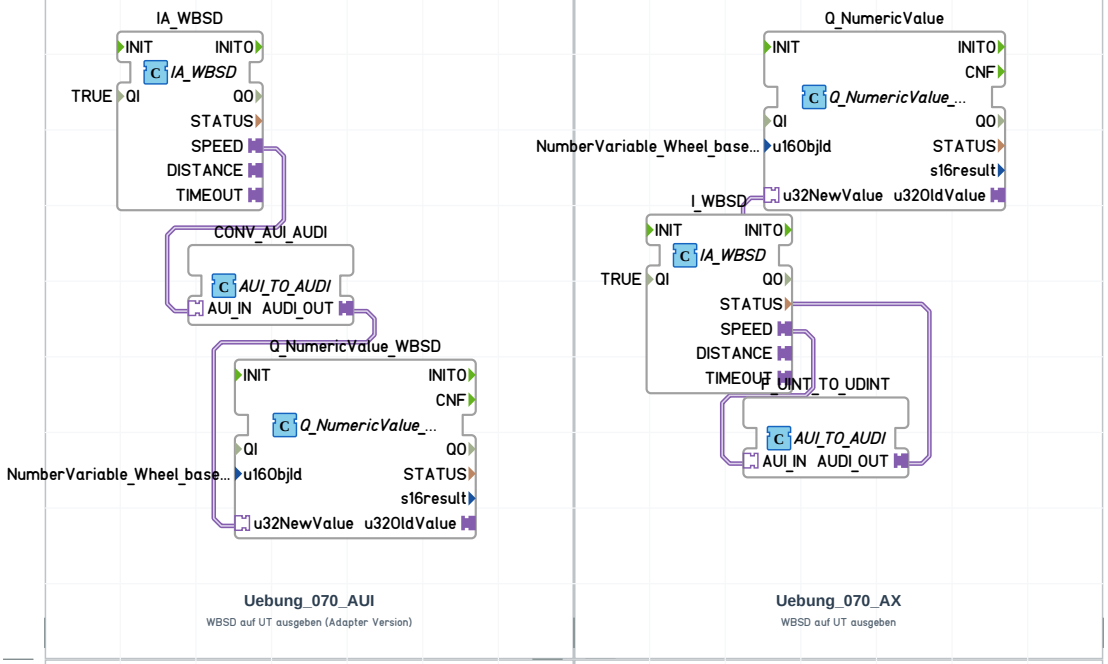

AX_X_TO_BOOL_1
I1
I2
I3
I4
C AX_X_TO_BOOL
CNF
IN
AX_IN
AX_X_TO_BOOL_2
AX_X_TO_BOOL_3
AX_X_TO_BOOL_4
STRUCT_MUX
REQ
CNF
Si STRUCT_MUX
X_00
X_01
X_02
X_03
OUT
Uebung_052_AX
DigitalInput_I1_..I4 auf DigitalOutput_Q1_..Q4, mit Plug and Socket
ASSEMBLE_AB_FROM_AX
Si ASSEMBLE_AB_FRO...
BIT_00
BIT_01
BIT_02
BIT_03
BIT_04
BIT_05
BIT_06
BIT_07
OUT
Q1
Q2
Q3
Q4
SPLIT_BYTE_INTO_BOOLS
Si SPLIT_AB_INTO_A...
IN
Uebung_053_AB
DigitalInput_I1_..I4 auf DigitalOutput_Q1_..Q4

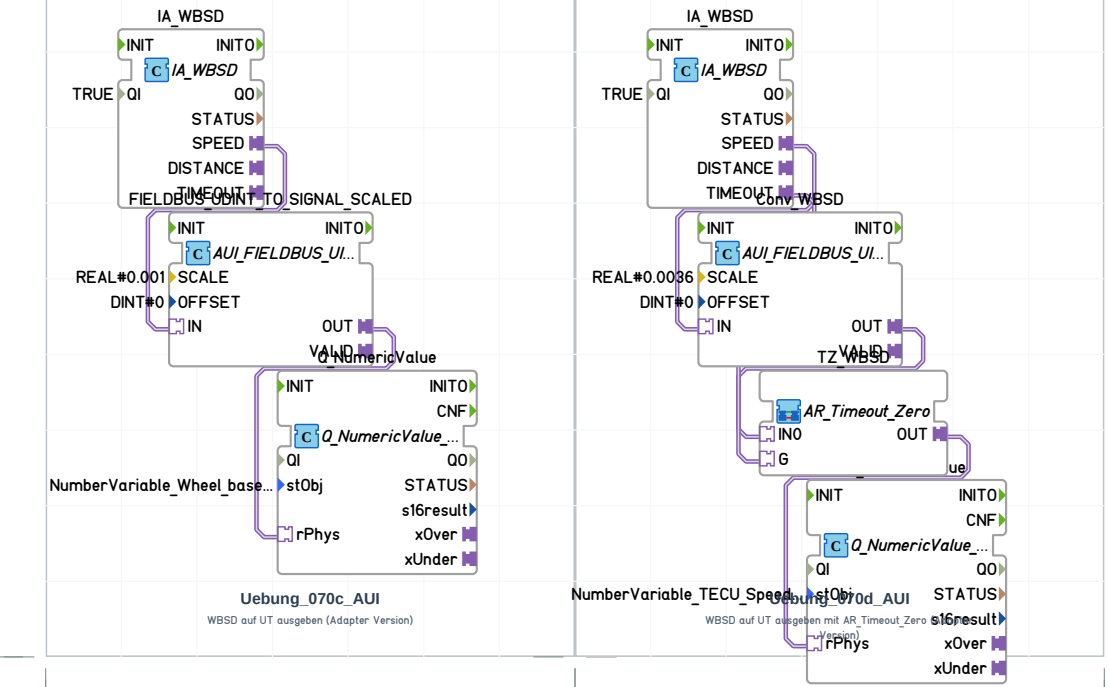

ASSEMBLE_BYTE_FROM_BOOLS
Si ASSEMBLE_AB_FRO...
BIT_00
BIT_01
BIT_02
BIT_03
BIT_04
BIT_05
BIT_06
BIT_07
OUT
SPLIT_BYTE_INTO_BOOLS
Si SPLIT_AB_INTO_A...
IN
Uebung_053_AX
DigitalInput_I1_..I4 auf DigitalOutput_Q1_..Q4
AX_X_TO_BOOL_1
AX_X_TO_BOOL_2
AX_X_TO_BOOL_3
AX_X_TO_BOOL_4
BOOLS_TO_ARR08X
REQ
CNF
IN_00
IN_01
IN_02
IN_03
OUT
Uebung_054_AX
DigitalInput_I1_..I4 auf DigitalOutput_Q1_..Q4, mit Plug and Socket

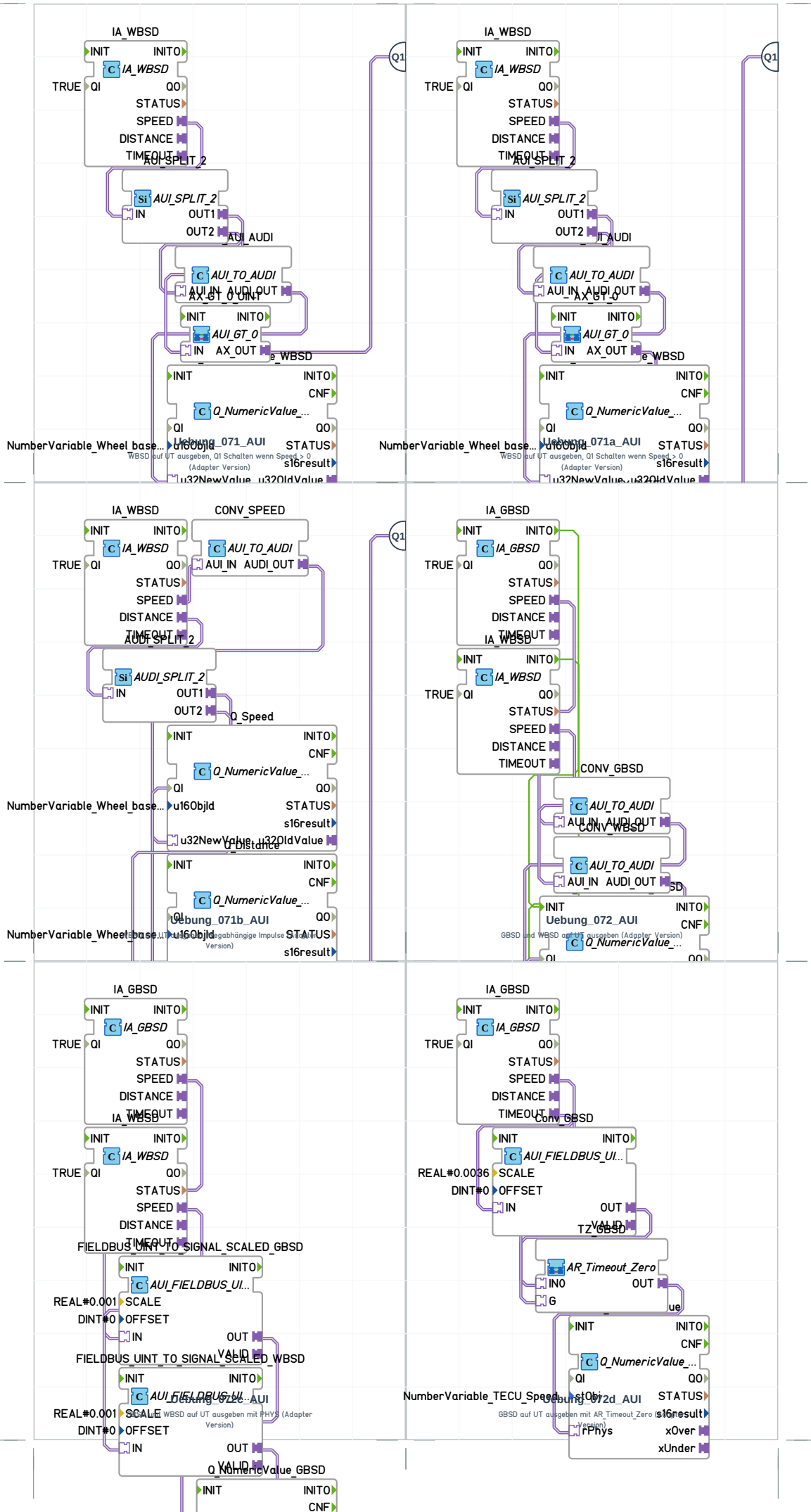

AX_X_TO_BOOL
BOOL_TO_Q
B BOOL_TO_QUARTER
Q_TO_BOOL
B QUARTER_TO_BOOL
Q_TO_STR_STATUS
B QUARTER_TO_STR_...
AX_BOOL_TO_X
C AX_BOOL_TO_X
AX_OUT
Uebung_055_AX
DigitalInput_I1 auf DigitalOutput_Q1, mit Plug and Socket
AX_X_TO_BOOL_1
AX_X_TO_BOOL_2
AX_X_TO_BOOL_3
AX_X_TO_BOOL_4
BOOL_TO_Q_I2
BOOL_TO_Q_I3
Uebung_056_AX
DigitalInput_I1_..I4 auf DigitalOutput_Q1_..Q4, mit Plug and Socket

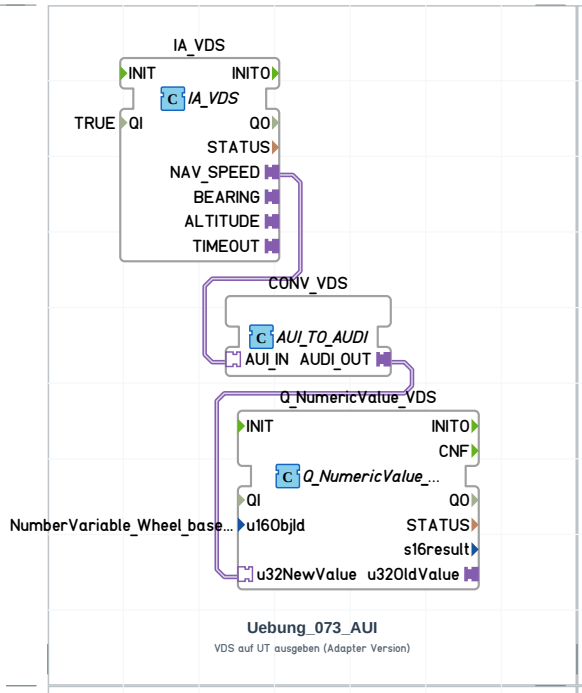

sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_START
RESET
sequence_E_05_A...
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
AS_TO_AUDI
AS_IN AUDI_OUT
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
Uebung_057_AX
s16result
u32NewValue u32OldValue
sequence_E_05_I...
S5_S1
Uebung_058_AX
initval_1
initval_AUI
UINT#1
INIT_VAL
OUT
AUI_SPLIT_2
IN
OUT1
OUT2
SETPOINT_CONDENSED_WORK_S...
NumericValue_AUDI
AUI_DEMUX_4
K
E_DEMUX_1
EO1
EO2
EO3
EO4
Uebung_059_AX
SETPOINT_CONDENSED_WORK_STATE_1_16
REQ
IND
TC_ID
TRUE
PARAMS
DE_NUM_DEVICE
wDevElementN...
wDDI
SECTION_CONTROL_STATE_IN
SECTION_CONTROL_STATE
SPLIT_DWORD
SPLIT_DWORD_INT...
QUARTER_BYTE...
Uebung_060_AX
sequence_T_04_A...
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_START
timeOut
E_TimeOut
TimeOutSocke...
Uebung_061_AX
sequence_T_05_A...
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_START
Uebung_062_AX
IB_07
IB_08
IB_09
IB_10

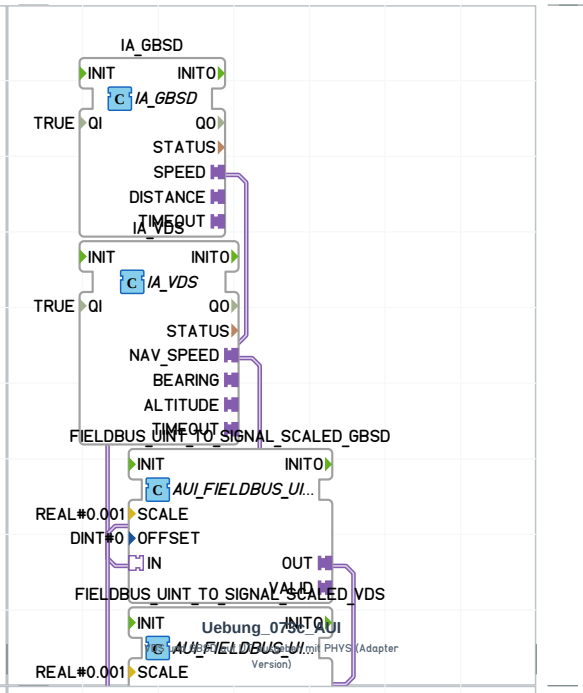

sequence_block
START_S1
RESET
sequence_T_08_A...
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_S6
DT_S6_S7
DT_S7_S8
DT_S8_START
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
DO_S6
DO_S7
DO_S8
timeOut
E_TimeOut
AS_IN
AUDI_OUT
sequence_block
START_S1
S1_S2
S2_S3
S3_S4
S4_S5
S5_S6
S6_S7
S7_S8
S8_S1
RESET
sequence_Patter...
DT_S1_S2
DT_S2_S3
DT_S3_S4
DT_S4_S5
DT_S5_S6
DT_S6_S7
DT_S7_S8
DT_S8_S1
P_S1
P_S2
P_S3
P_S4
P_S5
P_S6
Uebung_064_AX
Musterschnittkette 8-Kanal ENDLOS (Adapter Version)

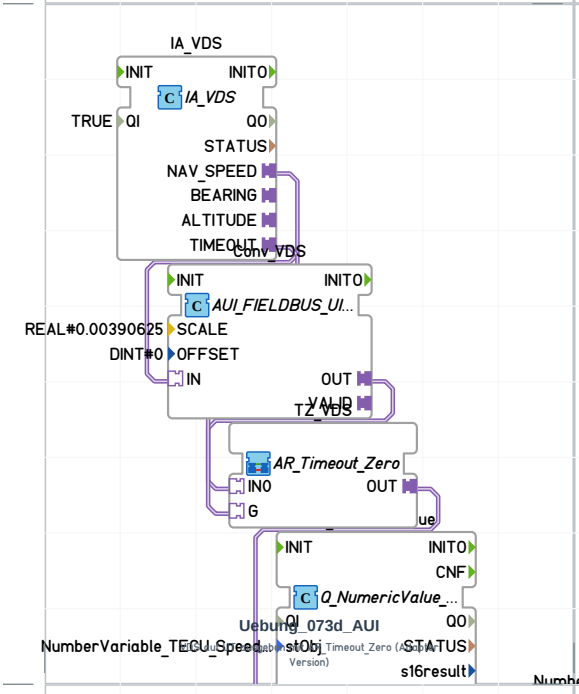

IA_WBSD
TRUE
STATUS
SPEED
DISTANCE
TIMEOUT
CONV_AUI_AUDI
AUI_TO_AUDI
Q_NumericValue_WBSD
NumberVariable_Wheel_base...
u16ObjId
u32NewValue
u32OldValue
Uebung_070_AUI
WBSD auf UT ausgeben (Adapter Version)
Q_NumericValue
CNF
Q_NumericValue_...
I_WBSD
UINT_TO_UDINT
Uebung_070_AX
WBSD auf UT ausgeben

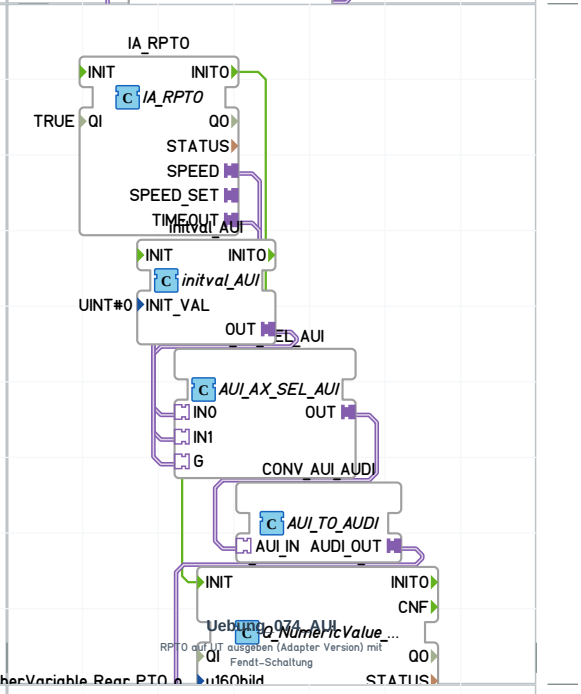

IA_WBSD
FIELDBUS_UDINT_TO_SIGNAL_SCALED
AUI_FIELDBUS_UI...
REAL#0.001
SCALE
DINT#0
OFFSET
IN
OUT
VALID
Q_NumericValue
NumberVariable_Wheel_base...
stObj
rPhys
xOver
xUnder
Uebung_070c_AUI
WBSD auf UT ausgeben (Adapter Version)
Conv_WBSD
REAL#0.0036
TZ_WBSD
AR_Timeout_Zero
IN0
G
NumberVariable_TECU_Speed
Uebung_070d_AUI
WBSD auf UT ausgeben mit AR_Timeout_Zero (Adapter Version)

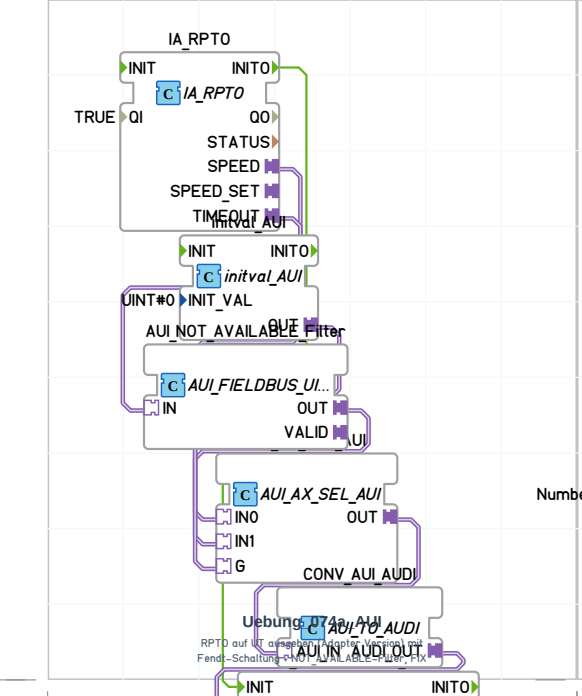

IA_WBSD
INIT
INITO
TRUE
QI
QO
STATUS
SPEED
DISTANCE
TIMEOUT
AUI_SPLIT_2
IN
OUT1
OUT2
AUI_AUDI
AUI_TO_AUDI
AUI_IN
AUDI_OUT
AUI_GT_0
AX_OUT
Q1
CNF
Q_NumericValue_...
u160bjId
s16result
u32NewValue
u32OldValue
Uebung_071_AUI
Uebung_071a_AUI
NumberVariable_Wheel base...
CONV_SPEED
Q_Speed
Q_Distance
Uebung_071b_AUI
IA_GBSD
CONV_GBSD
CONV_WBSD
Uebung_072_AUI
FIELDBUS_UINT_TO_SIGNAL_SCALED_GBSD
AUI_FIELDBUS_UI...
REAL#0.001
DINT#0
SCALE
OFFSET
VALID
FIELDBUS_UINT_TO_SIGNAL_SCALED_WBSD
Uebung_072c_AUI
Q_NumericValue_GBSD
REAL#0.0036
TZ_GBSD
AR_Timeout_Zero
IN0
G
Uebung_072d_AUI
NumberVariable_TECU_Speed
rPhys
xOver
xUnder

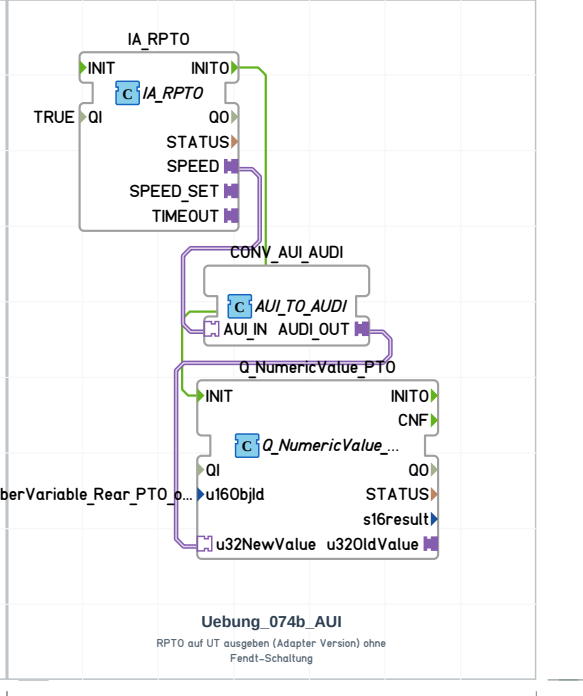

IA_VDS
INIT
INITO
IA_VDS
TRUE
QI
QO
STATUS
NAV_SPEED
BEARING
ALTITUDE
TIMEOUT
CONV_VDS
AUI_TO_AUDI
AUI_IN
AUDI_OUT
Q_NumericValue_VDS
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
NumberVariable_Wheel_base...
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_073_AUI
VDS auf UT ausgeben (Adapter Version)

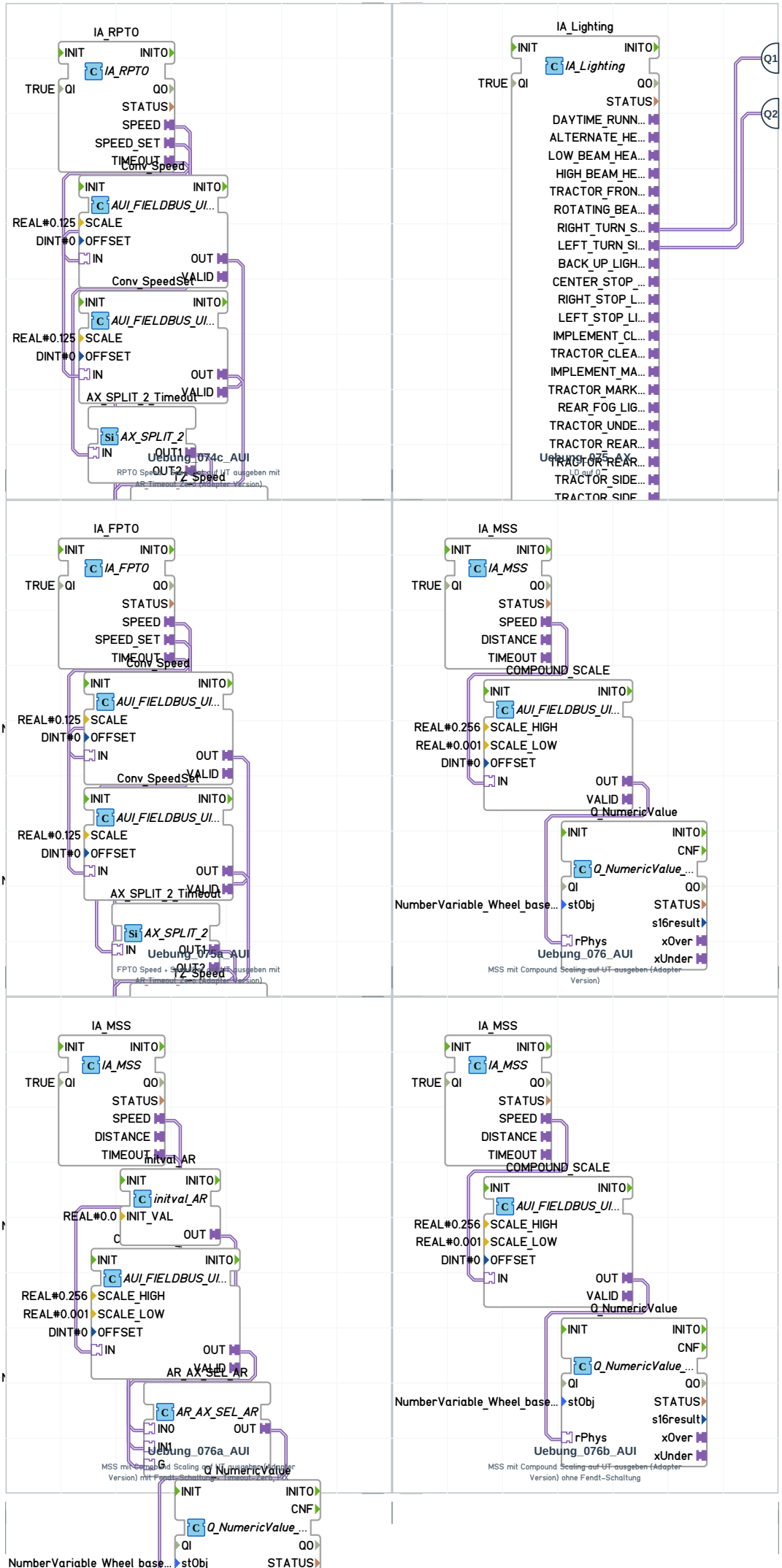

IA_GBSD
INIT
INITO
IA_GBSD
TRUE
QI
QO
STATUS
SPEED
DISTANCE
TIMEOUT
IA_VDS
IA_VDS
NAV_SPEED
BEARING
ALTITUDE
FIELDBUS_UINT_TO_SIGNAL_SCALED_GBSD
AUI_FIELDBUS_UI...
REAL#0.001
SCALE
DINT#0
OFFSET
IN
OUT
VALID
FIELDBUS_UINT_TO_SIGNAL_SCALED_VDS
REAL#0.001
SCALE

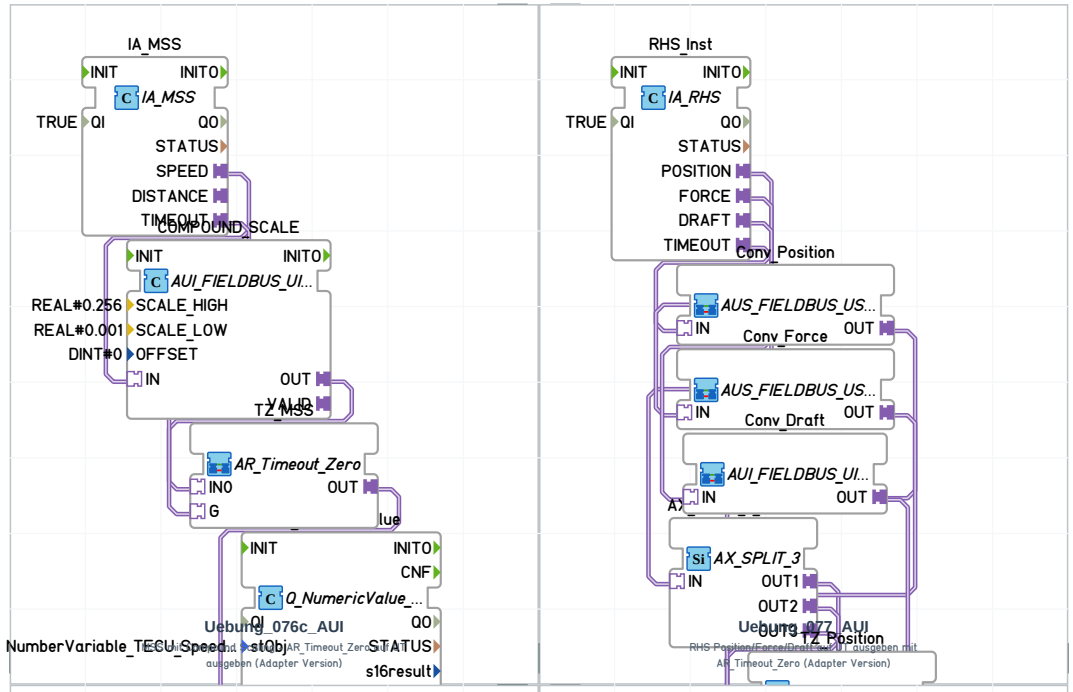

IA_VDS
INIT
INITO
IA_VDS
TRUE
QI
QO
STATUS
NAV_SPEED
BEARING
ALTITUDE
TIMEOUT
Conv_VDS
AUI_FIELDBUS_UI...
REAL#0.00390625
SCALE
DINT#0
OFFSET
IN
OUT
TZ_VDS
AR_Timeout_Zero
IN0
OUT
G
Q_NumericValue_...
QI
QO
STATUS
s16result
NumberVariable_TECU_Speed
Uebung_073d_AUI

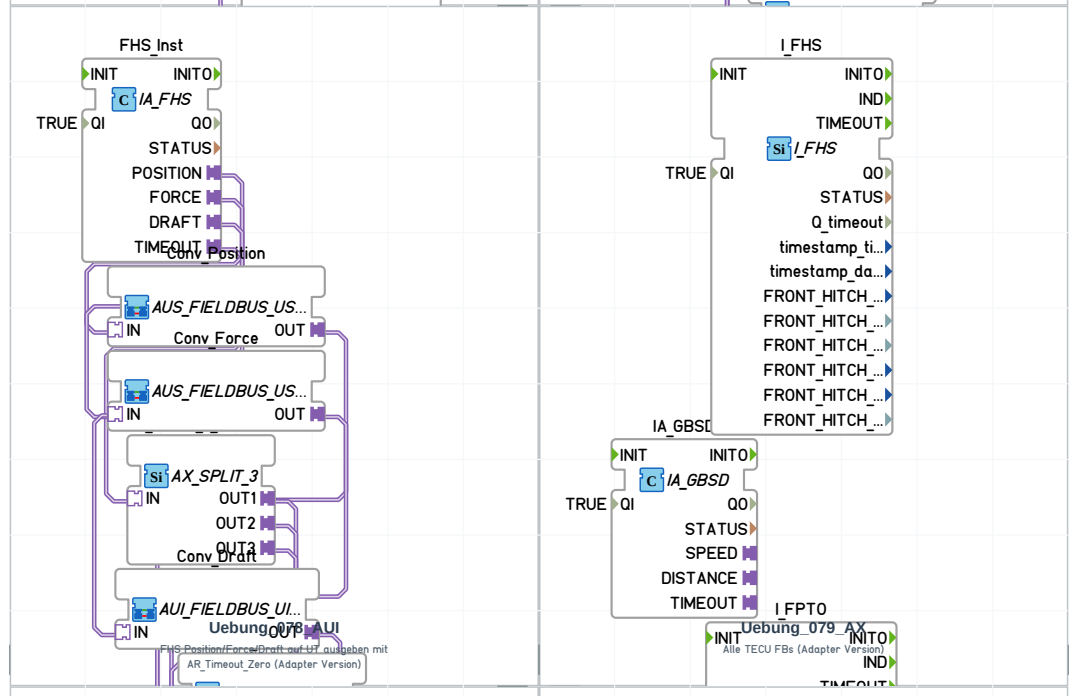

IA_RPTO
INIT
INITO
IA_RPTO
TRUE
QI
QO
STATUS
SPEED
SPEED_SET
TIMEOUT
initval_AUI
initval_AUI
UINT#0
INIT_VAL
OUT
AUI_AX_SEL_AUI
IN0
OUT
IN1
G
CONV_AUI_AUDI
AUI_TO_AUDI
AUI_IN
AUDI_OUT
Q_NumericValue_...
QI
QO
STATUS
s16result

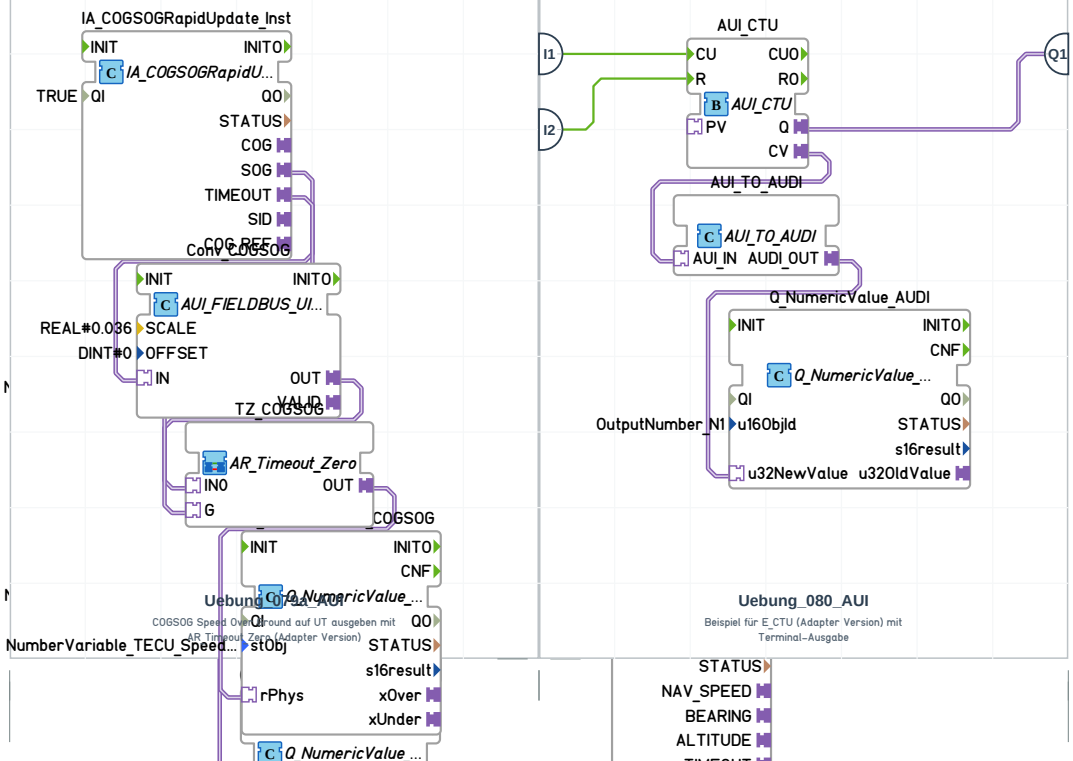

IA_RPTO
INIT
INITO
IA_RPTO
TRUE
QI
QO
STATUS
SPEED
SPEED_SET
TIMEOUT
initval_AUI
initval_AUI
UINT#0
INIT_VAL
OUT
AUI_NOT_AVAILABLE_Filter
AUI_FIELDBUS_UI...
IN
OUT
VALID
AUI_AX_SEL_AUI
IN0
OUT
IN1
G
CONV_AUI_AUDI
AUI_TO_AUDI
AUI_IN
AUDI_OUT
Uebung_074a_AUI
Q_NumericValue_...
QI
QO

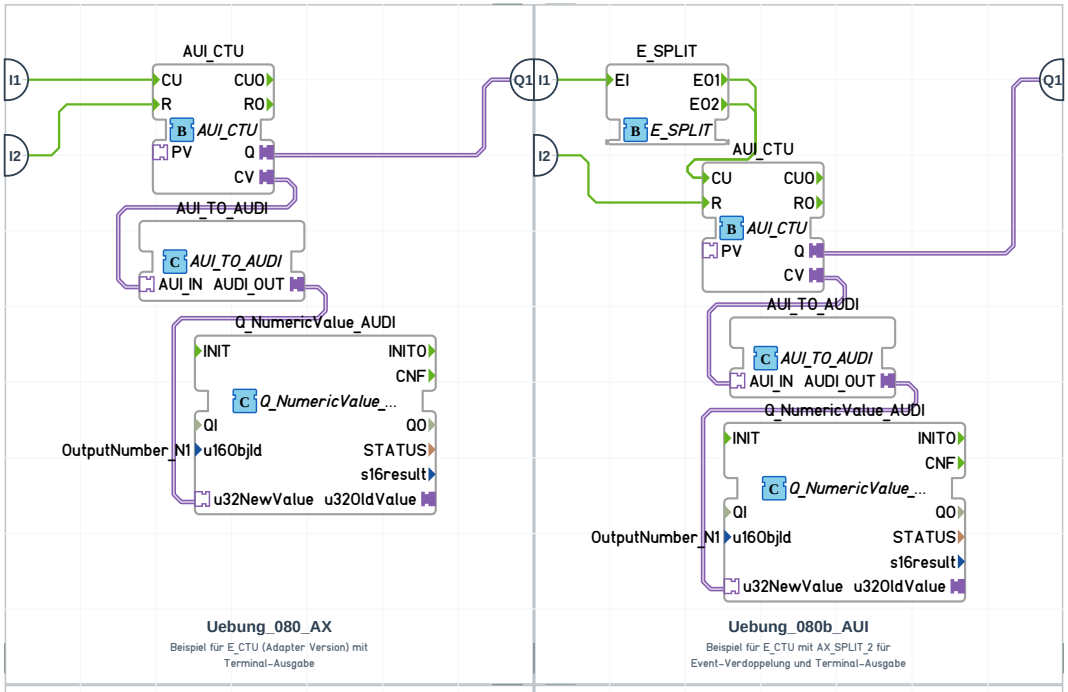

IA_RPTO
INIT
INITO
IA_RPTO
TRUE
QI
QO
STATUS
SPEED
SPEED_SET
TIMEOUT
CONV_AUI_AUDI
AUI_TO_AUDI
AUI_IN
AUDI_OUT
Q_NumericValue_PTO
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
NumberVariable_Rear_PTO_o...
u16ObjId
STATUS
s16result
u32NewValue
u32OldValue
Uebung_074b_AUI
RPTO auf UT ausgeben (Adapter Version) ohne Fendt-Schaltung

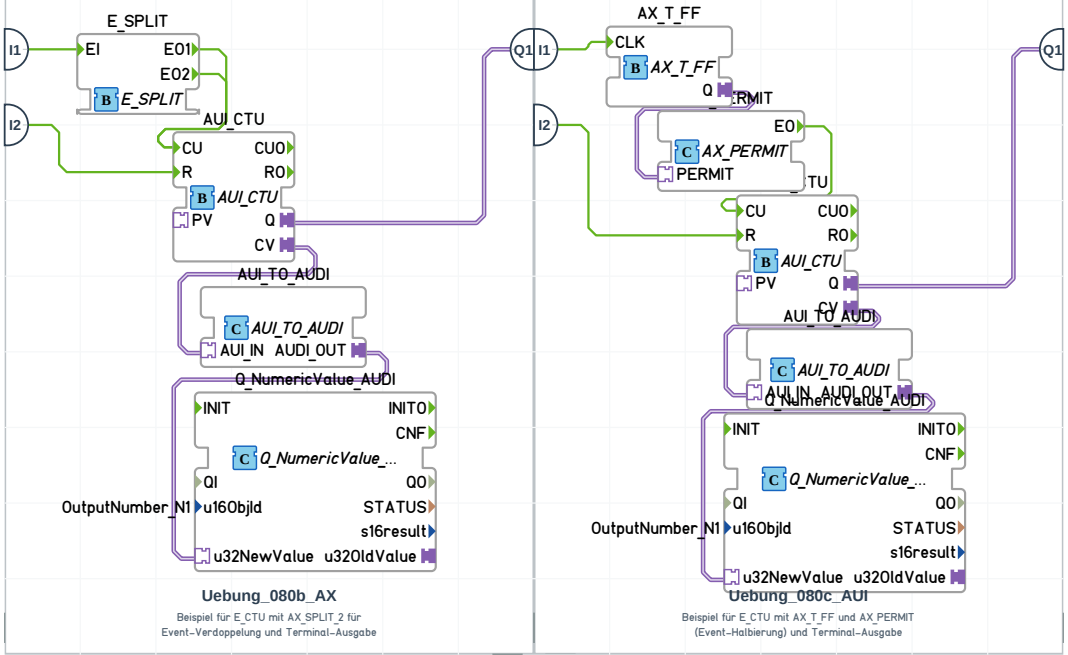

IA_RPTO
INIT
INITO
IA_RPTO
TRUE
QI
QO
STATUS
SPEED
SPEED_SET
TIMEOUT
Conv_Speed
AUI_FIELDBUS_UI...
REAL#0.125
SCALE
DINT#0
OFFSET
IN
OUT
VALID
Conv_SpeedSet
AX_SPLIT_2_Timeout
AX_SPLIT_2
OUT1
OUT2
TZ_Speed
Uebung_074c_AUI
IA_Lighting
Q1
Q2
DAYTIME_RUNN...
ALTERNATE_HE...
LOW_BEAM_HEA...
HIGH_BEAM_HE...
TRACTOR_FRON...
ROTATING_BEA...
RIGHT_TURN_S...
LEFT_TURN_SI...
BACK_UP_LIGH...
CENTER_STOP_...
RIGHT_STOP_L...
LEFT_STOP_LI...
IMPLEMENT_CL...
TRACTOR_CLEA...
IMPLEMENT_MA...
TRACTOR_MARK...
REAR_FOG_LIG...
TRACTOR_UNDE...
TRACTOR_REAR...
TRACTOR_SIDE...
Uebung_075_AX
IA_FPTO
Uebung_075a_AUI
IA_MSS
DISTANCE
COMPOUND_SCALE
REAL#0.256
SCALE_HIGH
REAL#0.001
SCALE_LOW
Q_NumericValue
CNF
Q_NumericValue_...
NumberVariable_Wheel_base...
stObj
s16result
xOver
xUnder
rPhys
Uebung_076_AUI
initval_AR
REAL#0.0
INIT_VAL
AR_AX_SEL_AR
IN0
IN1
G
Uebung_076a_AUI
Uebung_076b_AUI

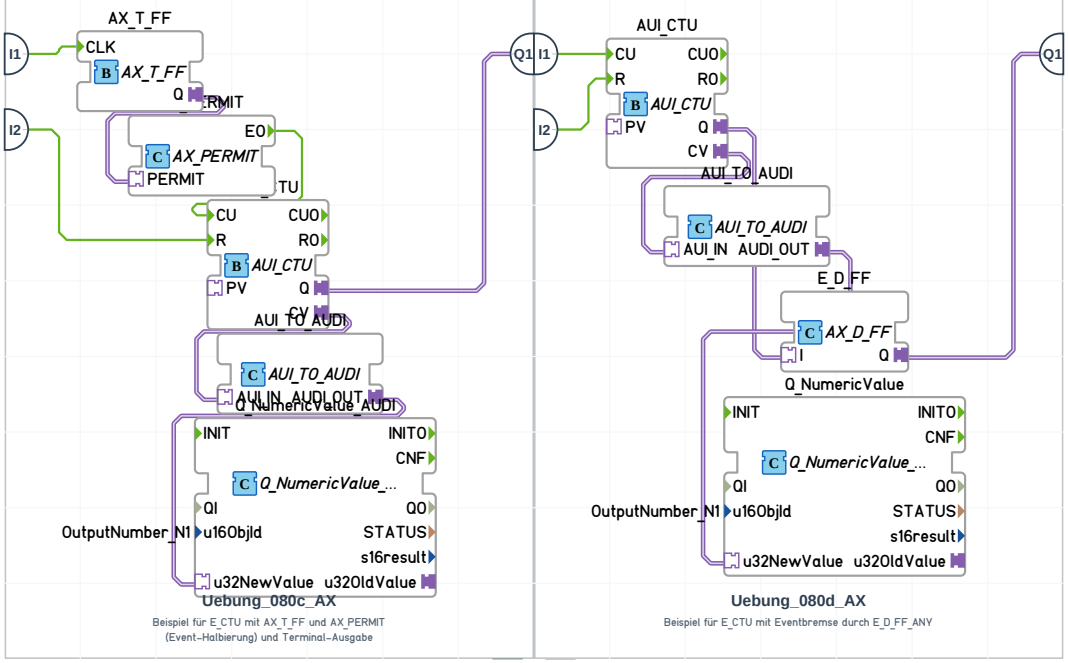

IA_MSS
INIT
INITO
IA_MSS
TRUE
QI
QO
STATUS
SPEED
DISTANCE
TIMEOUT
COMPOUND_SCALE
AUI_FIELDBUS_UI...
REAL#0.256
SCALE_HIGH
REAL#0.001
SCALE_LOW
DINT#0
OFFSET
IN
OUT
VALID
TZ_MSS
AR_Timeout_Zero
IN0
G
Q_NumericValue_...
Uebung_076c_AUI
NumberVariable_TECU_Speed...
stObj
s16result
CNF
RHS_Inst
IA_RHS
POSITION
FORCE
DRAFT
Conv_Position
AUS_FIELDBUS_US...
Conv_Force
Conv_Draft
AX_SPLIT_3
OUT1
OUT2
TZ_Position

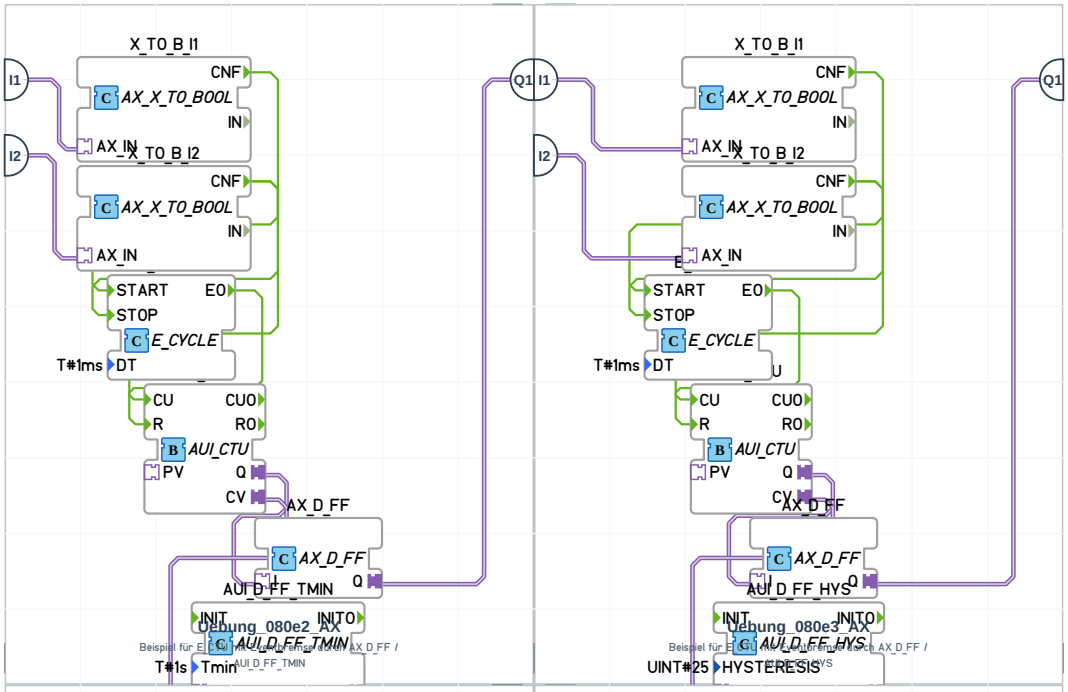

FHS_Inst
IA_FHS
I_FHS
IND
TIMEOUT
Q_timeout
timestamp_ti...
timestamp_da...
FRONT_HITCH_...
IA_GBSD
I_FPTO
Uebung_079_AX

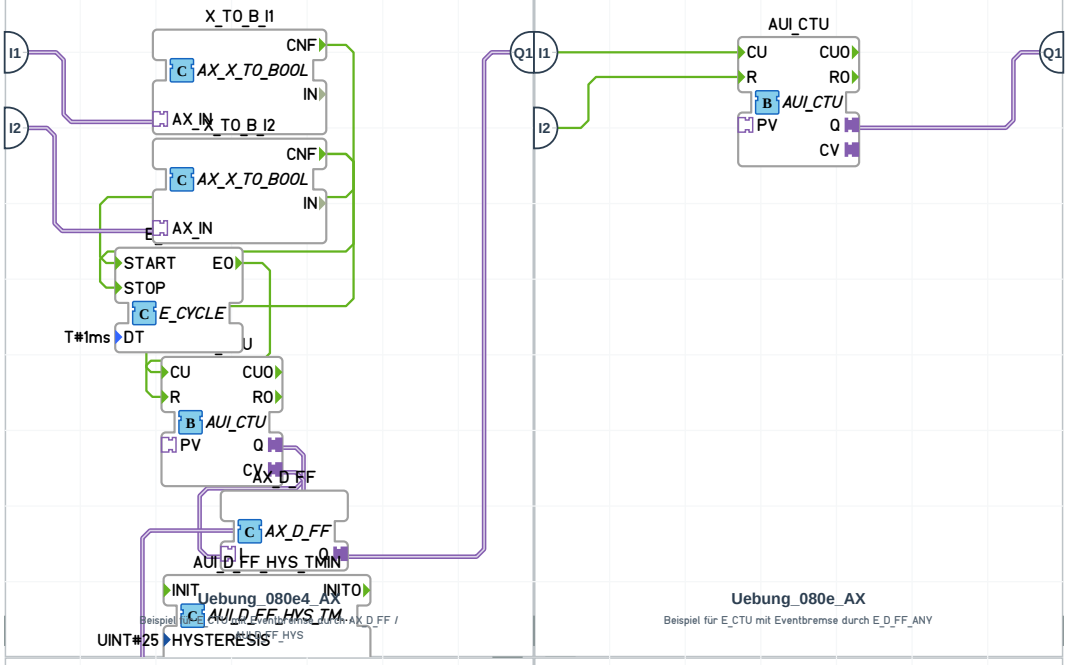

IA_COGSOGRapidUpdate_Inst
IA_COGSOGRapidU...
COG
SOG
SID
COG_REF
Conv_COGSOG
SCALE
OFFSET
TZ_COGSOG
COGSOG
Uebung_078_AUI
AUI_CTU
CU
CUO
R
RO
PV
Q
CV
Q1
I1
I2
AUI_TO_AUDI
AUI_IN
AUDI_OUT
Q_NumericValue_AUDI
u16ObjId
u32NewValue
u32OldValue
Uebung_080_AUI
Beispiel für E_CTU (Adapter Version) mit
Terminal-Ausgabe
NAV_SPEED
BEARING
ALTITUDE

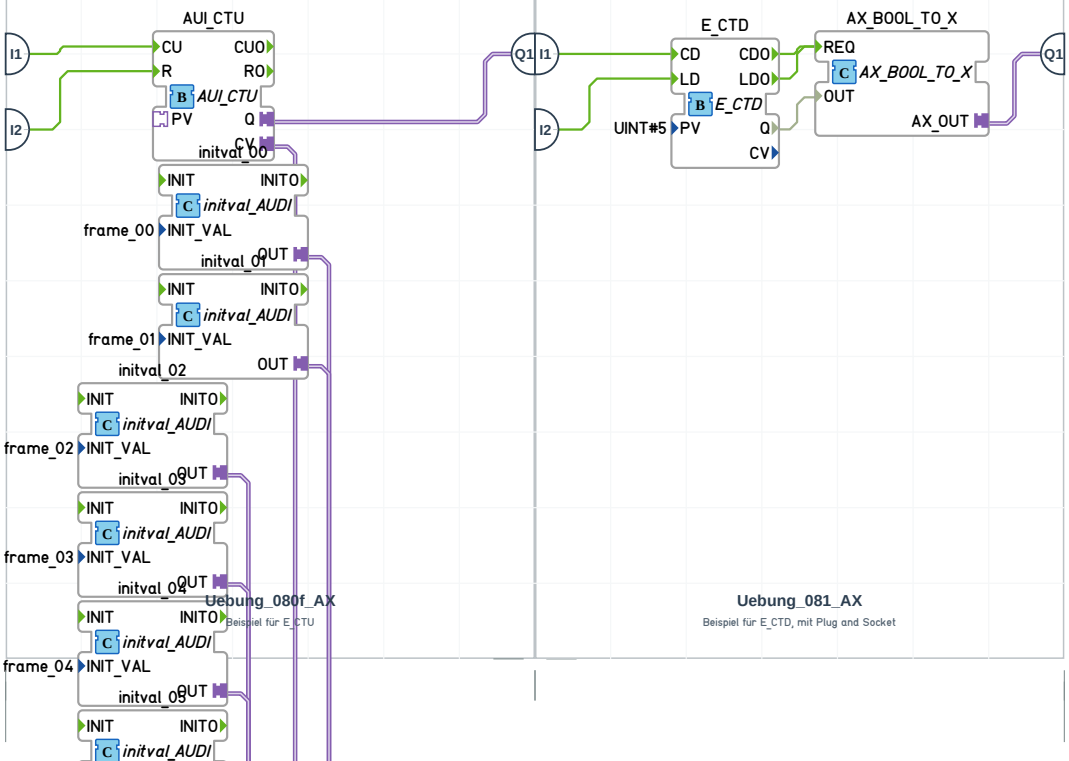

AUI_CTU
I1
I2
CU
CUO
R
RO
B AUI_CTU
PV
Q
CV
Q1
AUI_TO_AUDI
C AUI_TO_AUDI
AUI_IN AUDI_OUT
Q_NumericValue_AUDI
INIT
INITO
CNF
C Q_NumericValue_...
QI
QO
OutputNumber_N1
u16ObjId
STATUS
s16result
u32NewValue u32OldValue
Uebung_080_AX
Beispiel für E_CTU (Adapter Version) mit Terminal-Ausgabe
E_SPLIT
EI
EO1
EO2
B E_SPLIT
Uebung_080b_AUI
Beispiel für E_CTU mit AX_SPLIT_2 für Event-Verdoppelung und Terminal-Ausgabe

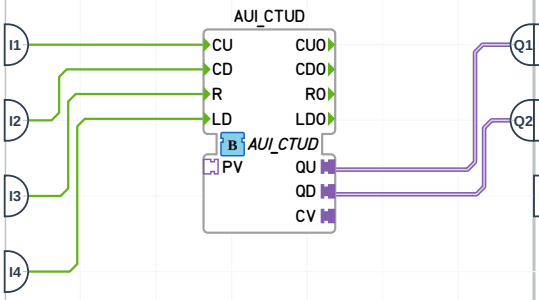

E_SPLIT
AUI_CTU
Uebung_080b_AX
Beispiel für E_CTU mit AX_SPLIT_2 für Event-Verdoppelung und Terminal-Ausgabe
AX_T_FF
CLK
B AX_T_FF
C AX_PERMIT
PERMIT
Uebung_080c_AUI
Beispiel für E_CTU mit AX_T_FF und AX_PERMIT (Event-Halbierung) und Terminal-Ausgabe


AX_T_FF
AUI_CTU
Uebung_080c_AX
Beispiel für E_CTU mit AX_T_FF und AX_PERMIT (Event-Halbierung) und Terminal-Ausgabe
E_D_FF
C AX_D_FF
Q_NumericValue
Uebung_080d_AX
Beispiel für E_CTU mit Eventbremse durch E_D_FF_ANY


X_TO_B_I1
I1
I2
CNF
AX_X_TO_BOOL
IN
AX_IN
X_TO_B_I2
START
EO
STOP
E_CYCLE
T#1ms
DT
CU
CUO
R
RO
AUI_CTU
PV
Q
CV
AX_D_FF
AX_D_FF
AUI_D_FF_TMIN
INIT
INITO
AUI_D_FF_TMIN
T#1s
Tmin
Q1
Uebung_080e2_AX
Beispiel für E_CTU mit Eventbremse durch AX_D_FF / AUI_D_FF_TMIN
X_TO_B_I1
AUI_D_FF_HYS
UINT#25
HYSTERESIS
Uebung_080e3_AX
Beispiel für E_CTU mit Eventbremse durch AX_D_FF / AUI_D_FF_HYS

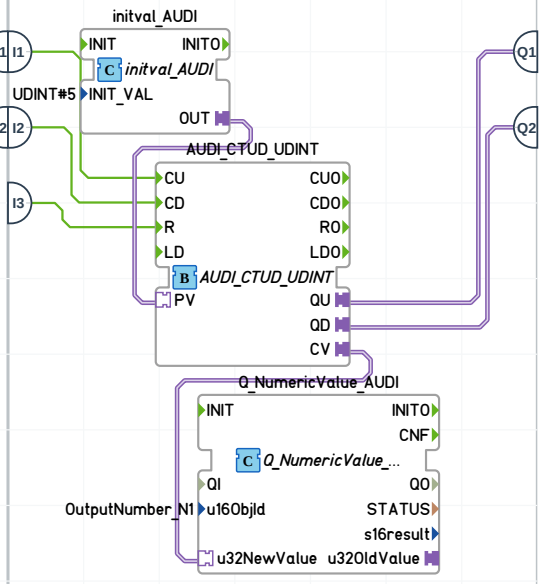

X_TO_B_I1
AUI_D_FF_HYS_TMIN
AUI_D_FF_HYS_TM
UINT#25
HYSTERESIS
Uebung_080e4_AX
Beispiel für E_CTU mit Eventbremse durch AX_D_FF / AUI_D_FF_HYS
AUI_CTU
Uebung_080e_AX
Beispiel für E_CTU mit Eventbremse durch E_D_FF_ANY


AUI_CTU
initval_00
initval_AUDI
frame_00
INIT_VAL
initval_01
frame_01
initval_02
frame_02
initval_03
frame_03
initval_04
frame_04
initval_05
OUT
Uebung_080f_AX
Beispiel für E_CTU
E_CTD
CD
CDO
LD
LDO
UINT#5
AX_BOOL_TO_X
REQ
AX_OUT
Uebung_081_AX
Beispiel für E_CTD, mit Plug and Socket



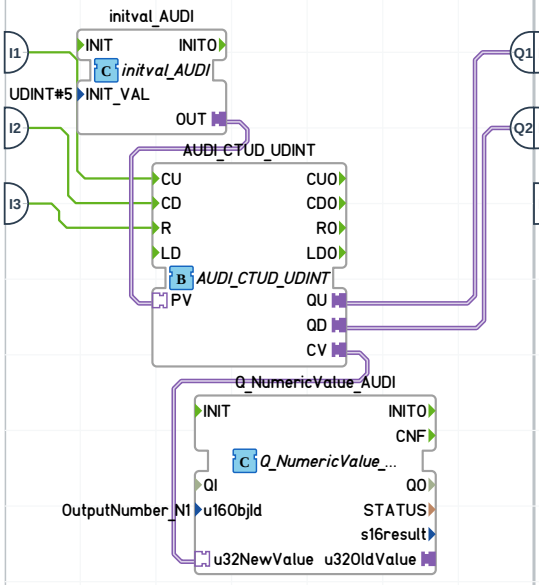
Uebung_082_AX

Beispiel für E_CTUD, mit Plug and Socket



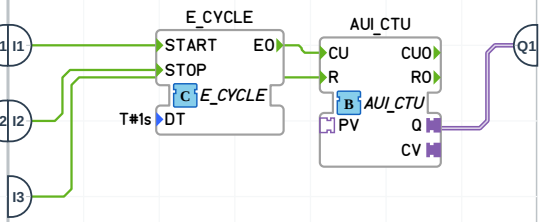
Uebung_083_AUDI

Beispiel für AUDI_CTUD_UDINT (Adapter-basiert)



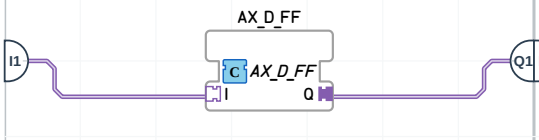
Uebung_083_AX

Beispiel für AUDI_CTUD_UDINT (Adapter-basiert)



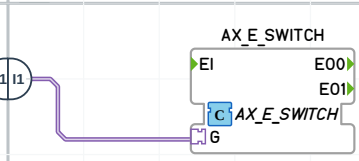
Uebung_084_AX

Beispiel für E_CYCLE



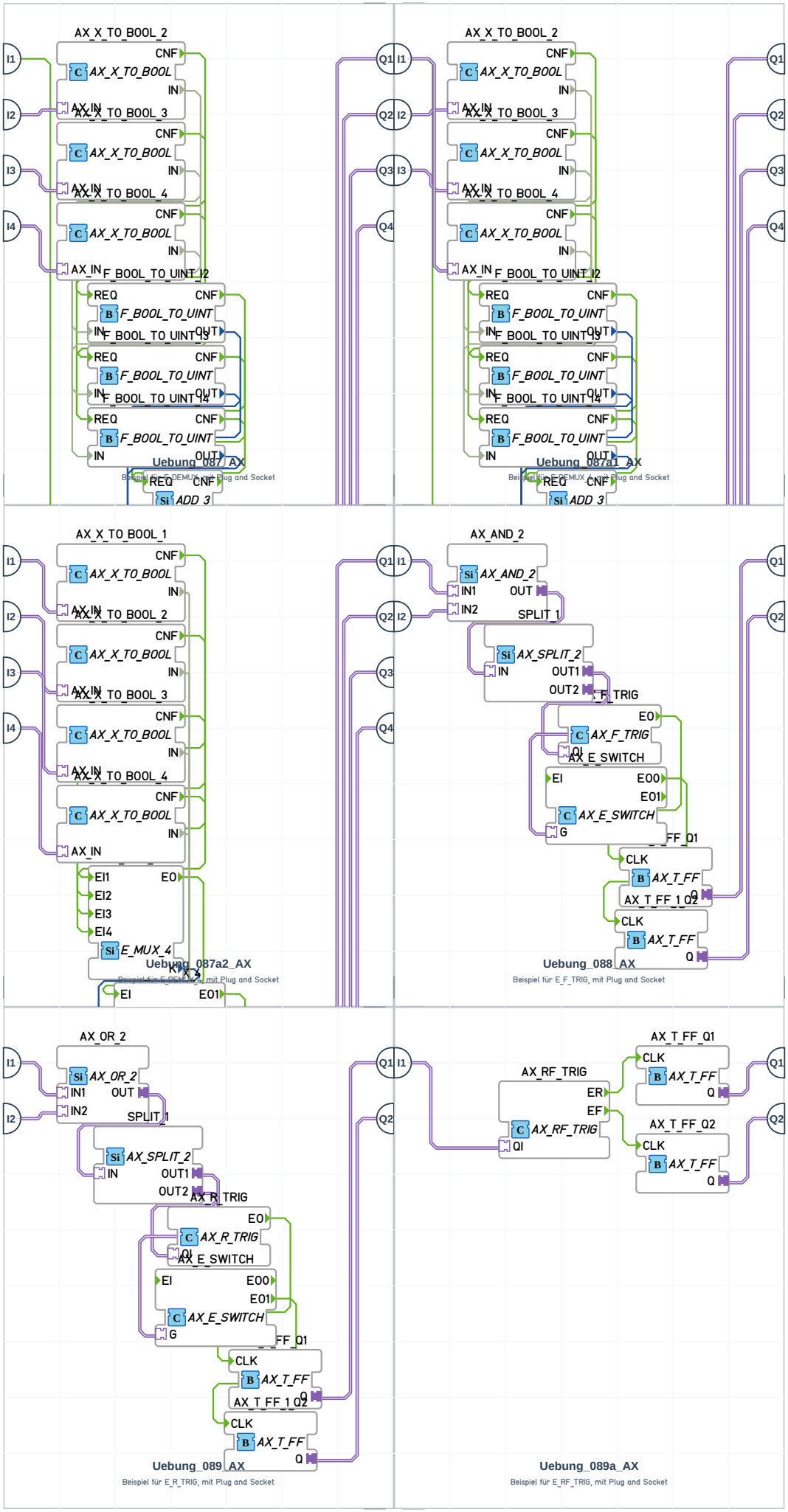
Uebung_085_AX

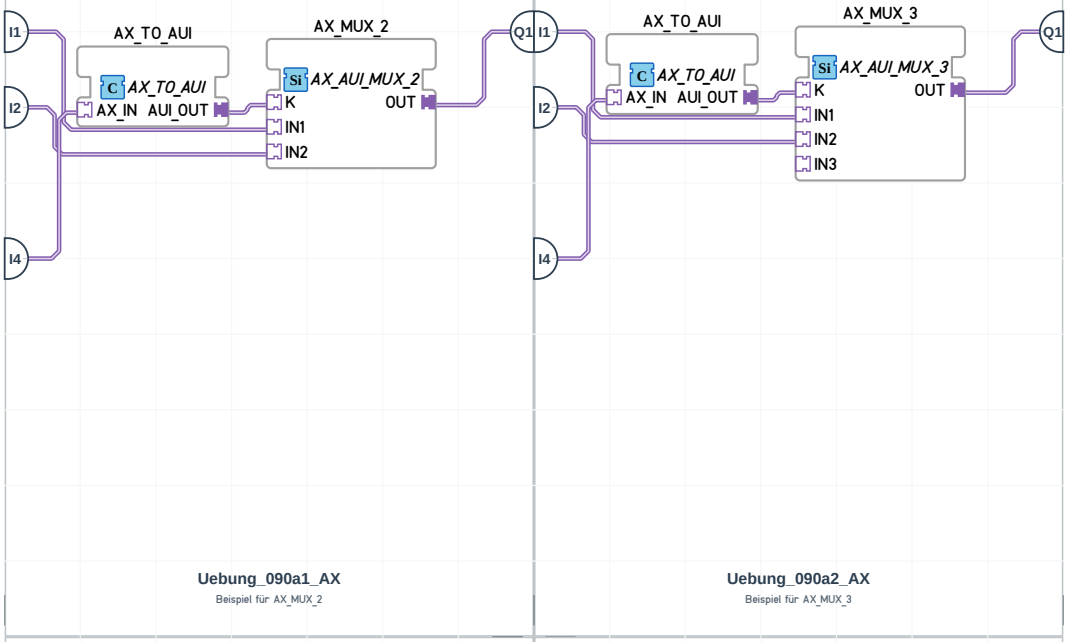
Beispiel für E_D_FF, mit Plug and Socket



Uebung_086_AX

Beispiel für E_SWITCH, mit Plug and Socket


AX_X_TO_BOOL_2
AX_X_TO_BOOL_3
AX_X_TO_BOOL_4
F_BOOL_TO_UINT_2
F_BOOL_TO_UINT_3
F_BOOL_TO_UINT_4
ADD_3
Uebung_087_AX
Beispiel für E_DEMUX, mit Plug and Socket
Uebung_087a1_AX
AX_X_TO_BOOL_1
E_MUX_4
Uebung_087a2_AX
AX_AND_2
AX_SPLIT_2
AX_F_TRIG
AX_E_SWITCH
AX_T_FF
Uebung_088_AX
Beispiel für E_F_TRIG, mit Plug and Socket
AX_OR_2
AX_R_TRIG
Uebung_089_AX
Beispiel für E_R_TRIG, mit Plug and Socket
AX_RF_TRIG
AX_T_FF_Q1
AX_T_FF_Q2
Uebung_089a_AX
Beispiel für E_RF_TRIG, mit Plug and Socket

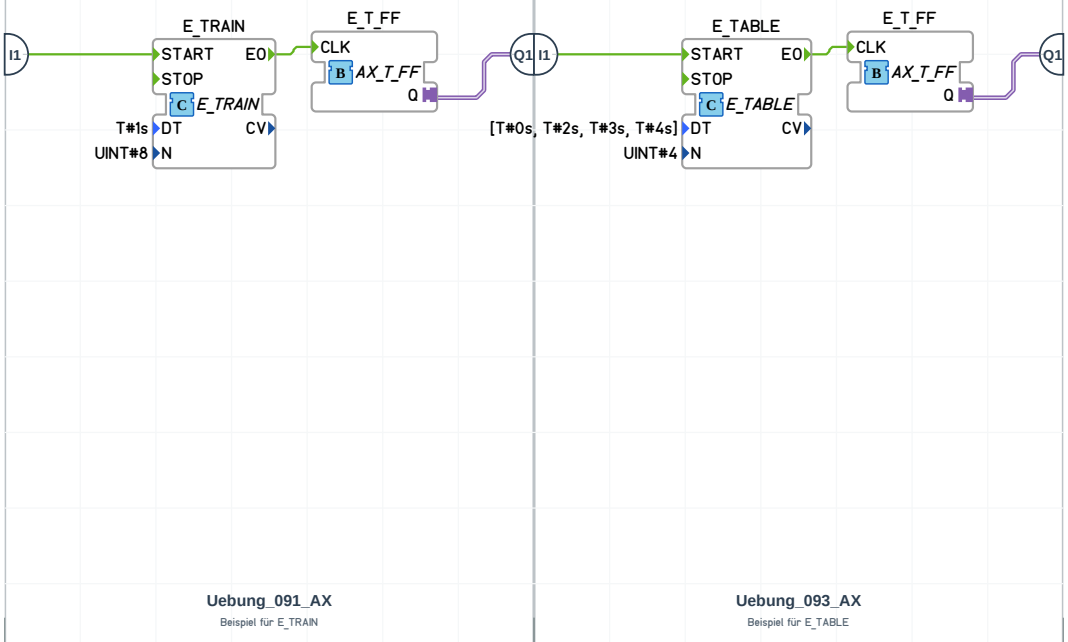


Uebung_090a1_AX

Beispiel für AX_MUX_2

Uebung_090a2_AX

Beispiel für AX_MUX_3

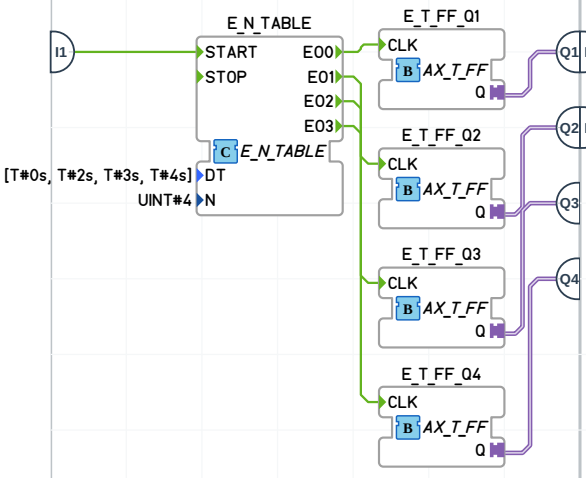


Uebung_091_AX

Beispiel für E_TRAIN

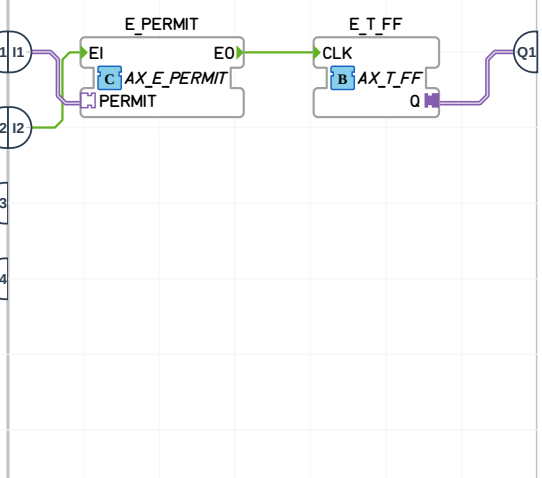
Uebung_093_AX

Beispiel für E_TABLE



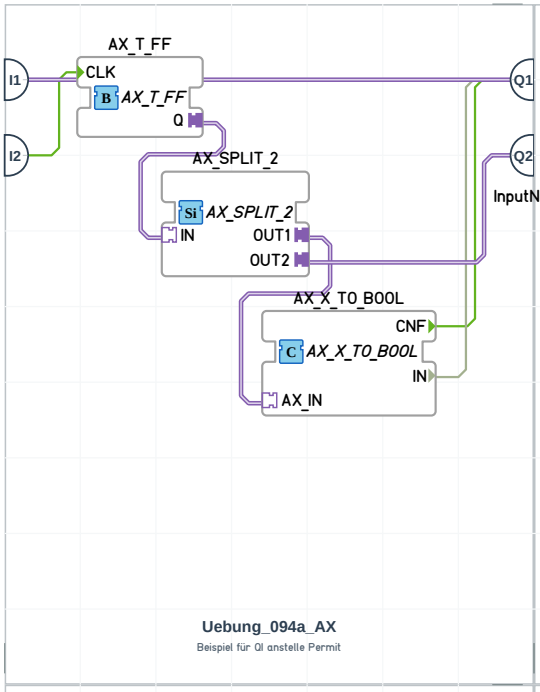
Uebung_093b_AX

Beispiel für E_N_TABLE



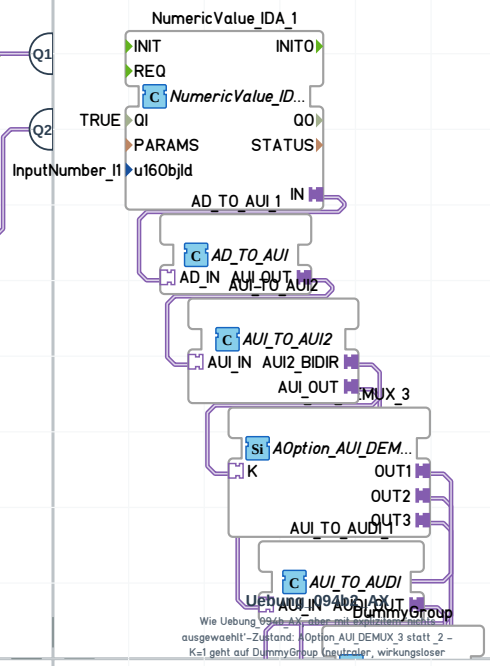
Uebung_094_AX

Beispiel für E_PERMIT



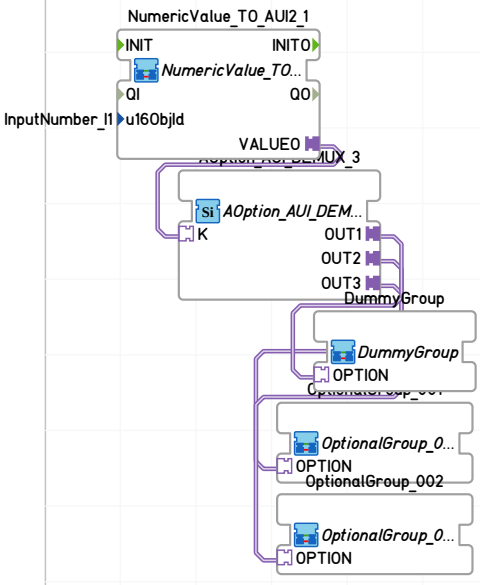
Uebung_094a_AX

Beispiel für QI anstelle Permit



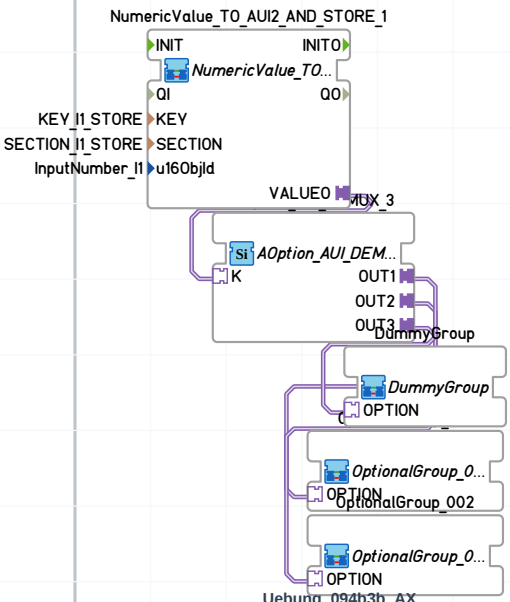
Uebung_094b_AX

Wie Uebung_094b_AX, aber mit explizitem 'nichts in ausgewaehlt'-Zustand: AOption_AUI_DEMUX_3 statt _2 – K=1 geht auf DummyGroup (neutraler, wirkungsloser



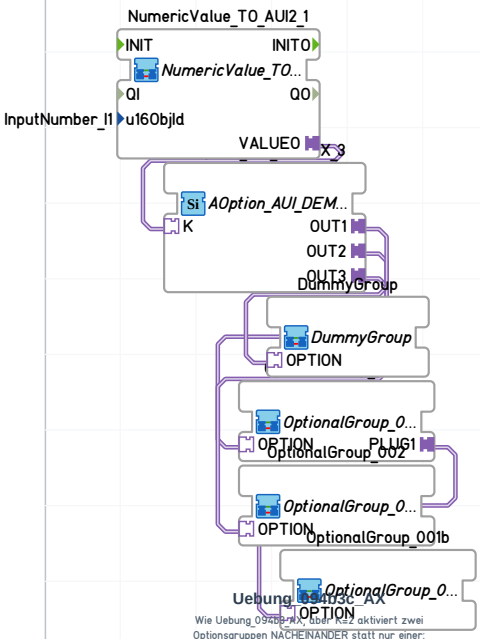
Uebung_094b3_AX

Wie Uebung_094b2_AX, aber mit MyLib::sys::NumericValue_TO_AUI2 als wiederverwendbarem Baustein statt der inline



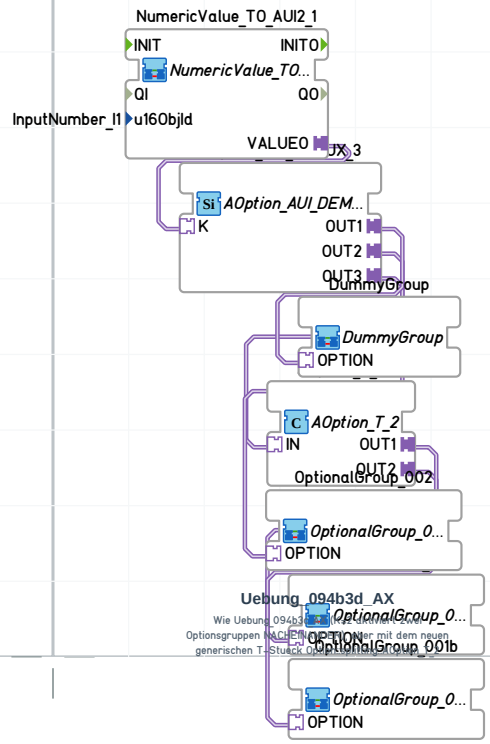
Uebung_094b3b_AX

Wie Uebung_094b3_AX, aber mit MyLib::sys::NumericValue_TO_AUI2_AND_STORE statt NumericValue_TO_AUI2: der Index (K) wird zusaetzlich



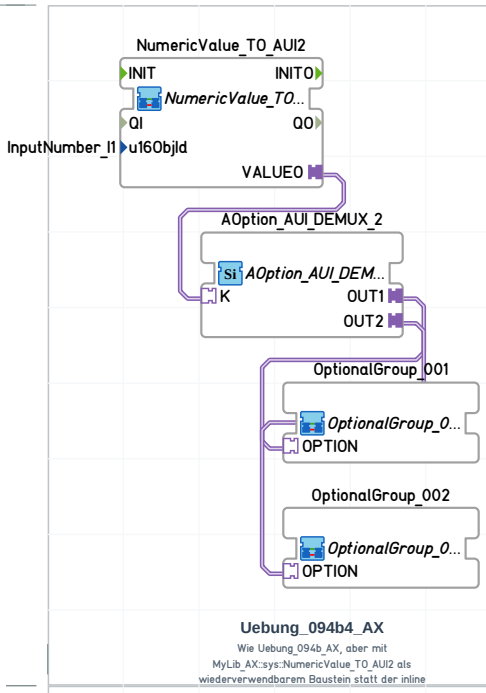
Uebung_094b3c_AX

Wie Uebung_094b3_AX, aber PLG2 aktiviert zwei Optionsgruppen NACHEINANDER statt nur einer: OptionalGroup_001a wird vom DEMUX aktiviert und



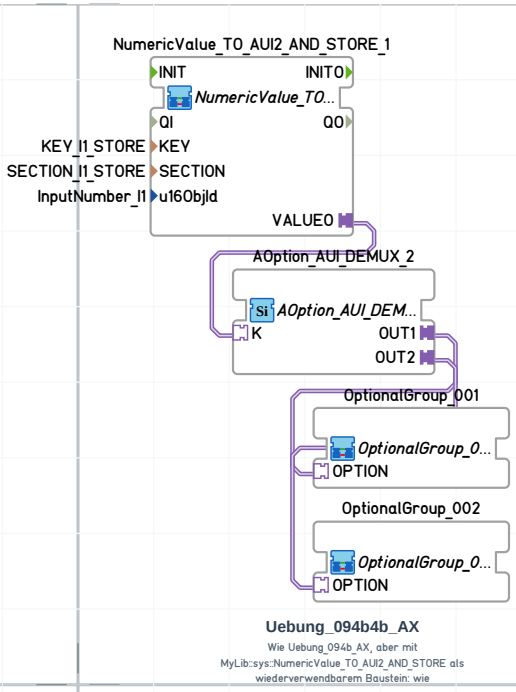
Uebung_094b3d_AX

Wie Uebung_094b3_AX, aber mit dem neuen generischen T-Stueck (AOption_T_2) statt zweier Optionsgruppen NACHEINANDER



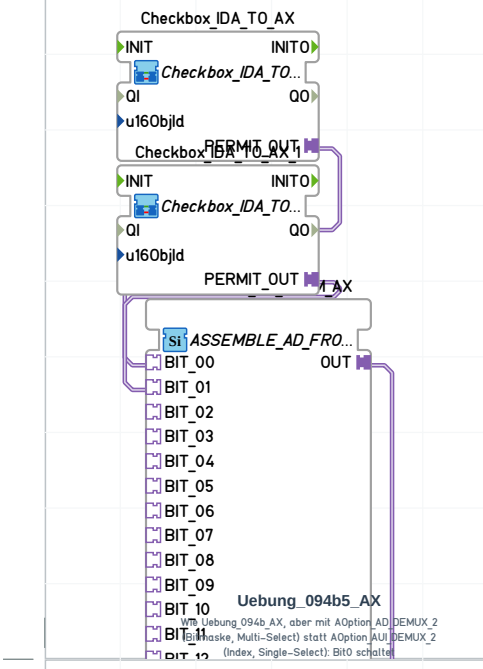
Uebung_094b4_AX

Wie Uebung_094b_AX, aber mit MyLib_AX::sys::NumericValue_TO_AUI2 als wiederverwendbarem Baustein statt der inline



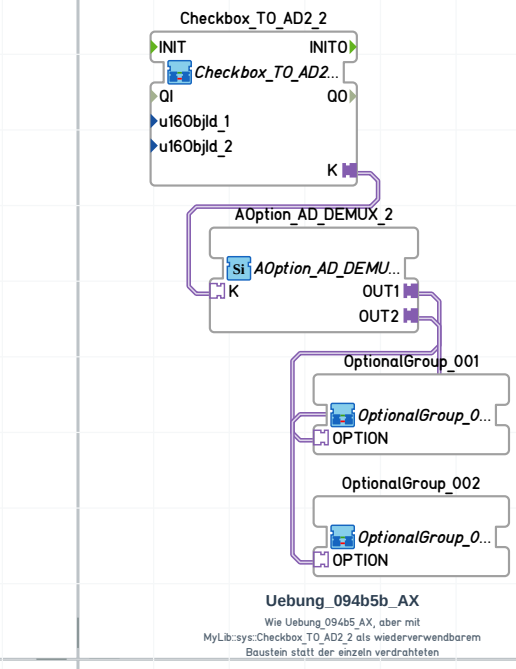
Uebung_094b4b_AX

Wie Uebung_094b_AX, aber mit MyLib::sys::NumericValue_TO_AUI2_AND_STORE als wiederverwendbarem Baustein: wie



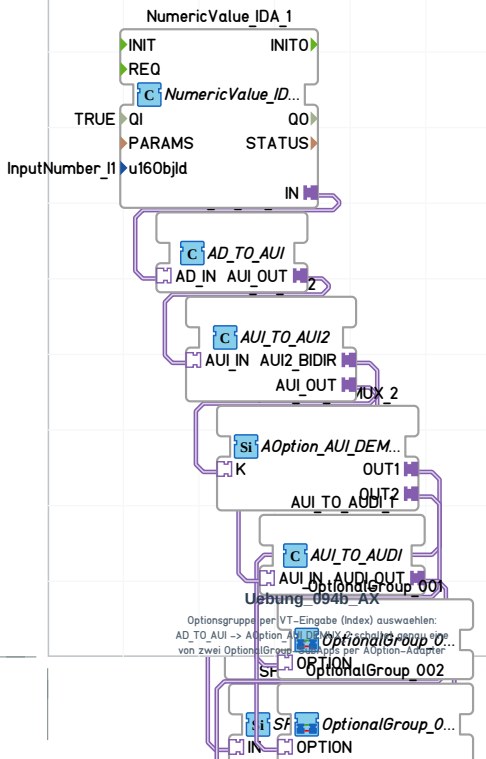
Uebung_094b5_AX

wie Uebung_094b_AX, aber mit AOption_AD_DEMUX_2 (Bitmaske, Multi-Select) statt AOption_AUI_DEMUX_2 (Index, Single-Select): Bit0 schaltet



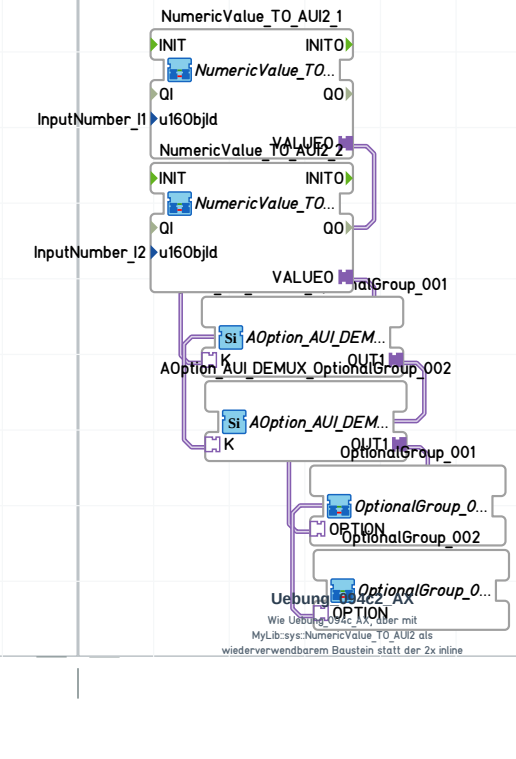
Uebung_094b5b_AX

Wie Uebung_094b5_AX, aber mit MyLib::sys::Checkbox_TO_AD2_2 als wiederverwendbarem Baustein statt der einzeln verdrahteten



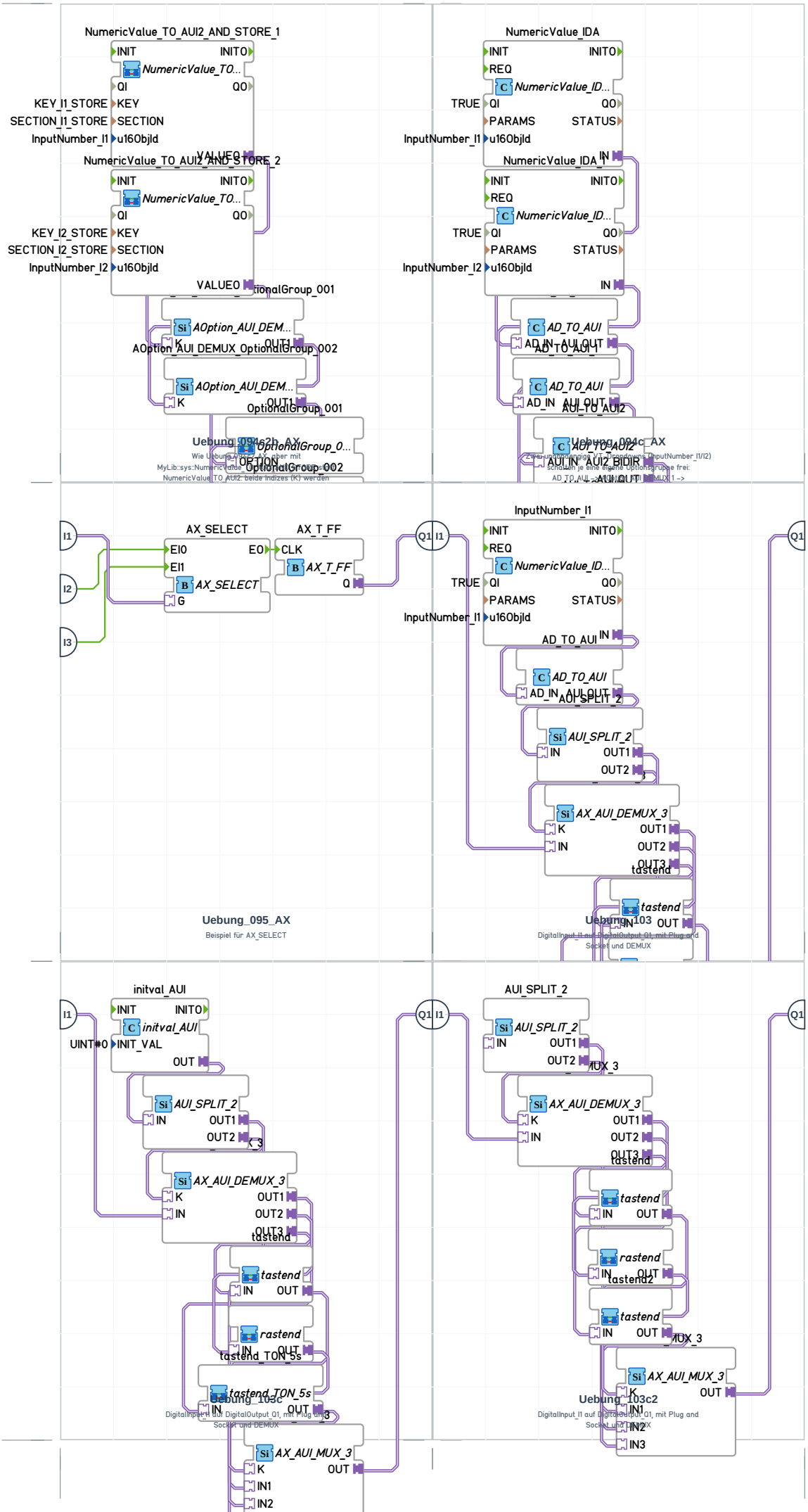
Uebung_094b_AX

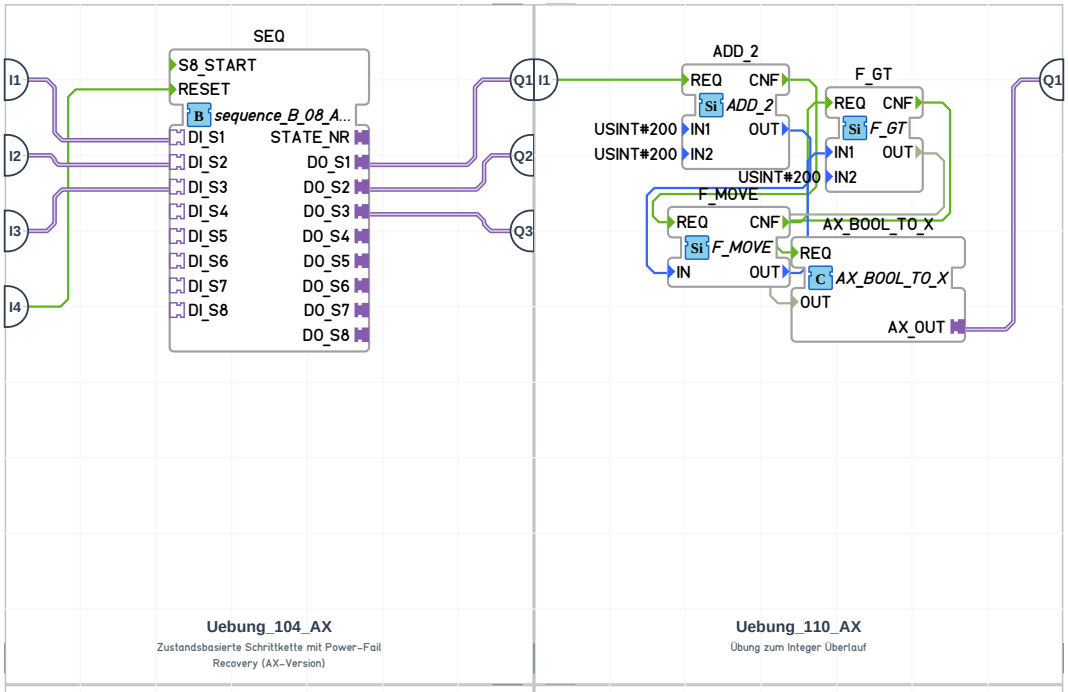
Optionsgruppe per VT-Eingabe (Index) auswaehlen: AD_TO_AUI -> AOption_AUI_DEMUX schaltet genau eine von zwei OptionalGroups - Gegenpart per AOption-Adapter

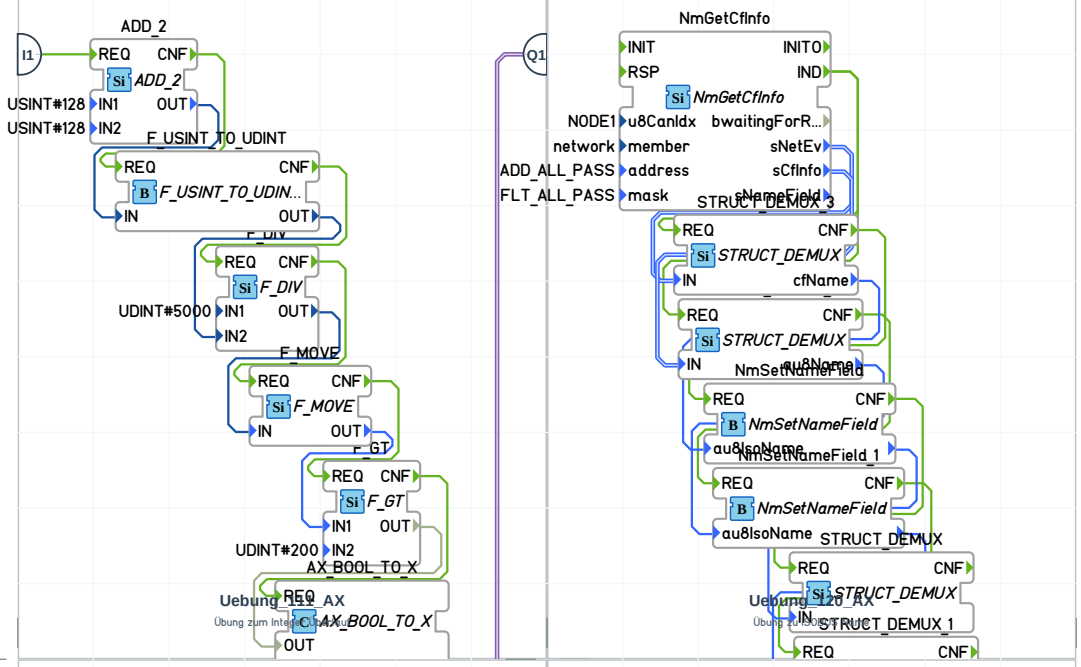


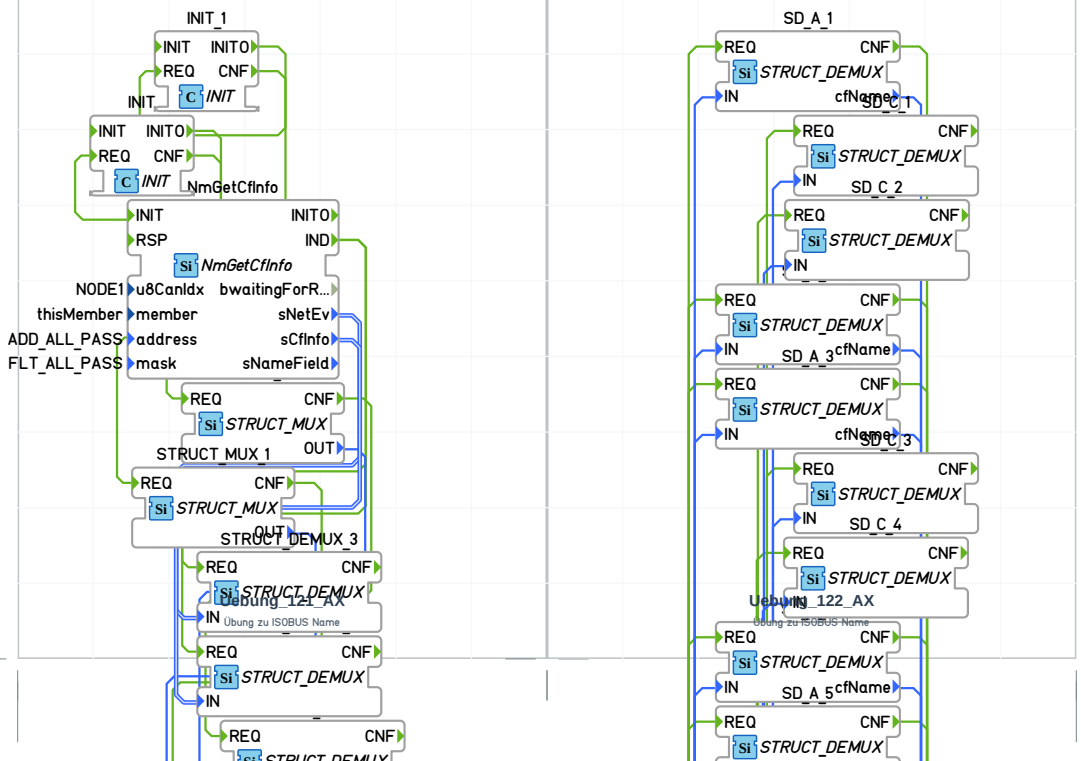
Uebung_094c_AX

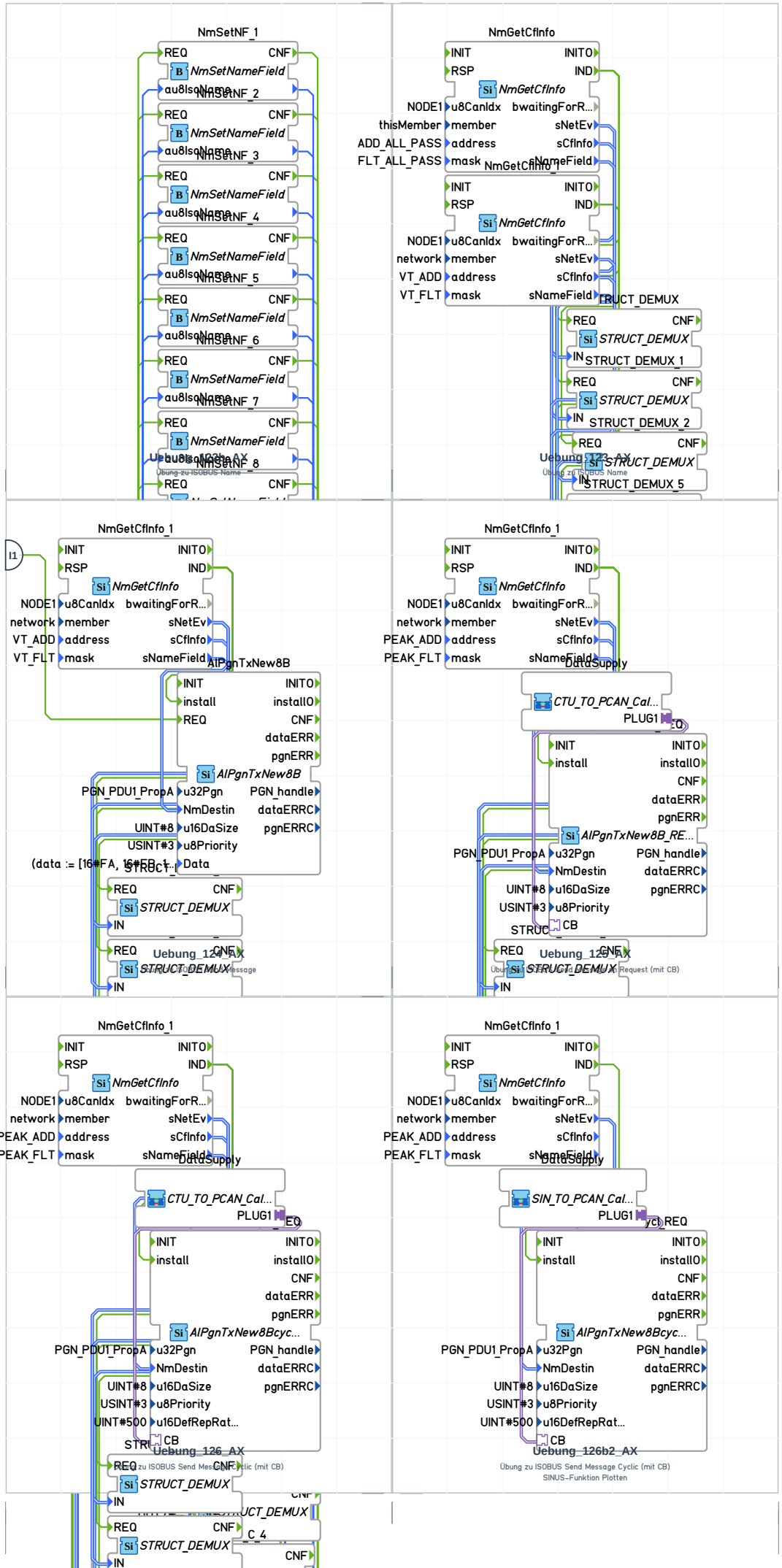
Wie Uebung_094c_AX, aber mit MyLib::sys::NumericValue_TO_AUI2 als wiederverwendbarem Baustein statt der 2x inline

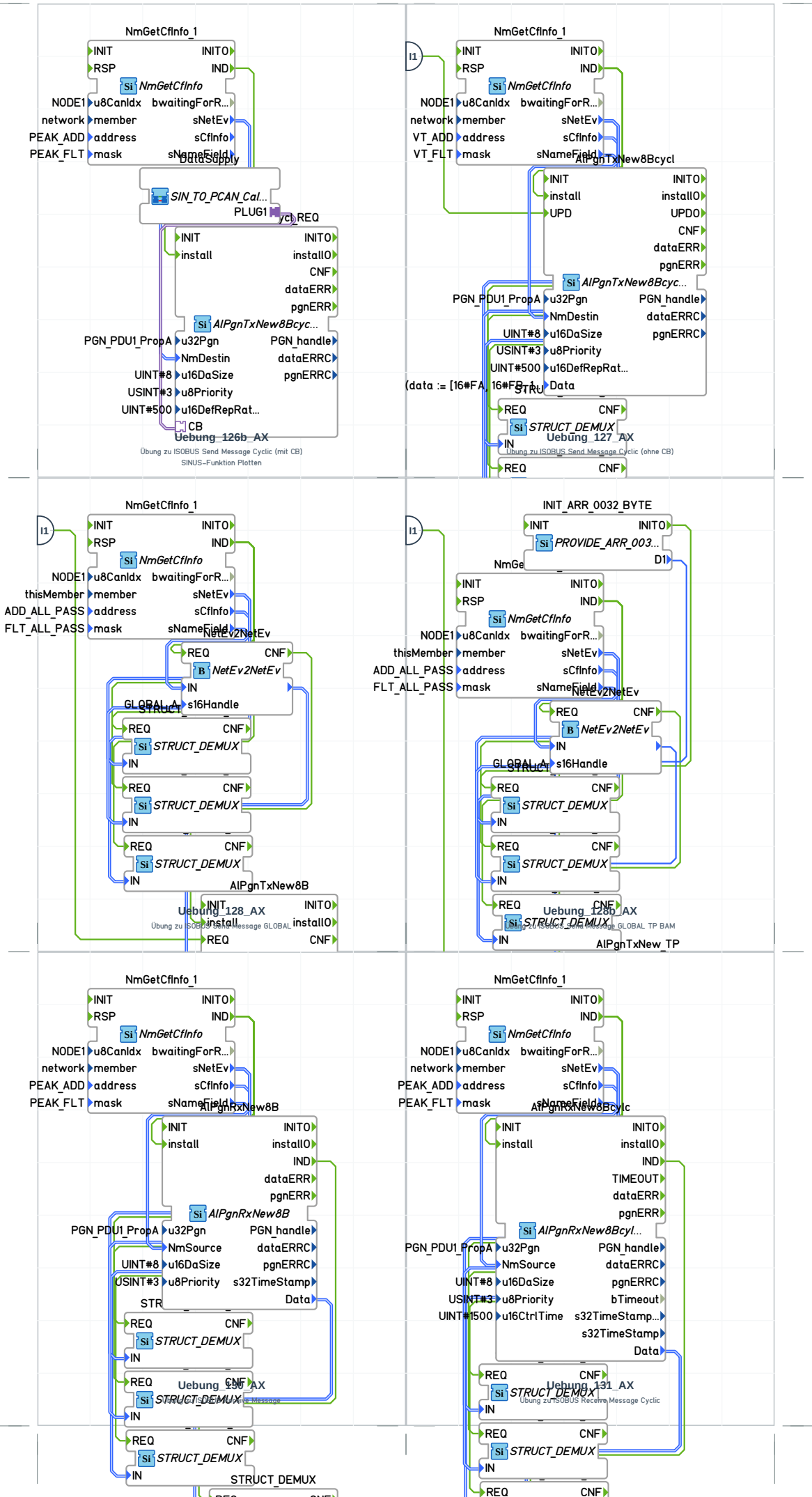

NumericValue_TO_AUI2_AND_STORE_1
INIT
INITO
NumericValue_TO...
QI
QO
KEY_I1_STORE
KEY
SECTION_I1_STORE
SECTION
InputNumber_I1
u160bjId
VALUEO
NumericValue_TO_AUI2_AND_STORE_2
KEY_I2_STORE
SECTION_I2_STORE
InputNumber_I2
AOption_AUI_DEM...
K
OUT1
AOption_AUI_DEMUX_OptionalGroup_002
OptionalGroup_001
OPTION
NumericValue_IDA
REQ
NumericValue_ID...
TRUE
PARAMS
STATUS
IN
NumericValue_IDA_1
AD_TO_AUI
AD_IN
AUI_OUT
AUI_TO_AUI2
AUI_IN
AUI2_BIDIR
Uebung_094c_AX
Uebung_094c2b_AX
AX_SELECT
AX_T_FF
I1
I2
I3
EI0
EO
EI1
G
CLK
Q
Q1
Uebung_095_AX
Beispiel für AX_SELECT
InputNumber_I1
AUI_SPLIT_2
AX_AUI_DEMUX_3
tastend
Uebung_103
initval_AUI
UINT#0
INIT_VAL
rastend
tastend_TON_5s
tastend2
AX_AUI_MUX_3
IN1
IN2
IN3
Uebung_103c
Uebung_103c2

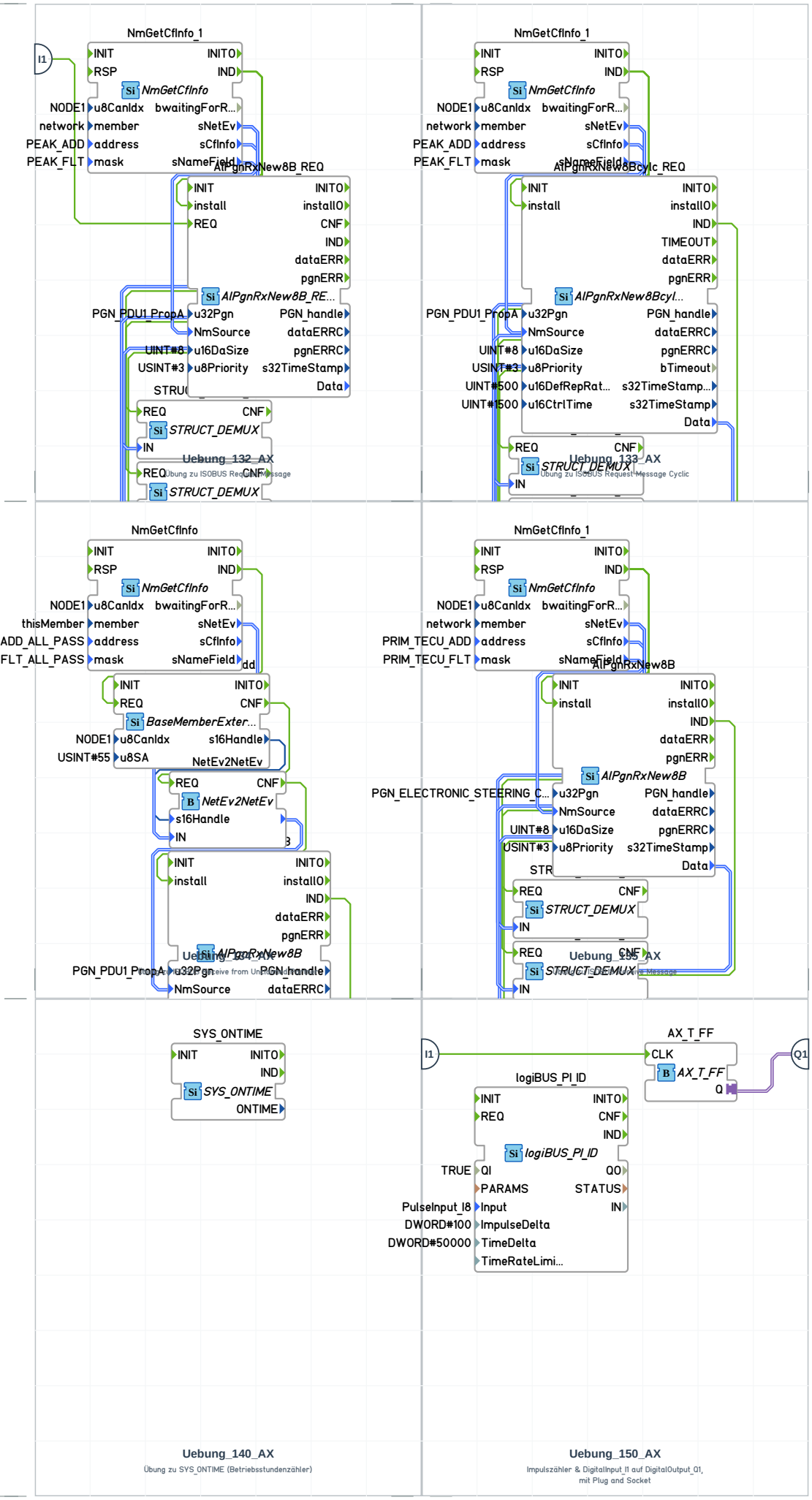

SEQ
S8_START
RESET
sequence_B_08_A...
DI_S1
DI_S2
DI_S3
DI_S4
DI_S5
DI_S6
DI_S7
DI_S8
STATE_NR
DO_S1
DO_S2
DO_S3
DO_S4
DO_S5
DO_S6
DO_S7
DO_S8
I1
I2
I3
I4
Q1
Q2
Q3
ADD_2
F_GT
F_MOVE
AX_BOOL_TO_X
USINT#200
AX_OUT
Uebung_104_AX
Zustandsbasierte Schrittkette mit Power-Fail Recovery (AX-Version)
Uebung_110_AX
Übung zum Integer Überlauf

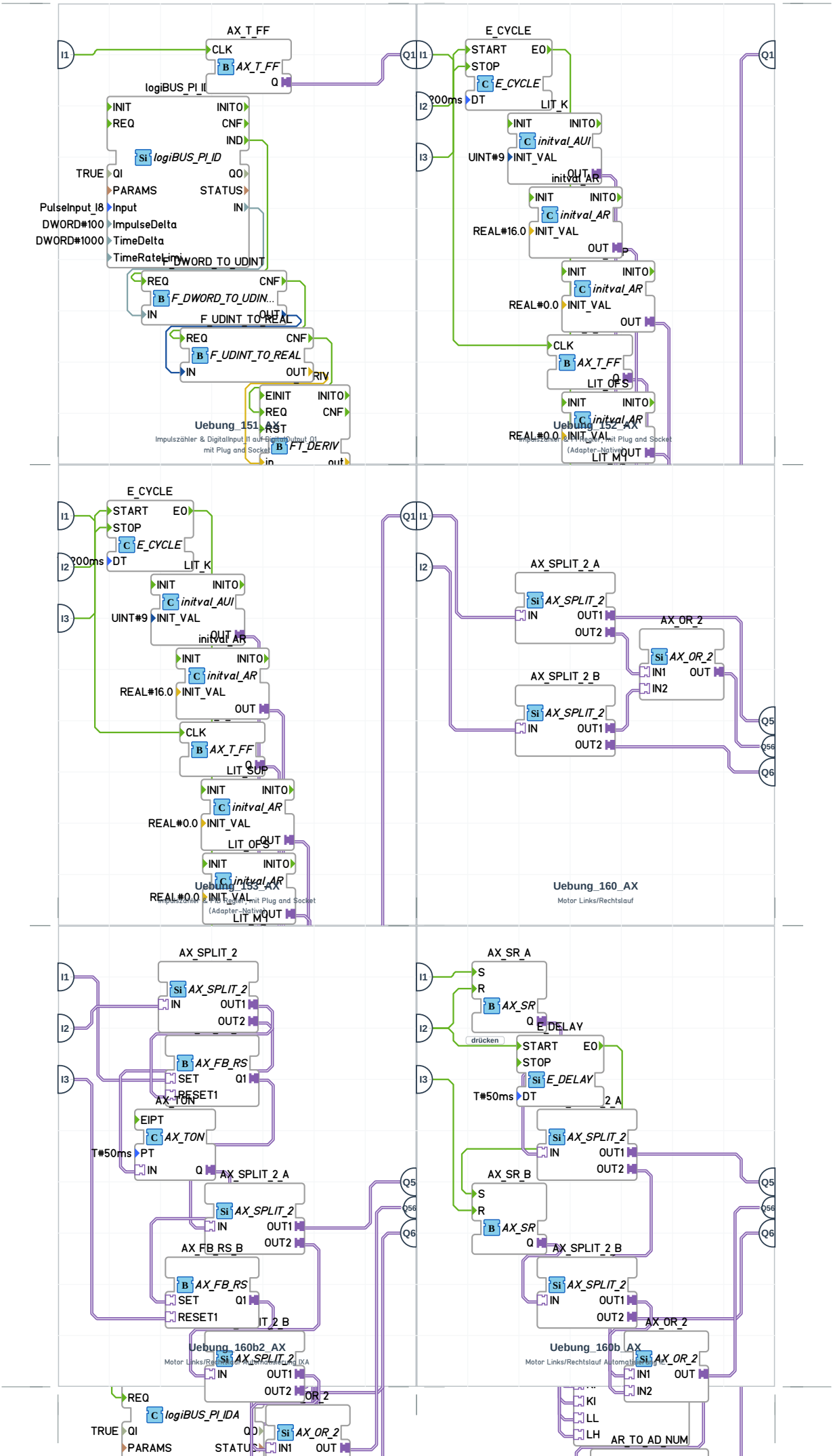

ADD_2
F_USINT_TO_UDINT
F_DIV
F_MOVE
F_GT
AX_BOOL_TO_X
USINT#128
UDINT#5000
UDINT#200
NmGetCfInfo
NmSetNameField
STRUCT_DEMUX
STRUCT_DEMUX_1
STRUCT_DEMUX_3
Uebung_119_AX
Uebung_120_AX

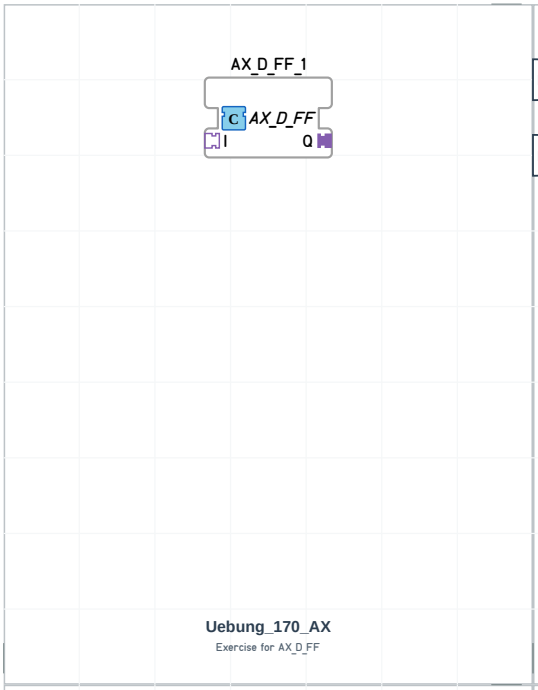

INIT_1
NmGetCfInfo
STRUCT_MUX
STRUCT_MUX_1
STRUCT_DEMUX_3
SD_A_1
SD_C_1
SD_C_2
SD_A_3
SD_C_3
SD_C_4
SD_A_5
Uebung_121_AX
Uebung_122_AX
Übung zu ISOBUS Name


NmSetNF_1
REQ
CNF
NmSetNameField
au8IsoName
NmSetNF_2
NmSetNF_3
NmSetNF_4
NmSetNF_5
NmSetNF_6
NmSetNF_7
NmSetNF_8
NmGetCfInfo
INIT
INITO
RSP
IND
NmGetCfInfo
NODE1
u8CanIdx
bwaitingForR...
thisMember
member
sNetEv
ADD_ALL_PASS
address
sCfInfo
FLT_ALL_PASS
mask
sNameField
NmGetCfInfo_1
network
VT_ADD
VT_FLT
STRUCT_DEMUX
IN
STRUCT_DEMUX_1
STRUCT_DEMUX_2
STRUCT_DEMUX_5
Uebung_123_AX
Übung zu ISOBUS Name
I1
AIPgnTxNew8B
install
installO
dataERR
pgnERR
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
UINT#8
u16DaSize
USINT#3
u8Priority
Data
(data := [16#FA, 16#FB, 1...
DataSupply
CTU_TO_PCAN_Cal...
PLUG1
CB
AIPgnTxNew8B_RE...
Uebung_124_AX
Übung zu ISOBUS Send Message Request (mit CB)
AIPgnTxNew8Bcyc...
UINT#500
u16DefRepRat...
Uebung_126_AX
Übung zu ISOBUS Send Message Cyclic (mit CB)
SIN_TO_PCAN_Cal...
Uebung_126b2_AX
Übung zu ISOBUS Send Message Cyclic (mit CB)
SINUS-Funktion Plotten


NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
u8CanIdx
network
member
PEAK_ADD
address
PEAK_FLT
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
DataSupply
SIN_TO_PCAN_Cal...
PLUG1
REQ
install
installO
CNF
dataERR
pgnERR
AlPgnTxNew8Bcyc...
PGN_PDU1_PropA
u32Pgn
PGN_handle
NmDestin
UINT#8
u16DaSize
dataERRC
USINT#3
u8Priority
pgnERRC
UINT#500
u16DefRepRat...
CB
Uebung_126b_AX
Übung zu ISOBUS Send Message Cyclic (mit CB)
SINUS-Funktion Plotten
I1
VT_ADD
VT_FLT
UPD
UPDO
Data
Uebung_127_AX
Übung zu ISOBUS Send Message Cyclic (ohne CB)
thisMember
ADD_ALL_PASS
FLT_ALL_PASS
NetEv2NetEv
B
IN
GLOBAL_A
s16Handle
STRUCT_DEMUX
AlPgnTxNew8B
Uebung_128_AX
Übung zu ISOBUS Send Message GLOBAL
INIT_ARR_0032_BYTE
PROVIDE_ARR_003...
D1
AlPgnTxNew_TP
AlPgnRxNew8B
AlPgnRxNew8Bcyl...
TIMEOUT
bTimeout
s32TimeStamp...
s32TimeStamp
s16CtrlTime
Uebung_131_AX
Übung zu ISOBUS Receive Message Cyclic

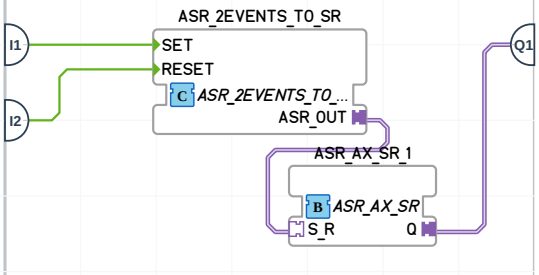

NmGetCfInfo_1
INIT
RSP
NmGetCfInfo
NODE1
network
PEAK_ADD
PEAK_FLT
u8CanIdx
member
address
mask
INITO
IND
bwaitingForR...
sNetEv
sCfInfo
sNameField
AIPgnRxNew8B_REQ
install
REQ
installO
CNF
dataERR
pgnERR
AIPgnRxNew8B_RE...
PGN_PDU1_PropA
u32Pgn
NmSource
UINT#8
u16DaSize
USINT#3
u8Priority
PGN_handle
dataERRC
s32TimeStamp
Data
STRU
STRUCT_DEMUX
IN
Uebung_132_AX
Übung zu ISOBUS Request Message
NmGetCfInfo_1
AIPgnRxNew8BCyIc_REQ
AIPgnRxNew8BCyI...
TIMEOUT
bTimeout
UINT#500
u16DefRepRat...
u16CtrlTime
s32TimeStamp...
Uebung_133_AX
Übung zu ISOBUS Request Message Cyclic
NmGetCfInfo
thisMember
ADD_ALL_PASS
FLT_ALL_PASS
BaseMemberExter...
s16Handle
USINT#55
u8SA
NetEv2NetEv
AIPgnRxNew8B
PGN_ELECTRONIC_STEERING_C...
Uebung_135_AX
NmGetCfInfo_1
PRIM_TECU_ADD
PRIM_TECU_FLT
SYS_ONTIME
ONTIME
Uebung_140_AX
Übung zu SYS_ONTIME (Betriebsstundenzähler)
AX_T_FF
CLK
Q
Q1
AX_T_FF
logiBUS_PI_ID
QI
QO
PARAMS
STATUS
TRUE
PulseInput_I8
DWORD#100
DWORD#50000
Input
ImpulseDelta
TimeDelta
TimeRateLimi...
Uebung_150_AX
Impulszähler & DigitalInput_I1 auf DigitalOutput_Q1,
mit Plug and Socket


AX_T_FF
CLK
AX_T_FF
Q
logiBUS_PI_II
INIT
INITO
REQ
CNF
IND
logiBUS_PI_ID
TRUE
QI
QO
PARAMS
STATUS
PulseInput_I8
Input
IN
DWORD#100
ImpulseDelta
DWORD#1000
TimeDelta
TimeRateUnit
F_DWORD_TO_UDINT
F_DWORD_TO_UDIN...
OUT
F_UDINT_TO_REAL
RIV
EINIT
RST
FT_DERIV
Uebung_151_AX
Impulszähler & DigitalInput I1 auf DigitalOutput Q1
mit Plug and Socket
E_CYCLE
START
EO
STOP
E_CYCLE
DT
LIT_K
?00ms
initval_AUI
UINT#9
INIT_VAL
initval_AR
REAL#16.0
REAL#0.0
CLK
AX_T_FF
LIT_OFS
LIT_M1
Uebung_152_AX
Impulszähler & PWM Regler mit Plug and Socket
(Adapter-Native)
Uebung_153_AX
Uebung_160_AX
Motor Links/Rechtslauf
AX_SPLIT_2_A
AX_SPLIT_2
IN
OUT1
OUT2
AX_OR_2
IN1
IN2
AX_SPLIT_2_B
AX_SPLIT_2
AX_FB_RS
SET
RESET1
Q1
AX_TON
EIPT
AX_TON
T#50ms
PT
AX_FB_RS_B
Uebung_160b2_AX
Motor Links/Rechtslauf Automatisch mit IXA
Uebung_160b_AX
REQ
logiBUS_PI_IDA
AX_SR_A
AX_SR
E_DELAY
drücken
AX_SR_B
AR_TO_AD_NUM
KI
LL
LH



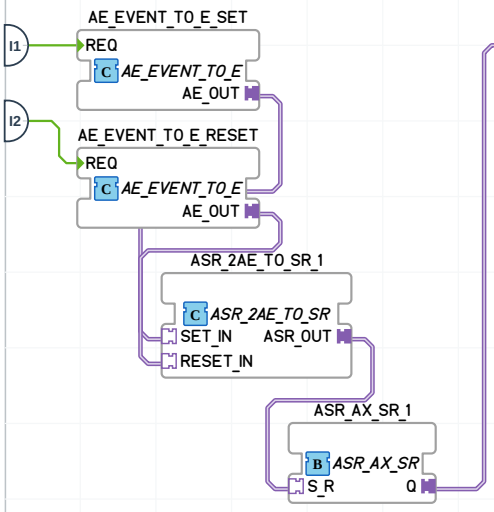
Uebung_170_AX

Exercise for AX_D_FF



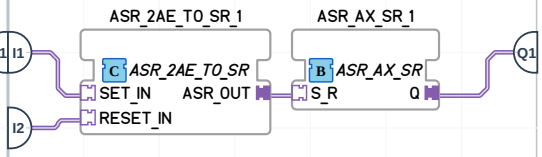
Uebung_171_ASR

Exercise for ASR_AX_SR



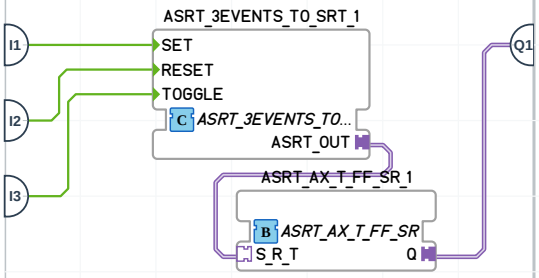
Uebung_171b_ASR

Exercise for ASR_AX_SR, ueber AE-Adapter statt rohe Events (ASR_2AE_TO_SR statt ASR_2EVENTS_TO_SR)



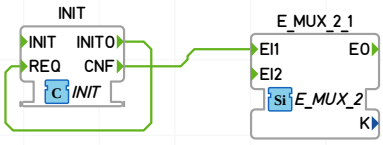
Uebung_171c_ASR

Exercise for ASR_AX_SR, ueber AE-Adapter mit logiBUS_IEA statt logiBUS_IE+AE_EVENT_TO_E



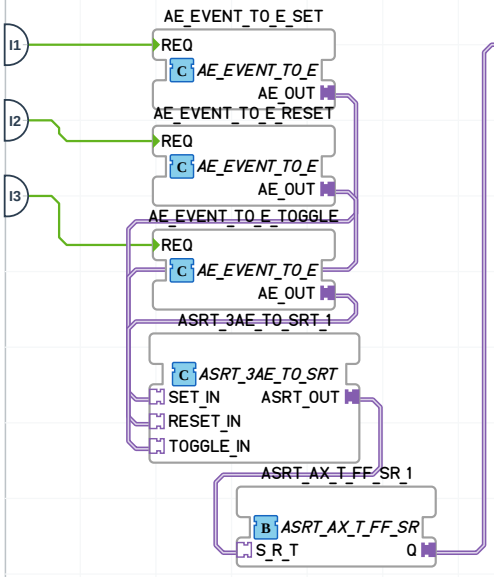
Uebung_172_ASRT

Exercise for ASRT_AX_T_FF_SR



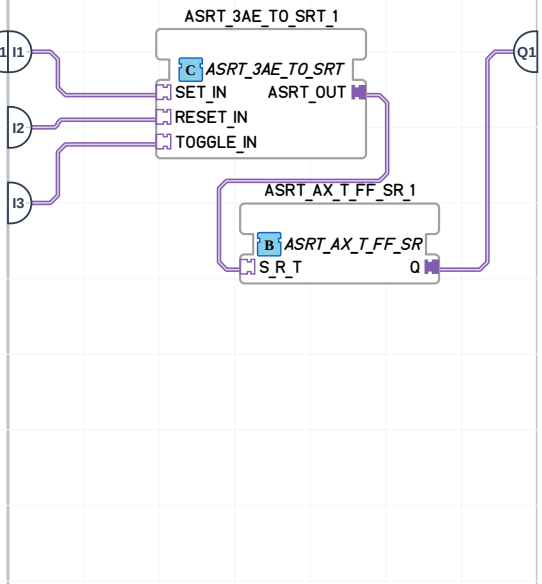
Uebung_172_AX

Exercise for E_MUX_2



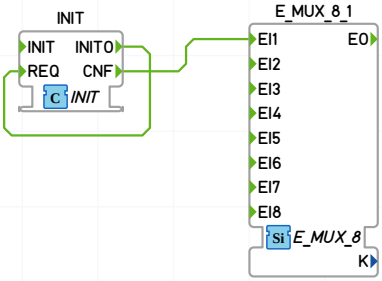
Uebung_172b_ASRT

Exercise for ASRT_AX_T_FF_SR, ueber AE-Adapter statt rohe Events (ASRT_3AE_TO_SRT statt ASRT_3EVENTS_TO_SRT)



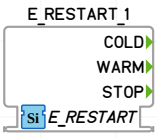
Uebung_172c_ASRT

Exercise for ASRT_AX_T_FF_SR, ueber AE-Adapter mit logiBUS_IEA statt logiBUS_IE+AE_EVENT_TO_E



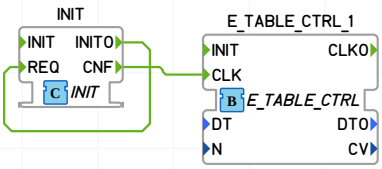
Uebung_173_AX

Exercise for E_MUX_8



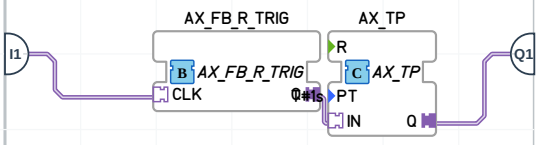
Uebung_174_AX

Exercise for E_RESTART



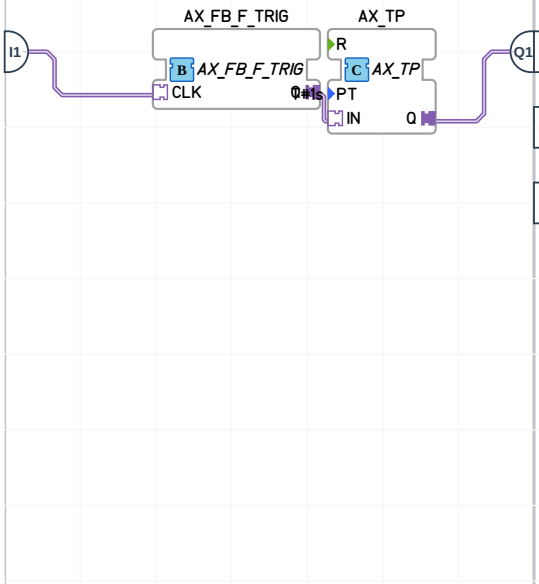
Uebung_175_AX

Exercise for E_TABLE_CTRL



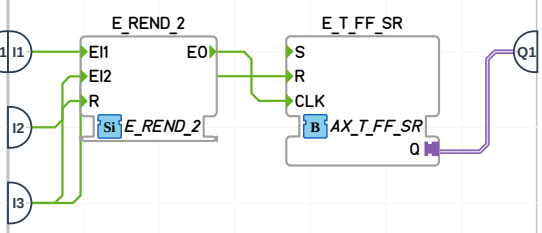
Uebung_177_AX

Beispiel für AX_FB_R_TRIG (Rising Edge Detection)



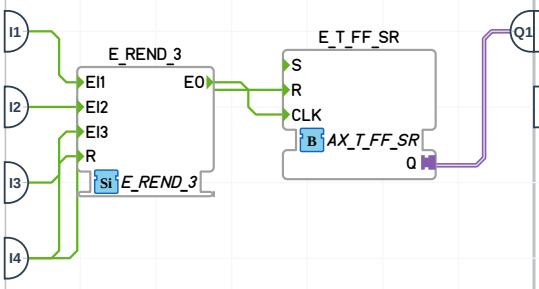
Uebung_178_AX

Beispiel für AX_FB_F_TRIG (Falling Edge Detection)



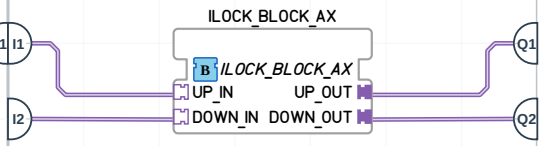
Uebung_179_AX

Beispiel für E_REND_2 (Rendezvous von zwei Events)



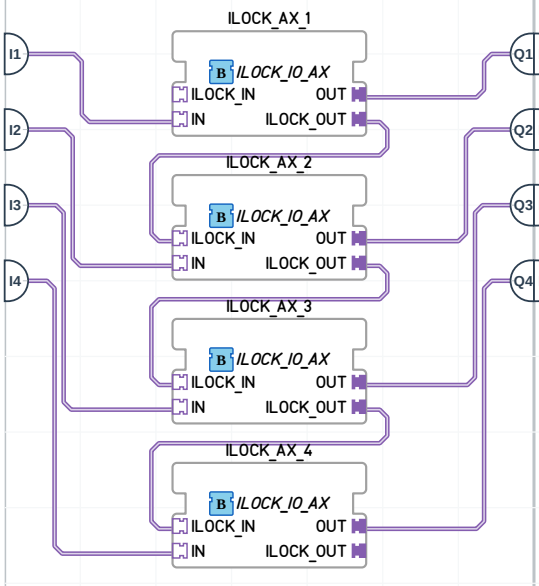
Uebung_180_AX

Beispiel für E_REND_3 (Rendezvous von drei Events)



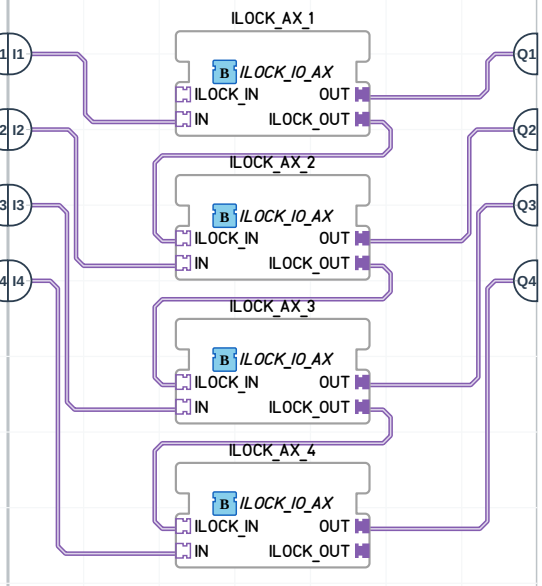
Uebung_201_AX

Interlock: ILOCK_BLOCK_AX (Gegenseitige Verriegelung via Adapter)



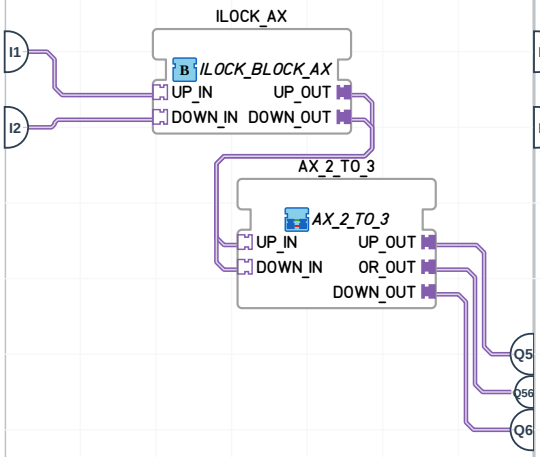
Uebung_201_Interlock_AX

Interlock advanced exercise (AX)



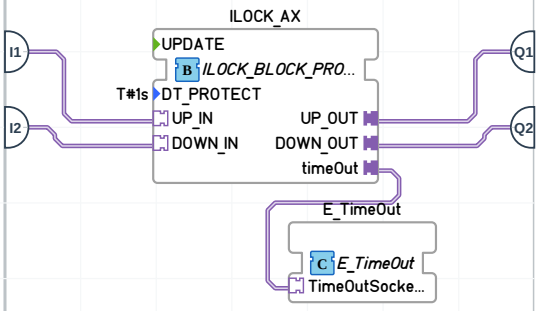
Uebung_201_Interlock_BOOL_AX

Interlock basic exercise (BOOL)



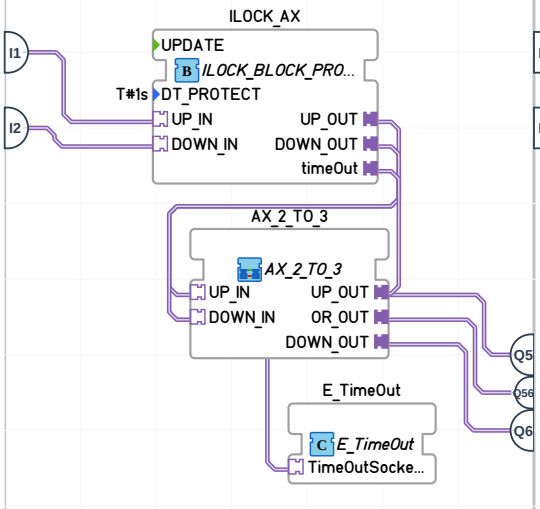
Uebung_201b_AX

Interlock: ILOCK_BLOCK_AX (Motor Rechts/Linkslauf via Adapter)



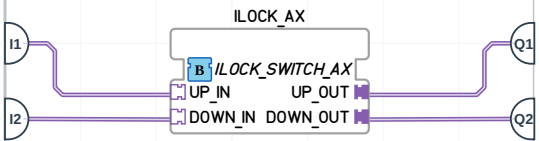
Uebung_202_AX

Interlock: ILOCK_BLOCK_PROTECT_AX (Verriegelung mit Schutzzeit via Adapter)



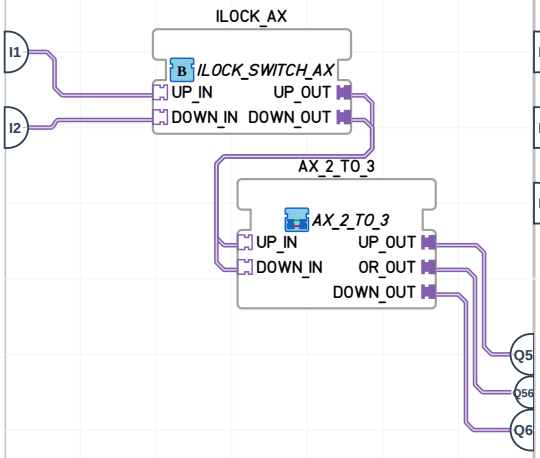
Uebung_202b_AX

Interlock: ILOCK_BLOCK_PROTECT_AX (Motor Reversierung mit Schutzzeit via Adapter)



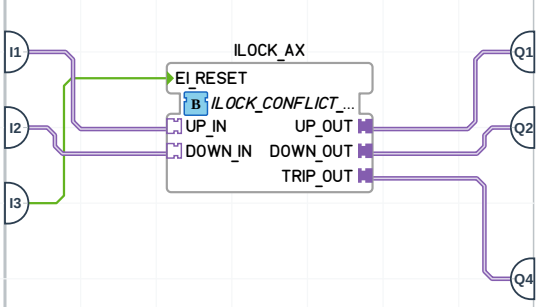
Uebung_203_AX

Interlock: ILOCK_SWITCH_AX (Umschalt-Priorität via Adapter)



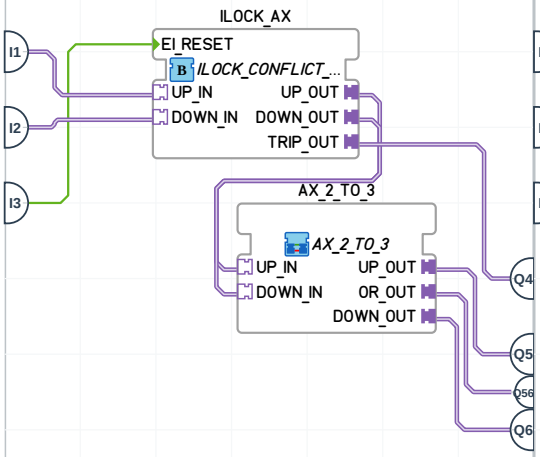
Uebung_203b_AX

Interlock: ILOCK_SWITCH_AX (Motor Reversierung Priorität via Adapter)



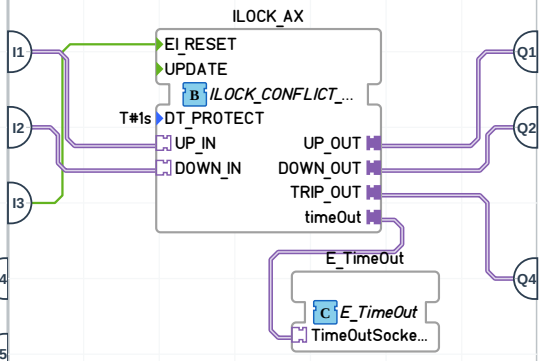
Uebung_204_AX

Interlock: ILOCK_CONFLICT_TRIP_AX (Trip bei Konflikt via Adapter)



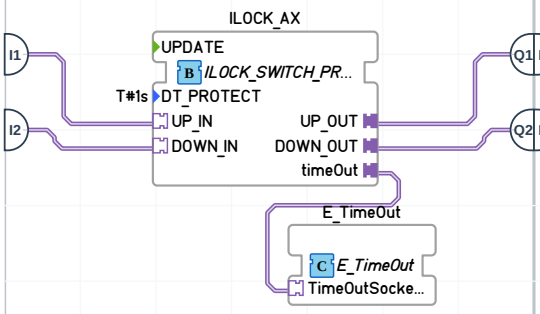
Uebung_204b_AX

Interlock: ILOCK_CONFLICT_TRIP_AX
(Motor-Sicherheitsabschaltung via Adapter)



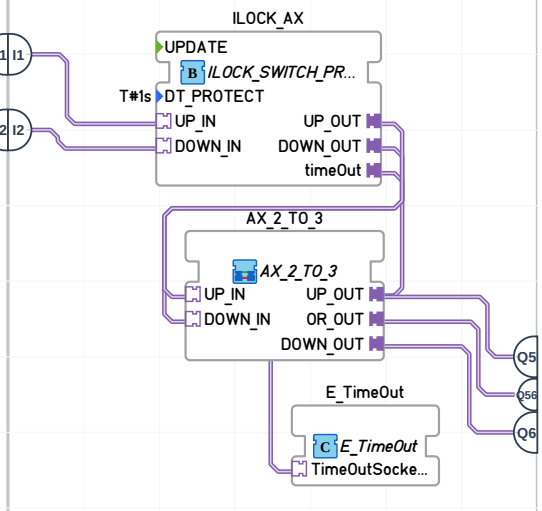
Uebung_204c_AX

Interlock: ILOCK_CONFLICT_TRIP_PROTECT_AX (Trip bei
Konflikt UND Schutzzeit nach Freigabe, via Adapter)



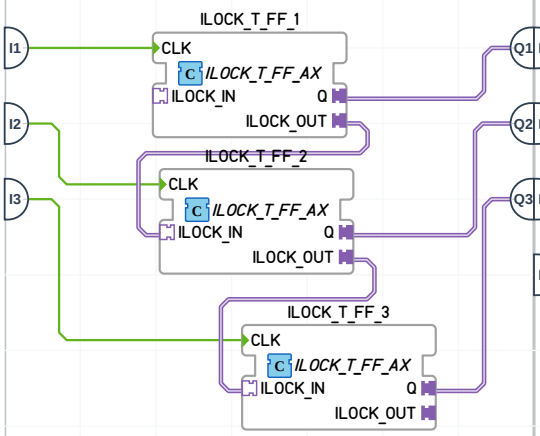
Uebung_205_AX

Interlock: ILOCK_SWITCH_PROTECT_AX
(Umschalt-Priorität mit Schutzzeit via Adapter)



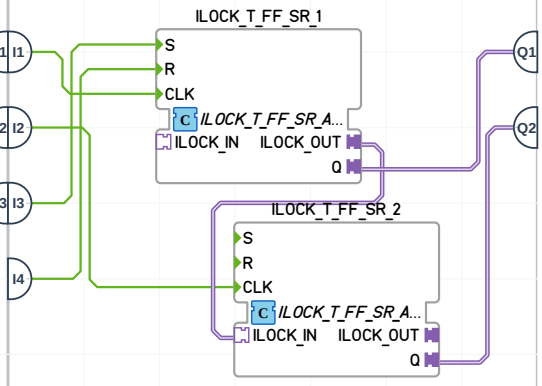
Uebung_205b_AX

Interlock: ILOCK_SWITCH_PROTECT_AX (Motor
Reversierung mit Schutzzeit via Adapter)



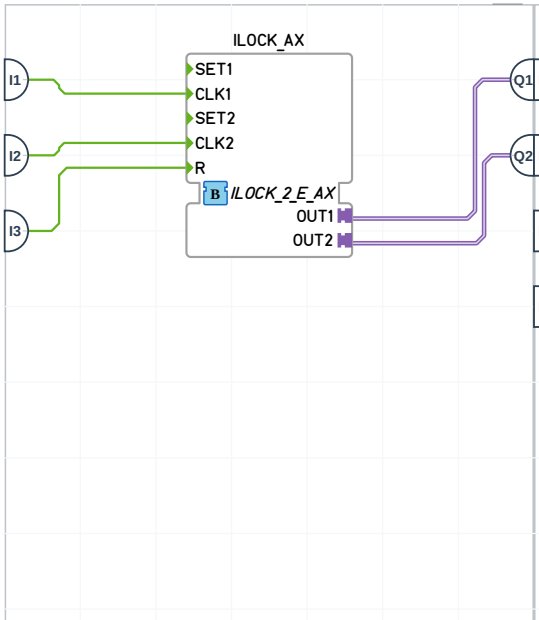
Uebung_206_AX

Interlock: ILOCK_T_FF_AX (Drei gegenseitig
verriegelte Toggle-Flip-Flops via Adapter-Kette)



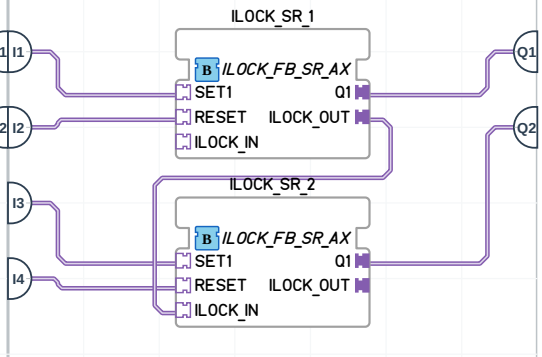
Uebung_206b_AX

Interlock: ILOCK_T_FF_SR_AX (2 gegenseitig
verriegelte Toggle-Flip-Flops via Adapter-Kette,
zusaetzlich direktes Set/Reset auf FF1)



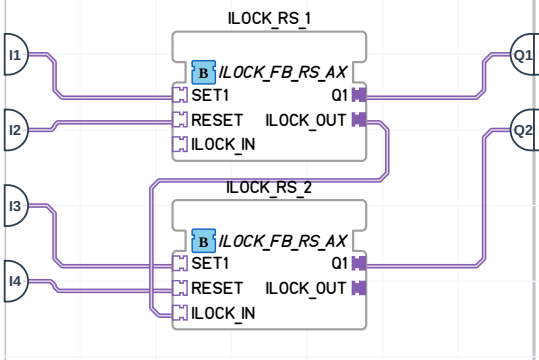
Uebung_207_AX

Interlock: ILOCK_2_E_AX (Event-gesteuertes bistabiles Relais mit gegenseitiger Verriegelung via Adapter)



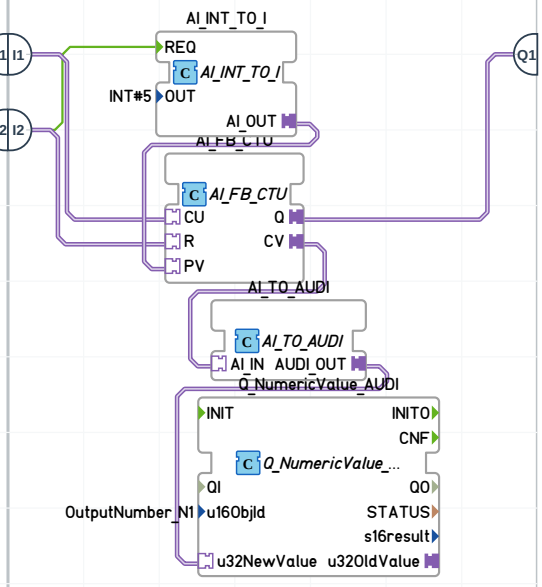
Uebung_208_AX

Interlock: ILOCK_FB_SR_AX (Zwei gegenseitig verriegelte Set-Dominante Latches via AX/AX2-Adapter)



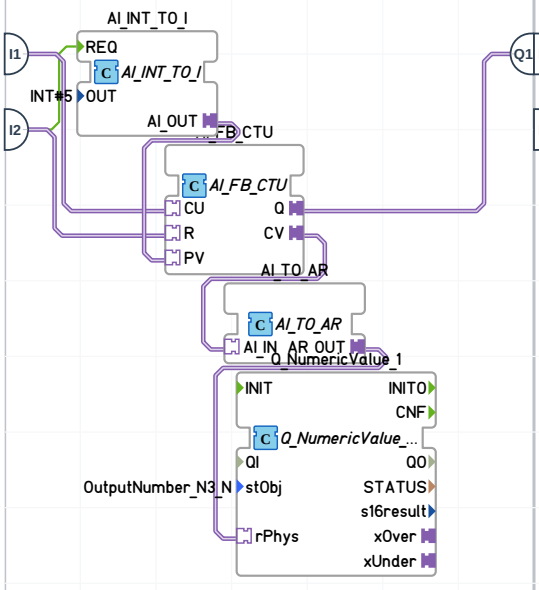
Uebung_209_AX

Interlock: ILOCK_FB_RS_AX (Zwei gegenseitig verriegelte Reset-Dominante Latches via AX/AX2-Adapter)



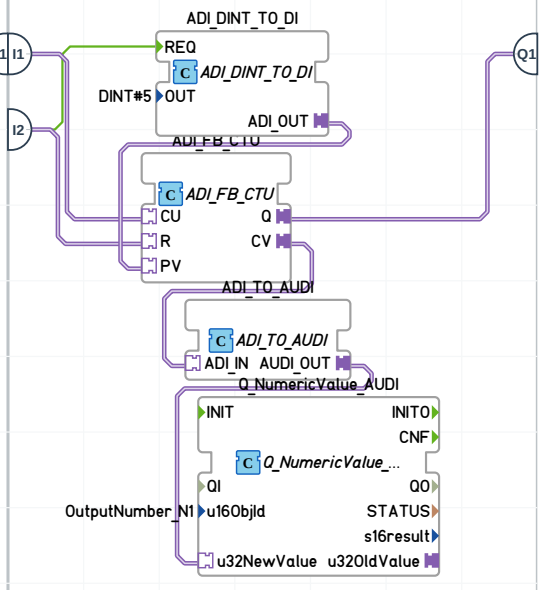
Uebung_210_AI

Standard IEC 61131-3 AI_FB_CTU (Adapter Version, Vorwärtszähler, INT) mit Terminal-Ausgabe



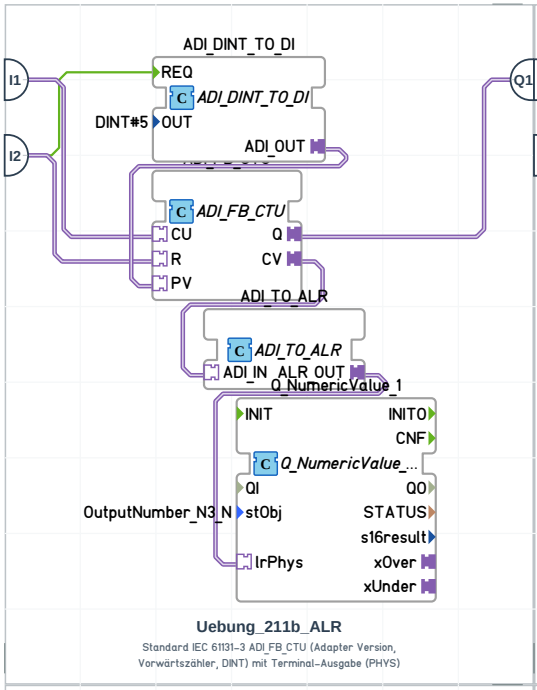
Uebung_210b_AR

Standard IEC 61131-3 AI_FB_CTU (Adapter Version, Vorwärtszähler, INT) mit Terminal-Ausgabe (PHYS)



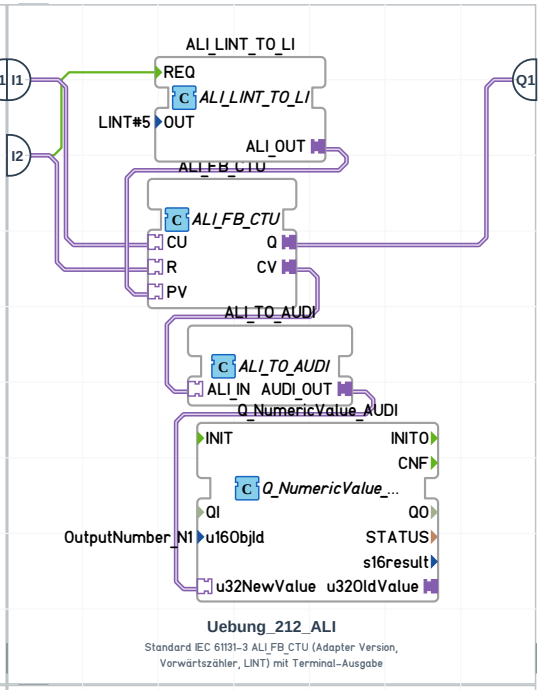
Uebung_211_ADI

Standard IEC 61131-3 ADI_FB_CTU (Adapter Version, Vorwärtszähler, DINT) mit Terminal-Ausgabe



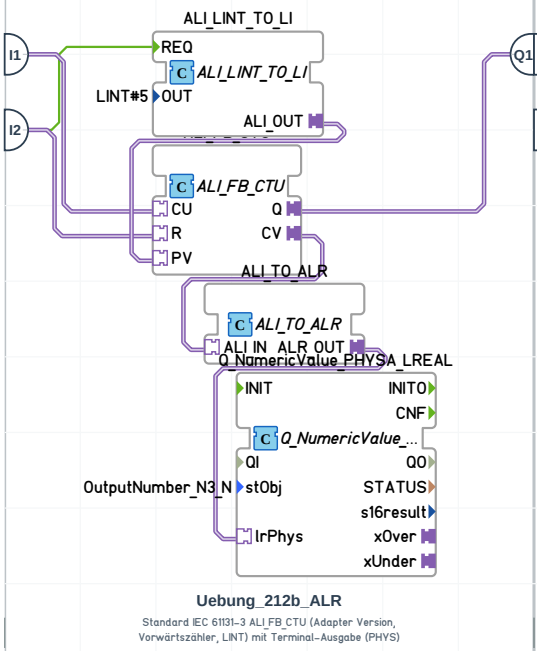
Uebung_211b_ALR

Standard IEC 61131-3 ADI_FB_CTU (Adapter Version, Vorwärtszähler, DINT) mit Terminal-Ausgabe (PHYS)



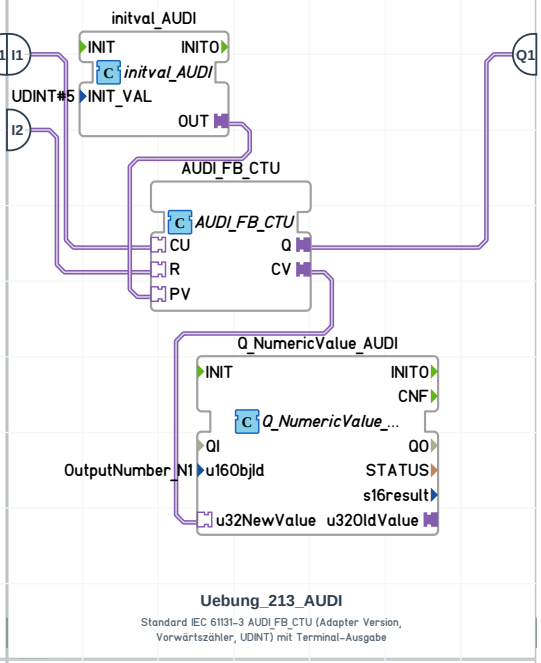
Uebung_212_ALI

Standard IEC 61131-3 ALI_FB_CTU (Adapter Version, Vorwärtszähler, LINT) mit Terminal-Ausgabe



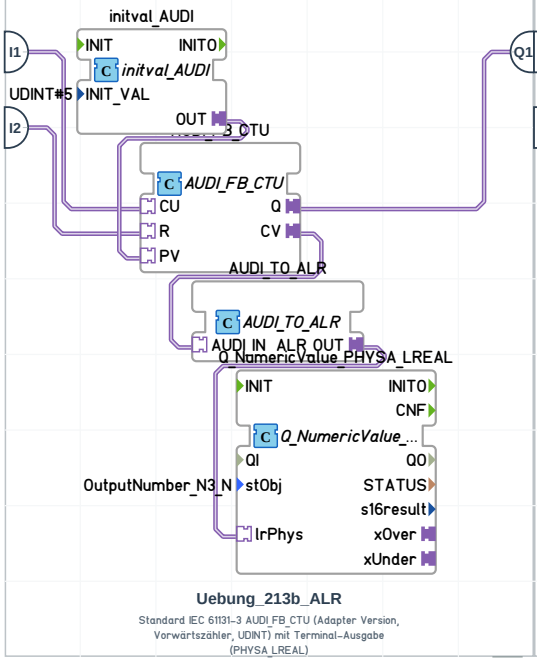
Uebung_212b_ALR

Standard IEC 61131-3 ALI_FB_CTU (Adapter Version, Vorwärtszähler, LINT) mit Terminal-Ausgabe (PHYS)



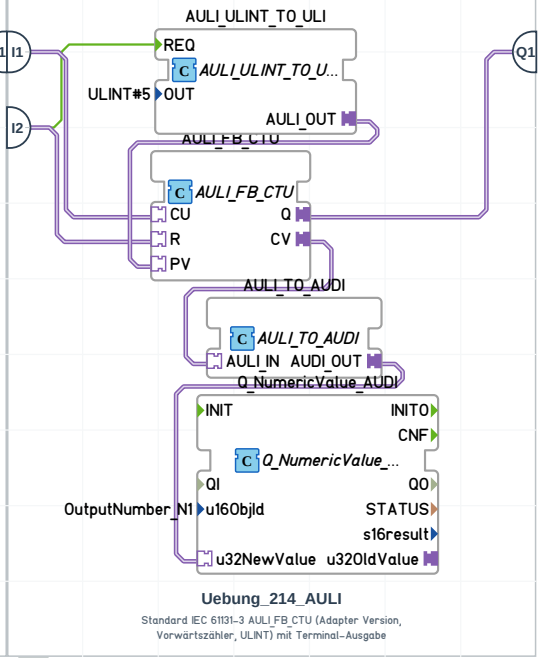
Uebung_213_AUDI

Standard IEC 61131-3 AUDI_FB_CTU (Adapter Version, Vorwärtszähler, UDINT) mit Terminal-Ausgabe



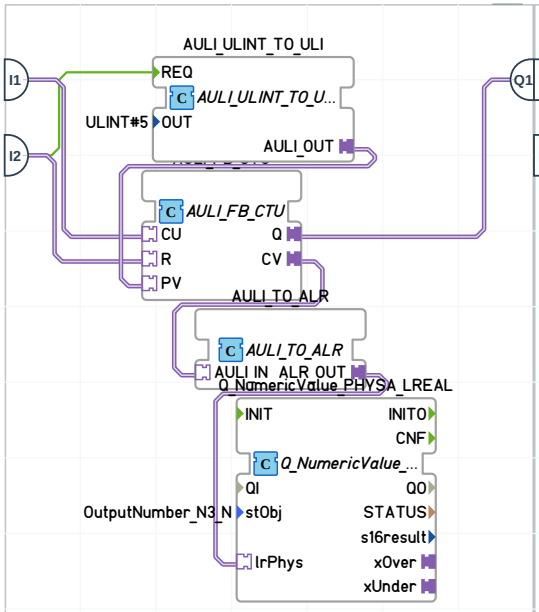
Uebung_213b_ALR

Standard IEC 61131-3 AUDI_FB_CTU (Adapter Version, Vorwärtszähler, UDINT) mit Terminal-Ausgabe (PHYSA_LREAL)



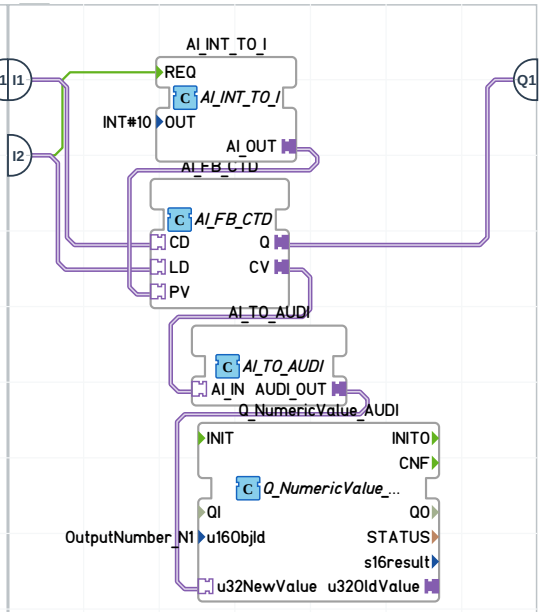
Uebung_214_AULI

Standard IEC 61131-3 AULI_FB_CTU (Adapter Version, Vorwärtszähler, ULINT) mit Terminal-Ausgabe



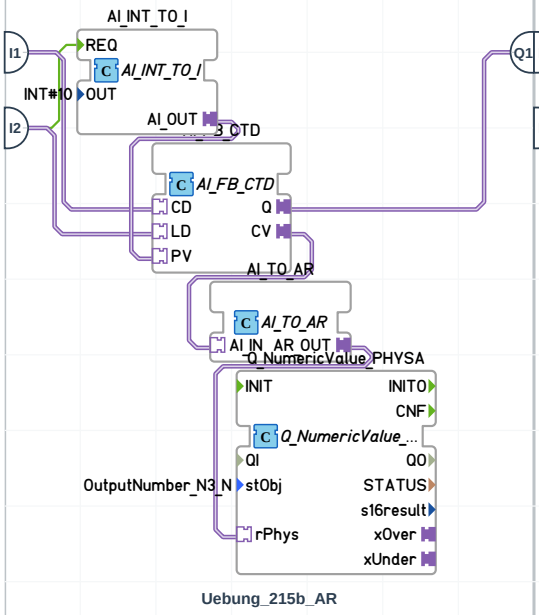
Uebung_214b_ALR

Standard IEC 61131-3 AULI_FB_CTU (Adapter Version, Vorwärtszähler, ULINT) mit Terminal-Ausgabe (PHYSA_LREAL)



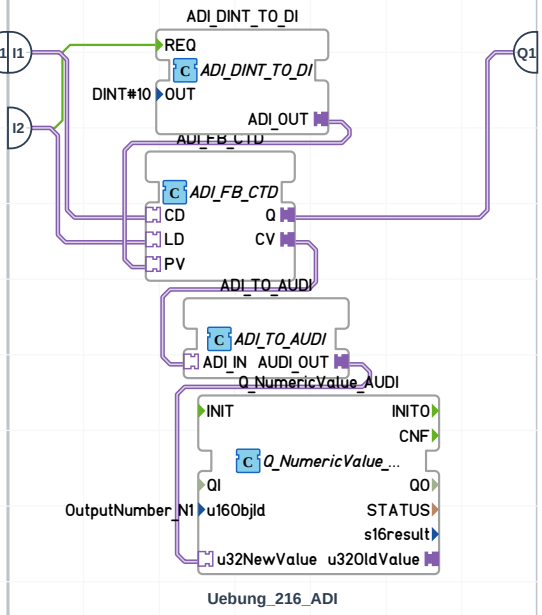
Uebung_215_AI

Standard IEC 61131-3 AI_FB_CTD (Adapter Version, Rückwärtszähler, INT) mit Terminal-Ausgabe



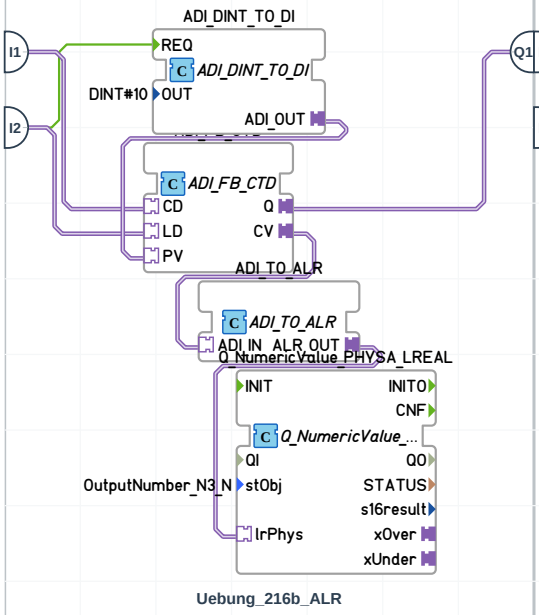
Uebung_215b_AR

Standard IEC 61131-3 AI_FB_CTD (Adapter Version, INT) mit Terminal-Ausgabe (PHYS)



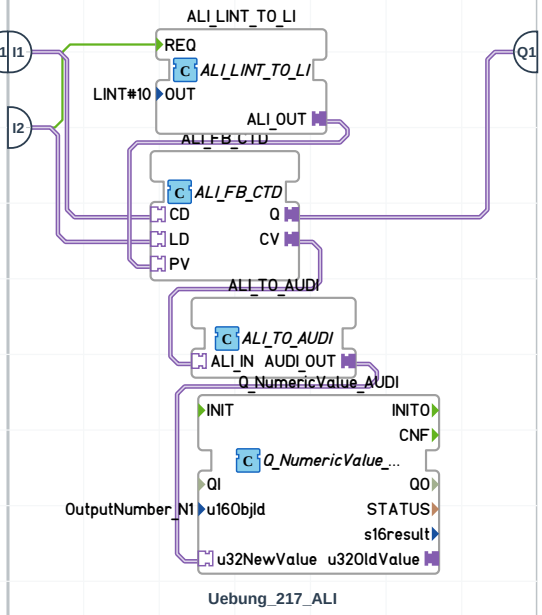
Uebung_216_ADI

Standard IEC 61131-3 ADI_FB_CTD (Adapter Version, Rückwärtszähler, DINT) mit Terminal-Ausgabe



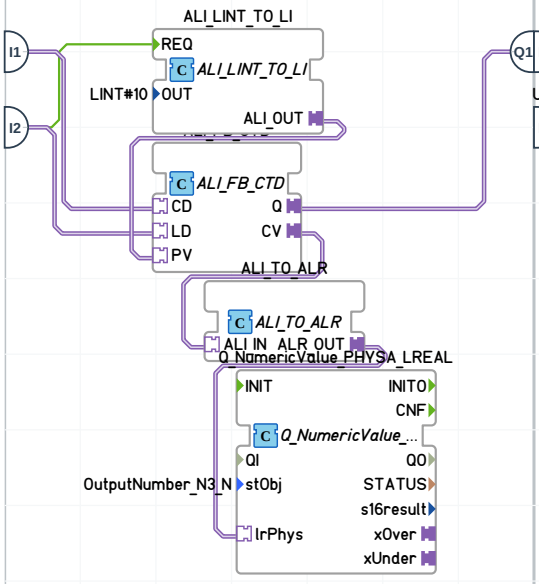
Uebung_216b_ALR

Standard IEC 61131-3 ADI_FB_CTD (Adapter Version, DINT) mit Terminal-Ausgabe (PHYS)



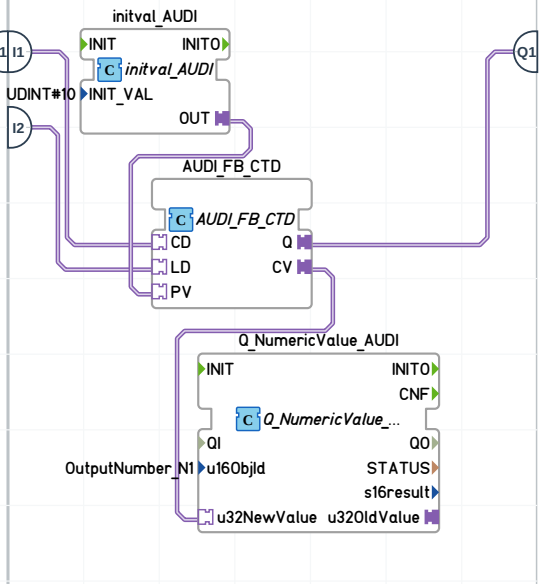
Uebung_217_ALI

Standard IEC 61131-3 ALI_FB_CTD (Adapter Version, Rückwärtszähler, LINT) mit Terminal-Ausgabe



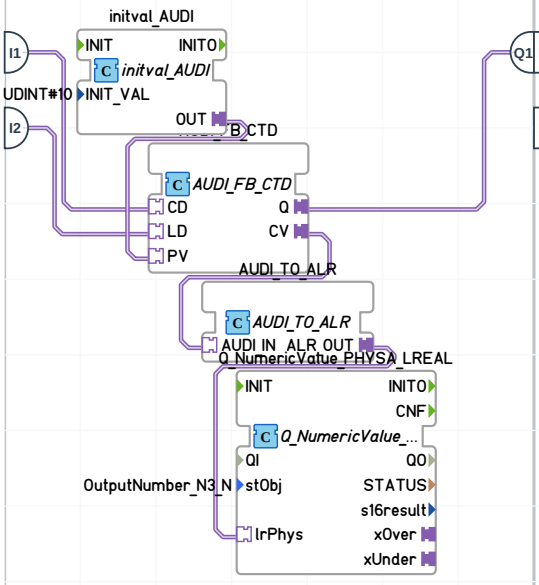
Uebung_217b_ALR

Standard IEC 61131-3 ALI_FB_CTD (Adapter Version, LINT) mit Terminal-Ausgabe (PHYSA_LREAL)



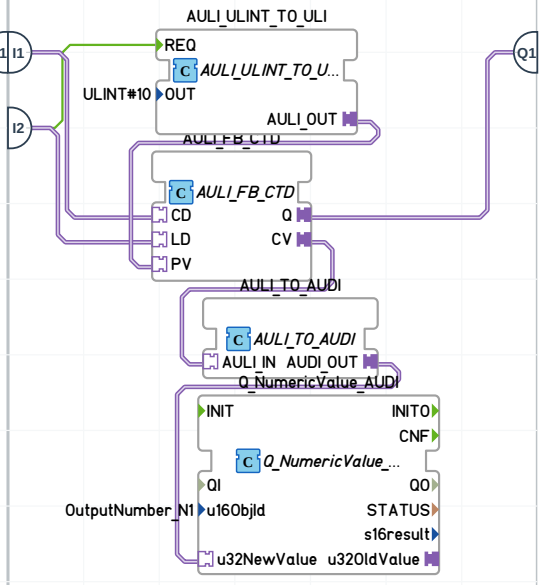
Uebung_218_AUDI

Standard IEC 61131-3 AUDI_FB_CTD (Adapter Version, Rückwärtszähler, UDINT) mit Terminal-Ausgabe



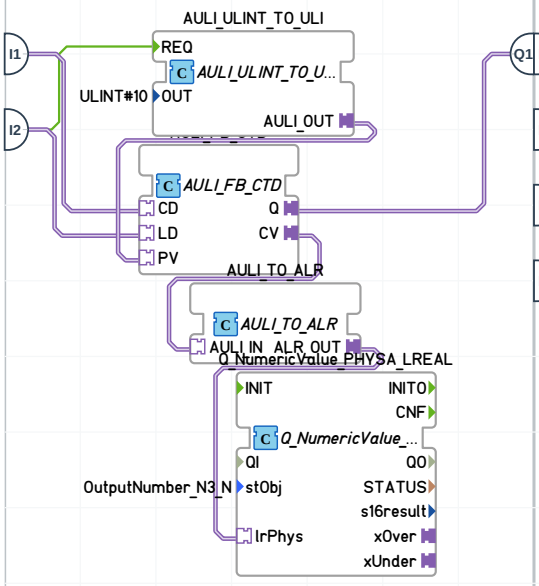
Uebung_218b_ALR

Standard IEC 61131-3 AUDI_FB_CTD (Adapter Version, Rückwärtszähler, UDINT) mit Terminal-Ausgabe (PHYSA_LREAL)



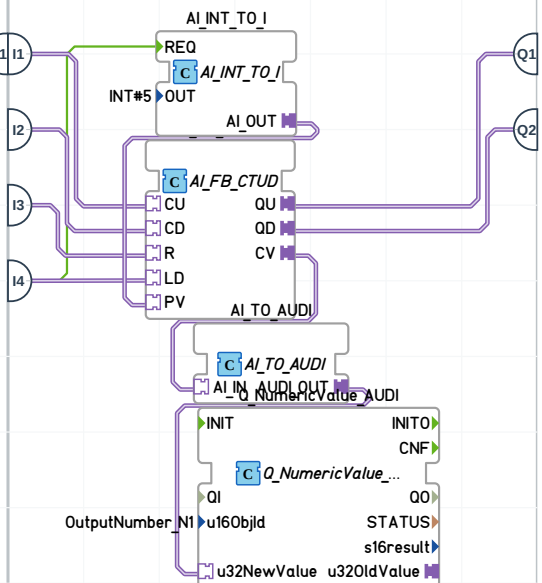
Uebung_219_AULI

Standard IEC 61131-3 AULI_FB_CTD (Adapter Version, Rückwärtszähler, ULINT) mit Terminal-Ausgabe



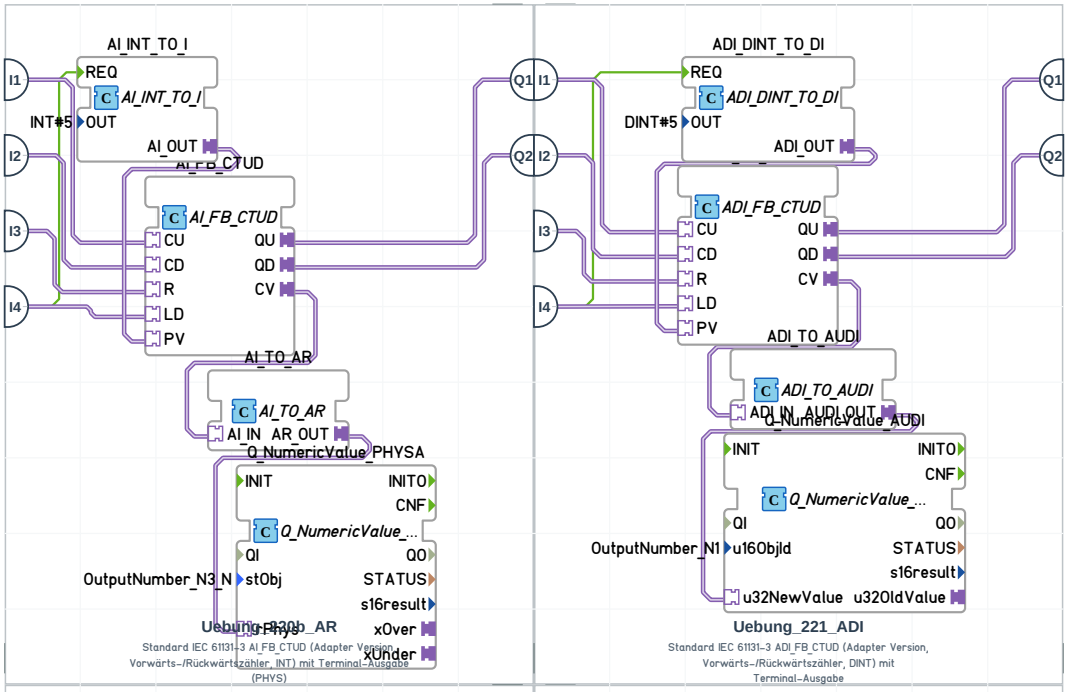
Uebung_219b_ALR

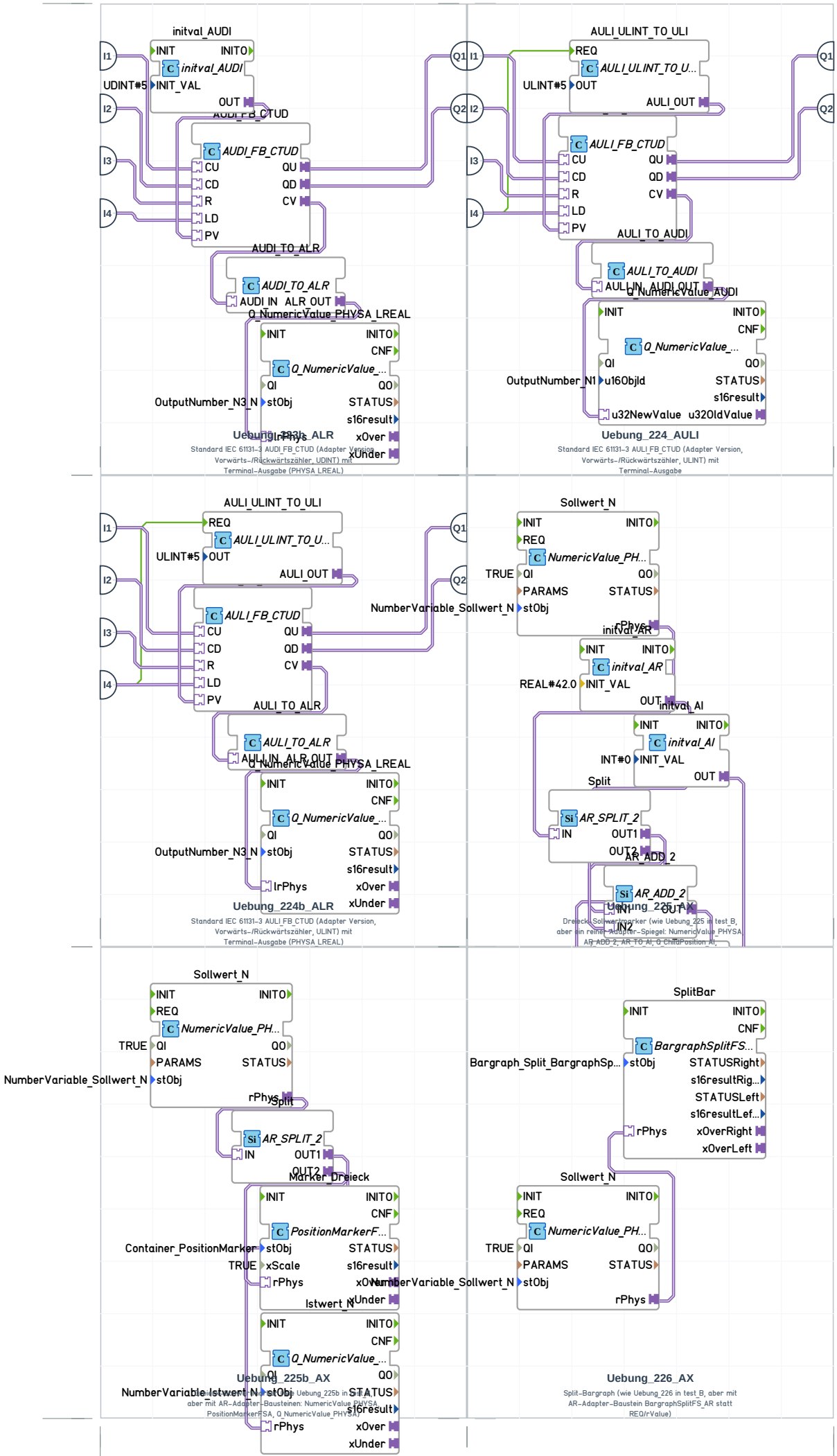
Standard IEC 61131-3 AULI_FB_CTD (Adapter Version, Rückwärtszähler, ULINT) mit Terminal-Ausgabe (PHYSA_LREAL)

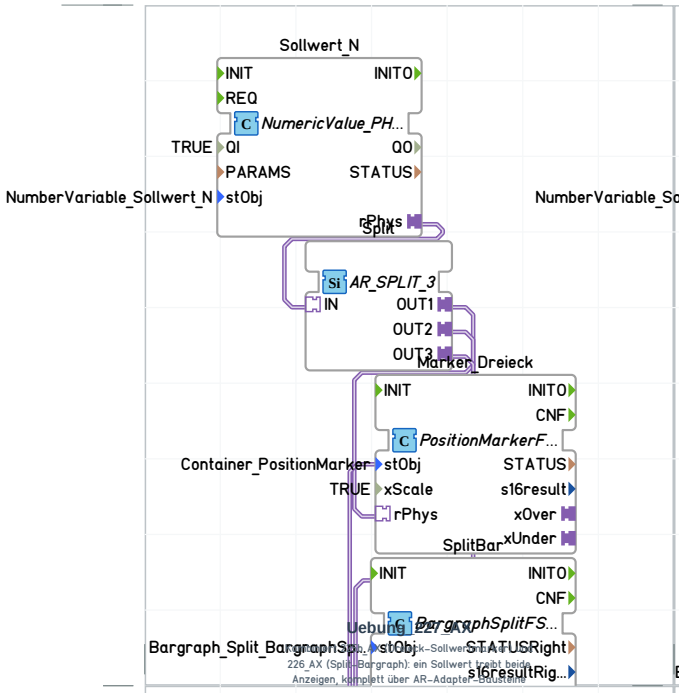


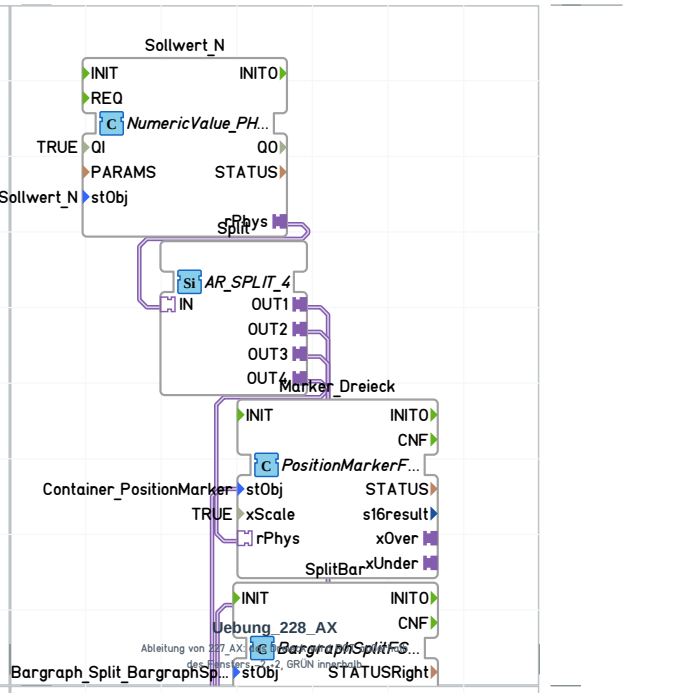
Uebung_220_AI

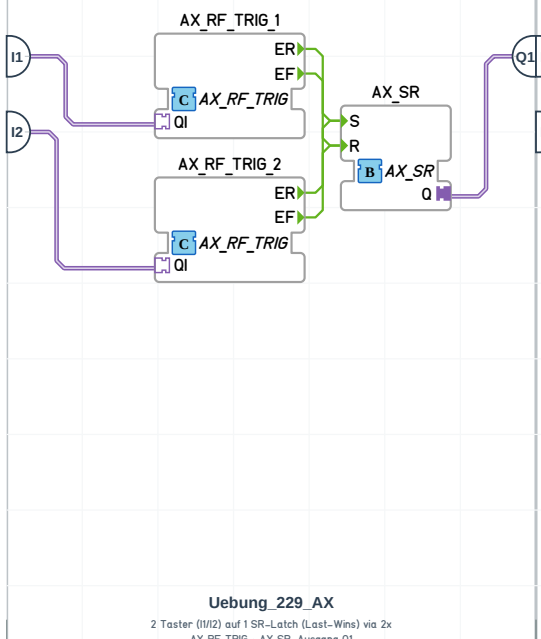
Standard IEC 61131-3 AI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, INT) mit Terminal-Ausgabe


AI_INT_TO_I
REQ
AI_INT_TO_I
INT#5
OUT
AI_OUT
AI_FB_CTUD
AI_FB_CTUD
CU
CD
R
LD
PV
QU
QD
CV
AI_TO_AR
AI_TO_AR
AI_IN
AR_OUT
Q_NumericValue_PHYSA
INIT
INITO
CNF
Q_NumericValue_...
QI
QO
OutputNumber_N3_N
stObj
STATUS
s16result
lrPhys
xOver
xUnder
Uebung_220b_AR
Standard IEC 61131-3 AI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, INT) mit Terminal-Ausgabe (PHYS)
ADI_DINT_TO_DI
ADI_DINT_TO_DI
DINT#5
ADI_OUT
ADI_FB_CTUD
ADI_TO_AUDI
ADI_TO_AUDI
ADI_IN
AUDI_OUT
Q_NumericValue_AUDI
OutputNumber_N1
u16ObjId
u32NewValue
u32OldValue
Uebung_221_ADI
Standard IEC 61131-3 ADI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, DINT) mit Terminal-Ausgabe
ADI_DINT_TO_DI
ADI_TO_ALR
ADI_TO_ALR
ADI_IN
ALR_OUT
Q_NumericValue_PHYSA_LREAL
Uebung_221b_ALR
Standard IEC 61131-3 ADI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, DINT) mit Terminal-Ausgabe (PHYS)
ALI_LINT_TO_LI
ALI_LINT_TO_LI
LINT#5
ALI_OUT
ALI_FB_CTUD
ALI_TO_AUDI
ALI_TO_AUDI
ALI_IN
AUDI_OUT
Q_NumericValue_AUDI
Uebung_222_ALI
Standard IEC 61131-3 ALI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, LINT) mit Terminal-Ausgabe
ALI_LINT_TO_LI
ALI_TO_ALR
ALI_TO_ALR
ALI_IN
ALR_OUT
Q_NumericValue_PHYSA_LREAL
Uebung_222b_ALR
Standard IEC 61131-3 ALI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, LINT) mit Terminal-Ausgabe (PHYSA_LREAL)
initval_AUDI
initval_AUDI
UDINT#5
INIT_VAL
OUT
AUDI_FB_CTUD
Q_NumericValue_AUDI
Uebung_223_AUDI
Standard IEC 61131-3 AUDI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, UDINT) mit Terminal-Ausgabe


initval_AUDI
INIT
INITO
initval_AUDI
UDINT#5
INIT_VAL
OUT
AUDI_FB_CTUD
CU
QU
CD
QD
R
CV
LD
PV
AUDI_TO_ALR
AUDI_TO_ALR
AUDI IN
ALR OUT
Q_NumericValue_PHYSA_LREAL
Q_NumericValue_...
QI
QO
OutputNumber_N3_N
stObj
STATUS
s16result
lrPhys
xOver
xUnder
Uebung_223b_ALR
Standard IEC 61131-3 AUDI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, UDINT) mit Terminal-Ausgabe (PHYSA LREAL)
AULI_ULINT_TO_ULI
REQ
AULI_ULINT_TO_U...
ULINT#5
OUT
AULI_OUT
AULI_FB_CTUD
AULI_TO_AUDI
AULI_TO_AUDI
AULI IN
AUDI OUT
Q_NumericValue_AUDI
u16ObjId
u32NewValue
u32OldValue
Uebung_224_AULI
Standard IEC 61131-3 AUDI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, ULINT) mit Terminal-Ausgabe
AULI_ULINT_TO_ULI
AULI_TO_ALR
AULI_TO_ALR
AULI IN
ALR OUT
Uebung_224b_ALR
Standard IEC 61131-3 AUDI_FB_CTUD (Adapter Version, Vorwärts-/Rückwärtszähler, ULINT) mit Terminal-Ausgabe (PHYSA LREAL)
Sollwert_N
NumericValue_PH...
TRUE
PARAMS
NumberVariable_Sollwert_N
rPhys
initval_AR
REAL#42.0
initval_AI
INT#0
Split
AR_SPLIT_2
IN
OUT1
OUT2
AR_ADD_2
IN1
IN2
Uebung_225_AX
Dreieck-Sollwertmarker (wie Uebung_225 in test_B, aber in reiner AR-Adapter-Spiegel: NumericValue_PHYSA, AR_ADD_2, AR_TO_AI, Q_ChildPosition_AI,
Sollwert_N
Marker_Dreieck
PositionMarkerF...
Container_PositionMarker
xScale
xUnder
Istwert_N
Q_NumericValue_...
Uebung_225b_AX
NumberVariable_Istwert_N
SplitBar
BargraphSplitFS...
Bargraph_Split_BargraphSp...
STATUSRight
s16resultRig...
STATUSLeft
s16resultLef...
xOverRight
xOverLeft
Uebung_226_AX
Split-Bargraph (wie Uebung_226 in test_B, aber mit AR-Adapter-Baustein BargraphSplitFS_AR statt REQ/rValue)

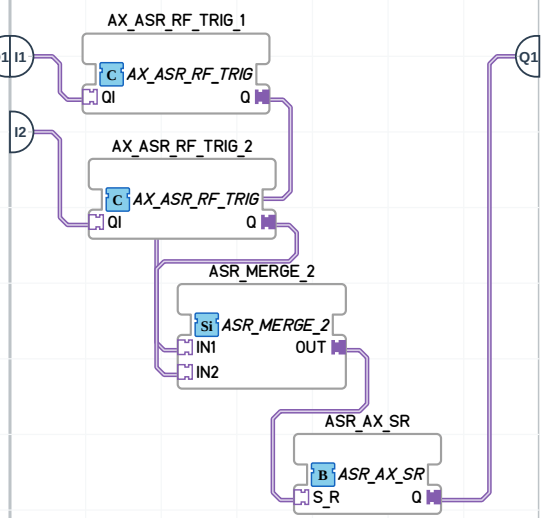






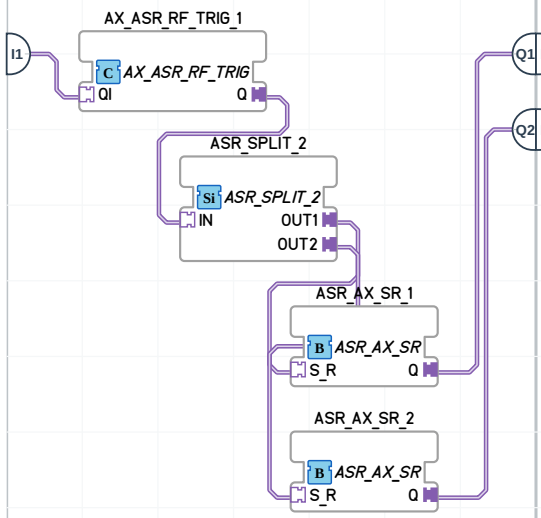
Uebung_229_AX

2 Taster (I1/I2) auf 1 SR-Latch (Last-Wins) via 2x AX_RF_TRIG + AX_SR, Ausgang Q1



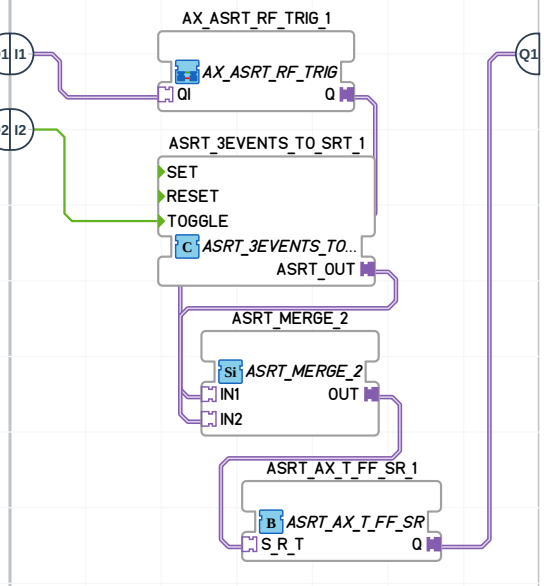
Uebung_230_AX

Wie Uebung_229_AX, aber mit ASR-Adapter-Umbau: 2x AX_ASR_RF_TRIG + ASR_MERGE_2 + ASR_AX_SR statt loser Event-Verkabelung



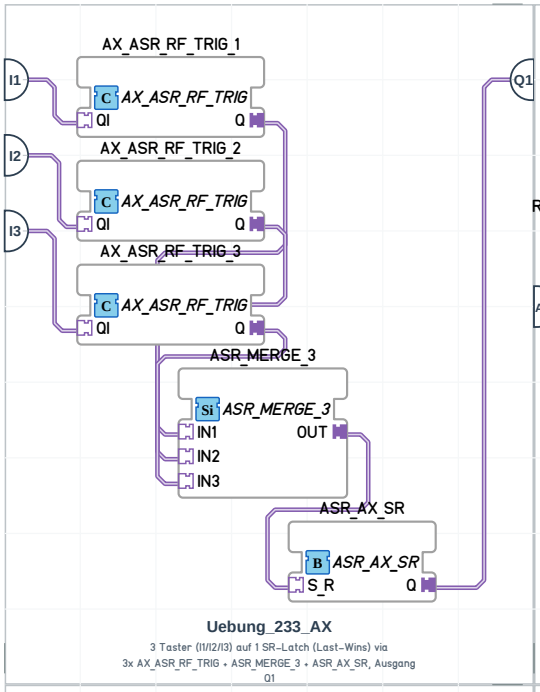
Uebung_231_AX

1 Taster via AX_ASR_RF_TRIG + ASR_SPLIT_2 auf 2 unabhaengige ASR_AX_SR-Latches (Q1+Q2) – Spiegelbild zu Uebung_230_AX (dort MERGE N->1, hier SPLIT 1->N)



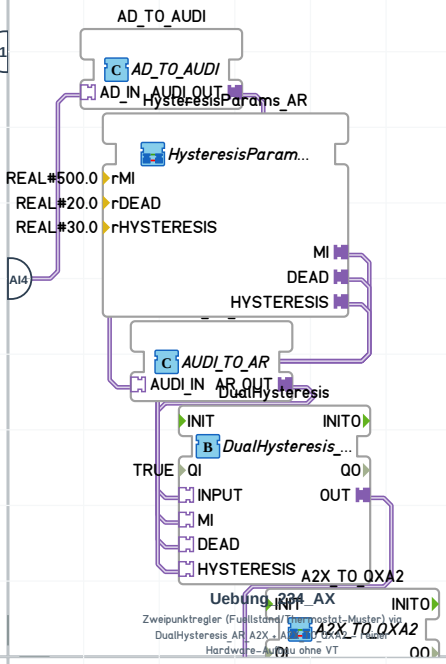
Uebung_232_AX

2 ASRT-Quellen (Taster Last-Wins + echter Toggle-Taster) auf 1 ASRT-Latch via ASRT_MERGE_2, Ausgang Q1



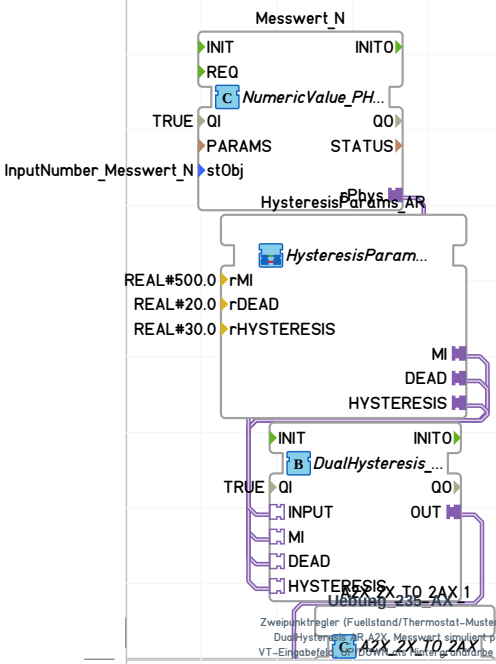
Uebung_233_AX

3 Taster (I1/I2/I3) auf 1 SR-Latch (Last-Wins) via 3x AX_ASR_RF_TRIG + ASR_MERGE_3 + ASR_AX_SR, Ausgang Q1



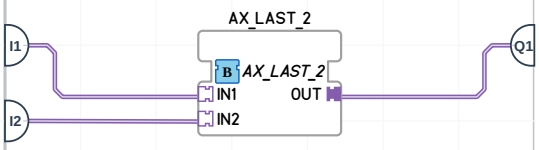
Uebung_234_AX

Zweipunktregler (Fuellstand/Thermostat-Muster) via DualHysteresis_AR_A2X + IA... Hardware-Ausgang ohne VT



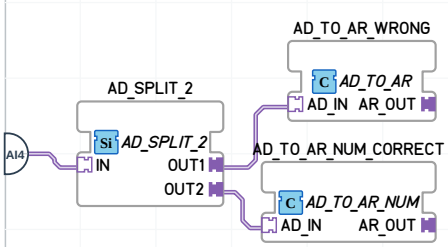
Uebung_235_AX

Zweipunktregler (Fuellstand/Thermostat-Muster) via DualHysteresis_AR_A2X, Messwert simuliert per VT-Eingabefeld, GUI/DUWA-Last-Hysterese-Simulation auf 2



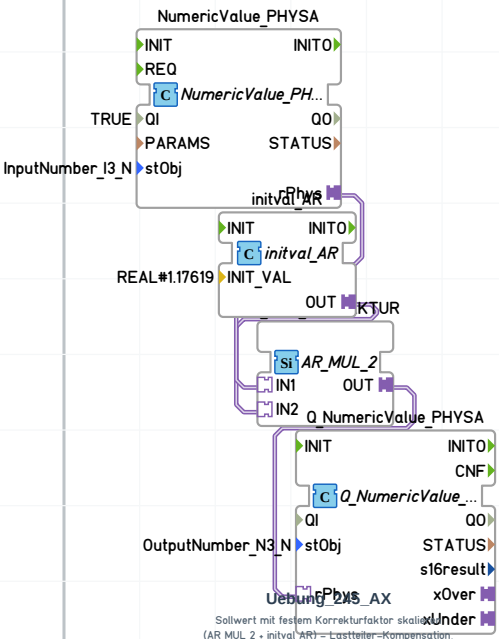
Uebung_236_AX

AX_LAST_2: derselbe Last-Wins-auf-jeder-Flanke-Effekt wie Uebung 229/230/233 AX, aber ohne Edge-Detektor-Latch



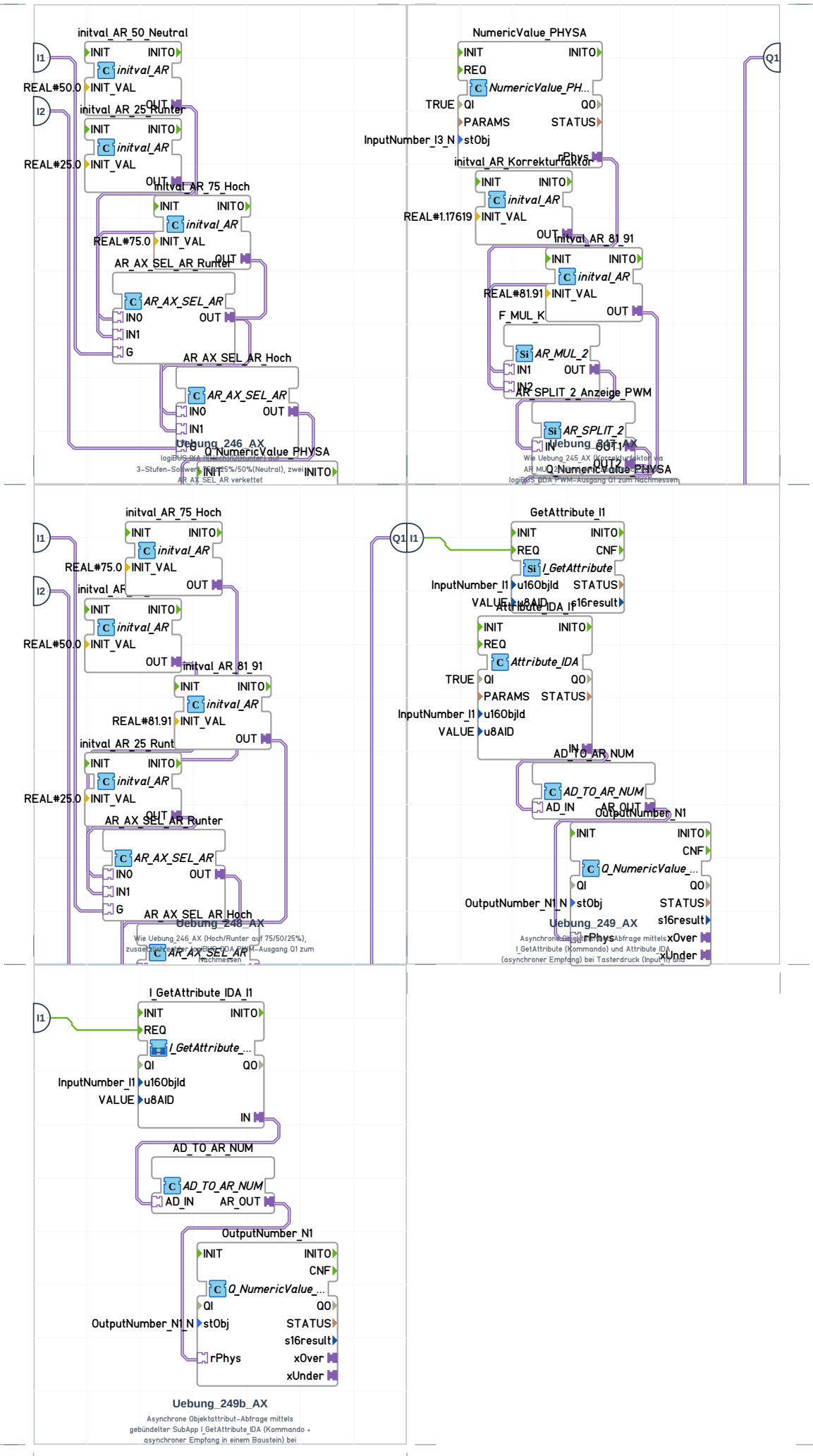
Uebung_237_AX

AD_TO_AR-Bitfalle: derselbe Analogwert einmal FALSCH ueber AD_TO_AR (Bit-Reinterpretation) und einmal RICHTIG ueber AD_TO_AR_NUM (echter numerischer Cast)



Uebung_245_AX

Sollwert mit festem Korrekturfaktor skaliert (AR_MUL_2 + initval_AR) = Lastteiler-Kompensation, I3 nach N3


initval_AR_50_Neutral
I1
INIT
INITO
initval_AR
REAL#50.0
INIT_VAL
OUT
I2
initval_AR_25_Runter
REAL#25.0
initval_AR_75_Hoch
REAL#75.0
AR_AX_SEL_AR_Runter
AR_AX_SEL_AR
IN0
IN1
G
AR_AX_SEL_AR_Hoch
Uebung_246_AX
logiBUS_IXA_Hilfsschnittstelle (Runter) auf 3-Stufen-Sollwert 25%/50%(Neutral), zwei AR_AX_SEL_AR verkettet
Q_NumericValue_PHYSA
NumericValue_PHYSA
REQ
NumericValue_PH...
TRUE
QI
QO
PARAMS
STATUS
InputNumber_I3_N
stObj
rPhys
initval_AR_Korrekturfaktor
REAL#1.17619
initval_AR_81_91
REAL#81.91
F_MUL_K
AR_MUL_2
IN2
AR_SPLIT_2_Anzeige_PWM
AR_SPLIT_2
OUT1
OUT2
Uebung_247_AX
Wie Uebung_245_AX (Korrekturfaktor via AR_MUL_2), zusätzlich logiBUS_PDA PWM-Ausgang Q1 zum Nachmessen
Q1
GetAttribute_I1
REQ
CNF
I_GetAttribute
InputNumber_I1
u16ObjId
VALUE
u8AID
STATUS
s16result
Attribute_IDA_I1
Attribute_IDA
AD_TO_AR_NUM
AD_IN
AR_OUT
OutputNumber_N1
Q_NumericValue_...
OutputNumber_N1_N
xOver
xUnder
Uebung_249_AX
Asynchrone Objektattribut-Abfrage mittels I_GetAttribute (Kommando) und Attribute_IDA (asynchroner Empfang) bei Tasterdruck (Input I1) und
I_GetAttribute_IDA_I1
I_GetAttribute_...
Uebung_249b_AX
Asynchrone Objektattribut-Abfrage mittels gebündelter SubApp I_GetAttribute_IDA (Kommando + asynchroner Empfang in einem Baustein) bei